

60. Electrical Characteristics

Unless otherwise specified, minimum and maximum values are guaranteed by either design simulation, characterization results or test in production.

Supported peripheral functions and pins differ from one product name to another.

Unless otherwise specified, the electrical characteristics of the MCU are defined under the following conditions:

- $VCC = VCC_DCDC = VBATT = 1.62$ to 3.63 V
- $VCC2 = 1.62$ to 3.63 V for Standard Product
- $VCC2 = 1.70$ to 2.00 V for SiP Product
- $AVCC0 = 1.62$ to 3.63 V
- $VCC_USB = 3.0$ to 3.6 V
- $VREFH0/VREFH = 1.62$ V to $AVCC0$
- $VSS = VSS_DCDC = AVSS0 = VREFL0/VREFL = VSS_USB = 0$ V
- VCC voltage is lower than 2.7 V : $LVOCR.LVO0E = 1$, otherwise $LVOCR.LVO0E = 0$
- $VCC2$ voltage is lower than 2.7 V : $LVOCR.LVO1E = 1$, otherwise $LVOCR.LVO1E = 0$
- $T_j = T_{opj}$
When not specified otherwise, typical values are measured at room temperature of 25 °C and $VCC = VCC_DCDC = VCC_USB = VBATT = AVCC0 = VREFH0 = VREFH = 3.3$ V

Figure 60.1 shows the timing conditions.

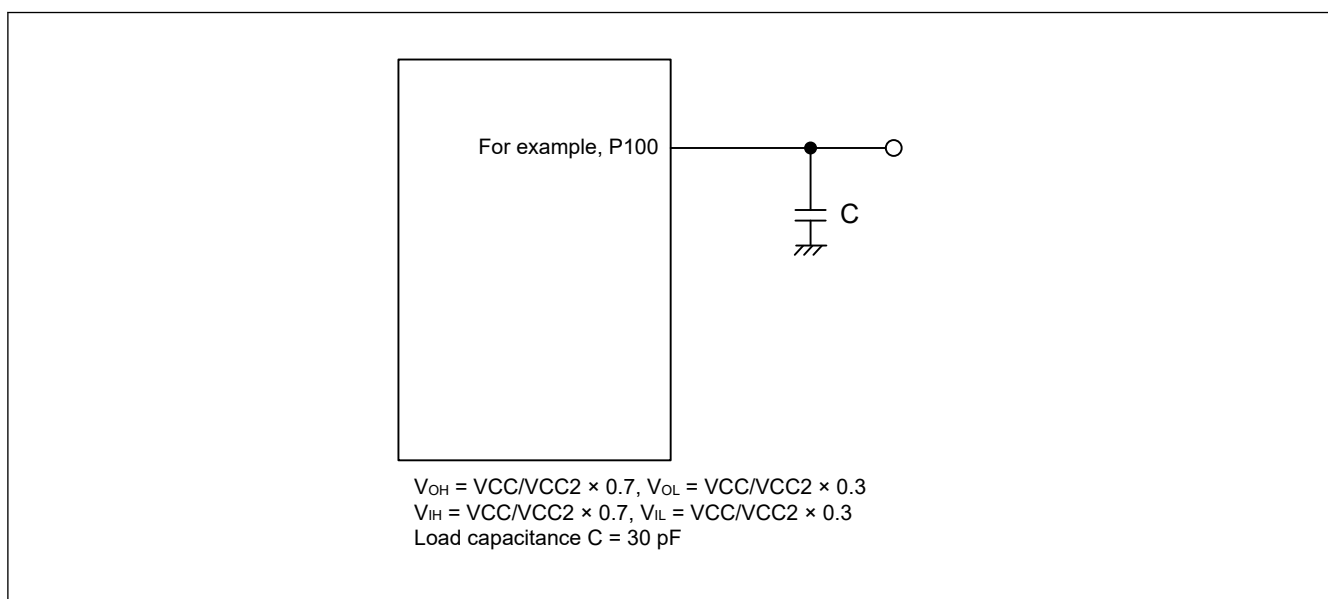


Figure 60.1 Input or output timing measurement conditions

The recommended measurement conditions for the timing specification of each peripheral provided are for the best peripheral operation. Make sure to adjust the driving abilities of each pin to meet your conditions.

60.1 Absolute Maximum Ratings

Table 60.1 Absolute maximum ratings (1 of 2)

Parameter	Symbol	Value	Unit
Power supply voltage	VCC, VCC_DCDC^{*2}	-0.3 to $+4.0$	V
	$VCC2$ Standard Product	-0.3 to $+4.0$	V
	$VCC2$ SiP Product	-0.3 to $+2.5$	V

Table 60.1 Absolute maximum ratings (2 of 2)

Parameter	Symbol	Value	Unit
External power supply voltage ^{*6}	VCL	-0.3 to +1.2	V
VBATT power supply voltage	VBATT	-0.3 to +4.0	V
Input voltage (except for 5 V-tolerant ports ^{*1})	V _{in}	-0.3 to VCC + 0.3, -0.3 to VCC2 + 0.3, -0.3 to VCC_USB + 0.3 or -0.3 to VBATT_R + 0.3	V
Input voltage (5 V-tolerant ports ^{*1})	V _{in}	-0.3 to + VCC + 4.0 (max. 5.8) or -0.3 to + VCC2 + 4.0 (max. 5.8)	V
Reference power supply voltage	VREFH/VREFH0	-0.3 to AVCC0 + 0.3	V
USBFS power supply voltage	VCC_USB	-0.3 to +4.0	V
Analog power supply voltage	AVCC0	-0.3 to +4.0	V
Analog input voltage	V _{AN}	-0.3 to AVCC0 + 0.3	V
Operating junction temperature ^{*3 *4 *5}	T _{opj}	0 to 95 or -40 to +105 or -40 to +125	°C
Storage temperature	T _{stg}	-55 to +125	°C

Note 1. Ports P204, P205, P303, P407 to P413, P511, P512, P514, P515 and P708 to P715 are 5 V tolerant.

Note 2. Connect VCC_DCDC to VCC.

Note 3. See [section 60.2.1. Tj/Ta Definition](#).

Note 4. Contact a Renesas Electronics sales office for information on derating operation when Tj = +95 °C to +125 °C. Derating is the systematic reduction of load for improved reliability.

Note 5. The lower and upper limit of operating junction temperature depend on the product. For details, see [section 1.3. Part Numbering](#).

Note 6. External VDD mode does not support in HLQFP package.

Caution: Permanent damage to the MCU might result if absolute maximum ratings are exceeded.

Table 60.2 Recommended operating conditions

Parameter	Symbol		Min	Typ	Max	Unit	
Power supply voltages	VCC, VCC_DCDC	Other than the following		1.62	—	3.63	V
		When ESC is used		2.70	—	3.60	V
		When ESWM is used		2.30	—	3.63	V
		When SDRAM is used		3.00	—	3.63	V
	VCC2	Standard Product	Other than the following	1.62	—	3.63	V
			When 32bit SDRAM is used	3.00	—	3.63	V
		SiP Product		1.70	—	2.00	V
	VCL	When external VDD is used*2*3	voltage range 1	0.92	—	0.99	V
			voltage range 2	0.87	—	0.99	V
		When DCDC is used (High-speed mode)	VSCR_1	—	0.95	—	V
			VSCR_2	—	0.925	—	V
		When DCDC is used (Software Standby mode)	SVSCR_1	—	0.95	—	V
			SVSCR_2	—	0.925	—	V
			SVSCR_3	—	0.825	—	V
			SVSCR_4	—	0.765	—	V
			SVSCR_5	—	0.715	—	V
	VSS, VSS_DCDC			—	0	—	V
USB power supply voltages	VCC_USB	When USB is not used		1.62	—	3.63	V
		When USB is used		3.00	—	3.60	V
	VSS_USB			—	0	—	V
VBATT power supply voltage	VBATT			1.62	—	3.63	V
Analog power supply voltages	AVCC0*1		Other than the following	1.62	—	3.63	V
			When Channel dedicated sample-and-hold circuit is used	2.70	—	3.63	V
	AVSS0			—	0	—	V

Note 1. When the A/D converter, the D/A converter and the High-Speed Analog Comparator are not in use, do not leave the AVCC0, VREFH/VREFH0, AVSS0, and VREFL/VREFL0 pins open. Connect the AVCC0 and VREFH/VREFH0 pins to VCC, and the AVSS0 and VREFL/VREFL0 pins to VSS, respectively.

Note 2. VCL voltage must never be higher than VCC voltage.

Note 3. External VDD mode does not support in HLQFP package.

60.2 DC Characteristics

60.2.1 Tj/Ta Definition

Table 60.3 DC characteristics

Parameter	Symbol	Typ	Max	Unit	Test conditions
Permissible operating junction temperature	T _j	—	125*1	°C	High-speed mode

Note: Make sure that $T_j = T_a + \theta_{ja} \times \text{total power consumption (W)}$, where total power consumption = $(V_{CC} - V_{OH}) \times \Sigma I_{OH} + V_{OL} \times \Sigma I_{OL} + (I_{CCmax} + I_{CC_DCDCmax}) \times V_{CC}$.

Note: Minimum Ambient Temperature (Ta) is -40°C or 0°C, depending on the product. For details, see [section 1.3. Part Numbering](#).

Note 1. The upper limit of operating junction temperature is 95°C, 105°C or 125°C, depending on the product. For details, see [section 1.3. Part Numbering](#).

60.2.2 I/O V_{IH} , V_{IL}

Table 60.4 I/O V_{IH} , V_{IL} except for Schmitt trigger input pins (1 of 2)

Parameter		VCC/VCC2/ AVCC0/ VCC_USB	Symbol	Min	Typ	Max	Unit
Peripheral function pin	EXTAL (external clock input), WAIT, SPI* ¹ (except RSPCK)	1.62 V or above	V_{IH}	$VCC \times 0.8$	—	—	V
			V_{IL}	—	—	$VCC \times 0.2$	
	SPI* ² (except RSPCK)	1.62 V or above	V_{IH}	$VCC2 \times 0.8$	—	—	
			V_{IL}	—	—	$VCC2 \times 0.2$	
	OSPI (except OM_0_RSTO1, OM_0_ECSINT ¹ , OM_1_RSTO1 and OM_1_ECSINT ¹)	2.70 V or above	V_{IH}	$VCC2 \times 0.8$	—	—	
			V_{IL}	—	—	$VCC2 \times 0.2$	
		1.62 to 2.00V	V_{IH}	$VCC2 \times 0.7$	—	$VCC2 + 0.3$	
			V_{IL}	$VSS - 0.3$	—	$VCC2 \times 0.3$	
	SD* ³	2.70 V or above	V_{IH}	$VCC \times 0.625$	—	$VCC + 0.3$	
			V_{IL}	$VSS - 0.3$	—	$VCC \times 0.25$	
		1.70 to 1.95 V	V_{IH}	1.27	—	2	
			V_{IL}	$VSS - 0.3$	—	0.58	
	SD* ⁴	2.70 V or above	V_{IH}	$VCC2 \times 0.625$	—	$VCC2 + 0.3$	
			V_{IL}	$VSS - 0.3$	—	$VCC2 \times 0.25$	
		1.70 to 1.95 V	V_{IH}	1.27	—	2	
			V_{IL}	$VSS - 0.3$	—	0.58	
	MMC* ⁵	2.70 V or above	V_{IH}	$VCC \times 0.625$	—	$VCC + 0.3$	
			V_{IL}	$VSS - 0.3$	—	$VCC \times 0.25$	
		1.70 to 1.95 V	V_{IH}	$VCC \times 0.65$	—	$VCC + 0.3$	
			V_{IL}	$VSS - 0.3$	—	$VCC \times 0.35$	
	MMC* ⁶	2.70 V or above	V_{IH}	$VCC2 \times 0.625$	—	$VCC2 + 0.3$	
			V_{IL}	$VSS - 0.3$	—	$VCC2 \times 0.25$	
		1.70 to 1.95 V	V_{IH}	$VCC2 \times 0.65$	—	$VCC2 + 0.3$	
			V_{IL}	$VSS - 0.3$	—	$VCC2 \times 0.35$	
	D00 to D19, TMS, TDI, SWDIO	1.62 V or above	V_{IH}	$VCC \times 0.7$	—	—	
			V_{IL}	—	—	$VCC \times 0.3$	
	D20 to D31	1.62 V or above	V_{IH}	$VCC2 \times 0.7$	—	—	
			V_{IL}	—	—	$VCC2 \times 0.3$	
	DQ00 to DQ19	3.00 V or above	V_{IH}	$VCC \times 0.7$	—	—	
			V_{IL}	—	—	$VCC \times 0.3$	
	DQ20 to DQ31	3.00 V or above	V_{IH}	$VCC2 \times 0.7$	—	—	
			V_{IL}	—	—	$VCC2 \times 0.3$	

Table 60.4 I/O V_{IH} , V_{IL} except for Schmitt trigger input pins (2 of 2)

Parameter		VCC/VCC2/ AVCC0/ VCC_USB	Symbol	Min	Typ	Max	Unit
Peripheral function pin	ESWM* ⁹ , ESC* ¹³	2.30 V to 3.60 V	V_{IH}	$VCC \times 0.7$	—	—	V
			V_{IL}	—	—	$VCC \times 0.3$	
			V_{IH}	$VCC2 \times 0.7$	—	—	
			V_{IL}	—	—	$VCC2 \times 0.3$	
	ESWM (MII)* ¹⁰ , ESWM (RMII)* ¹¹ , ESC* ¹⁴	2.70 V to 3.60 V	V_{IH}	2.3	—	—	
			V_{IL}	—	—	$VCC \times 0.2$	
	ESWM (GMII)* ¹⁰ , ESWM (RGMII)* ¹²	3.00 to 3.60 V	V_{IH}	2	—	—	
			V_{IL}	—	—	0.8	
		2.30 to 2.70 V	V_{IH}	1.7	—	—	
			V_{IL}	—	—	0.7	
	IIC (SMBus)* ⁷	2.70 V or above	V_{IH}	2.1	—	$VCC + 3.6$ (max 5.8)	
			V_{IL}	—	—	0.8	
	IIC (SMBus)* ⁸	2.70 V or above	V_{IH}	2.1	—	$VCC2 + 3.6$ (max 5.8)	
			V_{IL}	—	—	0.8	
	I3C (SMBus)	2.70 V or above	V_{IH}	2.1	—	$VCC + 0.3$	
			V_{IL}	—	—	0.8	
	RTCIC0, RTCIC1, RTCIC2, when VCC power supply is selected	1.62 V or above	V_{IH}	0.9	—	3.9	
			V_{IL}	—	—	0.3	
	RTCIC0, RTCIC1, RTCIC2 when VBATT power supply is selected		V_{IH}	0.9	—	3.9	
			V_{IL}	—	—	0.3	
	EXCIN when VCC power supply is selected	1.62 V or above	V_{IH}	0.9	—	VCC	
			V_{IL}	—	—	0.3	
	EXCIN when VBATT power supply is selected		V_{IH}	0.9	—	VBATT	
			V_{IL}	—	—	0.3	

Note 1. SPI0_B, SPI0_C and SPI1_B

Note 2. SPI0_A, SPI1_A

Note 3. SD_A ch0, SD_B ch0, SD_C ch0 and SD_B ch1

Note 4. SD_A ch1

Note 5. MMC_A ch0, MMC_B ch0, MM_C ch0 and MMC_B ch1

Note 6. MMC_A ch1

Note 7. IIC0_A, IIC0_B, IIC1_A, IIC2_A and IIC2_B

Note 8. IIC1_B

Note 9. GPTP_CAPTUREn, ETn_LINKSTA, ETn_MDIO and ETn_INT (n = 0, 1)

Note 10. ETn_RX_CLK, ETn_RX_DV, ETn_RXD7 to ETn_RXD0, ETn_RX_ER and ETn_TX_CLK (n = 0, 1)

Note 11. RMIIIn_REF50CK, RMIIIn_CRS_DV, RMIIIn_RXD1 to RMIIIn_RXD0 and RMIIIn_RX_ER (n = 0, 1)

Note 12. RGMIIIn_RXC, RGMIIIn_RX_CTL and RGMIIIn_RXD3 to RGMIIIn_RXD0 (n = 0, 1)

Note 13. CATn_LINKSTA, CAT0_MDIO, CATI2CDATA and CATLATCHn (n = 0, 1)

Note 14. CATn_RX_CLK, CATn_RX_DV, CATn_ERXD3 to CATn_ERXD0, CATn_RX_ER and CATn_TX_CLK (n = 0, 1)

Table 60.5 I/O V_{IH} , V_{IL} of Schmitt trigger input pins (1 of 2)

Parameter		VCC/VCC2/ AVCC0/ VCC_USB	Symbol	Min	Typ	Max	Unit
Peripheral function pin	IIC (except for SMBus)*7	1.62 V or above	V_{IH}	$VCC \times 0.7$	—	$VCC + 3.6$ (max 5.8)	V
			V_{IL}	—	—	$VCC \times 0.3$	
			ΔV_T	$VCC \times 0.05$	—	—	
	IIC (except for SMBus)*8	1.62 V or above	V_{IH}	$VCC2 \times 0.7$	—	$VCC2 + 3.6$ (max 5.8)	
			V_{IL}	—	—	$VCC2 \times 0.3$	
			ΔV_T	$VCC2 \times 0.05$	—	—	
	I3C (except for SMBus)	1.65 V or above	V_{IH}	$VCC \times 0.7$	—	$VCC + 0.3$	
			V_{IL}	—	—	$VCC \times 0.3$	
			ΔV_T	$VCC \times 0.1$	—	—	
	5 V-tolerant ports*1*6	1.62 V or above	V_{IH}	$VCC \times 0.8$	—	$VCC + 3.6$ (max 5.8)	
			V_{IL}	—	—	$VCC \times 0.2$	
			ΔV_T	$VCC \times 0.05$	—	—	
	5 V-tolerant ports*2*6	1.62 V or above	V_{IH}	$VCC2 \times 0.8$	—	$VCC2 + 3.6$ (max 5.8)	
			V_{IL}	—	—	$VCC2 \times 0.2$	
			ΔV_T	$VCC2 \times 0.05$	—	—	
	Other VCC input pins*3	1.62 V or above	V_{IH}	$VCC \times 0.8$	—	—	
			V_{IL}	—	—	$VCC \times 0.2$	
			ΔV_T	$VCC \times 0.05$	—	—	
	Other VCC2 input pins*3	1.62 V or above	V_{IH}	$VCC2 \times 0.8$	—	—	
			V_{IL}	—	—	$VCC2 \times 0.2$	
			ΔV_T	$VCC2 \times 0.05$	—	—	
	Other AVCC0 input pins*3	1.62 V or above	V_{IH}	$AVCC0 \times 0.8$	—	—	
			V_{IL}	—	—	$AVCC0 \times 0.2$	
			ΔV_T	$AVCC0 \times 0.05$	—	—	
	Other VCC_USB input pins*3	1.62 V or above	V_{IH}	$VCC_USB \times 0.8$	—	—	
			V_{IL}	—	—	$VCC_USB \times 0.2$	
			ΔV_T	$VCC_USB \times 0.05$	—	—	
	Other VBATT_R input pins when VCC power supply is selected*3	1.62 V or above	V_{IH}	$VCC \times 0.8$	—	VCC	
			V_{IL}	—	—	$VCC \times 0.2$	
			ΔV_T	$VCC \times 0.05$	—	—	

Table 60.5 I/O V_{IH} , V_{IL} of Schmitt trigger input pins (2 of 2)

Parameter		VCC/VCC2/ AVCC0/ VCC_USB	Symbol	Min	Typ	Max	Unit
Ports	5 V-tolerant port*4*6	1.62 V or above	V_{IH}	$VCC \times 0.8$	—	$VCC + 3.6$ (max 5.8)	V
			V_{IL}	—	—	$VCC \times 0.2$	
	5 V-tolerant port*2*6	1.62 V or above	V_{IH}	$VCC2 \times 0.8$	—	$VCC2 + 3.6$ (max 5.8)	
			V_{IL}	—	—	$VCC2 \times 0.2$	
	Other VCC input pins*5	1.62 V or above	V_{IH}	$VCC \times 0.8$	—	—	
			V_{IL}	—	—	$VCC \times 0.2$	
	Other VCC2 input pins*5	1.62 V or above	V_{IH}	$VCC2 \times 0.8$	—	—	
			V_{IL}	—	—	$VCC2 \times 0.2$	
	Other AVCC0 input pins*5	1.62 V or above	V_{IH}	$AVCC0 \times 0.8$	—	—	
			V_{IL}	—	—	$AVCC0 \times 0.2$	
	Other VCC_USB input pins*5	1.62 V or above	V_{IH}	$VCC_USB \times 0.8$	—	—	
			V_{IL}	—	—	$VCC_USB \times 0.2$	
	Other VBATT_R input pins when VCC power supply is selected*5	1.62 V or above	V_{IH}	$VCC \times 0.8$	—	VCC	
			V_{IL}	—	—	$VCC \times 0.2$	

Note 1. RES and peripheral function pins associated with P303, P407 to P413, P511, P512, P514, P515, P708 to P715 (total 21 pins).

Note 2. P204 and P205 (total 2 pins)

Note 3. All input pins except for the peripheral function pins already described in the table. There is an item for each power supply voltage for each port. Refer to the IO chapter for the power supply of the port.

Note 4. P303, P407 to P413, P511, P512, P514, P515, P708 to P715 (total 20 pins).

Note 5. All input pins except for the ports already described in the table. There is an item for each power supply voltage for each port. Refer to the IO chapter for the power supply of the port.

Note 6. When VCC or VCC2 is less than 1.62 V, the input voltage of 5 V-tolerant ports should be less than 3.6 V, otherwise breakdown may occur because 5 V-tolerant ports are electrically controlled so as not to violate the break down voltage.

60.2.3 I/O I_{OH} , I_{OL} Table 60.6 I/O I_{OH} , I_{OL} (1 of 5)

Parameter			VCC/VCC2/ AVCC0/ VCC_USB	Symbol	Min	Typ	Max	Unit
Permissible output current (average value per pin)	Ports P000 to P015, P201	—	—	I_{OH}	—	—	−2.0	mA
				I_{OL}	—	—	2.0	mA
	Ports P204, P205, P303, P407 to P413, P511, P512, P514, P515, P708 to P715, PA15 (total 23 pins)	Low drive*1	—	I_{OH}	—	—	−2.0	mA
				I_{OL}	—	—	2.0	mA
		Middle drive*2	—	I_{OH}	—	—	−4.0	mA
				I_{OL}	—	—	4.0	mA
		High drive*3	—	I_{OH}	—	—	−16	mA
				I_{OL}	—	—	20.0	mA
		High-speed high drive*4	—	I_{OH}	—	—	−20	mA
				I_{OL}	—	—	20.0	mA
	Other output pins*5	Low drive*1	—	I_{OH}	—	—	−2.0	mA
				I_{OL}	—	—	2.0	mA
		Middle drive*2	—	I_{OH}	—	—	−4.0	mA
				I_{OL}	—	—	4.0	mA
		High drive*3	—	I_{OH}	—	—	−16	mA
				I_{OL}	—	—	16.0	mA
		High-speed high drive*4	—	I_{OH}	—	—	−20	mA
				I_{OL}	—	—	20.0	mA
Permissible output current (max value per pin)	Ports P000 to P015, P201	—	—	I_{OH}	—	—	−4.0	mA
				I_{OL}	—	—	4.0	mA
	Ports P204, P205, P303, P407 to P413, P511, P512, P514, P515, P708 to P715, PA15 (total 23 pins)	Low drive*1	—	I_{OH}	—	—	−4.0	mA
				I_{OL}	—	—	4.0	mA
		Middle drive*2	—	I_{OH}	—	—	−8.0	mA
				I_{OL}	—	—	8.0	mA
		High drive*3	—	I_{OH}	—	—	−32	mA
				I_{OL}	—	—	40.0	mA
		High-speed high drive*4	—	I_{OH}	—	—	−40	mA
				I_{OL}	—	—	40.0	mA
	Other output pins*5	Low drive*1	—	I_{OH}	—	—	−4.0	mA
				I_{OL}	—	—	4.0	mA
		Middle drive*2	—	I_{OH}	—	—	−8.0	mA
				I_{OL}	—	—	8.0	mA
		High drive*3	—	I_{OH}	—	—	−32	mA
				I_{OL}	—	—	32.0	mA
		High-speed high drive*4	—	I_{OH}	—	—	−40	mA
				I_{OL}	—	—	40.0	mA

Table 60.6 I/O I_{OH} , I_{OL} (2 of 5)

Parameter				VCC/VCC2/ AVCC0/ VCC_USB	Symbol	Min	Typ	Max	Unit
Permissible output current (maxvalue of total of all pins)	Maximum of all output pins	VCC I/O	Ports P411 to P415, P511 to P515, P708 to P715, P805 to P807 (total 21 pins)	1.62 V or above	$\Sigma I_{OH(max)}$	—	—	−40	mA
			Ports P212, P213, P400 to P410, PB08 to PB14 (total 20 pins)	1.62 V or above		—	—	−40	
			Ports P700 to P707, PB00 to PB04 (total 13 pins)	1.62 V or above		—	—	−40	
			Ports PB05 to PB07, PD06, PD07 (total 5 pins)	1.62 V or above		—	—	−40	
			Ports P207, PD01 to PD05 (total 6 pins)	1.62 V or above		—	—	−40	
			Ports P904, P910 to P913 (total 5 pins)	1.62 V or above		—	—	−40	
			Ports P206, P304 to P312, P902, P903, P905 to P909 (total 17 pins)	1.62 V or above		—	—	−40	
			Ports P108 to P115, P201, P208 to P211, P300 to P303, P609, P914, P915, PA11 (total 21 pins)	1.62 V or above		—	—	−40	
			Ports P610 to P615, P813, PA04 to PA10, PA12 to PA15 (total 18 pins)	1.62 V or above		—	—	−40	
			Ports P503 to P510, P608, PA00 to PA03, PC11 to PC15, PD00 (total 19 pins)	1.62 V or above		—	—	−40	

Table 60.6 I/O I_{OH} , I_{OL} (3 of 5)

Parameter				VCC/VCC2/ AVCC0/ VCC_USB	Symbol	Min	Typ	Max	Unit
Permissible output current (maxvalue of total of all pins)	Maximum of all output pins	VCC2 I/O	Ports PC00 to PC10 (total 11 pins)	1.62 V or above	$\Sigma I_{OH(max)}$	—	—	-40	mA
			Ports P204, P205, P600 to P607 (total 10 pins)	1.62 V or above		—	—	-40	
			Ports P202, P203, P313 to P315, P900, P901 (total 7 pins)	1.62 V or above		—	—	-40	
			Ports P100 to P107, P800, P801 (total 10 pins)	1.62 V or above		—	—	-40	
			Ports P500 to P502, P802 to P804, P808 to P812 (total 11 pins)	1.62 V or above		—	—	-40	
		AVCC0 I/O		1.62 V or above		—	—	-33	
		VCC_USB I/O		1.62 V or above		—	—	-33	

Table 60.6 I/O I_{OH} , I_{OL} (4 of 5)

Parameter				VCC/VCC2/ AVCC0/ VCC_USB	Symbol	Min	Typ	Max	Unit
Permissible output current (maxvalue of total of all pins)	Maximum of all output pins	VCC and VCC2 I/O	Ports P411 to P415, P511 to P515, P708 to P715, P805 to P807 (total 21 pins)	1.62 V or above	$\Sigma I_{OL(max)}$	—	—	40	mA
			Ports P212, P213, P400 to P410, PB08 to PB14 (total 20 pins)	1.62 V or above		—	—	40	
			Ports P700 to P707, PB00 to PB04 (total 13 pins)	1.62 V or above		—	—	40	
			Ports PB05 to PB07, PD06, PD07 (total 5 pins)	1.62 V or above		—	—	40	
			Ports P207, PD01 to PD05 (total 6 pins)	1.62 V or above		—	—	40	
			Ports P904, P910 to P913 (total 5 pins)	1.62 V or above		—	—	40	
			Ports P206, P304 to P312, P902, P903, P905 to P909 (total 17 pins)	1.62 V or above		—	—	40	
			Ports P108 to P115, P201, P208 to P211, P300 to P303, P609, P914, P915, PA11 (total 21 pins)	1.62 V or above		—	—	40	
			Ports P610 to P615, P813, PA04 to PA10, PA12 to PA15 (total 18 pins)	1.62 V or above		—	—	40	
			Ports P503 to P510, P608, PA00 to PA03, PC11 to PC15, PD00 (total 19 pins)	1.62 V or above		—	—	40	
			Ports PC00 to PC10 (total 11 pins)	1.62 V or above		—	—	40	
			Ports P204, P205, P600 to P607 (total 10 pins)	1.62 V or above		—	—	40	

Table 60.6 I/O I_{OH} , I_{OL} (5 of 5)

Parameter				VCC/VCC2/ AVCC0/ VCC_USB	Symbol	Min	Typ	Max	Unit
Permissible output current (maxvalue of total of all pins)	Maximum of all output pins	VCC and VCC2 I/O	Ports P202, P203, P313 to P315, P900, P901 (total 7 pins)	1.62 V or above	$\Sigma I_{OL(max)}$	—	—	40	mA
			Ports P100 to P107, P800, P801 (total 10 pins)	1.62 V or above		—	—	40	
			Ports P500 to P502, P802 to P804, P808 to P812 (total 11 pins)	1.62 V or above		—	—	40	
		AVCC0 I/O		1.62 V or above		—	—	33	
		VCC_USB I/O		1.62 V or above		—	—	33	

Note 1. This is the value when low driving ability is selected in the Port Drive Capability bit in the PmnPFS register. The selected driving ability except for P400 and P401 is retained in Deep Software Standby mode.

Note 2. This is the value when middle driving ability is selected in the Port Drive Capability bit in the PmnPFS register. The selected driving ability except for P400 and P401 is retained in Deep Software Standby mode.

Note 3. This is the value when high driving ability is selected in the Port Drive Capability bit in the PmnPFS register. The selected driving ability except for P400 and P401 is retained in Deep Software Standby mode.

Note 4. This is the value when high-speed high driving ability is selected in the Port Drive Capability in the PmnPFS register. The selected driving ability is retained in Deep Software Standby mode.

Note 5. Except for P200, P214 and P215, which is an input port.

Caution: To protect the reliability of the MCU, the output current values should not exceed the values in this table. The average output current indicates the average value of current measured during 100 μ s.

60.2.4 I/O V_{OH} , V_{OL} , and Other CharacteristicsTable 60.7 I/O V_{OH} , V_{OL} , and other characteristics (1 of 3)

Parameter		VCC/ VCC2/ AVCC0/ VCC_US B	Symbol	Min	Typ	Max	Unit	Test conditions
Output voltage	IIC	2.70 V or above	V_{OL}	—	—	0.4	V	$I_{OL} = 3.0 \text{ mA}$
			V_{OL}	—	—	0.6		$I_{OL} = 6.0 \text{ mA}$
		1.62 V to 1.95 V	V_{OL}	—	—	$VCC \times 0.2$		$I_{OL} = 2.0 \text{ mA}$
			V_{OL}	—	—	0.4		$I_{OL} = 3.0 \text{ mA}$
			V_{OL}	—	—	0.6^{*4}		$I_{OL} = 6.0 \text{ mA}$
			V_{OL}	—	—	$VCC2 \times 0.2$		$I_{OL} = 2.0 \text{ mA}$
	IIC*1	2.70 V or above	V_{OL}	—	—	0.4		$I_{OL} = 15.0 \text{ mA}$ (ICFER.FMPE = 1)
			V_{OL}	—	0.4	—		$I_{OL} = 20.0 \text{ mA}$ (ICFER.FMPE = 1)
		1.62 V to 1.95 V	V_{OL}	—	—	0.4		$I_{OL} = 15.0 \text{ mA}$ (ICFER.FMPE = 1)
			V_{OL}	—	0.4	—		$I_{OL} = 20.0 \text{ mA}$ (ICFER.FMPE = 1)
	I3C	2.70 V or above	V_{OL}	—	—	0.4		$I_{OL} = 3.0 \text{ mA}$ (PRTS.PRTMD = 1, BFCTL.FMPE = 0, BFCTL.HSME = 0)
			V_{OL}	—	—	0.6		$I_{OL} = 6.0 \text{ mA}$ (PRTS.PRTMD = 1, BFCTL.FMPE = 0, BFCTL.HSME = 0)
			V_{OL}	—	—	0.4		$I_{OL} = 15.0 \text{ mA}$ (PRTS.PRTMD = 1, BFCTL.FMPE = 1, BFCTL.HSME = 0)
			V_{OL}	—	0.4	—		$I_{OL} = 20.0 \text{ mA}$ (PRTS.PRTMD = 1, BFCTL.FMPE = 1, BFCTL.HSME = 0)
		3.00 V or above	V_{OL}	—	—	0.4		$I_{OL} = 3.0 \text{ mA}$ (PRTS.PRTMD = 1, BFCTL.FMPE = 0, BFCTL.HSME = 1)
			V_{OH}	$VCC - 0.27$	—	—		$I_{OH} = 3.0 \text{ mA}$ (PRTS.PRTMD = 0, BFCTL.FMPE = 0, BFCTL.HSME = 0)
			V_{OL}	—	—	0.27		$I_{OL} = 3.0 \text{ mA}$ (PRTS.PRTMD = 0, BFCTL.FMPE = 0, BFCTL.HSME = 0)
		1.65 V to 1.95 V	V_{OL}	—	—	$VCC \times 0.2$		$I_{OL} = 2.0 \text{ mA}$ (PRTS.PRTMD = 1, BFCTL.FMPE = 0, BFCTL.HSME = 0)
			V_{OL}	—	—	0.4		$I_{OL} = 3.0 \text{ mA}$ (PRTS.PRTMD = 1, BFCTL.FMPE = 0, BFCTL.HSME = 0)
			V_{OL}	—	—	0.6		$I_{OL} = 6.0 \text{ mA}$ (PRTS.PRTMD = 1, BFCTL.FMPE = 0, BFCTL.HSME = 0)
			V_{OL}	—	—	$VCC \times 0.2$		$I_{OL} = 2.0 \text{ mA}$ (PRTS.PRTMD = 1, BFCTL.FMPE = 1, BFCTL.HSME = 0)
			V_{OL}	—	—	0.4		$I_{OL} = 15.0 \text{ mA}$ (PRTS.PRTMD = 1, BFCTL.FMPE = 1, BFCTL.HSME = 0)
			V_{OL}	—	0.4	—		$I_{OL} = 20.0 \text{ mA}$ (PRTS.PRTMD = 1, BFCTL.FMPE = 1, BFCTL.HSME = 0)
			V_{OL}	—	—	$VCC \times 0.2$		$I_{OL} = 3.0 \text{ mA}$ (PRTS.PRTMD = 1, BFCTL.FMPE = 0, BFCTL.HSME = 1)
			V_{OH}	$VCC - 0.27$	—	—		$I_{OH} = 3.0 \text{ mA}$ (PRTS.PRTMD = 0, BFCTL.FMPE = 0, BFCTL.HSME = 0)
			V_{OL}	—	—	0.27		$I_{OL} = 3.0 \text{ mA}$ (PRTS.PRTMD = 0, BFCTL.FMPE = 0, BFCTL.HSME = 0)

Table 60.7 I/O V_{OH} , V_{OL} , and other characteristics (2 of 3)

Parameter		VCC/ VCC2/ AVCC0/ VCC_USB B	Symbol	Min	Typ	Max	Unit	Test conditions
Output voltage	ESWM, ESC	2.70 V to 3.60 V	V_{OH}	$VCC - 0.5$	—	—	V	$I_{OH} = -1.0 \text{ mA}$
			V_{OL}	—	—	0.4		$I_{OL} = 1.0 \text{ mA}$
			V_{OH}	$VCC2 - 0.5$	—	—		$I_{OH} = -1.0 \text{ mA}$
		2.30 V to 2.70 V	V_{OH}	2	—	—		$I_{OH} = -1.0 \text{ mA}$
			V_{OL}	—	—	0.4		$I_{OL} = 1.0 \text{ mA}$
			V_{OL}	—	—	0.4		$I_{OL} = 1.0 \text{ mA}$
	SD	2.70 V or above	V_{OH}	$VCC \times 0.75$	—	—		$I_{OH} = -2.0 \text{ mA}$
			V_{OL}	—	—	$VCC \times 0.125$		$I_{OL} = 3.0 \text{ mA}$
			V_{OH}	$VCC2 \times 0.75$	—	—		$I_{OH} = -2.0 \text{ mA}$
			V_{OL}	—	—	$VCC2 \times 0.125$		$I_{OL} = 3.0 \text{ mA}$
		1.70 to 1.95 V	V_{OH}	1.4	—	—		$I_{OH} = -2.0 \text{ mA}$
			V_{OL}	—	—	0.45		$I_{OL} = 2.0 \text{ mA}$
			V_{OH}	$VCC \times 0.75$	—	—		$I_{OH} = -0.1 \text{ mA (VCC = 2.7 V)}$
			V_{OL}	—	—	$VCC \times 0.125$		$I_{OL} = 0.1 \text{ mA (VCC = 2.7 V)}$
	MMC	2.70 V or above	V_{OH}	$VCC \times 0.75$	—	—		$I_{OH} = -0.1 \text{ mA (VCC2 = 2.7 V)}$
			V_{OL}	—	—	$VCC \times 0.125$		$I_{OL} = 0.1 \text{ mA (VCC2 = 2.7 V)}$
			V_{OH}	$VCC2 \times 0.75$	—	—		$I_{OH} = -0.1 \text{ mA (VCC2 = 2.7 V)}$
			V_{OL}	—	—	$VCC2 \times 0.125$		$I_{OL} = 0.1 \text{ mA (VCC2 = 2.7 V)}$
		1.70 to 1.95 V	V_{OH}	$VCC - 0.45$	—	—		$I_{OH} = -2.0 \text{ mA}$
			V_{OL}	—	—	0.45		$I_{OL} = 2.0 \text{ mA}$
			V_{OH}	$VCC2 - 0.45$	—	—		$I_{OH} = -2.0 \text{ mA}$
			V_{OL}	—	—	0.45		$I_{OL} = 2.0 \text{ mA}$
	Ports P204, P205, P303, P407 to P413, P511, P512, P514, P515, P708 to P715, PA15 (total of 23 pins)*2	—	V_{OH}	$VCC - 1.0$	—	—		$I_{OH} = -16 \text{ mA (VCC = 3.3 V)}$
			V_{OL}	—	—	1		$I_{OL} = 20 \text{ mA (VCC = 3.3 V)}$
			V_{OH}	$VCC2 - 1.0$	—	—		$I_{OH} = -16 \text{ mA (VCC2 = 3.3 V)}$
			V_{OL}	—	—	1		$I_{OL} = 20 \text{ mA (VCC2 = 3.3 V)}$
		1.62 V or above	V_{OH}	$VCC - 0.5$	—	—	V	$I_{OH} = -1.0 \text{ mA}$
			V_{OL}	—	—	0.5		$I_{OL} = 1.0 \text{ mA}$
			V_{OH}	$VCC2 - 0.5$	—	—		$I_{OH} = -1.0 \text{ mA}$
			V_{OL}	—	—	0.5		$I_{OL} = 1.0 \text{ mA}$
	Other output pins	1.62 V or above	V_{OH}	$AVCC0 - 0.5$	—	—		$I_{OH} = -1.0 \text{ mA}$
			V_{OL}	—	—	0.5		$I_{OL} = 1.0 \text{ mA}$
			V_{OH}	$VCC_USB - 0.5$	—	—		$I_{OH} = -1.0 \text{ mA}$
			V_{OL}	—	—	0.5		$I_{OL} = 1.0 \text{ mA}$

Table 60.7 I/O V_{OH} , V_{OL} , and other characteristics (3 of 3)

Parameter		VCC/ VCC2/ AVCC0/ VCC_USB B	Symbol	Min	Typ	Max	Unit	Test conditions
Input leakage current	RES	1.62 V or above	$ I_{in} $	—	—	5	μA	$V_{in} = 0 V$ $V_{in} = 5.5 V$
	Port P200, P214, P215	1.62 V or above		—	—	1		$V_{in} = 0 V$ $V_{in} = VCC$
Three-state leakage current (off state)	5 V-tolerant ports	1.62 V or above	$ I_{TSI} $	—	—	5	μA	$V_{in} = 0 V$ $V_{in} = 5.5 V$
	Other ports (except for port P200, P214, P215)	1.62 V or above		—	—	1		$V_{in} = 0 V$ $V_{in} = VCC, VCC2, AVCC0, VCC_USB$
Input pull-up MOS current	Ports P0 to PD	2.70 V or above	I_p	-300	—	-10	μA	$VCC, VCC2, AVCC0, VCC_USB = 2.7$ to 3.63 V $V_{in} = 0 V$
		1.62 V or above		-300	—	-5		$VCC, VCC2, AVCC0, VCC_USB = 1.62$ to 3.63 V $V_{in} = 0 V$
Pull-up current serving as the SCL current source	I3C ^{*3}	3.00 to 3.63 V	I_{cs}	3	—	12	mA	$VCC = 3.0$ to 3.63 V $V_{in} = 0.3 \times VCC$ to $0.7 \times VCC$
		1.65 to 1.95 V						$VCC = 1.65$ to 1.95 V $V_{in} = 0.3 \times VCC$ to $0.7 \times VCC$
Input capacitance	Ports P014, P015	—	C_{in}	—	—	16	pF	$V_{bias} = 0 V$ $V_{amp} = 20 mV$ $f = 1 MHz$ $T_a = 25^\circ C$
	Ports P814/USB_DP, P815/USB_DM	—		—	—	12		
	Ports P400, P401, P409, P410, P511, P512, P708, P709	—		—	—	10		
	Other input pins	—		—	—	8		

Note 1. SCL0_A, SDA0_A, SCL1_A, SDA1_A, SCL2_A, SDA2_A (total 6 pins).

Note 2. This is the value when high speed high driving ability is selected in the Port Drive Capability bit in the PmnPFS register.

The selected driving ability is retained in Deep Software Standby mode.

Note 3. I3C_SCL0 (1 pin). This is the value when IIC high speed mode is selected.

Note 4. This is the value when high speed high driving ability is selected in the Port Drive Capability bit in the PmnPFS register for the following pins: SDA0_B, SCL0_B, SDA1_B, SCL1_B, SDA2_B, SCL2_B.

60.2.5 Operating and Standby Current

Current value of SiP Flash memory is not included in this section, refer to the datasheet of IS25WX064.

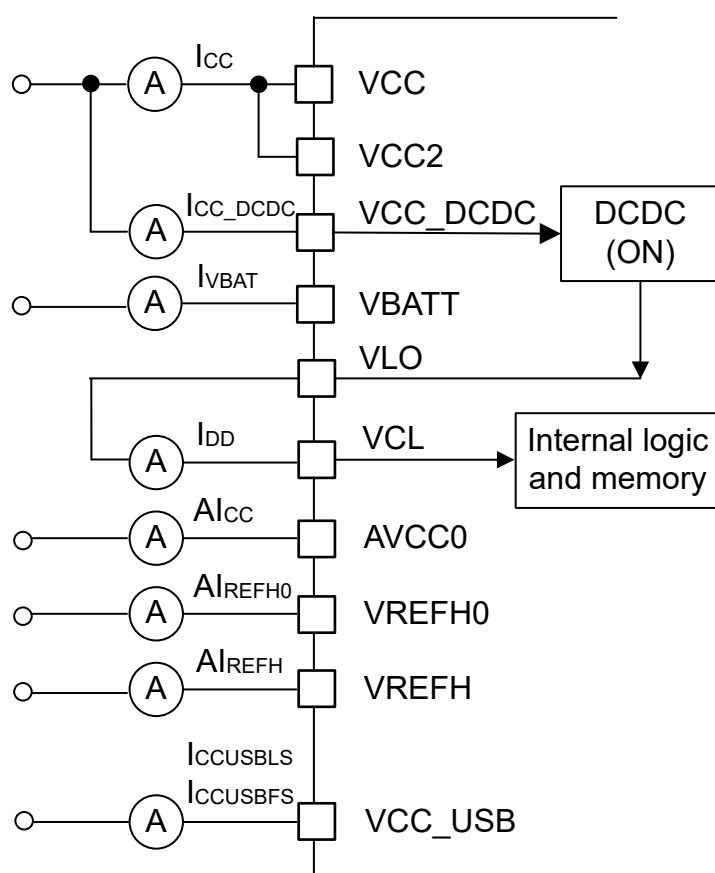


Figure 60.2 Consumption current measurement diagram (DCDC mode)

Table 60.8 Current of high-speed mode, maximum condition (MVE and peripheral operation) (DCDC mode)

Parameter			Symbol	Typ	Max			Unit	Test conditions
					95 °C	105 °C	125 °C		
Supply current *1 *2*6	—		I _{CC}	3.85	6.27	6.69	8.01	mA	—
	CPUC LK0 = 1 GHz CPUC LK1 = 250 MHz VSCR_1	VCC_DCD C ≥ 2.5 V	I _{CC_DCDC} *4	176	340	—	—	mA	VCC_DCDC = 3.3 V MRICK = 250 MHz, MRPCLK = 125 MHz, ICLK = 250 MHz, BCLK = 125 MHz, PCLKA = 125 MHz, PCLKB = 62.5 MHz, PCLKC = 125 MHz, PCLKD = 250 MHz, PCLKE = 250 MHz
			I _{DD} *3	452	873	—	—		
		VCC_DCD C < 2.5 V	I _{CC_DCDC} *4	343	664	—	—		VCC_DCDC = 1.8 V Clock settings are the same as above
			I _{DD}	452	873	—	—		
	CPUC LK0 = 800 MHz CPUC LK1 = 200 MHz VSCR_1	VCC_DCD C ≥ 2.5 V	I _{CC_DCDC} *4	148	—	356	—	mA	VCC_DCDC = 3.3 V MRICK = 200 MHz, MRPCLK = 100 MHz, ICLK = 200 MHz, BCLK = 100 MHz, PCLKA = 100 MHz, PCLKB = 50 MHz, PCLKC = 100 MHz, PCLKD = 200 MHz, PCLKE = 200 MHz
			I _{DD} *3	380	—	915	—		
		VCC_DCD C < 2.5 V	I _{CC_DCDC} *4	289	—	696	—		VCC_DCDC = 1.8 V Clock settings are the same as above
			I _{DD}	380	—	915	—		
	CPUC LK0 = 600 MHz CPUC LK1 = 200 MHz VSCR_1	VCC_DCD C ≥ 2.5 V	I _{CC_DCDC} *4	129	—	—	390	mA	VCC_DCDC = 3.3 V MRICK = 200 MHz, MRPCLK = 100 MHz, ICLK = 200 MHz, BCLK = 100 MHz, PCLKA = 100 MHz, PCLKB = 50 MHz, PCLKC = 100 MHz, PCLKD = 200 MHz, PCLKE = 200 MHz
			I _{DD} *3	330	—	—	1000 *5		
		VCC_DCD C < 2.5 V	I _{CC_DCDC} *4	251	—	—	760		VCC_DCDC = 1.8 V Clock settings are the same as above
			I _{DD}	330	—	—	1000 *5		
	CPUC LK0 = 600 MHz CPUC LK1 = 150 MHz VSCR_2	VCC_DCD C ≥ 2.5 V	I _{CC_DCDC} *4	115	282	309	366	mA	VCC_DCDC = 3.3 V MRICK = 150 MHz, MRPCLK = 75 MHz, ICLK = 150 MHz, BCLK = 75 MHz, PCLKA = 75 MHz, PCLKB = 37.5 MHz, PCLKC = 75 MHz, PCLKD = 150 MHz, PCLKE = 150 MHz
			I _{DD} *3	301	738	810	959		
		VCC_DCD C < 2.5 V	I _{CC_DCDC} *4	224	550	604	715		VCC_DCDC = 1.8 V Clock settings are the same as above
			I _{DD}	301	738	810	959		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Measured with clocks supplied to the peripheral functions and peripherals being operated. This does not include the BGO operation.

Note 3. I_{DD} depends on f (CPUCLK0 and ICLK) as follows. The ICLK items in the formula include CPU1 current.

I_{DD} Typ. = 0.25 × fCPUCLK0 + 0.77 × fICLK + 21 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 95°C) = 0.22 × fCPUCLK0 + 0.80 × fICLK + 477 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 105°C) = 0.22 × fCPUCLK0 + 0.80 × fICLK + 553 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 125°C) = 0.22 × fCPUCLK0 + 0.80 × fICLK + 709 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 95°C) = 0.22 × fCPUCLK0 + 0.93 × fICLK + 464 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 105°C) = 0.22 × fCPUCLK0 + 0.93 × fICLK + 537 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 125°C) = 0.22 × fCPUCLK0 + 0.93 × fICLK + 691 (unit : mA, fCPUCLK0 and fICLK are MHz)

Note 4. Typical DCDC efficiency and the voltage of the test conditions are applied.

Note 5. Do not let actual consumption current during operation exceed the current value described here.

Note 6. The power consumption is calculated as Power = VCC × I_{CC} + VCC_DCDC × I_{CC_DCDC}.

Table 60.9 Current of high-speed mode, maximum condition (MVE and peripheral operation) (External VDD mode)

Parameter		Symbol	Typ	Max			Unit	Test conditions
				95 °C	105 °C	125 °C		
Supply current *1 *2*5	—	I _{CC}	3.85	6.27	6.69	8.01	mA	—
	CPUCL K0 = 1 GHz CPUCL K1 = 250 MHz VCL = voltage range 1	I _{DD} *3	452	873	—	—	mA	MRICKL = 250 MHz, MRPCLK = 125 MHz, ICLK = 250 MHz, BCLK = 125 MHz, PCLKA = 125 MHz, PCLKB = 62.5 MHz, PCLKC = 125MHz, PCLKD = 250 MHz, PCLKE = 250 MHz
	CPUCL K0 = 800 MHz CPUCL K1 = 200 MHz VCL = voltage range 1	I _{DD} *3	380	—	915	—	mA	MRICKL = 200 MHz, MRPCLK = 100 MHz, ICLK = 200 MHz, BCLK = 100 MHz, PCLKA = 100 MHz, PCLKB = 50 MHz, PCLKC = 100 MHz, PCLKD = 200 MHz, PCLKE = 200 MHz
	CPUCL K0 = 600 MHz CPUCL K1 = 200 MHz VCL = voltage range 1	I _{DD} *3	330	—	—	1000 *4	mA	MRICKL = 200 MHz, MRPCLK = 100 MHz, ICLK = 200 MHz, BCLK = 100 MHz, PCLKA = 100 MHz, PCLKB = 50 MHz, PCLKC = 100 MHz, PCLKD = 200 MHz, PCLKE = 200 MHz
	CPUCL K0 = 600 MHz CPUCL K1 = 150 MHz VCL = voltage range 2	I _{DD} *3	301	738	810	959	mA	MRICKL = 150 MHz, MRPCLK = 75 MHz, ICLK = 150 MHz, BCLK = 75 MHz, PCLKA = 75 MHz, PCLKB = 37.5 MHz, PCLKC = 75 MHz, PCLKD = 150 MHz, PCLKE = 150 MHz

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Measured with clocks supplied to the peripheral functions and peripherals being operated. This does not include the BGO operation.

Note 3. I_{DD} depends on f (CPUCLK0 and ICLK) as follows. The ICLK items in the formula include CPU1 current.

I_{DD} Typ. = 0.25 × fCPUCLK0 + 0.77 × fICLK + 21 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 95°C) = 0.22 × fCPUCLK0 + 0.80 × fICLK + 477 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 105°C) = 0.22 × fCPUCLK0 + 0.80 × fICLK + 553 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 125°C) = 0.22 × fCPUCLK0 + 0.80 × fICLK + 709 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 95°C) = 0.22 × fCPUCLK0 + 0.93 × fICLK + 464 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 105°C) = 0.22 × fCPUCLK0 + 0.93 × fICLK + 537 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 125°C) = 0.22 × fCPUCLK0 + 0.93 × fICLK + 691 (unit : mA, fCPUCLK0 and fICLK are MHz)

Note 4. Do not let actual consumption current during operation exceed the current value described here.

Note 5. The power consumption is calculated as Power = VCC × I_{CC} + VCL × I_{DD}.

Table 60.10 Current of high-speed mode, maximum condition (MVE and peripheral operation), CPU0 active, CPU1 Deep Sleep (DCDC mode)

Parameter			Symbol	Typ	Max			Unit	Test conditions
					95 °C	105 °C	125 °C		
Supply current *1*2*5	—		I _{CC}	3.85	6.27	6.69	8.01	mA	—
	CPUC LK0 = 1 GHz CPUC LK1 = 250 MHz VSCR_1	VCC_DCD C ≥ 2.5V	I _{CC_DCDC} *4	168	329	—	—	mA	VCC_DCDC = 3.3 V MRICK = 250 MHz, MRPCLK = 125 MHz, ICLK = 250 MHz, BCLK = 125 MHz, PCLKA = 125 MHz, PCLKB = 62.5 MHz, PCLKC = 125 MHz, PCLKD = 250 MHz, PCLKE = 250 MHz CPU1 = Deep Sleep
			I _{DD} *3	430	844	—	—		
		VCC_DCD C < 2.5V	I _{CC_DCDC} *4	327	642	—	—		VCC_DCDC = 1.8 V Clock settings are the same as above
			I _{DD}	430	844	—	—		
	CPUC LK0 = 800 MHz CPUC LK1 = 200 MHz VSCR_1	VCC_DCD C ≥ 2.5V	I _{CC_DCDC} *4	141	—	347	—	mA	VCC_DCDC = 3.3 V MRICK = 200 MHz, MRPCLK = 100 MHz, ICLK = 200 MHz, BCLK = 100 MHz, PCLKA = 100MHz, PCLKB = 50 MHz, PCLKC = 100 MHz, PCLKD = 200 MHz, PCLKE = 200 MHz CPU1 = Deep Sleep
			I _{DD} *3	362	—	891	—		
		VCC_DCD C < 2.5V	I _{CC_DCDC} *4	275	—	678	—		VCC_DCDC = 1.8 V Clock settings are the same as above
			I _{DD}	362	—	891	—		
	CPUC LK0 = 600 MHz CPUC LK1 = 200 MHz VSCR_1	VCC_DCD C ≥ 2.5V	I _{CC_DCDC} *4	122	—	—	380	mA	VCC_DCDC = 3.3 V MRICK = 200 MHz, MRPCLK = 100 MHz, ICLK = 200 MHz, BCLK = 100 MHz, PCLKA = 100 MHz, PCLKB = 50 MHz, PCLKC = 100 MHz, PCLKD = 200 MHz, PCLKE = 200 MHz CPU1 = Deep Sleep
			I _{DD} *3	312	—	—	976		
		VCC_DCD C < 2.5V	I _{CC_DCDC} *4	237	—	—	742		VCC_DCDC = 1.8 V Clock settings are the same as above
			I _{DD}	312	—	—	976		
	CPUC LK0 = 600 MHz CPUC LK1 = 150 MHz VSCR_2	VCC_DCD C ≥ 2.5V	I _{CC_DCDC} *4	110	274	301	359	mA	VCC_DCDC = 3.3 V MRICK = 150 MHz, MRPCLK = 75 MHz, ICLK = 150 MHz, BCLK = 75 MHz, PCLKA = 75 MHz, PCLKB = 37.5 MHz, PCLKC = 75 MHz, PCLKD = 150 MHz, PCLKE = 150 MHz CPU1 = Deep Sleep
			I _{DD} *3	289	718	788	940		
		VCC_DCD C < 2.5V	I _{CC_DCDC} *4	216	536	588	701		VCC_DCDC = 1.8 V Clock settings are the same as above
			I _{DD}	289	718	788	940		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Measured with clocks supplied to the peripheral functions and peripherals being operated. This does not include the BGO operation.

Note 3. I_{DD} depends on f (CPUCLK0 and ICLK) as follows.

I_{DD} Typ. = 0.25 × fCPUCLK0 + 0.71 × fICLK + 21 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 95°C) = 0.22 × fCPUCLK0 + 0.70 × fICLK + 474 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 105°C) = 0.22 × fCPUCLK0 + 0.70 × fICLK + 549 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 125°C) = 0.22 × fCPUCLK0 + 0.70 × fICLK + 704 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 95°C) = 0.22 × fCPUCLK0 + 0.83 × fICLK + 461 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 105°C) = 0.22 × fCPUCLK0 + 0.83 × fICLK + 533 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 125°C) = 0.22 × fCPUCLK0 + 0.83 × fICLK + 686 (unit : mA, fCPUCLK0 and fICLK are MHz)

Note 4. Typical DCDC efficiency and the voltage of the test conditions are applied.

Note 5. The power consumption is calculated as Power = VCC × I_{CC} + VCC_DCDC × I_{CC_DCDC}.

Table 60.11 Current of high-speed mode, maximum condition (MVE and peripheral operation), CPU0 active, CPU1 Deep Sleep (External VDD mode)

Parameter		Symbol	Typ	Max			Unit	Test conditions
				95 °C	105 °C	125 °C		
Supply current *1*2*4	—	I _{CC}	3.85	6.27	6.69	8.01	mA	—
	CPUCL K0 = 1 GHz CPUCL K1 = 250 MHz VCL = voltage range 1	I _{DD} *3	430	844	—	—	mA	MRICKL = 250 MHz, MRPCLK = 125 MHz, ICLK = 250 MHz, BCLK = 125 MHz, PCLKA = 125 MHz, PCLKB = 62.5 MHz, PCLKC = 125 MHz, PCLKD = 250 MHz, PCLKE = 250 MHz CPU1 = Deep Sleep
	CPUCL K0 = 800 MHz CPUCL K1 = 200 MHz VCL = voltage range 1	I _{DD} *3	362	—	891	—	mA	MRICKL = 200 MHz, MRPCLK = 100 MHz, ICLK = 200 MHz, BCLK = 100 MHz, PCLKA = 100 MHz, PCLKB = 50 MHz, PCLKC = 100 MHz, PCLKD = 200 MHz, PCLKE = 200 MHz CPU1 = Deep Sleep
	CPUCL K0 = 600 MHz CPUCL K1 = 200 MHz VCL = voltage range 1	I _{DD} *3	312	—	—	976	mA	MRICKL = 200 MHz, MRPCLK = 100 MHz, ICLK = 200 MHz, BCLK = 100 MHz, PCLKA = 100 MHz, PCLKB = 50 MHz, PCLKC = 100 MHz, PCLKD = 200 MHz, PCLKE = 200 MHz CPU1 = Deep Sleep
	CPUCL K0 = 600 MHz CPUCL K1 = 150 MHz VCL = voltage range 2	I _{DD} *3	289	718	788	940	mA	MRICKL = 150 MHz, MRPCLK = 75 MHz, ICLK = 150 MHz, BCLK = 75 MHz, PCLKA = 75 MHz, PCLKB = 37.5 MHz, PCLKC = 75 MHz, PCLKD = 150 MHz, PCLKE = 150 MHz CPU1 = Deep Sleep

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Measured with clocks supplied to the peripheral functions and peripherals being operated. This does not include the BGO operation.

Note 3. I_{DD} depends on f (CPUCLK0 and ICLK) as follows.

I_{DD} Typ. = $0.25 \times \text{fCPUCLK0} + 0.71 \times \text{fCLK} + 21$ (unit : mA, fCPUCLK0 and fCLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 95°C) = $0.22 \times \text{fCPUCLK0} + 0.70 \times \text{fCLK} + 474$ (unit : mA, fCPUCLK0 and fCLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 105°C) = $0.22 \times \text{fCPUCLK0} + 0.70 \times \text{fCLK} + 549$ (unit : mA, fCPUCLK0 and fCLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 125°C) = $0.22 \times \text{fCPUCLK0} + 0.70 \times \text{fCLK} + 704$ (unit : mA, fCPUCLK0 and fCLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 95°C) = $0.22 \times \text{fCPUCLK0} + 0.83 \times \text{fCLK} + 461$ (unit : mA, fCPUCLK0 and fCLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 105°C) = $0.22 \times \text{fCPUCLK0} + 0.83 \times \text{fCLK} + 533$ (unit : mA, fCPUCLK0 and fCLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 125°C) = $0.22 \times \text{fCPUCLK0} + 0.83 \times \text{fCLK} + 686$ (unit : mA, fCPUCLK0 and fCLK are MHz)

Note 4. The power consumption is calculated as Power = VCC × I_{CC} + VCL × I_{DD}.

Table 60.12 Current of high-speed mode, maximum data processing (MVE operation), peripheral clock ON (DCDC mode)

Parameter	Symbol	Typ	Max			Unit	Test conditions
			95 °C	105 °C	125 °C		
Supply current *1*2	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VSCR_1	$I_{CC_DCDC}^{*4}$	136	322	—	mA	$V_{CC_DCDC} = 3.3 V^{*5}$
		I_{DD}^{*3}	350	827	—		
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VSCR_1	$I_{CC_DCDC}^{*4}$	113	—	340		
		I_{DD}^{*3}	290	—	874		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VSCR_1	$I_{CC_DCDC}^{*4}$	94	—	—		
		I_{DD}^{*3}	241	—	356		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VSCR_2	$I_{CC_DCDC}^{*4}$	86	264	305		
		I_{DD}^{*3}	226	692	798		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Measured with clocks supplied to the peripheral functions and peripherals being operated. This does not include the BGO operation.

Note 3. I_{DD} depends on f (CPUCLK0 and ICLK) as follows. The ICLK items in the formula include CPU1 current.

$I_{DD} \text{ Typ.} = 0.24 \times f_{CPUCLK0} + 0.37 \times f_{ICLK} + 21$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

$I_{DD} \text{ Max. (VSCR_1, 95°C)} = 0.27 \times f_{CPUCLK0} + 0.20 \times f_{ICLK} + 477$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

$I_{DD} \text{ Max. (VSCR_1, 105°C)} = 0.27 \times f_{CPUCLK0} + 0.20 \times f_{ICLK} + 553$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

$I_{DD} \text{ Max. (VSCR_1, 125°C)} = 0.27 \times f_{CPUCLK0} + 0.20 \times f_{ICLK} + 709$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

$I_{DD} \text{ Max. (VSCR_2, 95°C)} = 0.27 \times f_{CPUCLK0} + 0.18 \times f_{ICLK} + 464$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

$I_{DD} \text{ Max. (VSCR_2, 105°C)} = 0.27 \times f_{CPUCLK0} + 0.18 \times f_{ICLK} + 537$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

$I_{DD} \text{ Max. (VSCR_2, 125°C)} = 0.27 \times f_{CPUCLK0} + 0.18 \times f_{ICLK} + 691$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

Note 4. Typical DCDC efficiency and the voltage of the test conditions are applied.

Note 5. Same frequency condition is applied as in the maximum condition.

Table 60.13 Current of high-speed mode, maximum data processing (MVE operation), peripheral clock ON (External VDD mode)

Parameter	Symbol	Typ	Max			Unit	Test conditions
			95 °C	105 °C	125 °C		
Supply current *1*2	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VCL = voltage range 1	I_{DD}^{*3}	350	827	—	mA	*4
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I_{DD}^{*3}	290	—	874		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I_{DD}^{*3}	241	—	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VCL = voltage range 2	I_{DD}^{*3}	226	692	798		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Measured with clocks supplied to the peripheral functions and peripherals being operated. This does not include the BGO operation.

Note 3. I_{DD} depends on f (CPUCLK0 and ICLK) as follows. The ICLK items in the formula include CPU1 current.

I_{DD} Typ. = $0.24 \times f_{CPUCLK0} + 0.37 \times f_{ICLK} + 21$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 95°C) = $0.27 \times f_{CPUCLK0} + 0.20 \times f_{ICLK} + 477$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 105°C) = $0.27 \times f_{CPUCLK0} + 0.20 \times f_{ICLK} + 553$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 125°C) = $0.27 \times f_{CPUCLK0} + 0.20 \times f_{ICLK} + 709$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 95°C) = $0.27 \times f_{CPUCLK0} + 0.18 \times f_{ICLK} + 464$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 105°C) = $0.27 \times f_{CPUCLK0} + 0.18 \times f_{ICLK} + 537$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 125°C) = $0.27 \times f_{CPUCLK0} + 0.18 \times f_{ICLK} + 691$ (unit : mA, fCPUCLK0 and fICLK are MHz)

Note 4. Same frequency condition is applied as in the maximum condition.

Table 60.14 Current of high-speed mode, maximum data processing (MVE operation), CPU0 active, CPU1 Deep Sleep, peripheral clock ON (DCDC mode)

Parameter		Symbol	Typ	Max			Unit	Test conditions
				95 °C	105 °C	125 °C		
Supply current *1*2	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VSCR_1	I _{CC_DCDC} *4	131	313	—	—	mA	VCC_DCDC = 3.3 V CPU1 = Deep Sleep *5
		I _{DD} *3	335	804	—	—		
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VSCR_1	I _{CC_DCDC} *4	109	—	332	—		
		I _{DD} *3	279	—	853	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VSCR_1	I _{CC_DCDC} *4	89	—	—	342		
		I _{DD} *3	229	—	—	879		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VSCR_2	I _{CC_DCDC} *4	83	258	298	328		
		I _{DD} *3	216	676	780	858		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Measured with clocks supplied to the peripheral functions and peripherals being operated. This does not include the BGO operation.

Note 3. I_{DD} depends on f (CPUCLK0 and ICLK) as follows.

I_{DD} Typ. = 0.25 × fCPUCLK0 + 0.30 × fICLK + 21 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 95°C) = 0.27 × fCPUCLK0 + 0.05 × fICLK + 474 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 105°C) = 0.27 × fCPUCLK0 + 0.05 × fICLK + 549 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 125°C) = 0.27 × fCPUCLK0 + 0.05 × fICLK + 704 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 95°C) = 0.27 × fCPUCLK0 + 0.07 × fICLK + 461 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 105°C) = 0.27 × fCPUCLK0 + 0.07 × fICLK + 533 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 125°C) = 0.27 × fCPUCLK0 + 0.07 × fICLK + 686 (unit : mA, fCPUCLK0 and fICLK are MHz)

Note 4. Typical DCDC efficiency and the voltage of the test conditions are applied.

Note 5. Same frequency condition is applied as in the maximum condition.

Table 60.15 Current of high-speed mode, maximum data processing (MVE operation), CPU0 active, CPU1 Deep Sleep, peripheral clock ON (External VDD mode)

Parameter	Symbol	Typ	Max			Unit	Test conditions
			95 °C	105 °C	125 °C		
Supply current *1*2	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VCL = voltage range 1	I_{DD}^{*3}	335	804	—	mA	CPU1 = Deep Sleep *4
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I_{DD}^{*3}	279	—	853		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I_{DD}^{*3}	229	—	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VCL = voltage range 2	I_{DD}^{*3}	216	676	780		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Measured with clocks supplied to the peripheral functions and peripherals being operated. This does not include the BGO operation.

Note 3. I_{DD} depends on f (CPUCLK0 and ICLK) as follows.

I_{DD} Typ. = $0.25 \times f_{CPUCLK0} + 0.30 \times f_{ICLK} + 21$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 95°C) = $0.27 \times f_{CPUCLK0} + 0.05 \times f_{ICLK} + 474$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 105°C) = $0.27 \times f_{CPUCLK0} + 0.05 \times f_{ICLK} + 549$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 125°C) = $0.27 \times f_{CPUCLK0} + 0.05 \times f_{ICLK} + 704$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 95°C) = $0.27 \times f_{CPUCLK0} + 0.07 \times f_{ICLK} + 461$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 105°C) = $0.27 \times f_{CPUCLK0} + 0.07 \times f_{ICLK} + 533$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 125°C) = $0.27 \times f_{CPUCLK0} + 0.07 \times f_{ICLK} + 686$ (unit : mA, fCPUCLK0 and fICLK are MHz)

Note 4. Same frequency condition is applied as in the maximum condition.

Table 60.16 Current of high-speed mode, maximum data processing, CPU0 Deep Sleep, CPU1 active, peripheral clock ON (DCDC mode)

Parameter	Symbol	Typ	Max			Unit	Test conditions
			95 °C	105 °C	125 °C		
Supply current *1*2	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VSCR_1 $I_{CC_DCDC}^{*4}$	44	175	—	—	mA	VCC_DCDC = 3.3 V CPU0 = Deep Sleep *3
	I_{DD}	112	450	—	—		
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VSCR_1 $I_{CC_DCDC}^{*4}$	38	—	190	—		
	I_{DD}	99	—	488	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VSCR_1 $I_{CC_DCDC}^{*4}$	38	—	—	224		
	I_{DD}	96	—	—	545		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VSCR_2 $I_{CC_DCDC}^{*4}$	31	155	179	208		
	I_{DD}	82	406	470	545		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Measured with clocks supplied to the peripheral functions and peripherals being operated. This does not include the BGO operation.

Note 3. Same frequency condition is applied as in the maximum condition.

Note 4. Typical DCDC efficiency and the voltage of the test conditions are applied.

Table 60.17 Current of high-speed mode, maximum data processing, CPU0 Deep Sleep, CPU1 active, peripheral clock ON (External VDD mode)

Parameter	Symbol	Typ	Max			Unit	Test conditions
			95 °C	105 °C	125 °C		
Supply current *1*2	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VCL = voltage range 1 I_{DD}	112	450	—	—	mA	CPU0 = Deep Sleep *3
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VCL = voltage range 1 I_{DD}	99	—	488	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VCL = voltage range 1 I_{DD}	96	—	—	545		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VCL = voltage range 2 I_{DD}	82	406	470	545		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Measured with clocks supplied to the peripheral functions and peripherals being operated. This does not include the BGO operation.

Note 3. Same frequency condition is applied as in the maximum condition.

Table 60.18 Current of high-speed mode, maximum data processing (MVE operation), peripheral clock OFF (DCDC mode)

Parameter		Symbol	Typ	Max			Unit	Test conditions
				95 °C	105 °C	125 °C		
Supply current *1*2	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VSCR_1	I _{CC_DCDC} *4	121	317	—	—	mA	VCC_DCDC = 3.3 V*5
		I _{DD} *3	310	814	—	—		
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VSCR_1	I _{CC_DCDC} *4	99	—	325	—		
		I _{DD} *3	255	—	833	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VSCR_1	I _{CC_DCDC} *4	80	—	—	358		
		I _{DD} *3	207	—	—	919		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VSCR_2	I _{CC_DCDC} *4	75	258	292	340		
		I _{DD} *3	197	675	764	889		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 3. I_{DD} depends on f (CPUCLK0 and ICLK) as follows. The ICLK items in the formula include CPU1 current.

I_{DD} Typ. = 0.24 × fCPUCLK0 + 0.20 × fICLK + 21 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 95°C) = 0.27 × fCPUCLK0 + 0.23 × fICLK + 477 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 105°C) = 0.27 × fCPUCLK0 + 0.23 × fICLK + 553 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 125°C) = 0.27 × fCPUCLK0 + 0.23 × fICLK + 709 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 95°C) = 0.27 × fCPUCLK0 + 0.25 × fICLK + 464 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 105°C) = 0.27 × fCPUCLK0 + 0.25 × fICLK + 537 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 125°C) = 0.27 × fCPUCLK0 + 0.25 × fICLK + 691 (unit : mA, fCPUCLK0 and fICLK are MHz)

Note 4. Typical DCDC efficiency and the voltage of the test conditions are applied.

Note 5. Same frequency condition is applied as in the maximum condition.

Table 60.19 Current of high-speed mode, maximum data processing (MVE operation), peripheral clock OFF (External VDD mode)

Parameter	Symbol	Typ	Max			Unit	Test conditions
			95 °C	105 °C	125 °C		
Supply current *1*2	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VCL = voltage range 1	I_{DD}^{*3}	310	814	—	mA	*4
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I_{DD}^{*3}	255	—	833		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I_{DD}^{*3}	207	—	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VCL = voltage range 2	I_{DD}^{*3}	197	675	764		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 3. I_{DD} depends on f (CPUCLK0 and ICLK) as follows. The ICLK items in the formula include CPU1 current.

I_{DD} Typ. = $0.24 \times f_{CPUCLK0} + 0.20 \times f_{ICLK} + 21$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 95°C) = $0.27 \times f_{CPUCLK0} + 0.23 \times f_{ICLK} + 477$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 105°C) = $0.27 \times f_{CPUCLK0} + 0.23 \times f_{ICLK} + 553$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 125°C) = $0.27 \times f_{CPUCLK0} + 0.23 \times f_{ICLK} + 709$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 95°C) = $0.27 \times f_{CPUCLK0} + 0.25 \times f_{ICLK} + 464$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 105°C) = $0.27 \times f_{CPUCLK0} + 0.25 \times f_{ICLK} + 537$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 125°C) = $0.27 \times f_{CPUCLK0} + 0.25 \times f_{ICLK} + 691$ (unit : mA, fCPUCLK0 and fICLK are MHz)

Note 4. Same frequency condition is applied as in the maximum condition.

Table 60.20 Current of high-speed mode, maximum data processing (MVE operation), CPU0 active, CPU1 Deep Sleep, peripheral clock OFF (DCDC mode)

Parameter		Symbol	Typ	Max			Unit	Test conditions
				95 °C	105 °C	125 °C		
Supply current *1*2	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VSCR_1	I _{CC_DCDC} *4	115	308	—	—	mA	VCC_DCDC = 3.3 V CPU1 = Deep Sleep *5
		I _{DD} *3	295	792	—	—		
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VSCR_1	I _{CC_DCDC} *4	95	—	317	—		
		I _{DD} *3	244	—	813	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VSCR_1	I _{CC_DCDC} *4	76	—	—	348		
		I _{DD} *3	194	—	—	895		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VSCR_2	I _{CC_DCDC} *4	71	251	285	331		
		I _{DD} *3	187	658	746	868		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 3. I_{DD} depends on f (CPUCLK0 and ICLK) as follows.

I_{DD} Typ. = 0.25 × fCPUCLK0 + 0.13 × fICLK + 21 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 95°C) = 0.27 × fCPUCLK0 + 0.13 × fICLK + 474 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 105°C) = 0.27 × fCPUCLK0 + 0.13 × fICLK + 549 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 125°C) = 0.27 × fCPUCLK0 + 0.13 × fICLK + 704 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 95°C) = 0.27 × fCPUCLK0 + 0.14 × fICLK + 461 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 105°C) = 0.27 × fCPUCLK0 + 0.14 × fICLK + 533 (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 125°C) = 0.27 × fCPUCLK0 + 0.14 × fICLK + 686 (unit : mA, fCPUCLK0 and fICLK are MHz)

Note 4. Typical DCDC efficiency and the voltage of the test conditions are applied.

Note 5. Same frequency condition is applied as in the maximum condition.

Table 60.21 Current of high-speed mode, maximum data processing (MVE operation), CPU0 active, CPU1 Deep Sleep, peripheral clock OFF (External VDD mode)

Parameter	Symbol	Typ	Max			Unit	Test conditions
			95 °C	105 °C	125 °C		
Supply current *1*2	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VCL = voltage range 1	I_{DD}^{*3}	295	792	—	mA	CPU1 = Deep Sleep *4
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I_{DD}^{*3}	244	—	813		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I_{DD}^{*3}	194	—	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VCL = voltage range 2	I_{DD}^{*3}	187	658	746		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 3. I_{DD} depends on f (CPUCLK0 and ICLK) as follows.

I_{DD} Typ. = $0.25 \times f_{CPUCLK0} + 0.13 \times f_{ICLK} + 21$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 95°C) = $0.27 \times f_{CPUCLK0} + 0.13 \times f_{ICLK} + 474$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 105°C) = $0.27 \times f_{CPUCLK0} + 0.13 \times f_{ICLK} + 549$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 125°C) = $0.27 \times f_{CPUCLK0} + 0.13 \times f_{ICLK} + 704$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 95°C) = $0.27 \times f_{CPUCLK0} + 0.14 \times f_{ICLK} + 461$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 105°C) = $0.27 \times f_{CPUCLK0} + 0.14 \times f_{ICLK} + 533$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 125°C) = $0.27 \times f_{CPUCLK0} + 0.14 \times f_{ICLK} + 686$ (unit : mA, fCPUCLK0 and fICLK are MHz)

Note 4. Same frequency condition is applied as in the maximum condition.

Table 60.22 Current of high-speed mode, maximum data processing, CPU0 Deep Sleep, CPU1 active, peripheral clock OFF (DCDC mode)

Parameter	Symbol	Typ	Max			Unit	Test conditions
			95 °C	105 °C	125 °C		
Supply current *1*2	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VSCR_1 $I_{CC_DCDC}^{*4}$	28	157	—	—	mA	VCC_DCDC = 3.3 V CPU0 = Deep Sleep *3
	I_{DD}	72	404	—	—		
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VSCR_1 $I_{CC_DCDC}^{*4}$	25	—	174	—		
	I_{DD}	64	—	447	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VSCR_1 $I_{CC_DCDC}^{*4}$	24	—	—	208		
	I_{DD}	61	—	—	533		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VSCR_2 $I_{CC_DCDC}^{*4}$	20	142	166	195		
	I_{DD}	53	372	435	511		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 3. Same frequency condition is applied as in the maximum condition.

Note 4. Typical DCDC efficiency and the voltage of the test conditions are applied.

Table 60.23 Current of high-speed mode, maximum data processing, CPU0 Deep Sleep, CPU1 active, peripheral clock OFF (External VDD mode)

Parameter	Symbol	Typ	Max			Unit	Test conditions
			95 °C	105 °C	125 °C		
Supply current *1*2	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VCL = voltage range 1 I_{DD}	72	404	—	—	mA	CPU0 = Deep Sleep *3
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VCL = voltage range 1 I_{DD}	64	—	447	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VCL = voltage range 1 I_{DD}	61	—	—	533		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VCL = voltage range 2 I_{DD}	53	372	435	511		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 3. Same frequency condition is applied as in the maximum condition.

Table 60.24 Current of high-speed mode, CPU Sleep mode (DCDC mode)

Parameter		Symbol	Typ	Max			Unit	Test conditions
				95 °C	105 °C	125 °C		
Supply current *1*3*4	CPUCLK0 = 1 GHz	$I_{CC_DCDC}^{*5}$	22	200	—	—	mA	VCC_DCDC = 3.3 V
	CPUCLK1 = 250 MHz VSCR_1	I_{DD}^{*2}	56	513	—	—		
	CPUCLK0 = 800 MHz	$I_{CC_DCDC}^{*5}$	20	—	228	—		
	CPUCLK1 = 200 MHz VSCR_1	I_{DD}^{*2}	50	—	586	—		
	CPUCLK0 = 600 MHz	$I_{CC_DCDC}^{*5}$	18	—	—	279		
	CPUCLK1 = 200 MHz VSCR_1	I_{DD}^{*2}	46	—	—	716		
	CPUCLK0 = 600 MHz	$I_{CC_DCDC}^{*5}$	16	185	219	264		
	CPUCLK1 = 150 MHz VSCR_2	I_{DD}^{*2}	42	484	573	690		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. I_{DD} depends on f (ICLK) as follows.

I_{DD} Typ. = $0.02 \times f_{CPUCLK0} + 0.06 \times f_{ICLK} + 57$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 95°C) = $0.02 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 477$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 105°C) = $0.02 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 553$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_1, 125°C) = $0.02 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 709$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 95°C) = $0.02 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 464$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 105°C) = $0.02 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 537$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VSCR_2, 125°C) = $0.02 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 691$ (unit : mA, fCPUCLK0 and fICLK are MHz)

Note 3. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 4. MRCLK, MRPCLK, ICLK, PCLKA, PCLKB, PCLKC, PCLKD, PCLKE and BCLK are set to divided by 64.

Note 5. Typical DCDC efficiency and the voltage of the test conditions are applied.

Table 60.25 Current of high-speed mode, CPU Sleep mode (External VDD mode)

Parameter		Symbol	Typ	Max			Unit	Test conditions
				95 °C	105 °C	125 °C		
Supply current *1*3*4	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VCL = voltage range 1	I_{DD}^{*2}	56	513	—	—	mA	—
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I_{DD}^{*2}	50	—	586	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I_{DD}^{*2}	46	—	—	716		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VCL = voltage range 2	I_{DD}^{*2}	42	484	573	690		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. I_{DD} depends on f (ICLK) as follows.

I_{DD} Typ. = $0.02 \times f_{CPUCLK0} + 0.06 \times f_{ICLK} + 57$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 95°C) = $0.02 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 477$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 105°C) = $0.02 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 553$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 1, 125°C) = $0.02 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 709$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 95°C) = $0.02 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 464$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 105°C) = $0.02 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 537$ (unit : mA, fCPUCLK0 and fICLK are MHz)

I_{DD} Max.(VCL = voltage range 2, 125°C) = $0.02 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 691$ (unit : mA, fCPUCLK0 and fICLK are MHz)

Note 3. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 4. MRICKL, MRPCLK, ICLK, PCLKA, PCLKB, PCLKC, PCLKD, PCLKE and BCLK are set to divided by 64.

Table 60.26 Current of high-speed mode, CPU0 Sleep, CPU1 Deep Sleep (DCDC mode)

Parameter		Symbol	Typ	Max			Unit	Test conditions
				95 °C	105 °C	125 °C		
Supply current *1*3*4	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VSCR_1	$I_{CC_DCDC}^{*5}$	21	197	—	—	mA	VCC_DCDC = 3.3 V CPU1 = Deep Sleep
		I_{DD}^{*2}	54	505	—	—		
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VSCR_1	$I_{CC_DCDC}^{*5}$	19	—	229	—		
		I_{DD}^{*2}	48	—	576	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VSCR_1	$I_{CC_DCDC}^{*5}$	17	—	—	274		
		I_{DD}^{*2}	44	—	—	703		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VSCR_2	$I_{CC_DCDC}^{*5}$	15	182	216	259		
		I_{DD}^{*2}	40	477	565	678		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. I_{DD} depends on f (ICLK) as follows.

I_{DD} Typ. = $0.02 \times f_{CPUCLK0} + 0.05 \times f_{ICLK} + 55$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

I_{DD} Max.(VSCR_1, 95°C) = $0.004 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 474$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

I_{DD} Max.(VSCR_1, 105°C) = $0.004 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 549$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

I_{DD} Max.(VSCR_1, 125°C) = $0.004 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 704$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

I_{DD} Max.(VSCR_2, 95°C) = $0.004 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 461$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

I_{DD} Max.(VSCR_2, 105°C) = $0.004 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 533$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

I_{DD} Max.(VSCR_2, 125°C) = $0.004 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 686$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

Note 3. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 4. MRICKL, MRPCLK, ICLK, PCLKA, PCLKB, PCLKC, PCLKD, PCLKE and BCLK are set to divided by 64.

Note 5. Typical DCDC efficiency and the voltage of the test conditions are applied.

Table 60.27 Current of high-speed mode, CPU0 Sleep, CPU1 Deep Sleep (External VDD mode)

Parameter	Symbol	Typ	Max			Unit	Test conditions
			95 °C	105 °C	125 °C		
Supply current *1*3*4	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VCL = voltage range 1	I_{DD}^{*2}	54	505	—	mA	CPU1 = Deep Sleep
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I_{DD}^{*2}	48	—	576		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I_{DD}^{*2}	44	—	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VCL = voltage range 2	I_{DD}^{*2}	40	477	565		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. I_{DD} depends on f (ICLK) as follows.

I_{DD} Typ. = $0.02 \times f_{CPUCLK0} + 0.05 \times f_{ICLK} + 55$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

I_{DD} Max.(VCL = voltage range 1, 95°C) = $0.004 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 474$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

I_{DD} Max.(VCL = voltage range 1, 105°C) = $0.004 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 549$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

I_{DD} Max.(VCL = voltage range 1, 125°C) = $0.004 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 704$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

I_{DD} Max.(VCL = voltage range 2, 95°C) = $0.004 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 461$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

I_{DD} Max.(VCL = voltage range 2, 105°C) = $0.004 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 533$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

I_{DD} Max.(VCL = voltage range 2, 125°C) = $0.004 \times f_{CPUCLK0} + 0.01 \times f_{ICLK} + 686$ (unit : mA, $f_{CPUCLK0}$ and f_{ICLK} are MHz)

Note 3. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 4. MRICKL, MRPCLK, ICLK, PCLKA, PCLKB, PCLKC, PCLKD, PCLKE and BCLK are set to divided by 64.

Table 60.28 Current of high-speed mode, CPU0 Deep Sleep, CPU1 Sleep (DCDC mode)

Parameter		Symbol	Typ	Max			Unit	Test conditions
				95 °C	105 °C	125 °C		
Supply current *1*2*3	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VSCR_1	I _{CC_DCDC} *4	12	141	—	—	mA	VCC_DCDC = 3.3 V CPU0 = Deep Sleep
		I _{DD}	32	363	—	—		
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VSCR_1	I _{CC_DCDC} *4	12	—	161	—		
		I _{DD}	30	—	414	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VSCR_1	I _{CC_DCDC} *4	11	—	—	196		
		I _{DD}	29	—	—	504		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VSCR_2	I _{CC_DCDC} *4	10	132	157	184		
		I _{DD}	26	347	411	481		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 3. MRICKL, MRPCLK, ICLK, PCLKA, PCLKB, PCLKC, PCLKD, PCLKE and BCLK are set to divided by 64.

Note 4. Typical DCDC efficiency and the voltage of the test conditions are applied.

Table 60.29 Current of high-speed mode, CPU0 Deep Sleep, CPU1 Sleep (External VDD mode)

Parameter		Symbol	Typ	Max			Unit	Test conditions
				95 °C	105 °C	125 °C		
Supply current *1*2*3	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VCL = voltage range 1	I _{DD}	32	363	—	—	mA	CPU0 = Deep Sleep
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I _{DD}	30	—	414	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I _{DD}	29	—	—	504		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VCL = voltage range 2	I _{DD}	26	347	411	481		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 3. MRICKL, MRPCLK, ICLK, PCLKA, PCLKB, PCLKC, PCLKD, PCLKE and BCLK are set to divided by 64.

Table 60.30 Current of high-speed mode, CPU Deep Sleep mode (DCDC mode)

Parameter		Symbol	Typ	Max			Unit	Test conditions
				95 °C	105 °C	125 °C		
Supply current *1*2*3	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VSCR_1	I _{CC_DCDC} *4	13	138	—	—	mA	VCC_DCDC = 3.3 V
		I _{DD}	32	355	—	—		
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VSCR_1	I _{CC_DCDC} *4	12	—	158	—		
		I _{DD}	30	—	405	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VSCR_1	I _{CC_DCDC} *4	11	—	—	192		
		I _{DD}	29	—	—	492		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VSCR_2	I _{CC_DCDC} *4	11	130	154	182		
		I _{DD}	28	340	402	476		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 3. MRICKL, MRPCLK, ICLK, PCLKA, PCLKB, PCLKC, PCLKD, PCLKE and BCLK are set to divided by 64.

Note 4. Typical DCDC efficiency and the voltage of the test conditions are applied.

Table 60.31 Current of high-speed mode, CPU Deep Sleep mode (External VDD mode)

Parameter		Symbol	Typ	Max			Unit	Test conditions
				95 °C	105 °C	125 °C		
Supply current *1*2*3	CPUCLK0 = 1 GHz CPUCLK1 = 250 MHz VCL = voltage range 1	I _{DD}	32	355	—	—	mA	—
	CPUCLK0 = 800 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I _{DD}	30	—	405	—		
	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VCL = voltage range 1	I _{DD}	29	—	—	492		
	CPUCLK0 = 600 MHz CPUCLK1 = 150 MHz VCL = voltage range 2	I _{DD}	28	340	402	476		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Note 3. MRICKL, MRPCLK, ICLK, PCLKA, PCLKB, PCLKC, PCLKD, PCLKE and BCLK are set to divided by 64.

Table 60.32 Current increase during BGO operation (Programming of MRAM OTP) (DCDC mode and External VDD mode)

Parameter		Symbol	Typ	Max			Unit	Test conditions
				95 °C	105 °C	125 °C		
Supply current ^{*1}	normal speed write mode	I _{CC}	—	—	—	20	mA	VCC ≥ 1.62V
		I _{DD}	—	—	—	0.50		
	high speed write mode 0	I _{CC}	—	—	—	25		VCC ≥ 2.5V
		I _{DD}	—	—	—	0.5		
	high speed write mode 1	I _{CC}	—	—	—	80		VCC ≥ 3.0V
		I _{DD}	—	—	—	0.5		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Table 60.33 Standby current (DCDC mode) (1 of 4)

Parameter			Symbol	Typ	Max			Unit	Test conditions		
					95 °C	105 °C	125 °C				
Supply current*1	Software Standby mode		I _{CC}	0.10	1.11	1.12	1.14	mA	—		
		SS2LP_0	SVSCR_1	Data of SRAM and TCM is retained	I _{CC_DCDC}	2.67	54.24		62.23	78.41	VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 1 (n = 0 to 12) PDRAMSCR1.RKEEPn = 1 (n = 0, 1)
				Data of SRAM and TCM is not retained	I _{CC_DCDC}	2.44	52.53		59.86	74.47	VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 0 (n = 0 to 12) PDRAMSCR1.RKEEPn = 0 (n = 0, 1)
			SVSCR_2	Data of SRAM and TCM is retained	I _{CC_DCDC}	2.52	51.58		59.19	74.81	VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 1 (n = 0 to 12) PDRAMSCR1.RKEEPn = 1 (n = 0, 1)
				Data of SRAM and TCM is not retained	I _{CC_DCDC}	2.33	50.03		57.09	71.10	VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 0 (n = 0 to 12) PDRAMSCR1.RKEEPn = 0 (n = 0, 1)
			SVSCR_3	Data of SRAM and TCM is retained	I _{CC_DCDC}	1.68	37.85		43.60	56.69	VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 1 (n = 0 to 12) PDRAMSCR1.RKEEPn = 1 (n = 0, 1)
				Data of SRAM and TCM is not retained	I _{CC_DCDC}	1.60	36.87		42.22	54.07	VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 0 (n = 0 to 12) PDRAMSCR1.RKEEPn = 0 (n = 0, 1)
			SVSCR_4	Data of SRAM and TCM is retained	I _{CC_DCDC}	1.47	32.41		38.22	49.79	VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 1 (n = 0 to 12) PDRAMSCR1.RKEEPn = 1 (n = 0, 1)
				Data of SRAM and TCM is not retained	I _{CC_DCDC}	1.42	31.61		37.07	47.62	VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 0 (n = 0 to 12) PDRAMSCR1.RKEEPn = 0 (n = 0, 1)
			SVSCR_5	Data of SRAM and TCM is retained	I _{CC_DCDC}	1.28	29.69		34.28	44.67	VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 1 (n = 0 to 12) PDRAMSCR1.RKEEPn = 1 (n = 0, 1)
				Data of SRAM and TCM is not retained	I _{CC_DCDC}	1.24	29.03		33.32	42.82	VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 0 (n = 0 to 12) PDRAMSCR1.RKEEPn = 0 (n = 0, 1)

Table 60.33 Standby current (DCDC mode) (2 of 4)

Parameter					Symbol	Typ	Max			Unit	Test conditions
							95 °C	105 °C	125 °C		
Supply current*1	Software Standby mode	SS2LP_1	SVSCR_2	Data of SRAM and TCM is retained	I _{CC_DCDC}	2.12	43.32	49.71	62.83	mA	VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 1 (n = 0 to 12) PDRAMSCR1.RKEEPn = 1 (n = 0, 1)
				Data of SRAM and TCM is not retained	I _{CC_DCDC}	1.95	42.01	47.94	59.71		VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 0 (n = 0 to 12) PDRAMSCR1.RKEEPn = 0 (n = 0, 1)
			SVSCR_3	Data of SRAM and TCM is retained	I _{CC_DCDC}	1.40	31.70	36.52	47.48		VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 1 (n = 0 to 12) PDRAMSCR1.RKEEPn = 1 (n = 0, 1)
				Data of SRAM and TCM is not retained	I _{CC_DCDC}	1.34	30.88	35.36	45.29		VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 0 (n = 0 to 12) PDRAMSCR1.RKEEPn = 0 (n = 0, 1)
			SVSCR_4	Data of SRAM and TCM is retained	I _{CC_DCDC}	1.22	26.41	31.14	40.57		VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 1 (n = 0 to 12) PDRAMSCR1.RKEEPn = 1 (n = 0, 1)
				Data of SRAM and TCM is not retained	I _{CC_DCDC}	1.18	25.76	30.20	38.80		VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 0 (n = 0 to 12) PDRAMSCR1.RKEEPn = 0 (n = 0, 1)
			SVSCR_5	Data of SRAM and TCM is retained	I _{CC_DCDC}	1.06	24.15	27.89	36.35		VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 1 (n = 0 to 12) PDRAMSCR1.RKEEPn = 1 (n = 0, 1)
				Data of SRAM and TCM is not retained	I _{CC_DCDC}	1.03	23.62	27.11	34.84		VCC_DCDC = 3.3 V PDRAMSCR0.RKEEPn = 0 (n = 0 to 12) PDRAMSCR1.RKEEPn = 0 (n = 0, 1)

Table 60.33 Standby current (DCDC mode) (3 of 4)

Parameter				Symbol	Typ	Max			Unit	Test conditions
						95 °C	105 °C	125 °C		
Supply current*1	Deep Software Standby mode 1			I _{CC}	10.04	207	297	498	μA	—
				I _{CC_DCDC}	0.16	0.85	1.24	2.45		—
		Increase when the function is activated	PVDn (n = 0 to 2, 4, 5) or Battery power supply switch	I _{CC}	See Table 60.36					—
			When the LOCO is in use		2.46	—	—	—		—
			Crystal oscillator and RTC		See Table 60.37					—
			IWDT and ULPT (all units) are operating		1.58	—	—	—		—
Supply current*1	Deep Software Standby mode 2			I _{CC}	3.04	98	122	175	μA	—
				I _{CC_DCDC}	0.16	0.85	1.24	2.45		—
		Increase when the function is activated	PVDn (n = 0 to 2, 4, 5) or Battery power supply switch	I _{CC}	See Table 60.36					—
			Crystal oscillator and RTC		See Table 60.37					—
	Deep Software Standby mode 3			I _{CC}	2.78	97	121	173		—
				I _{CC_DCDC}	0.16	0.85	1.24	2.45		—
		Increase when the function is activated	Crystal oscillator and RTC	I _{CC}	See Table 60.37					—

Table 60.33 Standby current (DCDC mode) (4 of 4)

Parameter				Symbol	Typ	Max			Unit	Test conditions		
						95 °C	105 °C	125 °C				
Supply current*1	RTC operating while VCC is off (with the battery backup function, only the RTC operate)	When a crystal oscillator with low power mode 3 is in use		I _{VBAT}	0.53	—	—	—	μA	VBATT = 1.8 V, VCC = 0 V		
					0.82	—	—	—		VBATT = 3.3 V, VCC = 0 V		
					When a crystal oscillator with low power mode 2 is in use		0.63	—		—	—	VBATT = 1.8 V, VCC = 0 V
		0.94	—				—	—		VBATT = 3.3 V, VCC = 0 V		
		When a crystal oscillator with low power mode 1 is in use					0.73	—		—	—	VBATT = 1.8 V, VCC = 0 V
					1.03	—	—	—		VBATT = 3.3 V, VCC = 0 V		
					0.99	—	—	—		VBATT = 1.8 V, VCC = 0 V		
		When a crystal oscillator with standard mode is in use			1.29	—	—	—		VBATT = 3.3 V, VCC = 0 V		
					When EXCIN is in use		0.30	—		—	—	VBATT = 1.8 V, VCC = 0 V
							0.52	—		—	—	VBATT = 3.3 V, VCC = 0 V
		Increase when the function is activated	RTCI _{Cn} (n = 0 to 2) input is in use per channel				0.01	—		—	—	VBATT = 1.8 V, VCC = 0 V
					0.01	—	—	—		VBATT = 3.3 V, VCC = 0 V		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Table 60.34 Coremark and normal mode current, CPU0 active, CPU1 Deep Sleep. (DCDC mode and External VDD mode) (1 of 2)

Parameter				Symbol	Typ	Max			Unit	Test conditions
						95 °C	105 °C	125 °C		
Supply current*1*2	CPUCLK0 = 1 GHz VSCR_1 VCL = voltage range 1	Coremark	Cache on	I _{DD}	151	—	—	—	μA/ MH z	CPU1 = Deep Sleep ESWM power domain off CPUCLK1 = 250 MHz, MRICKL = 250 MHz, MRPCLK = 15.6 MHz, ICLK = 250 MHz, PCLKA = 15.6 MHz, PCLKB = 15.6 MHz, PCLKC = 15.6 MHz, PCLKD = 15.6 MHz, PCLKE = 15.6 MHz, BCLK = 15.6 MHz
			Cache off, executing from ITCM		143	—	—	—		
			Cache off, executing from SRAM		92	—	—	—		
			Cache off, executing from MRAM		104	—	—	—		
		Normal mode	All peripheral disabled, Cache on, while (1) code		118	—	—	—		
			All peripheral disabled, Cache off, while (1) code executing from MRAM		121	—	—	—		
	CPUCLK0 = 800 MHz VSCR_1 VCL = voltage range 1	Coremark	Cache on	I _{DD}	157	—	—	—	μA/ MH z	CPU1 = Deep Sleep ESWM power domain off CPUCLK1 = 200 MHz, MRICKL = 200 MHz, MRPCLK = 12.5 MHz, ICLK = 200 MHz, PCLKA = 12.5 MHz, PCLKB = 12.5 MHz, PCLKC = 12.5 MHz, PCLKD = 12.5 MHz, PCLKE = 12.5 MHz, BCLK = 12.5 MHz
			Cache off, executing from ITCM		149	—	—	—		
			Cache off, executing from SRAM		98	—	—	—		
			Cache off, executing from MRAM		111	—	—	—		
		Normal mode	All peripheral disabled, Cache on, while (1) code		124	—	—	—		
			All peripheral disabled, Cache off, while (1) code executing from MRAM		127	—	—	—		

Table 60.34 Coremark and normal mode current, CPU0 active, CPU1 Deep Sleep. (DCDC mode and External VDD mode) (2 of 2)

Parameter				Symbol	Typ	Max			Unit	Test conditions
						95 °C	105 °C	125 °C		
Supply current*1*2	CPUCLK0 = 600 MHz CPUCLK1 = 200 MHz VSCR_1 VCL = voltage range 1	Coremark	Cache on	I _{DD}	172	—	—	—	μA/ MH z	CPU1 = Deep Sleep ESWM power domain off CPUCLK1 = 200 MHz, MRICLK = 200 MHz, MRPCLK = 25 MHz, ICLK = 200 MHz, PCLKA = 25 MHz, PCLKB = 25 MHz, PCLKC = 25 MHz, PCLKD = 25 MHz, PCLKE = 25 MHz, BCLK = 25 MHz
			Cache off, executing from ITCM		164	—	—	—		
			Cache off, executing from SRAM		118	—	—	—		
			Cache off, executing from MRAM		133	—	—	—		
		Normal mode	All peripheral disabled, Cache on, while (1) code		139	—	—	—		
			All peripheral disabled, Cache off, while (1) code executing from MRAM		142	—	—	—		
	CPUCLK0 = 600 MHz VSCR_2 VCL = voltage range 2	Coremark	Cache on	I _{DD}	164	—	—	—	μA/ MH z	CPU1 = Deep Sleep ESWM power domain off CPUCLK1 = 150 MHz, MRICLK = 150 MHz, MRPCLK = 9.4 MHz, ICLK = 150 MHz, PCLKA = 9.4 MHz, PCLKB = 9.4 MHz, PCLKC = 9.4 MHz, PCLKD = 9.4 MHz, PCLKE = 9.4 MHz, BCLK = 9.4 MHz
			Cache off, executing from ITCM		156	—	—	—		
			Cache off, executing from SRAM		106	—	—	—		
			Cache off, executing from MRAM		119	—	—	—		
		Normal mode	All peripheral disabled, Cache on, while (1) code		131	—	—	—		
			All peripheral disabled, Cache off, while (1) code executing from MRAM		135	—	—	—		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Table 60.35 Coremark and normal mode current, CPU0 Deep Sleep, CPU1 active. (DCDC mode and External VDD mode) (1 of 2)

Parameter				Symbol	Typ	Max			Unit	Test conditions
						95 °C	105 °C	125 °C		
Supply current*1 *2	CPUCLK1 = 250 MHz VSCR_1 VCL = voltage range 1	Coremark	Cache on	I _{DD}	215	—	—	—	μA/ MHz	CPU0 = Deep Sleep ESWM power domain off CPUCLK0 = 1 GHz, MRICKL = 250 MHz, MRPCLK = 15.6 MHz, ICLK = 250 MHz, PCLKA = 15.6 MHz, PCLKB = 15.6 MHz, PCLKC = 15.6 MHz, PCLKD = 15.6 MHz, PCLKE = 15.6 MHz, BCLK = 15.6 MHz
			Cache off, executing from ITCM		197	—	—	—		
			Cache off, executing from SRAM		194	—	—	—		
			Cache off, executing from MRAM		268	—	—	—		
		Normal mode	All peripheral disabled, Cache on, while (1) code		201	—	—	—		
			All peripheral disabled, Cache off, while (1) code executing from MRAM		264	—	—	—		
	CPUCLK1 = 200 MHz VSCR_1 VCL = voltage range 1	Coremark	Cache on	I _{DD}	236	—	—	—	μA/ MHz	CPU0 = Deep Sleep ESWM power domain off CPUCLK0 = 800 MHz, MRICKL = 200 MHz, MRPCLK = 12.5 MHz, ICLK = 200 MHz, PCLKA = 12.5 MHz, PCLKB = 12.5 MHz, PCLKC = 12.5 MHz, PCLKD = 12.5 MHz, PCLKE = 12.5 MHz, BCLK = 12.5 MHz
			Cache off, executing from ITCM		218	—	—	—		
			Cache off, executing from SRAM		215	—	—	—		
			Cache off, executing from MRAM		299	—	—	—		
		Normal mode	All peripheral disabled, Cache on, while (1) code		222	—	—	—		
			All peripheral disabled, Cache off, while (1) code executing from MRAM		297	—	—	—		

Table 60.35 Coremark and normal mode current, CPU0 Deep Sleep, CPU1 active. (DCDC mode and External VDD mode) (2 of 2)

Parameter				Symbol	Typ	Max			Unit	Test conditions
						95 °C	105 °C	125 °C		
Supply current*1 *2	CPUCLK1 = 200 MHz VSCR_1 VCL = voltage range 1	Coremark	Cache on	I _{DD}	226	—	—	—	μA/ MHz	CPU0 = Deep Sleep ESWM power domain off CPUCLK0 = 600 MHz, MRCLK = 200 MHz, MRPCLK = 25 MHz, ICLK = 200 MHz, PCLKA = 25 MHz, PCLKB = 25 MHz, PCLKC = 25 MHz, PCLKD = 25 MHz, PCLKE = 25 MHz, BCLK = 25MHz
			Cache off, executing from ITCM		208	—	—	—		
			Cache off, executing from SRAM		205	—	—	—		
			Cache off, executing from MRAM		289	—	—	—		
		Normal mode	All peripheral disabled, Cache on, while (1) code		213	—	—	—		
			All peripheral disabled, Cache off, while (1) code executing from MRAM		287	—	—	—		
	CPUCLK1 = 150 MHz VSCR_2 VCL = voltage range 2	Coremark	Cache on	I _{DD}	265	—	—	—	μA/ MHz	CPU0 = Deep Sleep ESWM power domain off CPUCLK0 = 600 MHz, MRCLK = 150 MHz, MRPCLK = 9.4 MHz, ICLK = 150 MHz, PCLKA = 9.4 MHz, PCLKB = 9.4 MHz, PCLKC = 9.4 MHz, PCLKD = 9.4 MHz, PCLKE = 9.4 MHz, BCLK = 9.4 MHz
			Cache off, executing from ITCM		247	—	—	—		
			Cache off, executing from SRAM		244	—	—	—		
			Cache off, executing from MRAM		326	—	—	—		
		Normal mode	All peripheral disabled, Cache on, while (1) code		252	—	—	—		
			All peripheral disabled, Cache off, while (1) code executing from MRAM		324	—	—	—		

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. Supply of the clock signal to peripherals is stopped in this state. This does not include the BGO operation.

Table 60.36 Increase when the PVD1, PVD2, PVD4, PVD5 or Battery power supply switch is enabled in Deep Software Standby mode 1 and 2.

Parameter		Symbol	Typ	Max			Unit	Test conditions
				95 °C	105 °C	125 °C		
Supply current	Common circuit when enabling PVDn (n=1, 2, 4, 5) or disabling low power consumption function of PVD0 for the battery power supply switch control (OFS1(_SEC).PVDLPSEL=1) in Deep Software Standby mode 1	I _{CC}	4.00	—	—	—	μA	—
	Common circuit when enabling PVDn (n=1, 2, 4, 5) or disabling low power consumption function of PVD0 for the battery power supply switch control (OFS1(_SEC).PVDLPSEL=1) in Deep Software Standby mode 2		4.00	—	—	—		—
	PVD1 enabled		2.00	—	—	—		—
	PVD2 enabled		2.00	—	—	—		—
	PVD4 enabled		2.00	—	—	—		—
	PVD5 enabled		2.00	—	—	—		—
	Battery power supply switch enabled with following conditions.*1 Battery power supply switch enable (VBTBPCR1.BPWSWSTP=0), and low power consumption function of PVD0 select at Deep Software Standby mode disable (OFS1(_SEC).PVDLPSEL=1).		2.00	—	—	—		—

Note 1. Consumption current is not increased in other condition.

Table 60.37 Increase when the sub-clock oscillator and RTC are enabled in Deep Software Standby mode 1, 2 and 3.

Parameter			Symbol	Typ	Max			Unit	Test conditions
					95 °C	105 °C	125 °C		
Supply current	When a crystal oscillator is in use	Low Power mode 3	I _{CC}	0.31	—	—	—	μA	—
		Low Power mode 2		0.43	—	—	—		—
		Low Power mode 1		0.52	—	—	—		—
		Standard mode		0.78	—	—	—		—
	RTC is operating			0.30	—	—	—		—

Table 60.38 Inrush current

Parameter				Symbol	Typ	Max			Unit	Test conditions
						95 °C	105 °C	125 °C		
Supply current	Inrush current on Cold Start		Inrush current of VCC_DCDC*1	I _{RUSH}	—	1330	1330	1330	mA	—
	Inrush current on returning from deep software standby mode	DPSBYCR.DC SSMODE=1			—	1270	1270	1270		—
		DPSBYCR.DC SSMODE=2			—	1170	1170	1170		—
		DPSBYCR.DC SSMODE=3			—	1160	1160	1160		—

Note 1. Reference value

Table 60.39 Operating current (Analog)

Parameter				Symbol	Typ	Max			Unit	Test conditions
						95 °C	105 °C	125 °C		
Supply current*1	Oscillator	Main clock oscillator		I _{CC}	0.65	—	—	—	mA	MOMCR.MODRV0 [2:0] = 000b
					0.76	—	—	—	mA	MOMCR.MODRV0 [2:0] = 011b
					0.88	—	—	—	mA	MOMCR.MODRV0 [2:0] = 101b
	Analog power supply current	During 16-bit A/D conversion	SAR mode, Oversampling mode and Hybrid mode	A _I CC	2.4	3.2	3.2	3.2	mA	—
					3.9	5.1	5.1	5.1	mA	—
		ACMPHS(1unit)			99	192	192	192	μA	—
		Temperature sensor			0.1	0.2	0.2	0.2	mA	—
		During D/A conversion (per unit)			1.2	1.6	1.6	1.6	mA	—
		Waiting for A/D, D/A conversion (all units)			3.4	4.1	4.1	4.1	mA	—
		ADC16H, DAC12 in standby modes (all units)*2			1	16.0	22.4	42	μA	—
		Reference power supply current (VREFH0)	During 16-bit A/D conversion (unit 0)		SAR mode	A _I REFH0	70	120	120	120
	Oversampling mode and Hybrid mode			200	310		310	310	μA	—
	Waiting for 12-bit A/D conversion (unit 0)		14.00	14.00	14.00	14.00	μA	—		
	ADC16H in standby modes (unit 0)		0.01	0.12	0.1	0.2	μA	—		
	Reference power supply current (VREFH)	During 16-bit A/D conversion (unit 1)	SAR mode	A _I REFH	70	120	120	120	μA	—
			Oversampling mode and Hybrid mode		200	310	310	310	μA	—
		During D/A conversion (per unit)			29	41.0	41.0	41.0	μA	—
		Waiting for 16-bit A/D (unit 1), D/A (all units) conversion			14	14	14	14	μA	—
		ADC16H in standby modes (unit 1)			0.1	0.1	0.2	0.3	μA	—
	USB operating current	Low speed	USBFS	I _{CC} USBLS	2.9	4.0	4.0	4.0	mA	VCC_USB
		Full speed	USBFS	I _{CC} USBFS	4.0	4.7	4.7	4.7	mA	VCC_USB

Note 1. Supply current values are with all output pins unloaded and all input pull-up MOSs in the off state.

Note 2. When the MCU is in Software Standby mode or the MSTPCRD.MSTPD21 (16-Bit A/D Converter Module Stop bit) is in the module-stop state.

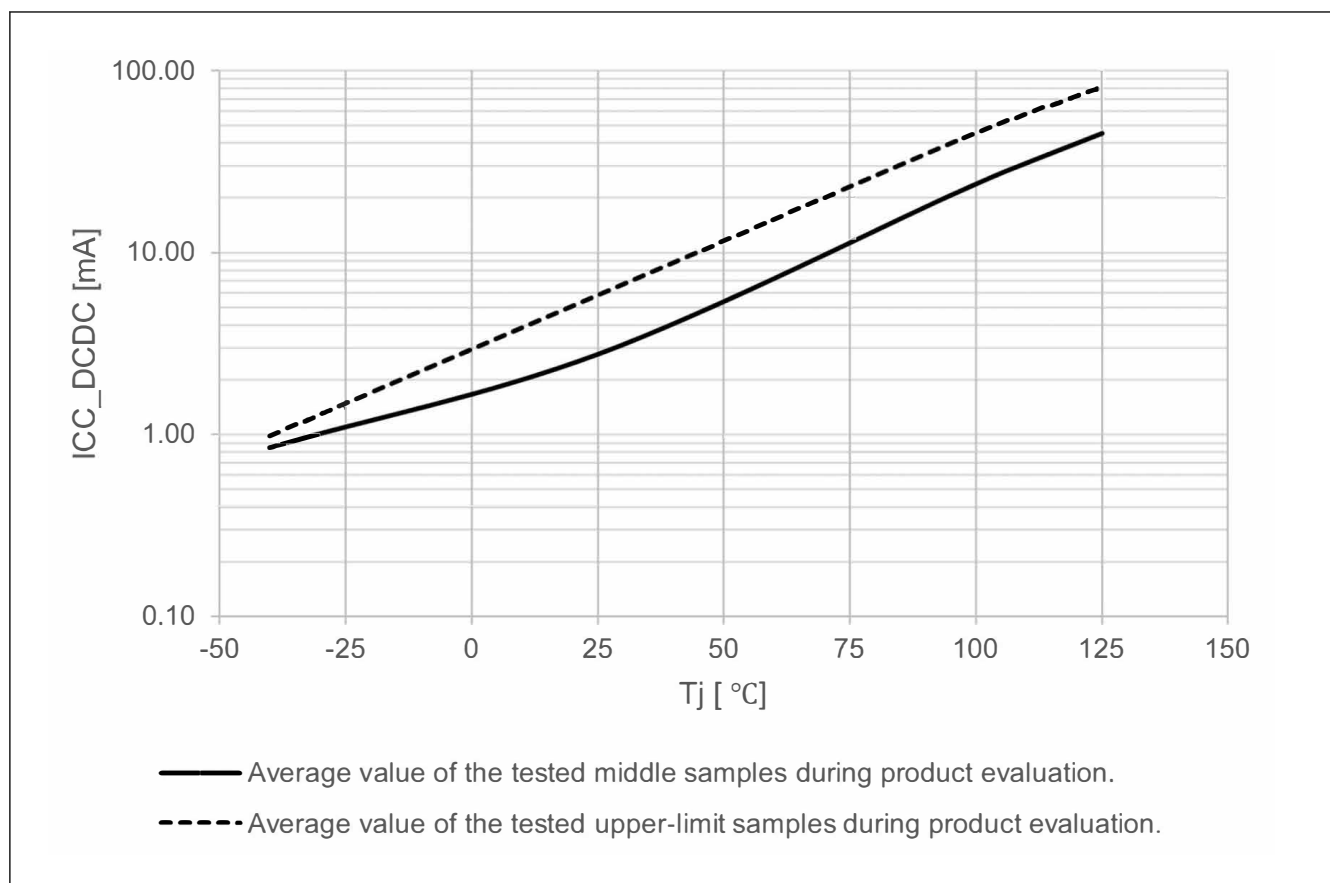


Figure 60.3 Temperature dependency in Software Standby mode (ICC_DCDC , $SS2LP_0$, $SVSCR_1$) (reference data)

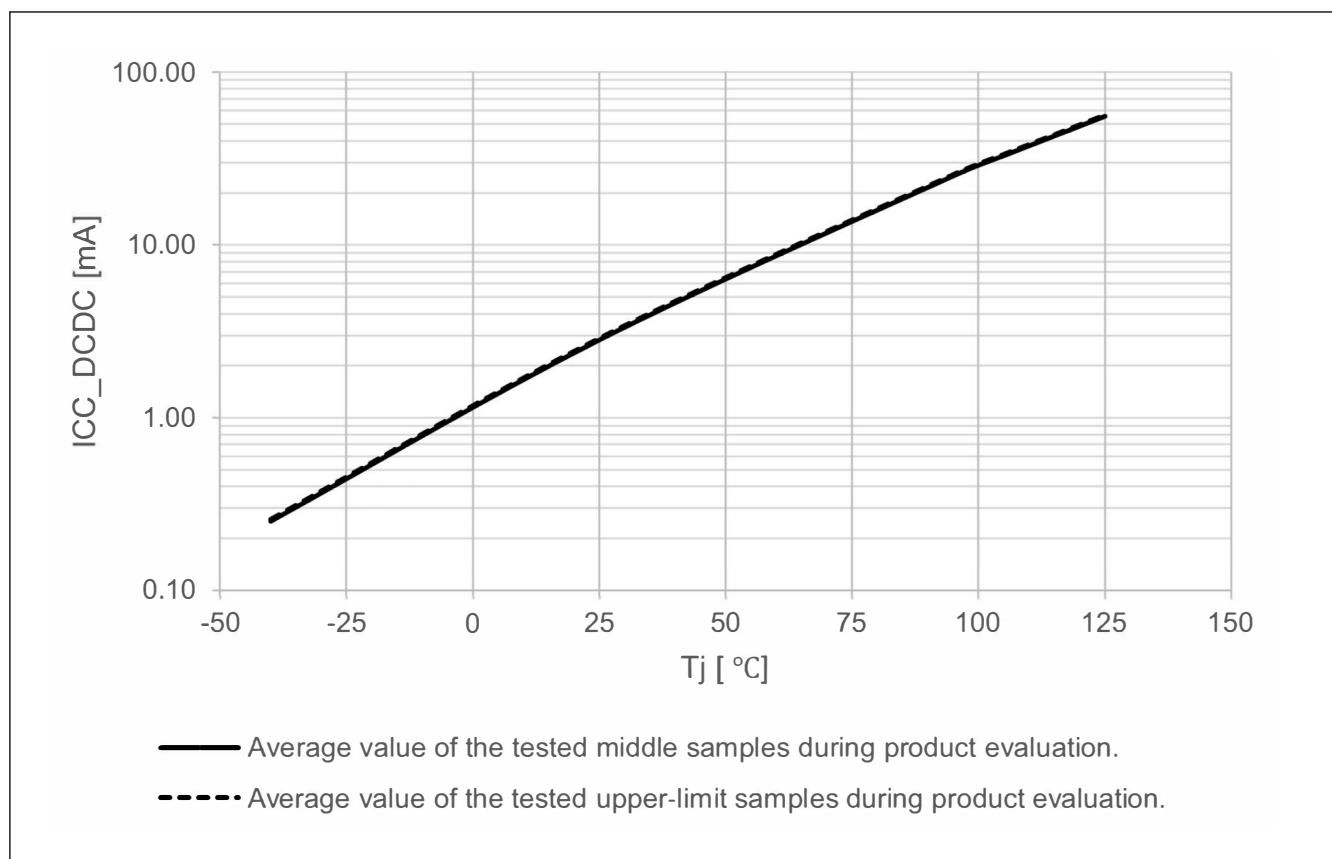


Figure 60.4 Temperature dependency in Software Standby mode (ICC_DCDC , $SS2LP_0$, $SVSCR_5$) (reference data)

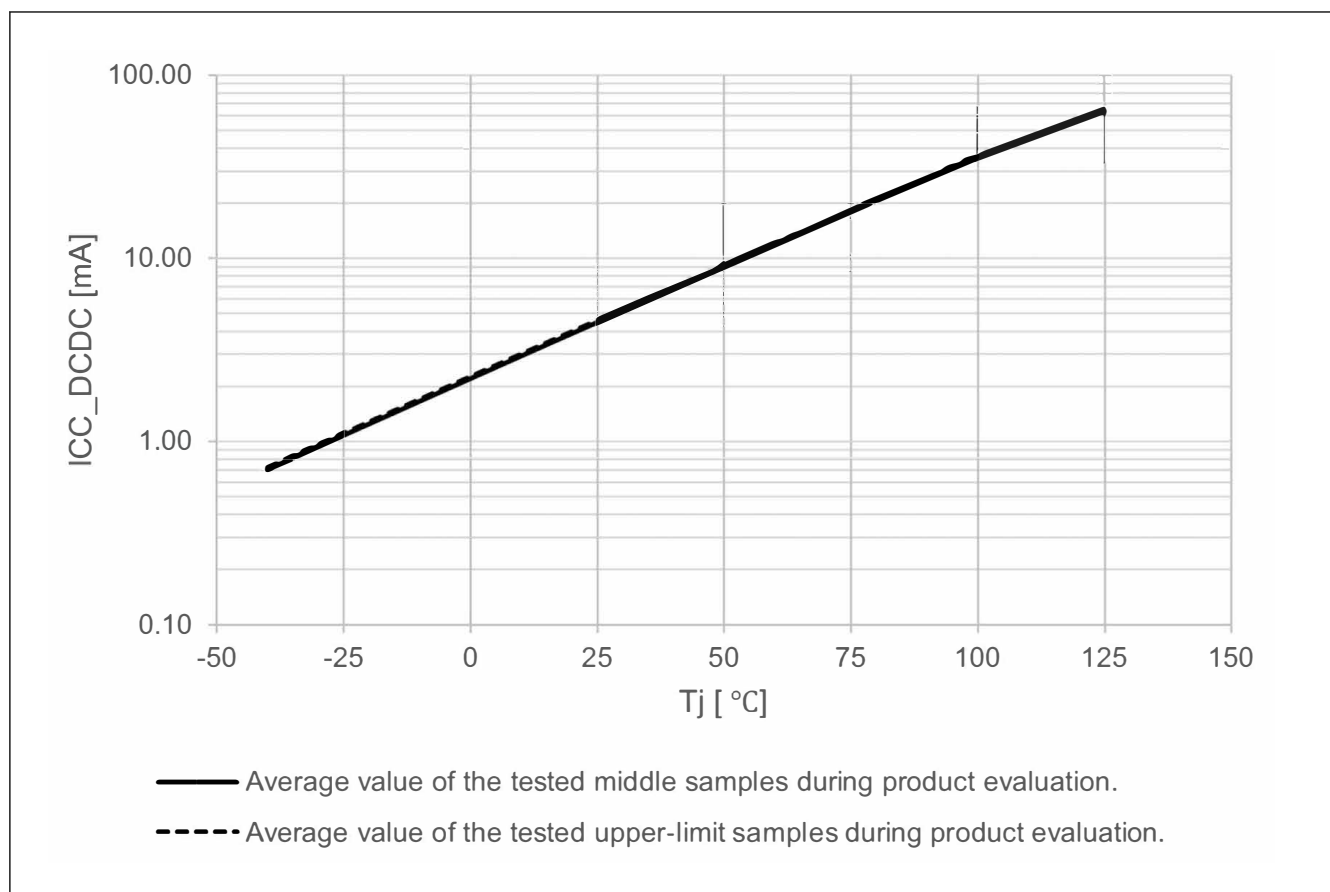


Figure 60.5 Temperature dependency in Software Standby mode (ICC_DCDC , $SS2LP_1$, $SVSCR_2$) (reference data)

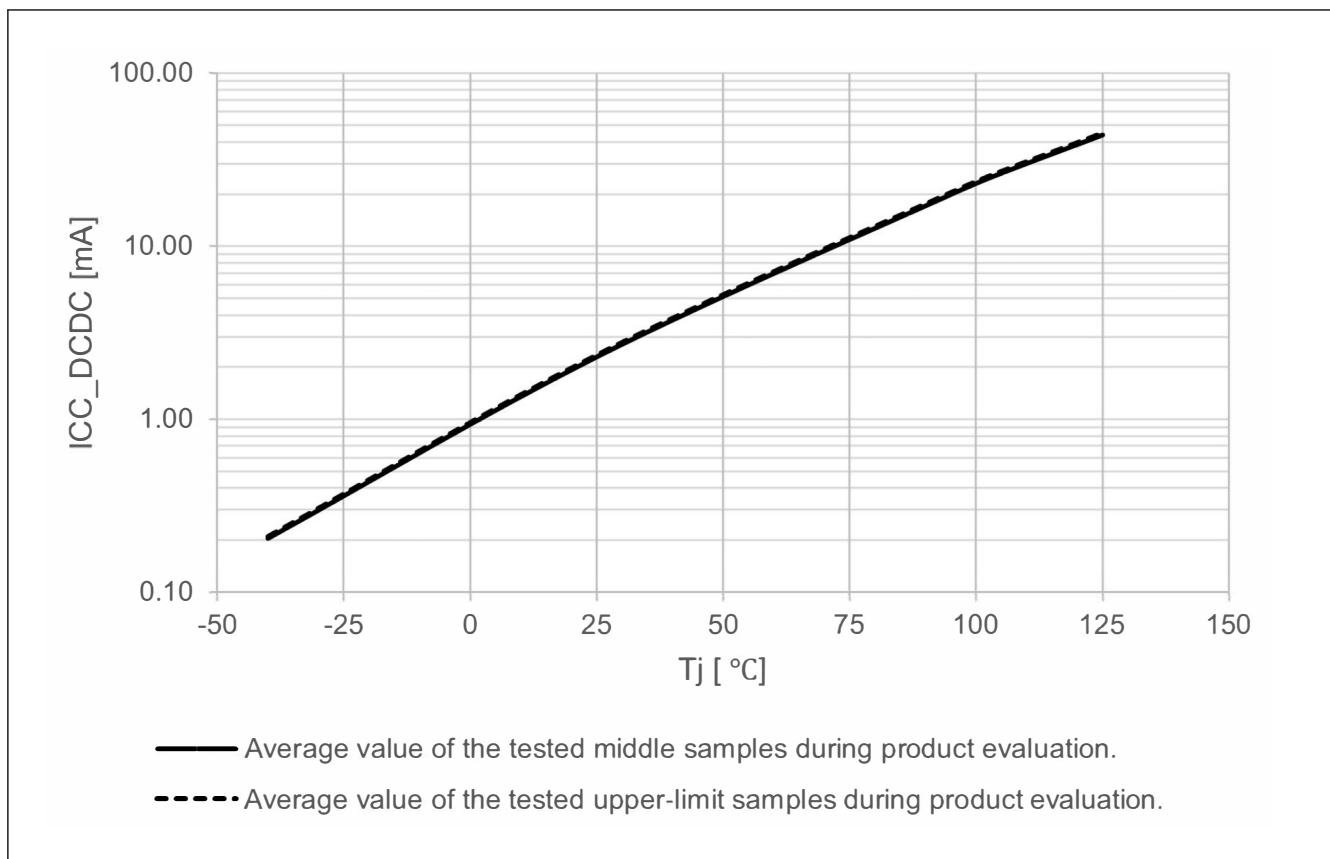


Figure 60.6 Temperature dependency in Software Standby mode (ICC_DCDC , $SS2LP_1$, $SVSCR_5$) (reference data)

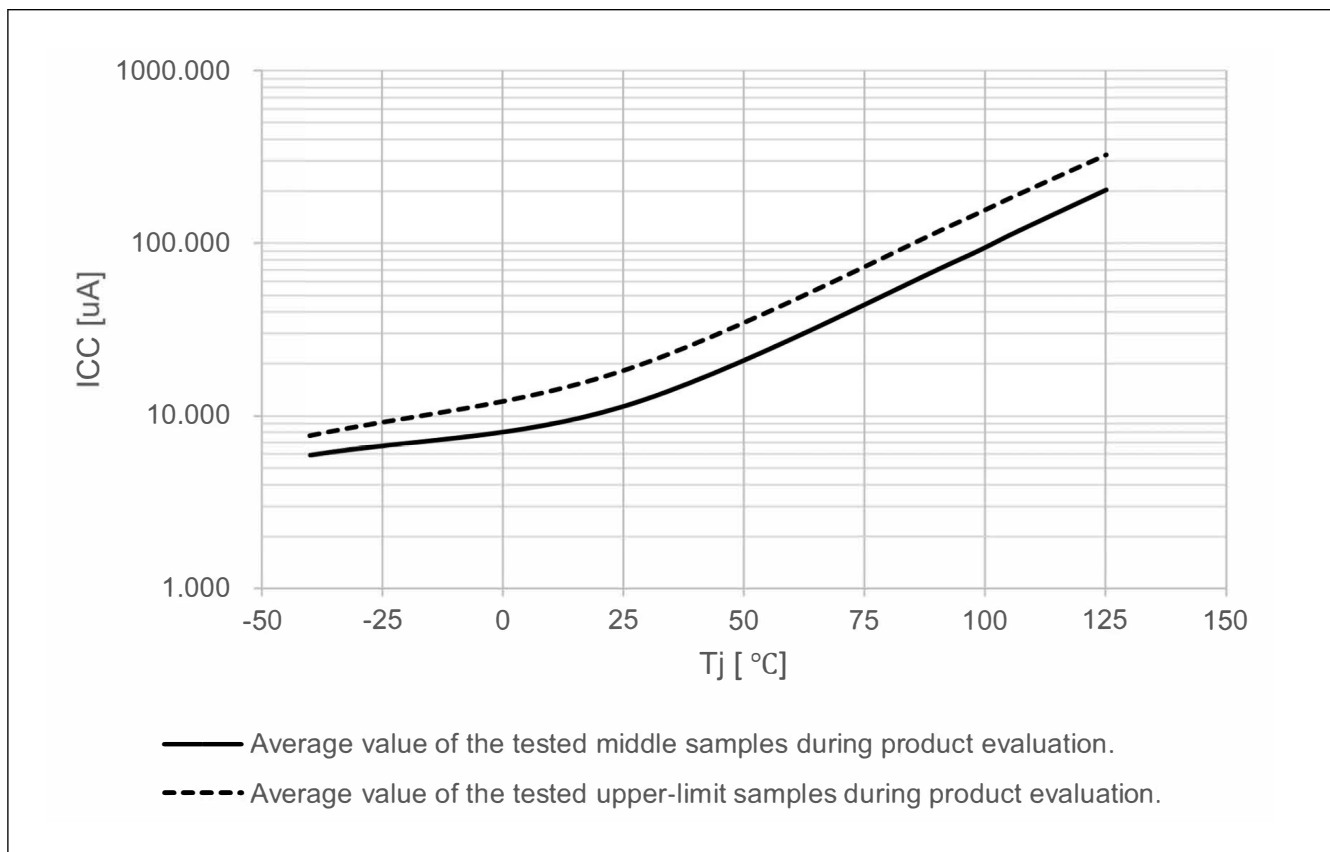


Figure 60.7 Temperature dependency in Deep Software Standby mode 1 (reference data)

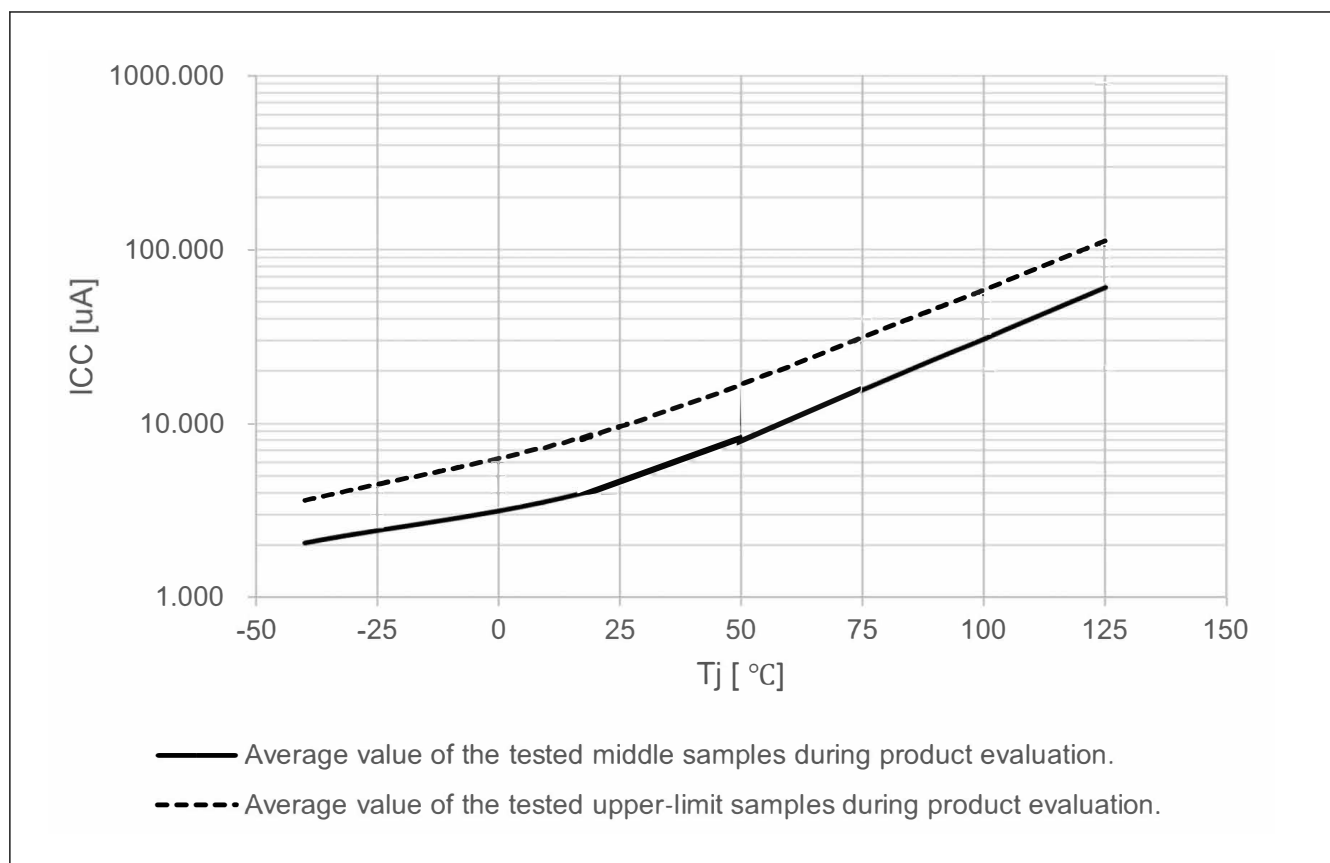


Figure 60.8 Temperature dependency in Deep Software Standby mode 2 (reference data)

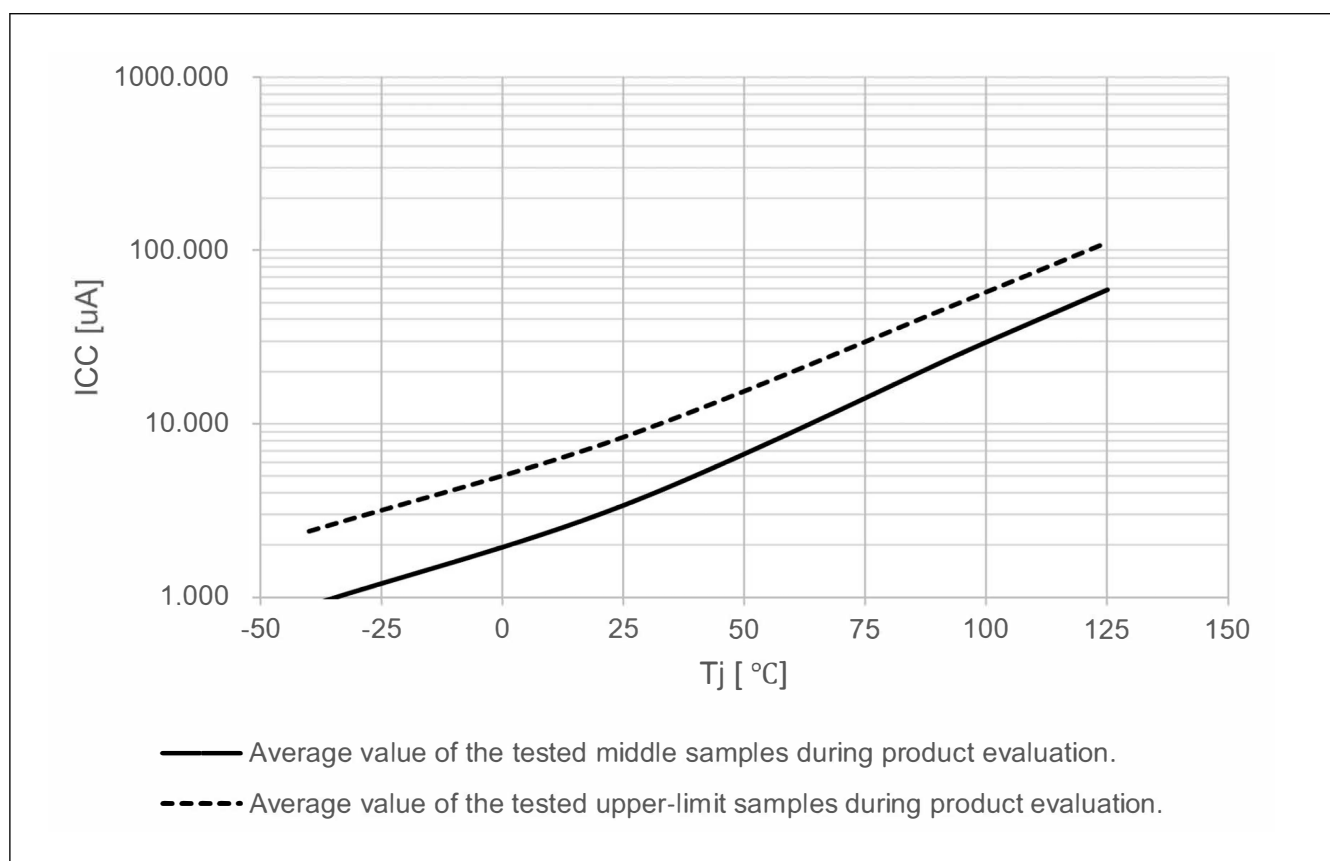


Figure 60.9 Temperature dependency in Deep Software Standby mode 3 (reference data)

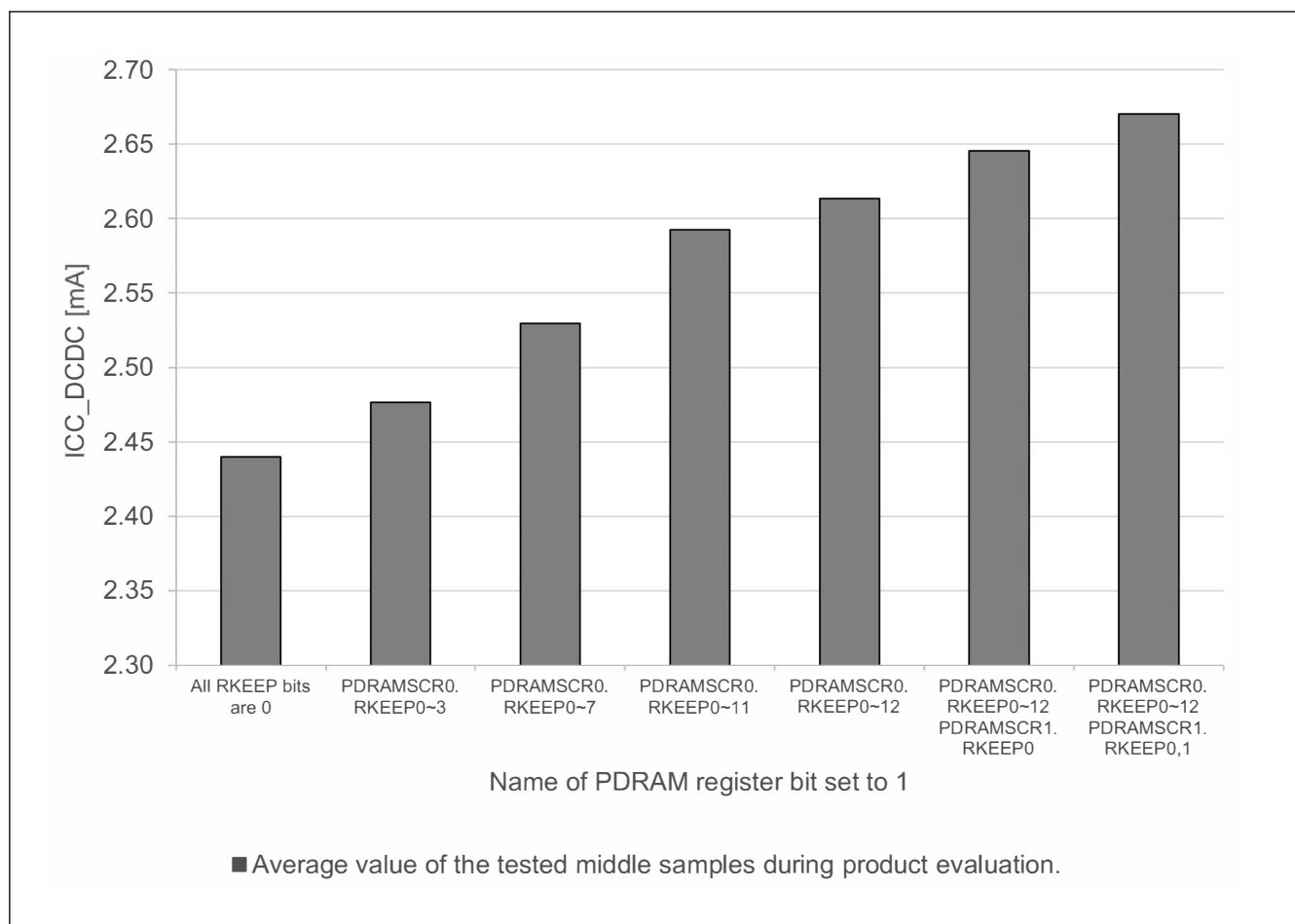


Figure 60.10 Software Standby current per SRAM state (ICC_DCDC , $SS2LP_0$, $SVSCR_1$) (reference data)

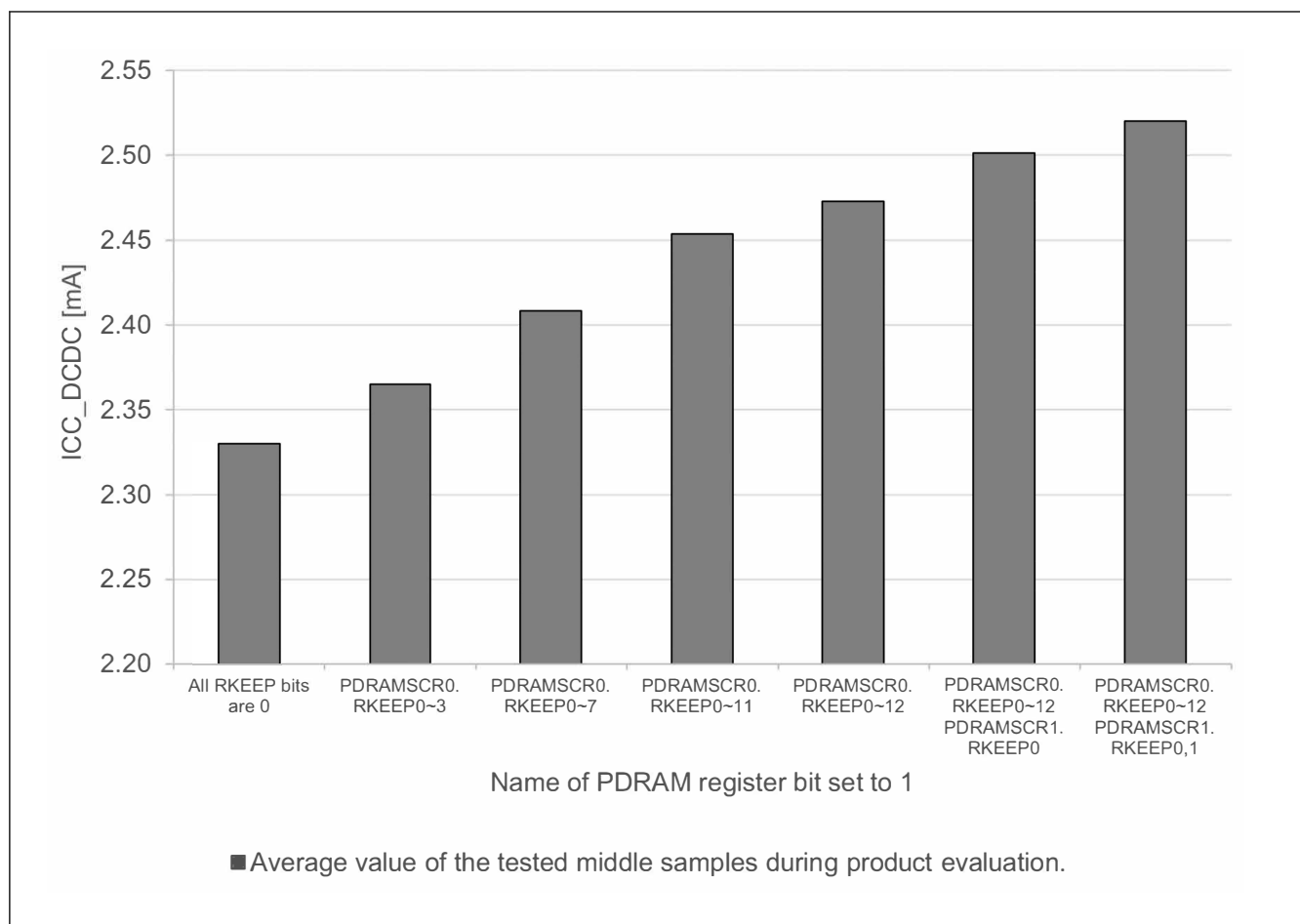


Figure 60.11 Software Standby current per SRAM state (ICC_DCDC, SS2LP_0, SVSCR_2) (reference data)

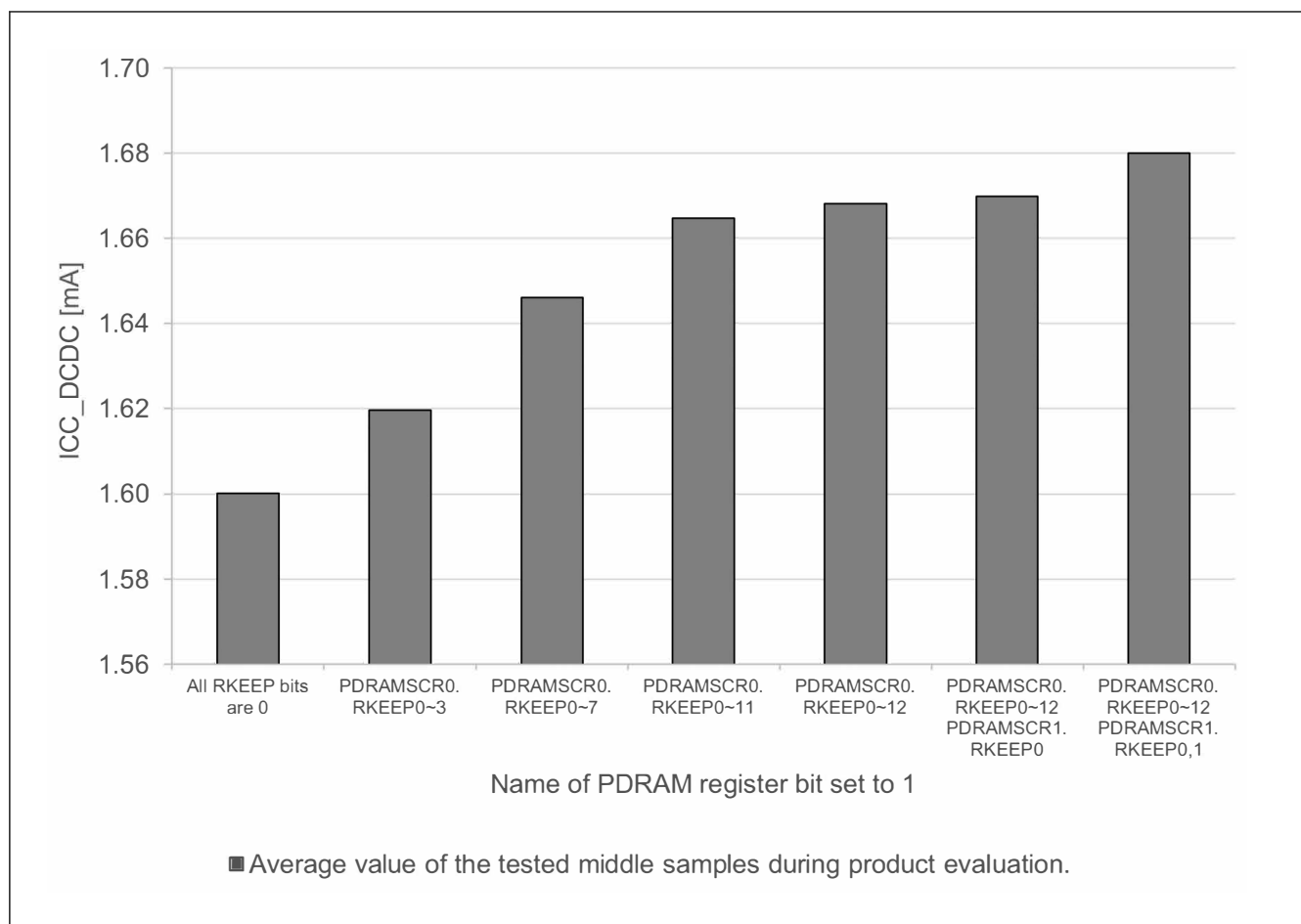


Figure 60.12 Software Standby current per SRAM state (ICC_DCDC, SS2LP_0, SVSCR_3) (reference data)

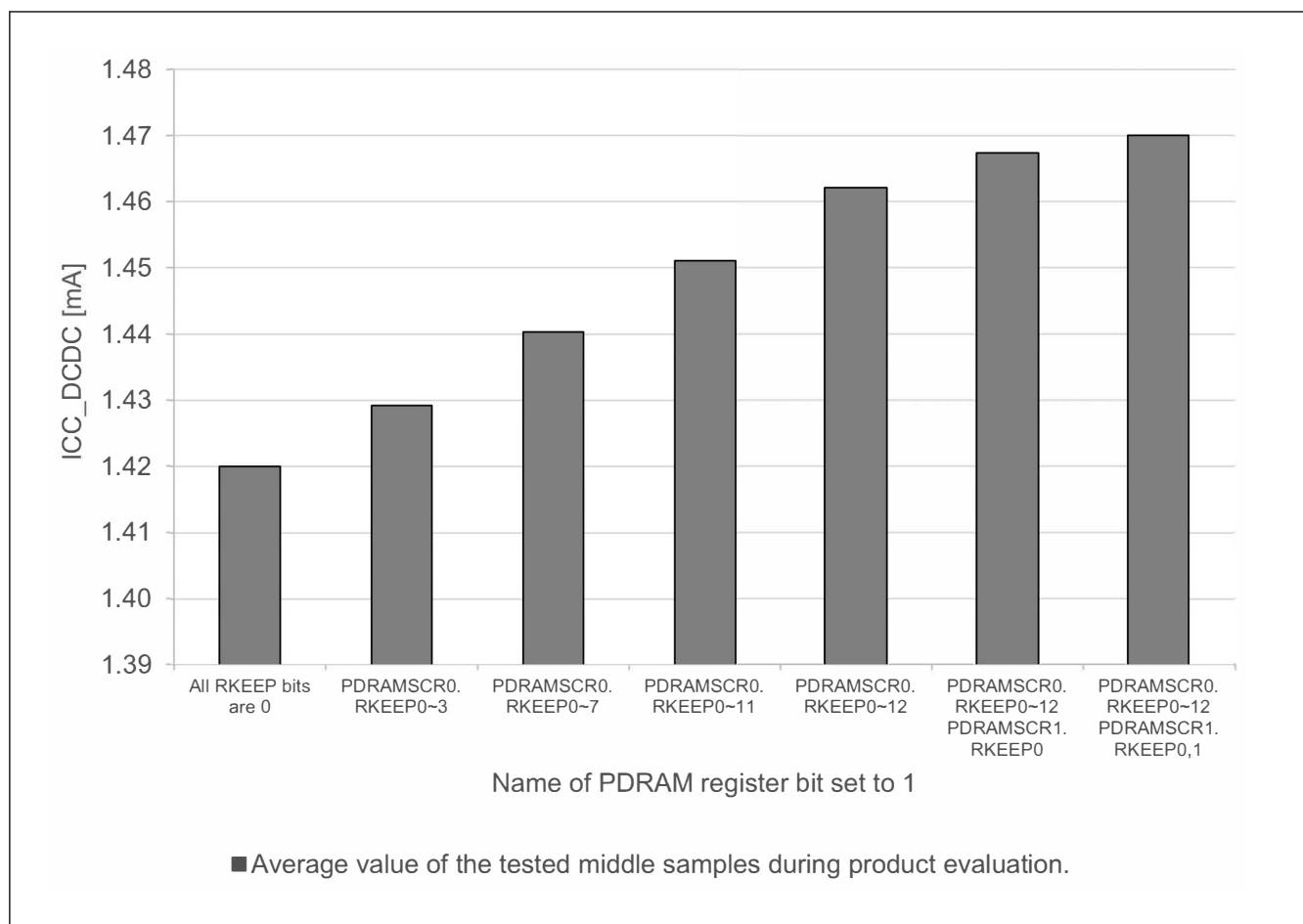


Figure 60.13 Software Standby current per SRAM state (ICC_DCDC, SS2LP_0, SVSCR_4) (reference data)

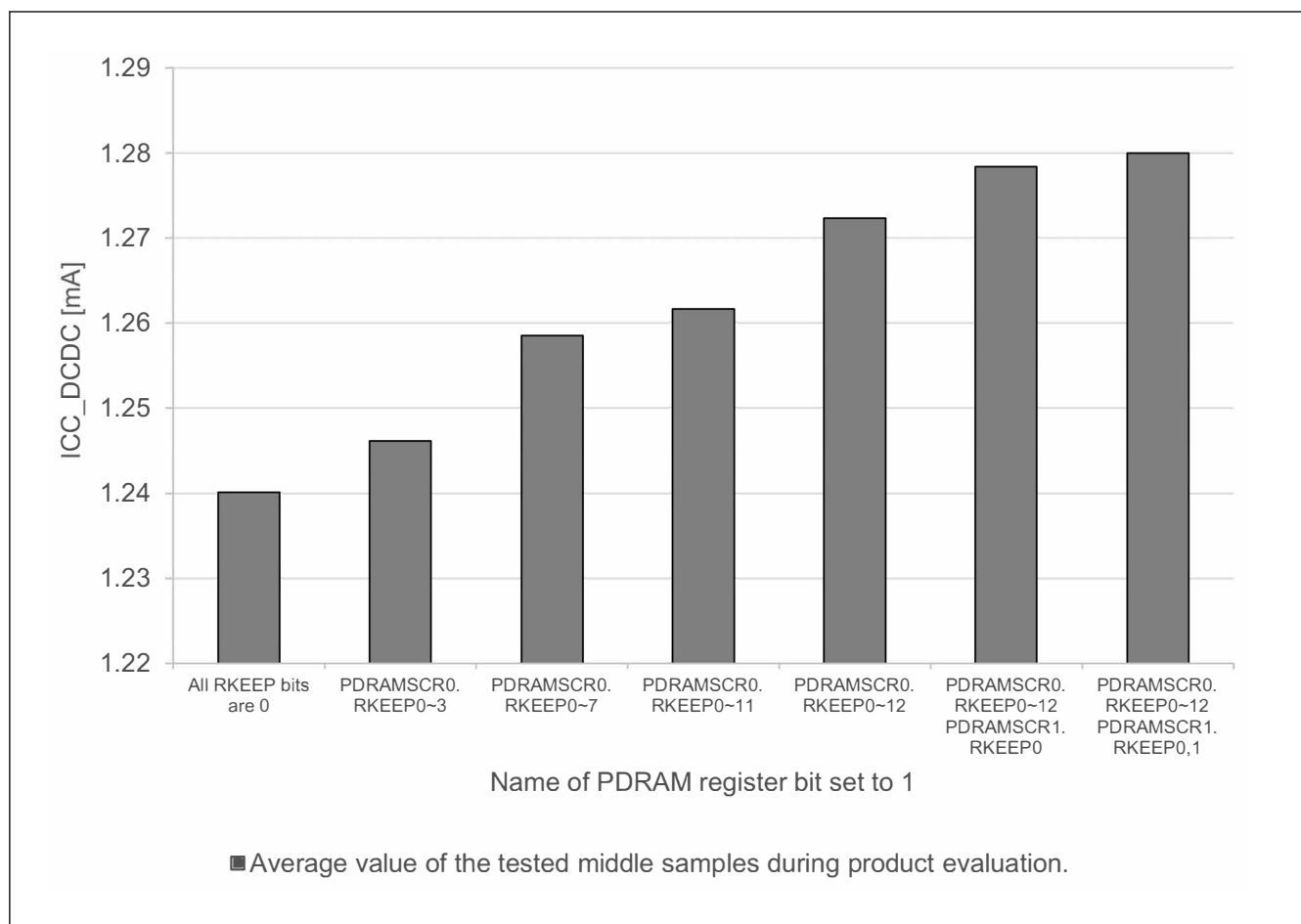


Figure 60.14 Software Standby current per SRAM state (ICC_DCDC, SS2LP_0, SVSCR_5) (reference data)

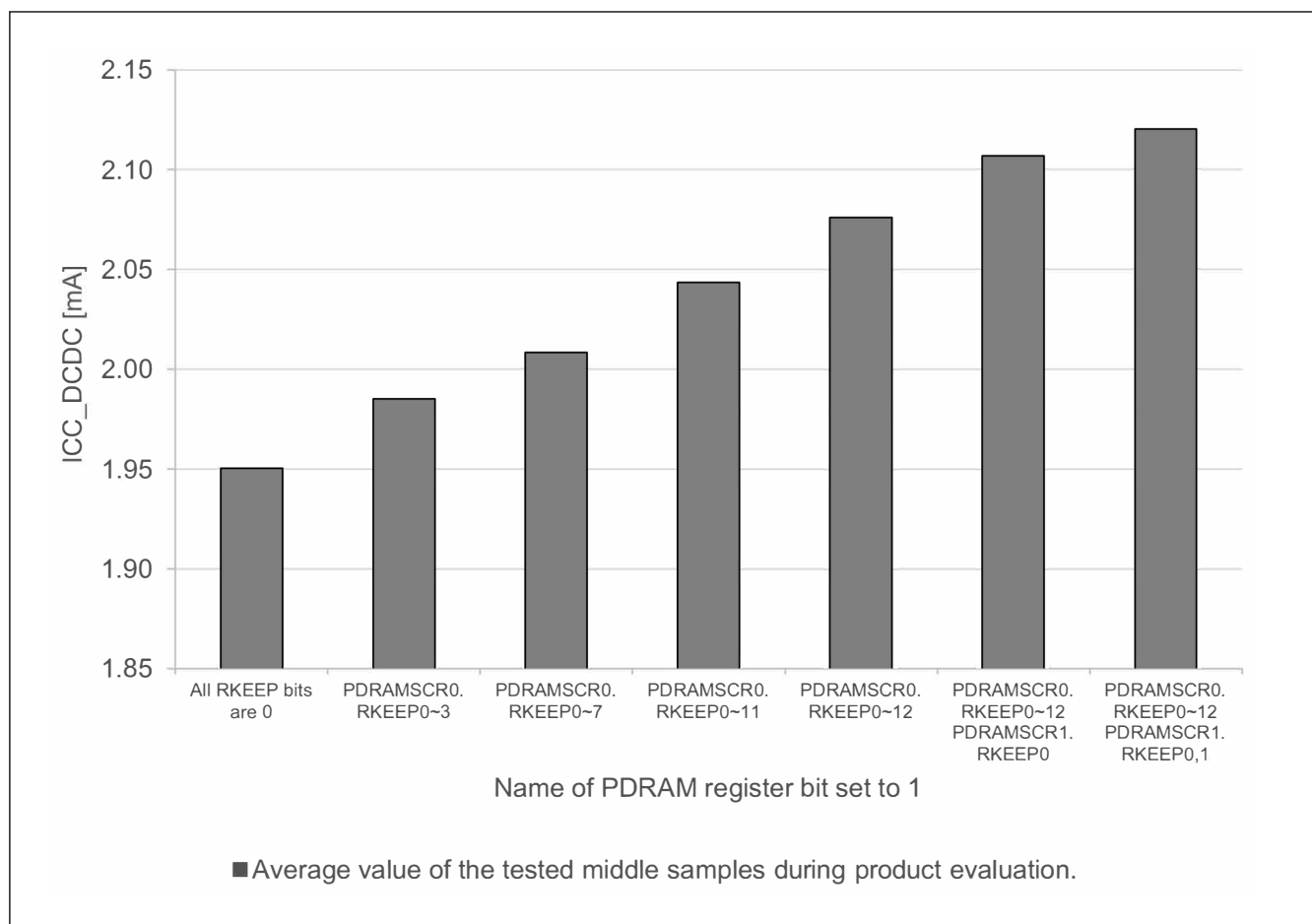


Figure 60.15 Software Standby current per SRAM state (ICC_DCDC , $SS2LP_1$, $SVSCR_2$) (reference data)

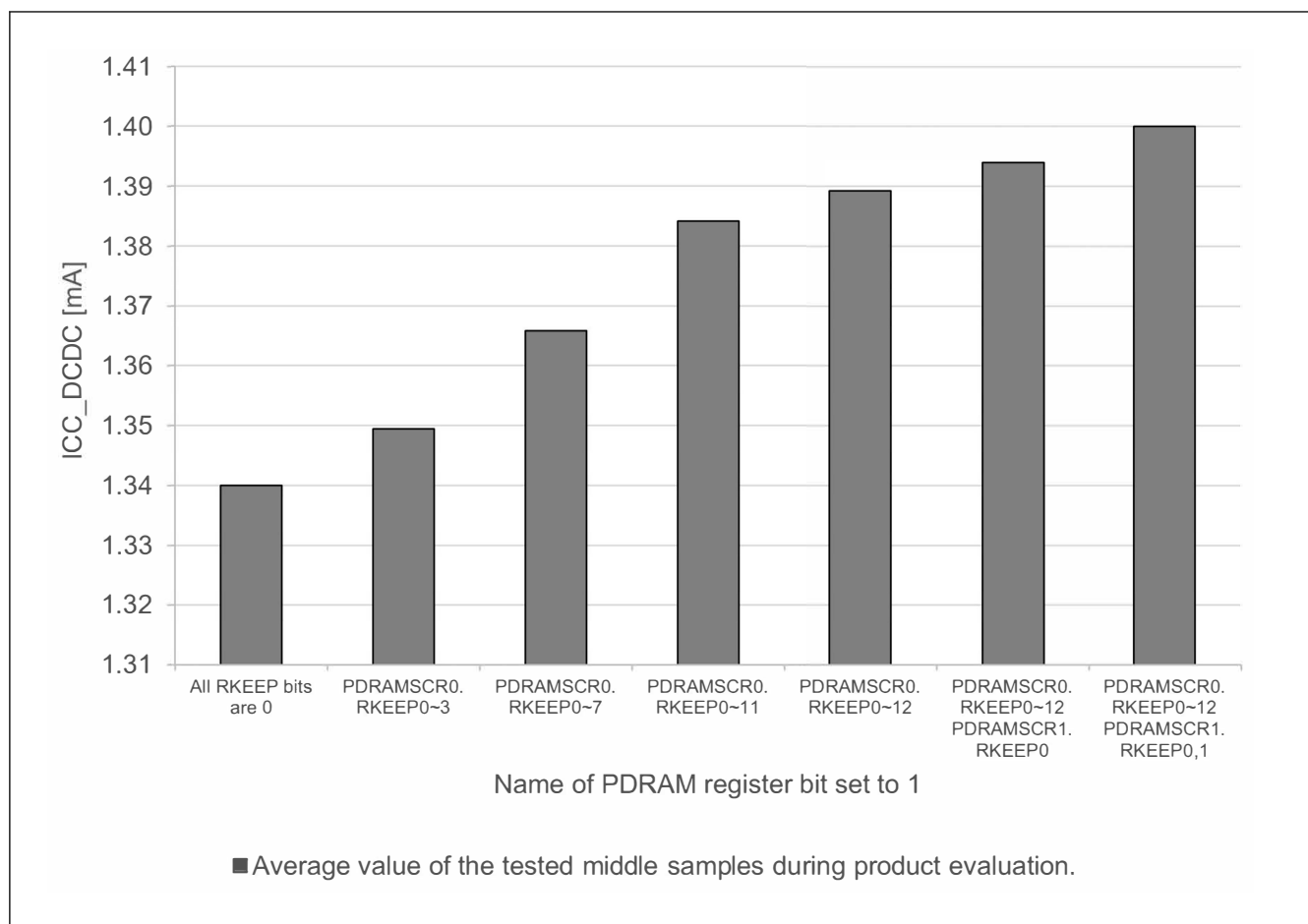


Figure 60.16 Software Standby current per SRAM state (ICC_DCDC, SS2LP_1, SVSCR_3) (reference data)

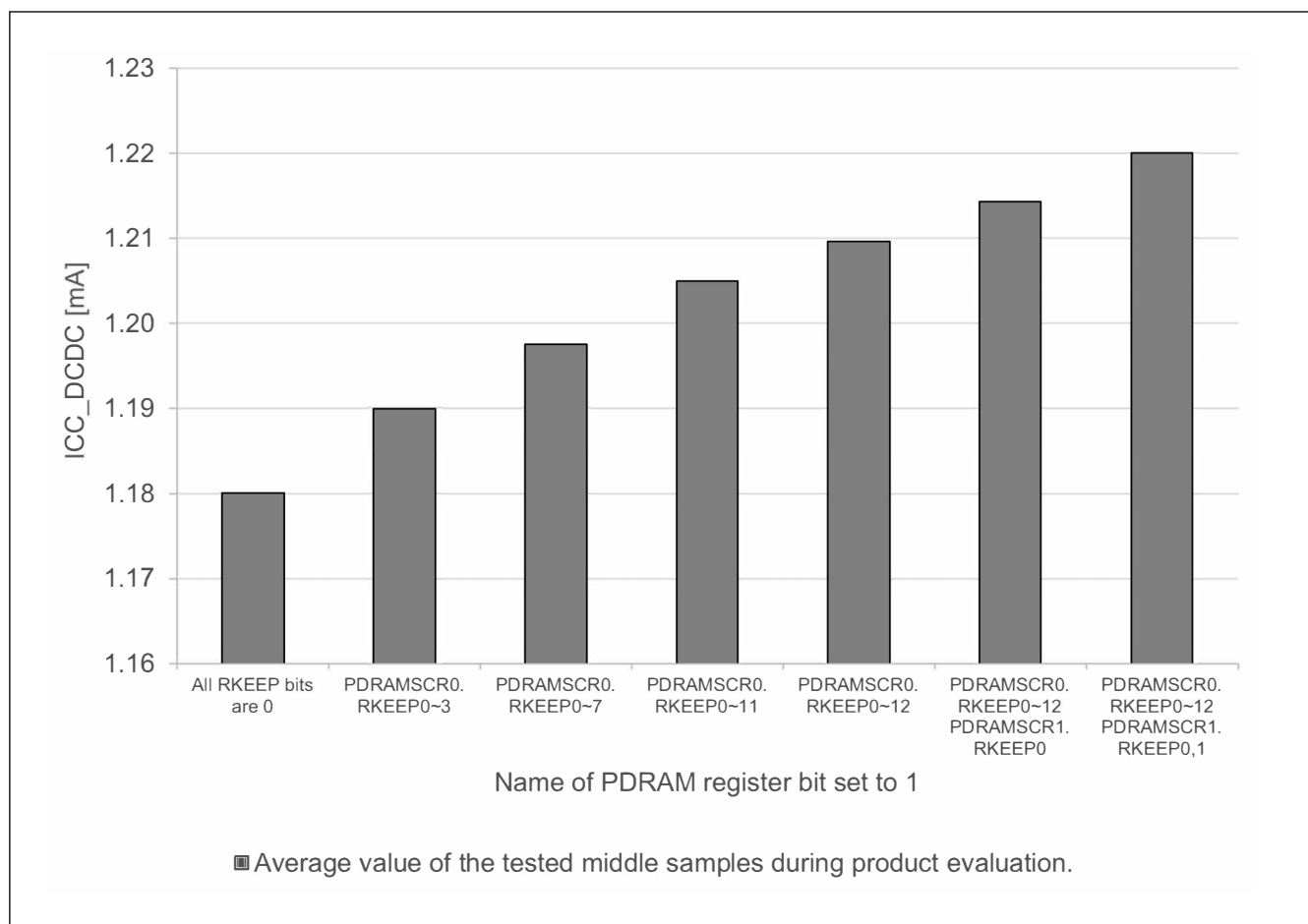


Figure 60.17 Software Standby current per SRAM state (ICC_DCDC , $SS2LP_1$, $SVSCR_4$) (reference data)

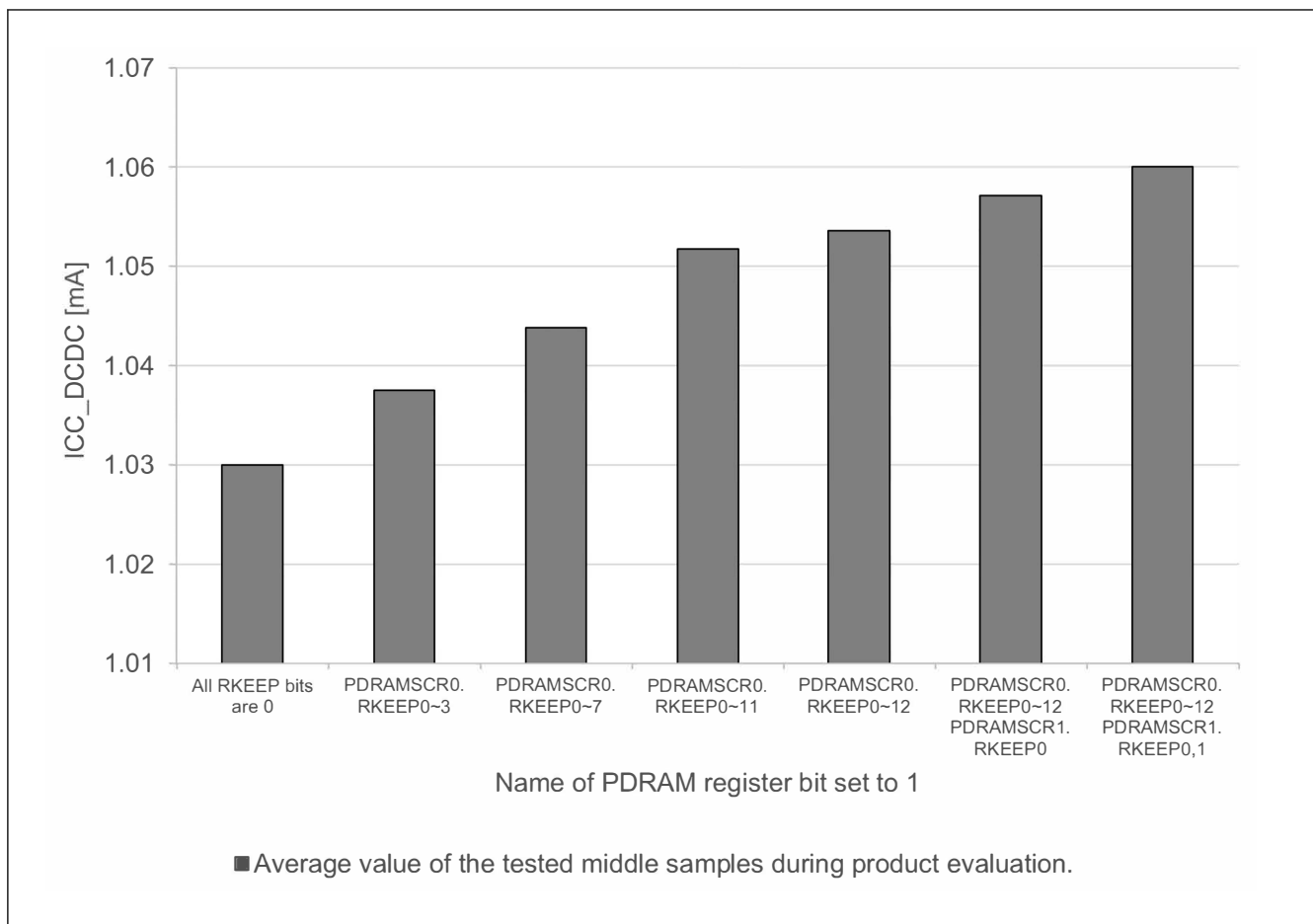


Figure 60.18 Software Standby current per SRAM state (ICC_DCDC, SS2LP_1, SVSCR_5) (reference data)

The more practical I_{CC_DCDC} value can be obtained with the following formula.

$$I_{CC_DCDC} = (I_{DD} \times V_{CL}) / (V_{CC_DCDC} \times \text{efficiency})$$

Where: V_{CL} and V_{CC} are the voltage of V_{CL} pin and V_{CC} pin respectively, and efficiency is shown in the following figures.

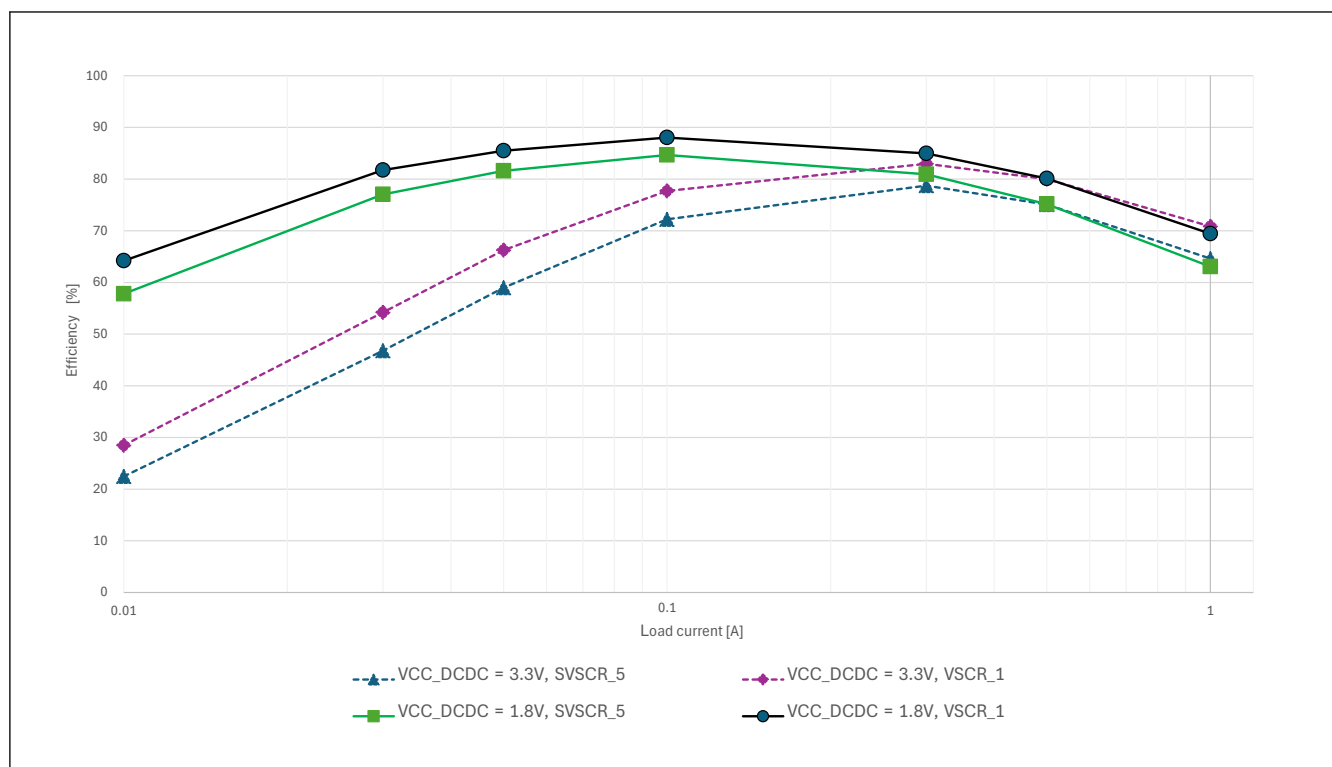


Figure 60.19 Typical DCDC efficiency (%) vs load current (A) in High-speed mode and Software Standby mode (SSCR1.SS2LP = SS2LP_1), Tj = 25 °C

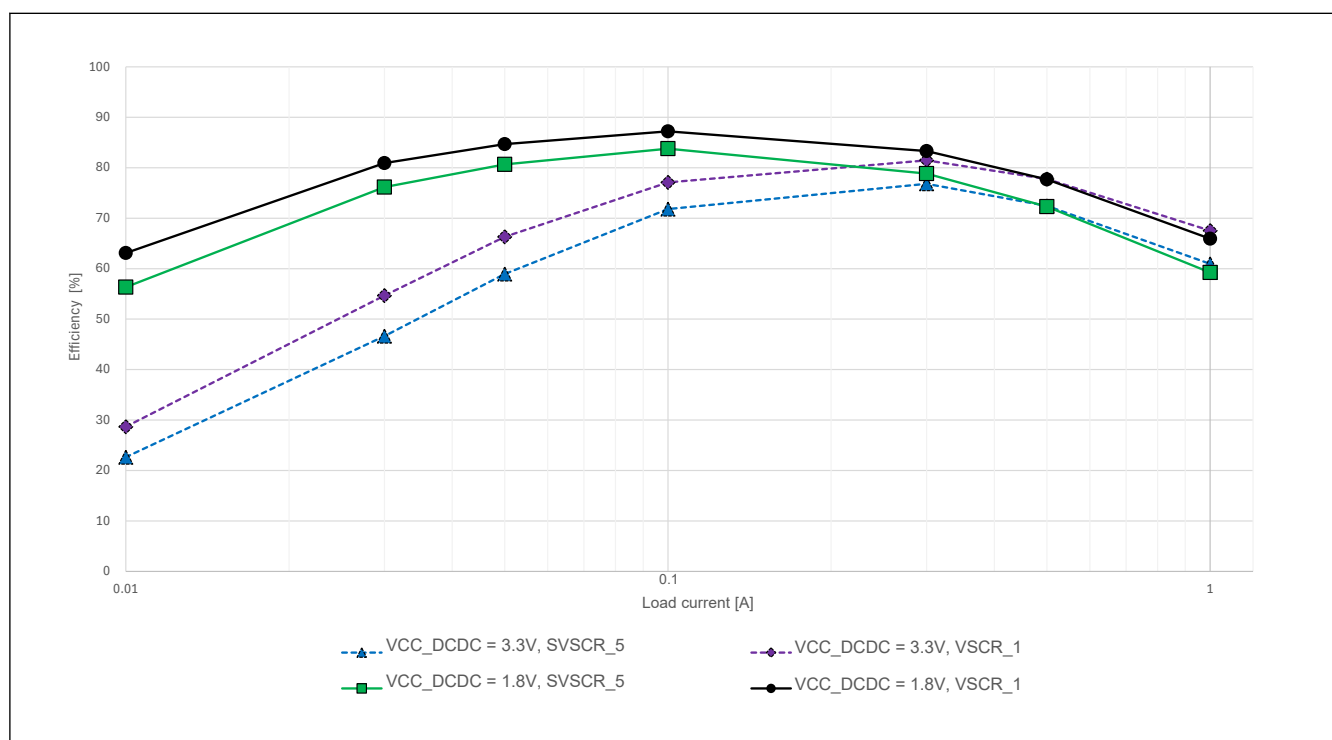


Figure 60.20 Typical DCDC efficiency (%) vs load current (A) in High-speed mode and Software Standby mode (SSCR1.SS2LP = SS2LP_0), Tj = 125 °C

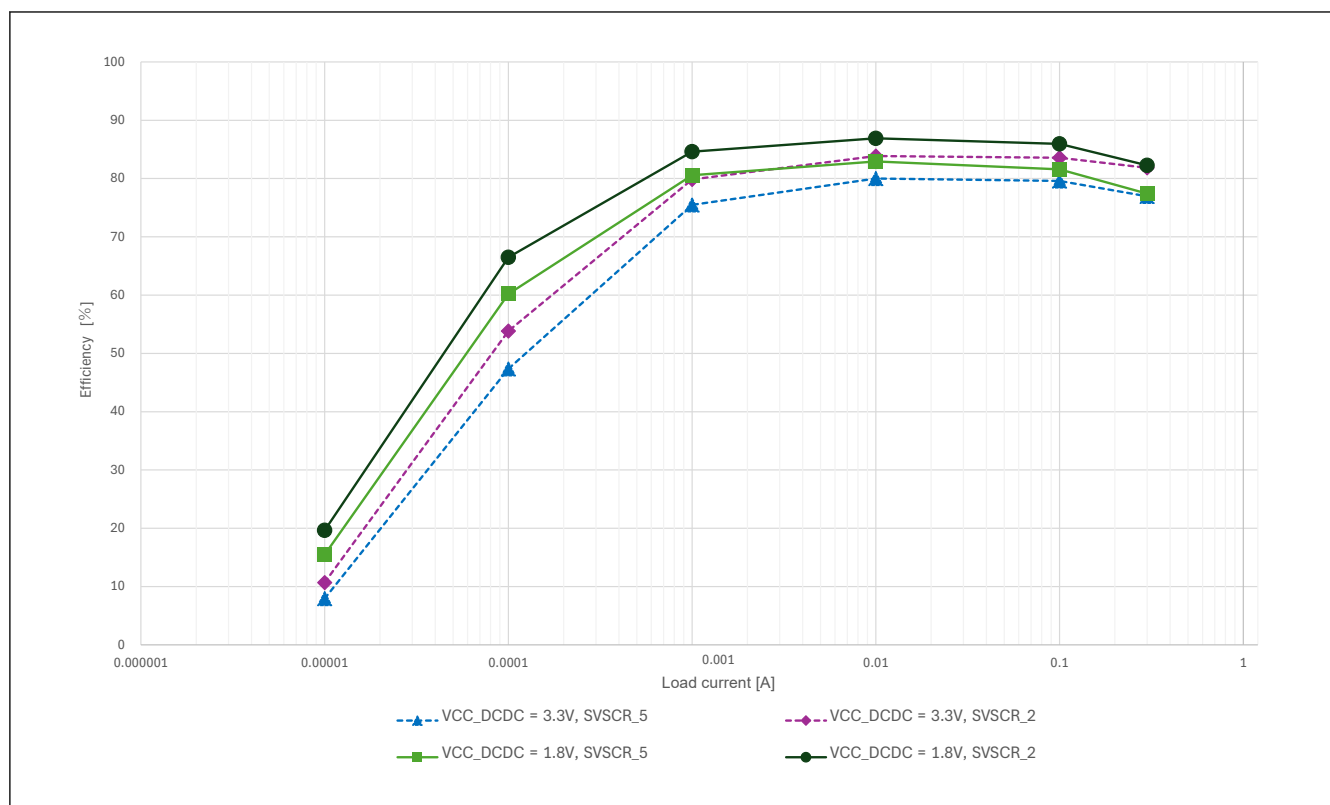


Figure 60.21 Typical DCDC efficiency (%) vs load current (A) in Software Standby mode (SSCR1.SS2LP = SS2LP_1), Tj = 25 °C

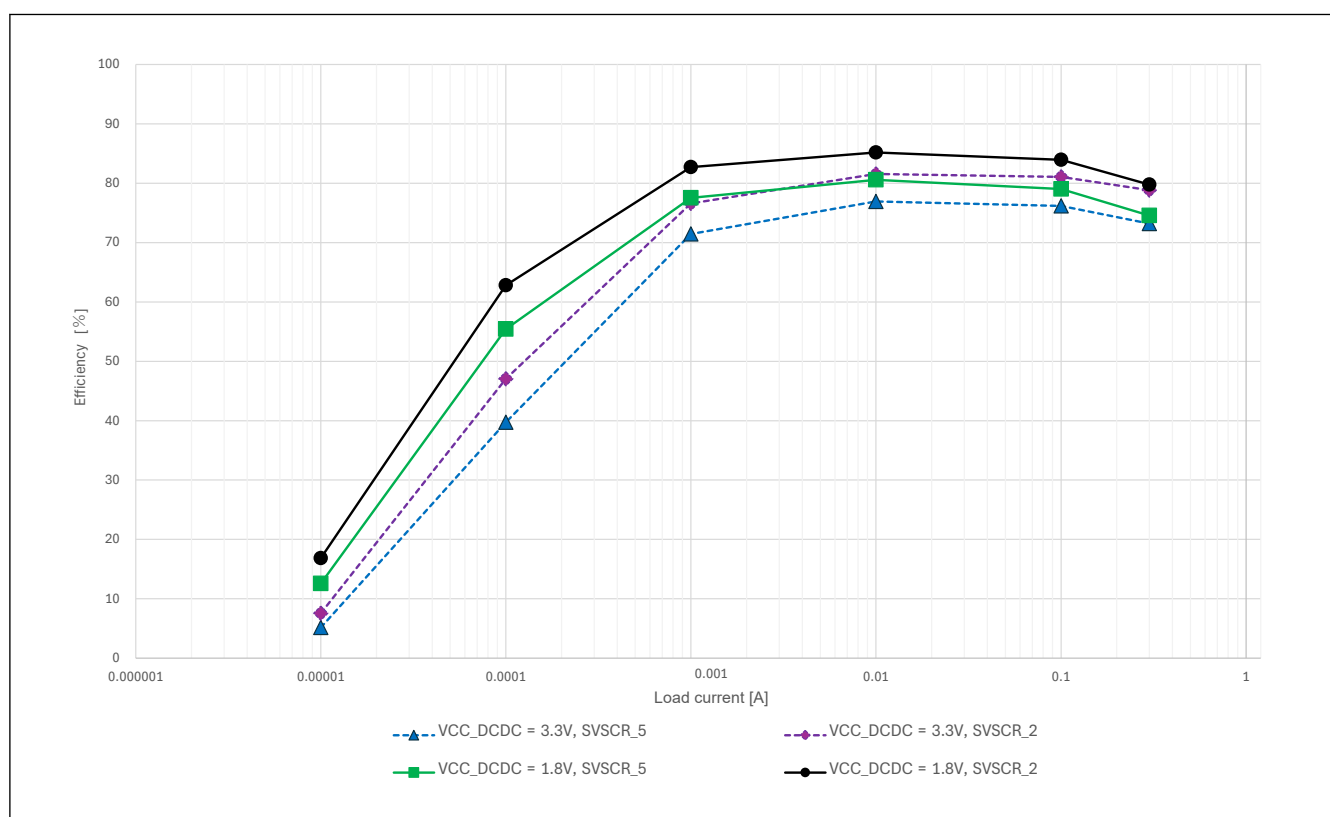


Figure 60.22 Typical DCDC efficiency (%) vs load current (A) in Software Standby mode (SSCR1.SS2LP = SS2LP_1), Tj = 125 °C

60.2.6 VCC Rise and Fall Gradient and Ripple Frequency

Table 60.40 VCC rise and fall gradient characteristics at power on/off

Parameter		Symbol	Min	Typ	Max	Unit	Test conditions
VCC rising gradient at power on ^{*1}		SrVCC	0.0084	—	20	ms/V	—
VCC falling gradient at power off	VBATT function is disabled ^{*1}	SfVCC1	0.0084	—	—	ms/V	—
	VBATT function is enabled.	SfVCC2	1.0000	—	—	—	—

Note 1. In case the VCC voltage crosses V_{POR1}

Table 60.41 VCC ripple frequency and gradient characteristics during operation

The ripple voltage must meet the allowable ripple frequency $f_{r(VCC)}$ within the range between the VCC upper limit (3.63 V) and lower limit (1.62 V). When the VCC change exceeds $VCC \pm 10\%$, the allowable voltage change rising and falling gradient $dt/dVCC$ must be met.

Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
Allowable ripple frequency	$f_{r(VCC)}$	—	—	10	kHz	Figure 60.23 $V_{r(VCC)} \leq VCC \times 0.2$
		—	—	1	MHz	Figure 60.23 $V_{r(VCC)} \leq VCC \times 0.08$
		—	—	10	MHz	Figure 60.23 $V_{r(VCC)} \leq VCC \times 0.06$
Allowable voltage change rising and falling gradient	$dt/dVCC^{*1}$	1.0	—	—	ms/V	When VCC change exceeds $VCC \pm 10\%$

Note 1. In case the VCC voltage does not cross V_{POR1} .

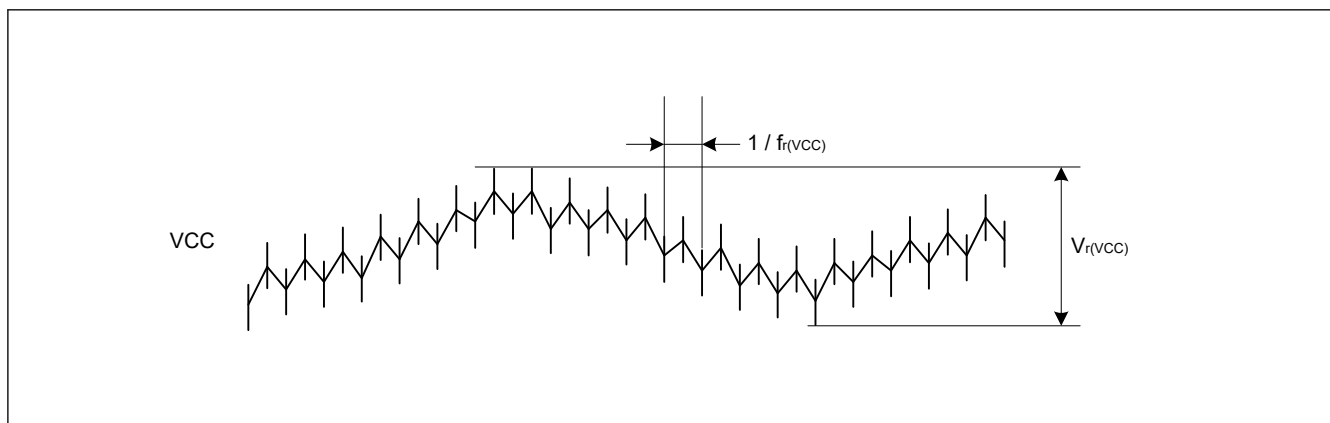


Figure 60.23 Ripple waveform

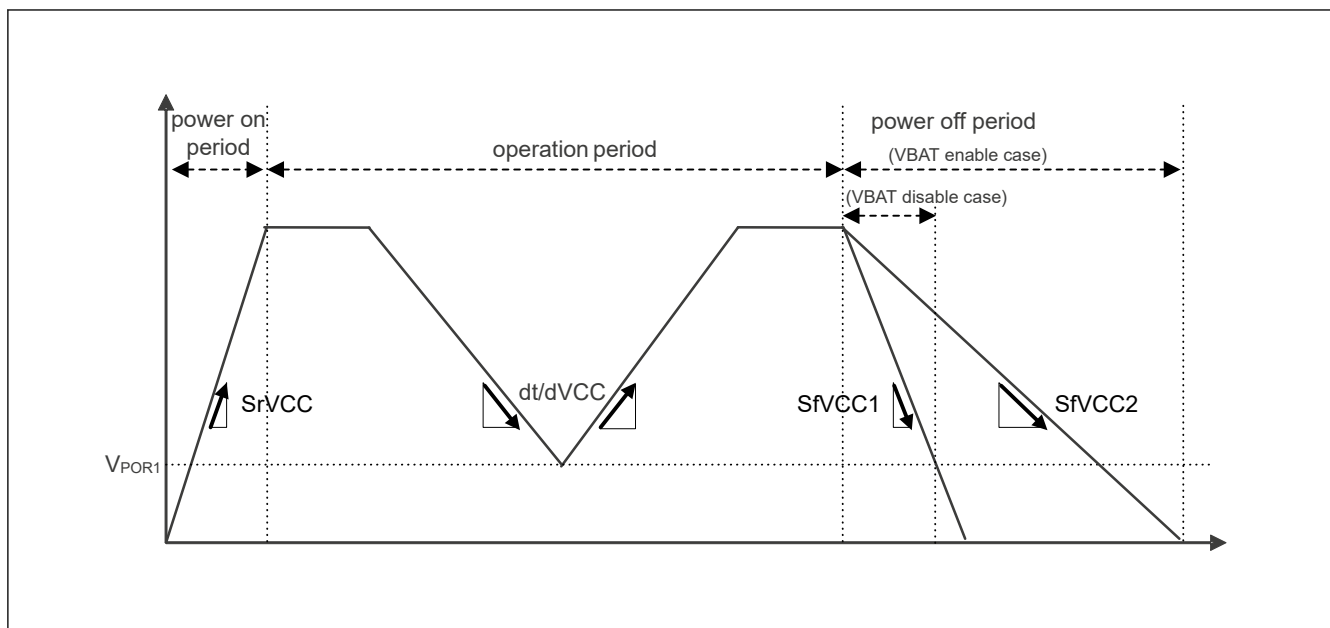


Figure 60.24 VCC rising and falling waveform

60.2.7 Power Supply Rise Gradient

Table 60.42 VCC_USB rise gradient characteristics at power on

Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
VCC_USB rising gradient at power on	SrVCC_USB	8.4	—	—	μs/V	—

60.2.8 Thermal Characteristics

Maximum value of junction temperature (T_j) must not exceed the value of [section 60.2.1. \$T_j/T_a\$ Definition.](#)

T_j is calculated by either of the following equations.

- $T_j = T_a + \theta_{ja} \times \text{Total power consumption}$
- $T_j = T_t + \Psi_{jt} \times \text{Total power consumption}$
 - T_j : Junction Temperature ($^{\circ}\text{C}$)
 - T_a : Ambient Temperature ($^{\circ}\text{C}$)
 - T_t : Top Center Case Temperature ($^{\circ}\text{C}$)
 - θ_{ja} : Thermal Resistance of “Junction”-to-“Ambient” ($^{\circ}\text{C}/\text{W}$)
 - Ψ_{jt} : Thermal Resistance of “Junction”-to-“Top Center Case” ($^{\circ}\text{C}/\text{W}$)
- Total power consumption = Voltage $\times$ (Leakage current + Dynamic current)
- Leakage current of IO = $\Sigma (I_{OL} \times V_{OL}) / \text{Voltage} + \Sigma (|I_{OH}| \times |V_{CC} - V_{OH}|) / \text{Voltage}$
- Dynamic current of IO = $\Sigma \text{IO} (C_{in} + C_{load}) \times \text{IO switching frequency} \times \text{Voltage}$
 - C_{in} : Input capacitance
 - C_{load} : Output capacitance

Regarding θ_{ja} and Ψ_{jt} , see [Table 60.43](#).

Table 60.43 Thermal Resistance

Parameter	Package	Symbol	Value*1	Unit	Test conditions
Thermal Resistance	176-pin HLQFP (PLQP0176K?-A)	θ_{ja}	15	°C/W	JESD 51-2 and 51-7 compliant
	224-pin BGA (PLBG0224J?-A)		21		JESD 51-2 and 51-9 compliant
	289-pin BGA (PLBG0289J?-A)		20		
	303-pin BGA (PLBG0303G?-A)		17		
	176-pin HLQFP (PLQP0176K?-A)	Ψ_{jt}	0.3	°C/W	JESD 51-2 and 51-7 compliant
	224-pin BGA (PLBG0224J?-A)		0.3		JESD 51-2 and 51-9 compliant
	289-pin BGA (PLBG0289J?-A)		0.3		
	303-pin BGA (PLBG0303G?-A)		0.5		

Note 1. The values are reference values when the 4-layer board is used. Thermal resistance depends on the number of layers or size of the board. For details, see the JEDEC standards.

60.2.8.1 Calculation Guide of Maximum Current

Table 60.44 Power consumption of each unit (DCDC mode) (1 of 3)

Dynamic current/ Leakage current	MCU Domain	Category	Item	Symbol	Frequency [MHz]	Current [μ A/MHz]	Current [mA]	Condition
Leakage current	Analog	Regulator and Leak*1	Tj = 95°C	I _{CC}	—	—	2.51	—
			Tj = 105°C		—	—	2.91	
			Tj = 115°C		—	—	3.45	
			Tj = 125°C		—	—	4.15	
			Tj = 95°C	I _{CC_DCDC}	—	—	186	VCC_DCDC = 3.3 V, VSCR_1, PDCTRESWM .PDDE = 0
			Tj = 105°C		—	—	216	
			Tj = 115°C		—	—	243	
			Tj = 125°C		—	—	277	
			Tj = 95°C		—	—	363	VCC_DCDC = 1.8 V, VSCR_1, PDCTRESWM .PDDE = 0
			Tj = 105°C		—	—	421	
			Tj = 115°C		—	—	474	
			Tj = 125°C		—	—	540	
			Tj = 95°C	I _{DD}	—	—	477	VSCR_1 PDCTRESWM .PDDE = 0
			Tj = 105°C		—	—	553	
			Tj = 115°C		—	—	622	
			Tj = 125°C		—	—	709	

Table 60.44 Power consumption of each unit (DCDC mode) (2 of 3)

Dynamic current/ Leakage current	MCU Domain	Category	Item	Symbol	Frequency [MHz]	Current [μ A/MHz]	Current [mA]	Condition
Dynamic current	CPU0	Operation with Cache	CoreMark	I_{DD}	1000	145	144	CPUCLK0 = 1000 MHz VSCR_1
	CPU1	Operation with Cache	CoreMark		250	166	41	CPUCLK1 = 250 MHz VSCR_1
	Peripheral Unit	Timer	RTC		62.5	1.229	0.077	VSCR_1
			GPT32 (14ch)*2		125	65.123	8.140	
			POEG (4 Groups)*2		62.5	1.539	0.096	
			PDG (4ch)*2		125	47.465	5.933	
			AGT (2ch)*2		62.5	1.518	0.095	
			ULPT (2ch)*2		62.5	2.373	0.148	
			WDT0		62.5	0.437	0.027	
			WDT1		62.5	0.446	0.028	
			IWDT		62.5	0.014	0.001	
Dynamic current	Peripheral Unit	Communication interfaces	ESWM	I_{DD}	125	294.026	36.753	VSCR_1
			ESC		125	14.095	1.762	
			USBFS		62.5	7.495	0.468	
			SCI (10ch)*2		125	32.336	4.042	
			IIC (3ch)*2		62.5	3.722	0.233	
			I3C		125	9.883	1.235	
			CANFD (2ch)*2		125	6.025	0.753	
			SPI (2ch)*2		125	11.36	1.420	
			OSPI (2ch)*2		62.5	100.8	6.300	
			SDHI (2ch)*2		62.5	9.858	0.616	
			DSMIF (2ch)*2		125	31.614	3.952	
		Analog	ADC16H (2 units)*2		125	66.267	8.283	VSCR_1
			DAC12 (2ch)*2		62.5	0.325	0.020	
			TSN		62.5	0.115	0.007	
			ACMPHS (4ch)*2		62.5	0.173	0.011	
		Event link	ELC		62.5	5.075	0.317	VSCR_1
		Security	RSIP-E50D		125	302.444	37.806	VSCR_1
			DOTF (2ch)*2		62.5	131.817	8.239	
		Data processing	CRC		125	1.455	0.182	VSCR_1
			DOC		125	0.241	0.030	
		System	CAC		62.5	0.946	0.059	VSCR_1
		DMA	DMAC0 (per 1ch)		250	7.278	1.819	VSCR_1
			DMAC1 (per 1ch)		250	6.858	1.715	VSCR_1
			DTC0		250	9.077	2.269	
			DTC1		250	8.716	2.179	

Table 60.44 Power consumption of each unit (DCDC mode) (3 of 3)

Dynamic current/ Leakage current	MCU Domain	Category	Item	Symbol	Frequency [MHz]	Current [μ A/MHz]	Current [mA]	Condition
Dynamic current	FSBL operation			I_{DD}	250	—	81.1	FSBLCLK[2:0] = 111
					200	—	67.0	FSBLCLK[2:0] = 110
					150	—	51.7	FSBLCLK[2:0] = 101
					133	—	47.2	FSBLCLK[2:0] = 100

Note 1. Regulator and Leak are Internal voltage regulator's current and MCU's leakage current.
It is selected according to the temperature of T_J .

Note 2. To determine the current consumption per channel or unit, divide Current [mA] by the number of channels, groups or units.

Table 60.45 Power consumption of each unit (External VDD mode) (1 of 3)

Dynamic current/ Leakage current	MCU Domain	Category	Item	Symbol	Frequency [MHz]	Current [μ A/MHz]	Current [mA]	Condition
Leakage Current	Analog	Regulator and Leak*1	$T_J = 95^\circ\text{C}$	I_{CC}	—	—	2.51	—
			$T_J = 105^\circ\text{C}$		—	—	2.91	
			$T_J = 115^\circ\text{C}$		—	—	3.45	
			$T_J = 125^\circ\text{C}$		—	—	4.15	
			$T_J = 95^\circ\text{C}$	I_{DD}	—	—	477	VCL = voltage range 1, PDCTRESWM. PDDE = 0
			$T_J = 105^\circ\text{C}$		—	—	553	
			$T_J = 115^\circ\text{C}$		—	—	622	
			$T_J = 125^\circ\text{C}$		—	—	709	
Dynamic Current	CPU0	Operation with Cache	CoreMark	I_{DD}	1000	145	144	CPUCLK0 = 1000 MHz, VCL = voltage range 1
	CPU1	Operation with Cache	CoreMark		250	166	41	CPUCLK1 = 250 MHz, VCL = voltage range 1

Table 60.45 Power consumption of each unit (External VDD mode) (2 of 3)

Dynamic current/ Leakage current	MCU Domain	Category	Item	Symbol	Frequency [MHz]	Current [μ A/MHz]	Current [mA]	Condition
Dynamic Current	Peripheral Unit	Timer	RTC	I_{DD}	62.5	1.229	0.077	VCL = voltage range 1
			GPT32 (14ch)* ²		125	65.123	8.140	
			POEG (4 Groups)* ²		62.5	1.539	0.096	
			PDG (4ch)* ²		125	47.465	5.933	
			AGT (2ch)* ²		62.5	1.518	0.095	
			ULPT (2ch)* ²		62.5	2.373	0.148	
			WDT0		62.5	0.437	0.027	
			WDT1		62.5	0.446	0.028	
			IWDT		62.5	0.014	0.001	
		Communication interfaces	ESWM	I_{DD}	125	294.026	36.753	VCL = voltage range 1
			ESC		125	14.095	1.762	
			USBFS		62.5	7.495	0.468	
			SCI (10ch)* ²		125	32.336	4.042	
			IIC (3ch)* ²		62.5	3.722	0.233	
			I3C		125	9.883	1.235	
			CANFD (2ch)* ²		125	6.025	0.753	
			SPI (2ch)* ²		125	11.36	1.420	
			OSPI (2ch)* ²		62.5	100.8	6.300	
			SDHI (2ch)* ²		62.5	9.858	0.616	
			DSMIF (2ch)* ²		125	31.614	3.952	
Dynamic Current	Peripheral Unit	Analog	ADC16H (2 units)* ²	I_{DD}	125	66.267	8.283	VCL = voltage range 1
			DAC12 (2ch)* ²		62.5	0.325	0.020	
			TSN		62.5	0.115	0.007	
			ACMPHS (4ch)* ²		62.5	0.173	0.011	
		Event link	ELC	I_{DD}	62.5	5.075	0.317	VCL = voltage range 1
		Security	RSIP-E50D	I_{DD}	125	302.444	37.806	VCL = voltage range 1
			DOTF (2ch)* ²		62.5	131.817	8.239	
		Data processing	CRC	I_{DD}	125	1.455	0.182	VCL = voltage range 1
			DOC		125	0.241	0.030	
		System	CAC	I_{DD}	62.5	0.946	0.059	VCL = voltage range 1
		DMA	DMAC0 (per 1ch)	I_{DD}	250	7.278	1.819	VCL = voltage range 1
			DMAC1 (per 1ch)		250	6.858	1.715	
			DTC0		250	9.077	2.269	
			DTC1		250	8.716	2.179	

Table 60.45 Power consumption of each unit (External VDD mode) (3 of 3)

Dynamic current/ Leakage current	MCU Domain	Category	Item	Symbol	Frequency [MHz]	Current [μ A/MHz]	Current [mA]	Condition
Dynamic Current	FSBL operation			I _{DD}	250	—	79.9	FSBLCLK[2:0] = 111
					200	—	66.1	FSBLCLK[2:0] = 110
					150	—	51.0	FSBLCLK[2:0] = 101
					133	—	46.6	FSBLCLK[2:0] = 100

Note 1. Regulator and Leak are Internal voltage regulator's current and MCU's leakage current.
It is selected according to the temperature of T_j.

Note 2. To determine the current consumption per channel or unit, divide Current [mA] by the number of channels, groups or units.

Table 60.46 Outline of operation for each unit (1 of 2)

Peripheral	Outline of operation
RTC	RTC is operating with LOCO.
GPT	Operating modes is set to saw-wave PWM mode. GPT is operating with PCLKD
POEG	Only clear module stop bit.
PDG	PDG is applying delay of 1/128 times GTCLK period.
AGT	AGT is operating with PCLKB.
ULPT	ULPT is operating with LOCO.
WDT	WDT is operating with PCLKB.
IWDT	IWDT is operating with IWDTCLK.
ESWM	Communication mode is set to 1 Gbps, MAC Loop back. gPTP timer is enabled. ESWM is operating continuous transmission and reception on two ports simultaneously.
ESC	ESC is operating to access USERRAM from EtherCAT network.
USBFS	Transfer types are set to bulk transfer. USBFS is operating using Full-speed transfer (12 Mbps).
SCI	SCI is transmitting data in clock synchronous mode.
IIC	Communication format is set to I2C-bus format. IIC is transmitting data in master mode.
I3C	Communication format is set to I3C SDR format. I3C is transmitting data in master mode (12.5MHz).
CANFD	CANFD is transmitting and receiving data in self-test mode 1.
SPI	SPI mode is set to SPI operation (4-wire method). SPI master/slave mode is set to master mode. SPI is transmitting 32-bit width data.
OSPI	OSPI is issuing memory write command to HyperRAM.
SDHI	Transfer bus mode is set to 8-bit wide bus mode. SDHI is issuing CMD24 (single-block write).
DSMIF	DSMIF is capturing current data of 3 channels by capture trigger A.
ADC16H	Resolution is set to 16-bit accuracy. Conversion Data Operation Control B Register is set to 16 times average mode. ADC is converting the analog input in continuous scan mode. ADC is operating with ADCCLK.
DAC12	DAC12 is outputting the conversion result while updating the value of data register.
TSN	TSN is operating.
ACMPHS	ACMPHS is operating.
ELC	Only clear module stop bit.

Table 60.46 Outline of operation for each unit (2 of 2)

Peripheral	Outline of operation
RSIP-E50D	RSIP is doing self-test operation.
DOTF	DOTF is doing decryption with AES.
CRC	CRC is generating CRC code using 32-bit CRC32-C polynomial.
DOC	DOC is operating in data comparison mode.
CAC	Measurement target clocks is set to PCLKB. Measurement reference clocks is set to PCLKB. CAC is measuring the clock frequency accuracy.
DMAC	Bit length of transfer data is set to 32 bits. Transfer mode is set to block transfer mode. DMAC is transferring data from SRAM0 to SRAM0.
DTC	Bit length of transfer data is set to 32 bits. Transfer mode is set to block transfer mode. DTC is transferring data from SRAM0 to SRAM0.

60.2.8.2 Example of Tj Calculation

Assumption :

- Package 289-pin BGA : $\theta_{ja} = 20\text{ }^{\circ}\text{C/W}$
- $T_a = 65\text{ }^{\circ}\text{C}$
- $I_{CC} + I_{CC_DCDC} = 320\text{ mA}$
- $V_{CC} = 3.5\text{ V}$ ($V_{CC} = V_{CC2} = AV_{CC0} = V_{CC_USB}$)
- $I_{OH} = 1\text{ mA}$, $V_{OH} = V_{CC} - 0.5\text{ V}$, 12 Outputs
- $I_{OL} = 20\text{ mA}$, $V_{OL} = 1.0\text{ V}$, 8 Outputs
- $I_{OL} = 1\text{ mA}$, $V_{OL} = 0.5\text{ V}$, 12 Outputs
- $C_{in} = 8\text{ pF}$, 32 pins, Input frequency = 10 MHz
- $C_{load} = 30\text{ pF}$, 32 pins, Output frequency = 10 MHz

$$\begin{aligned}
 \text{Static current of IO} &= \Sigma (V_{OL} \times I_{OL}) / \text{Voltage} + \Sigma ((V_{CC} - V_{OH}) \times I_{OH}) / \text{Voltage} \\
 &= (20\text{ mA} \times 1\text{ V}) \times 8 / 3.5\text{ V} + (1\text{ mA} \times 0.5\text{ V}) \times 12 / 3.5\text{ V} + ((V_{CC} - (V_{CC} - 0.5\text{ V})) \times 1\text{ mA}) \times 12 / 3.5\text{ V} \\
 &= 45.7\text{ mA} + 1.71\text{ mA} + 1.71\text{ mA} \\
 &= 49.1\text{ mA}
 \end{aligned}$$

$$\begin{aligned}
 \text{Dynamic current of IO} &= \Sigma I_O (C_{in} + C_{load}) \times \text{IO switching frequency} \times \text{Voltage} \\
 &= ((8\text{ pF} \times 32) \times 10\text{ MHz} + (30\text{ pF} \times 32) \times 10\text{ MHz}) \times 3.5\text{ V} \\
 &= 42.6\text{ mA}
 \end{aligned}$$

$$\begin{aligned}
 \text{Total power consumption} &= \text{Voltage} \times (\text{Static current} + \text{Dynamic current}) \\
 &= (320\text{ mA} \times 3.5\text{ V}) + (49.1\text{ mA} + 42.6\text{ mA}) \times 3.5\text{ V} \\
 &= 1441\text{ mW (1.441 W)}
 \end{aligned}$$

$$\begin{aligned}
 T_j &= T_a + \theta_{ja} \times \text{Total power consumption} \\
 &= 65\text{ }^{\circ}\text{C} + 20\text{ }^{\circ}\text{C/W} \times 1.441\text{ W} \\
 &= 93.82\text{ }^{\circ}\text{C}
 \end{aligned}$$

60.3 AC Characteristics

60.3.1 Frequency

Table 60.47 Operation frequency value in high-speed mode (1 of 8)

Parameter			Symbol	Min	Typ	Max	Unit
Operation frequency	PLL1 output clock P (PLL1P)	BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)	f	—	—	1000	MHz
		BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	800	
		BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	600	
		BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	600	
		BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	600	
		BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	600	
		HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode)		—	—	600	
		HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode)		—	—	600	
	Other PLL output clock (PLL1Q, PLL1R, PLL2P, PLL2Q, PLL2R)	BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	1200	
		BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	1200	
		BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	1200	
		BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	1200	
		BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	1200	
		BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	1200	
		HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode)		—	—	1200	
		HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode)		—	—	1200	

Table 60.47 Operation frequency value in high-speed mode (2 of 8)

Parameter			Symbol	Min	Typ	Max	Unit
Operation frequency	CPU0 clock (CPUCLK0)*1	BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)	f	—	—	1000	MHz
		BGA package, -40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	800	
		BGA package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	600	
		BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	600	
		BGA package, -40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	600	
		BGA package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	600	
		HLQFP package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode)		—	—	600	
		HLQFP package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode)		—	—	600	
	CPU1 clock (CPUCLK1)*1	BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	250	
		BGA package, -40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	200	
		BGA package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	200	
		BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		BGA package, -40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		BGA package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		HLQFP package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode)		—	—	200	
		HLQFP package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode)		—	—	150	

Table 60.47 Operation frequency value in high-speed mode (3 of 8)

Parameter			Symbol	Min	Typ	Max	Unit
Operation frequency	System clock (ICLK)*1	BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)	f	—	—	250	MHz
		BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	200	
		BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	200	
		BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode)		—	—	200	
		HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode)		—	—	150	

Table 60.47 Operation frequency value in high-speed mode (4 of 8)

Parameter			Symbol	Min	Typ	Max	Unit
Operation frequency	MRAM bus clock (MRICLK)*1	BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)	f	—	—	250	MHz
		BGA package, -40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	200	
		BGA package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	200	
		BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		BGA package, -40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		BGA package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		HLQFP package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode)		—	—	200	
		HLQFP package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode)		—	—	150	
	MRAM clock (MRPCLK)*1	BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	125	
		BGA package, -40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	100	
		BGA package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	100	
		BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	75	
		BGA package, -40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	75	
		BGA package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	75	
		HLQFP package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode)		—	—	100	
		HLQFP package, -40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode)		—	—	75	

Table 60.47 Operation frequency value in high-speed mode (5 of 8)

Parameter			Symbol	Min	Typ	Max	Unit
Operation frequency	Peripheral module clock (PCLKA)* ¹	BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)	f	—	—	125	MHz
		BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	100	
		BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	100	
		BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	75	
		BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	75	
		BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	75	
		HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode)		—	—	100	
		HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode)		—	—	75	
	Peripheral module clock (PCLKB)* ¹	BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	62.5	
		BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	50	
		BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	50	
		BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	37.5	
		BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	37.5	
		BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	37.5	
		HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode)		—	—	50	
		HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode)		—	—	37.5	

Table 60.47 Operation frequency value in high-speed mode (6 of 8)

Parameter			Symbol	Min	Typ	Max	Unit
Operation frequency	Peripheral module clock (PCLKD)* ¹	BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)	f	—	—	250	MHz
		BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	200	
		BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	200	
		BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode)		—	—	200	
		HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode)		—	—	150	
	Peripheral module clock (PCLKE)* ¹	BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	250	
		BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	200	
		BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	200	
		BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	150	
		HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode)		—	—	200	
		HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode)		—	—	150	

Table 60.47 Operation frequency value in high-speed mode (7 of 8)

Parameter				Symbol	Min	Typ	Max	Unit
Operation frequency	External bus clock (BCLK)*1	VCC ≥ 2.7 V	BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)	f	—	—	125	MHz
			BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	100	
			BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode), voltage range 1 (External VDD mode)		—	—	100	
			BGA package, 0 °C ≤ Tj ≤ 95 °C (Product group A), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	75	
			BGA package, −40 °C ≤ Tj ≤ 105 °C (Product group B), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	75	
			BGA package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode), voltage range 2 (External VDD mode)		—	—	75	
			HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_1 (DCDC mode)		—	—	100	
			HLQFP package, −40 °C ≤ Tj ≤ 125 °C (Product group C), VSCR_2 (DCDC mode)		—	—	75	
	EBCLK pin output	VCC ≥ 1.62 V			—	—	60	
		VCC ≥ 2.7 V			—	—	60	
		VCC ≥ 1.62 V			—	—	30	
	SDCLK pin output	VCC ≥ 3.0 V			—	—	133	

Table 60.47 Operation frequency value in high-speed mode (8 of 8)

Parameter			Symbol	Min	Typ	Max	Unit
Operation frequency	SCI clock (SCICLK)		f	—	—	120	MHz
	SPI clock (SPICLK)			—	—	333	
	Octal SPI clock (OCTACLK)			—	—	333	
	CANFD core clock (CANFDCLK)			—	—	80	
	ADC clock (ADCCLK)			25	—	120	
	GPT clock (GPTCLK)			—*2	—	300	
	USB clock (USBCLK)			—	—	48	
	I3C clock (I3CCLK)			—	—	200	
	Asynchronous external bus clock (BCLKA)			—	—	133	
	EtherSW clock (ESWCLK)	BGA package		—	—	250	
		HLQFP package		—	—	125	
	EtherSW-PHY clock (ESWPHYCLK)			—	—	500	
	EtherCAT clock (ESCCLK)			—	—	100	
	EtherCAT-PHY clock (ESCPHYCLK)			—	—	50	
	DSMIF clock (DSMIFCLK)			—	—	300	

Note 1. See [section 9, Clock Generation Circuit](#) for the relationship between the CPUCLK0, CPUCLK1, ICLK, MRICK, MRPCLK, PCLKA, PCLKB, PCLKC, PCLKD, PCLKE, and BCLK frequencies.

Note 2. When the GPTCLK is used for A/D conversion clock, the GPTCLK frequency must be at least 25 MHz.

60.3.2 Clock Timing

Table 60.48 Clock timing except for sub-clock oscillator (1 of 3)

Parameter		Symbol	Min	Typ	Max	Unit	Test conditions
EBCLK pin output cycle time	VCC = 2.70 V or above	t _{Bcyc}	16.6	—	—	ns	Figure 60.25
	VCC = 1.62 V or above		33.3	—	—		
EBCLK pin output high pulse width	VCC = 2.70 V or above	t _{CH}	3.3	—	—	ns	
	VCC = 1.62 V or above		9.6	—	—		
EBCLK pin output low pulse width	VCC = 2.70 V or above	t _{CL}	3.3	—	—	ns	
	VCC = 1.62 V or above		9.6	—	—		
EBCLK pin output rise time	VCC = 2.70 V or above	t _{Cr}	—	—	5.0	ns	
	VCC = 1.62 V or above		—	—	7.0		
EBCLK pin output fall time	VCC = 2.70 V or above	t _{Cf}	—	—	5.0	ns	
	VCC = 1.62 V or above		—	—	7.0		
SDCLK pin output cycle time		t _{SDcyc}	7.52	—	—	ns	
SDCLK pin output high pulse width		t _{CH}	1.0	—	—	ns	
SDCLK pin output low pulse width		t _{CL}	1.0	—	—	ns	
SDCLK pin output rise time		t _{Cr}	—	—	2.7	ns	
SDCLK pin output fall time		t _{Cf}	—	—	2.7	ns	
EXTAL external clock input cycle time		t _{EXcyc}	20.80	—	—	ns	Figure 60.26
EXTAL external clock input high pulse width		t _{EXH}	5.30	—	—	ns	
EXTAL external clock input low pulse width		t _{EXL}	5.30	—	—	ns	
EXTAL external clock rise time		t _{EXr}	—	—	3.0	ns	
EXTAL external clock fall time		t _{EXf}	—	—	3.0	ns	

Table 60.48 Clock timing except for sub-clock oscillator (2 of 3)

Parameter		Symbol	Min	Typ	Max	Unit	Test conditions
Main clock oscillator frequency		f _{MAIN}	8	—	48	MHz	—
Main clock oscillation stabilization wait time (crystal)* ¹		t _{MAINOSCW T}	—	—	—* ¹	ms	Figure 60.27
LOCO clock oscillation frequency		f _{LOCO}	29.4912	32.768	36.0448	kHz	—
LOCO clock oscillation stabilization wait time		t _{LOCOWT}	—	—	26.0	μs	Figure 60.28
MOCO clock oscillation frequency		F _{MOCO}	7.2	8.0	8.8	MHz	—
MOCO clock oscillation stabilization wait time		t _{MOCOWT}	—	—	3.5	μs	—
HOCO clock oscillator oscillation frequency	Without FLL (Standard Product)	f _{HOCO16}	15.776	16.000	16.224	MHz	–20 ≤ Tj ≤ 105 °C
		f _{HOCO18}	17.748	18.000	18.252		
		f _{HOCO20}	19.720	20.000	20.280		
		f _{HOCO32}	31.552	32.000	32.448		
		f _{HOCO48}	47.328	48.000	48.672		
		f _{HOCO16}	15.712	16.000	16.288		–40 ≤ Tj ≤ 125 °C
		f _{HOCO18}	17.676	18.000	18.324		
		f _{HOCO20}	19.640	20.000	20.360		
		f _{HOCO32}	31.424	32.000	32.576		
		f _{HOCO48}	47.136	48.000	48.864		
	Without FLL (SiP Product)	f _{HOCO16}	15.744	16.000	16.256	MHz	–20 ≤ Tj ≤ 105 °C
		f _{HOCO18}	17.712	18.000	18.288		
		f _{HOCO20}	19.680	20.000	20.320		
		f _{HOCO32}	31.488	32.000	32.512		
		f _{HOCO48}	47.232	48.000	48.768		
		f _{HOCO16}	15.680	16.000	16.320		–40 ≤ Tj ≤ 125 °C
		f _{HOCO18}	17.640	18.000	18.360		
		f _{HOCO20}	19.600	20.000	20.400		
		f _{HOCO32}	31.360	32.000	32.640		
		f _{HOCO48}	47.040	48.000	48.960		
	With FLL	f _{HOCO16}	15.960	16.000	16.040	MHz	–40 ≤ Tj ≤ 125 °C Sub-clock frequency accuracy is ±50 ppm.
		f _{HOCO18}	17.955	18.000	18.045		
		f _{HOCO20}	19.950	20.000	20.050		
		f _{HOCO32}	31.920	32.000	32.080		
		f _{HOCO48}	47.880	48.000	48.120		
HOCO clock oscillation stabilization wait time* ²		t _{HOCOWT}	—	—	15.0	μs	—
HOCO stop width time		t _{HOCOSTP}	1	—	—	μs	Figure 60.31
HOCO period jitter		—	-3	—	3	ps	—
FLL stabilization wait time (Standard Product)		t _{FLLWT}	—	—	1.92	ms	—
FLL stabilization wait time (SiP Product)		t _{FLLWT}	—	—	2.15	ms	—
PLL1/PLL2 clock frequency		f _{PLL}	60	—	1200	MHz	—
PLL1/PLL2 clock oscillation stabilization wait time		t _{PLLWT}	—	—	50	μs	Figure 60.29

Table 60.48 Clock timing except for sub-clock oscillator (3 of 3)

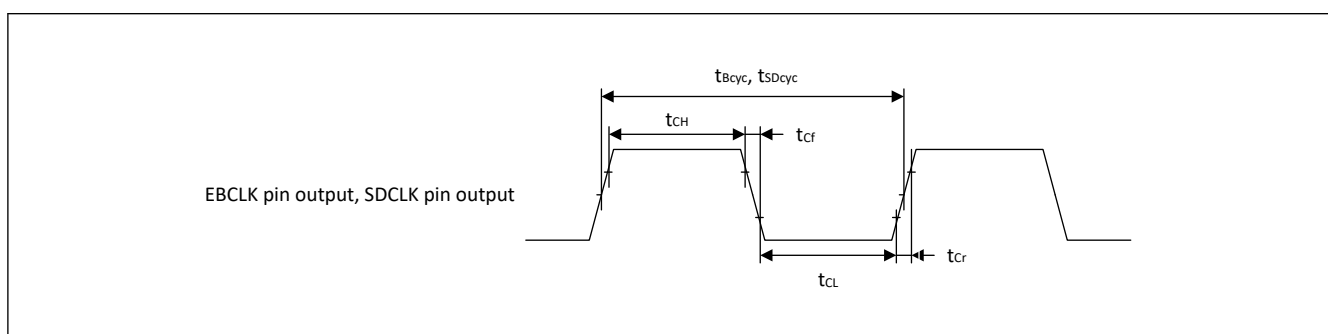
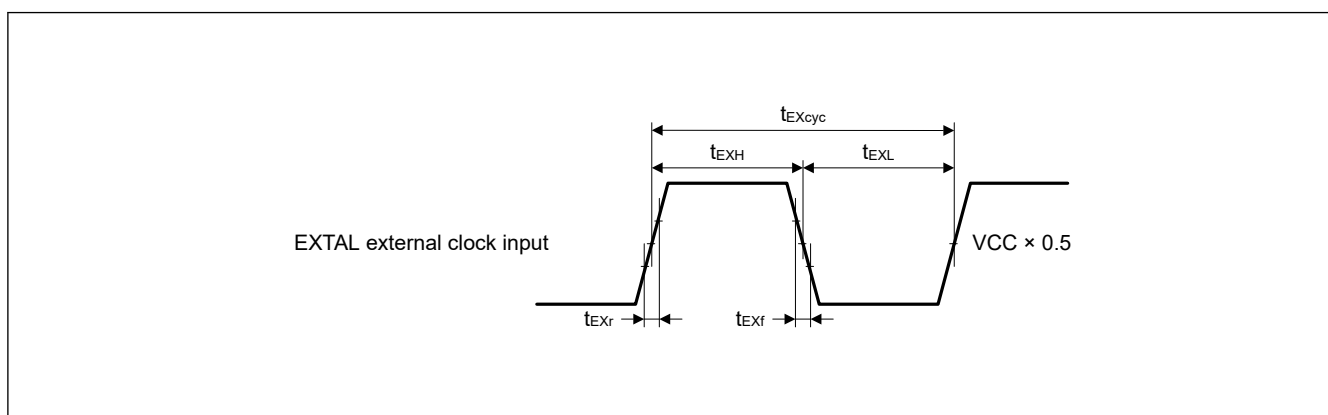
Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
PLL1/PLL2 period jitter	—	—	±52	—	ps	—
PLL1/PLL2 long term jitter	—	—	±300	—	ps	Term: 1 μs, 10 μs

- Note 1. When setting up the main clock oscillator, ask the oscillator manufacturer for an oscillation evaluation, and use the results as the recommended oscillation stabilization time. Set the MOSCWTCR register to a value equal to or greater than the recommended value.
After changing the setting in the MOSCCR.MOSTP bit to start main clock operation, read the OSCSF.MOSCSF flag to confirm that it is 1, and then start using the main clock oscillator.
- Note 2. This is the time from release from reset state until the HOCO oscillation frequency (f_{HOCO}) reaches the range for guaranteed operation.

Table 60.49 Clock timing for the sub-clock oscillator

Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
Sub-clock frequency	f_{SUB}	—	32.768	—	kHz	—
Sub-clock oscillation stabilization wait time	$t_{SUBOSCWT}$	—	—	—*1	s	Figure 60.30

- Note 1. When setting up the sub-clock oscillator, ask the oscillator manufacturer for an oscillation evaluation and use the results as the recommended oscillation stabilization time.
After changing the setting in the SOSCCR.SOSTP bit to start sub-clock operation, only start using the sub-clock oscillator after the sub-clock oscillation stabilization time elapses with an adequate margin. A value that is two times the value shown is recommended.

**Figure 60.25 EBCLK and SDCLK output timing****Figure 60.26 EXTAL external clock input timing**

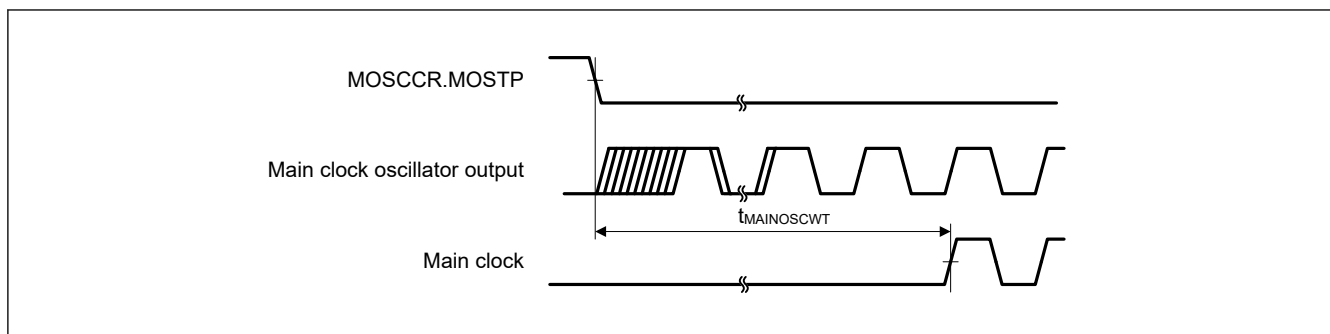


Figure 60.27 Main clock oscillation start timing

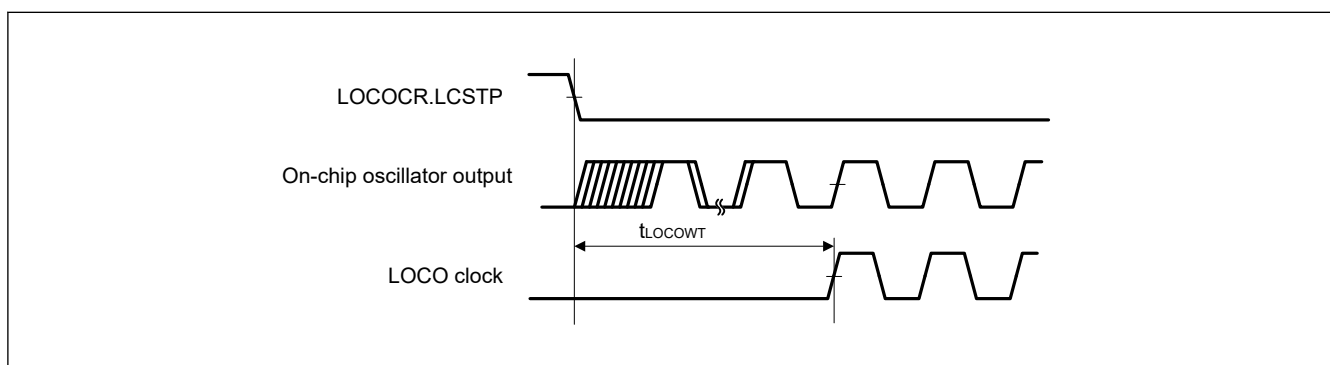


Figure 60.28 LOCO clock oscillation start timing

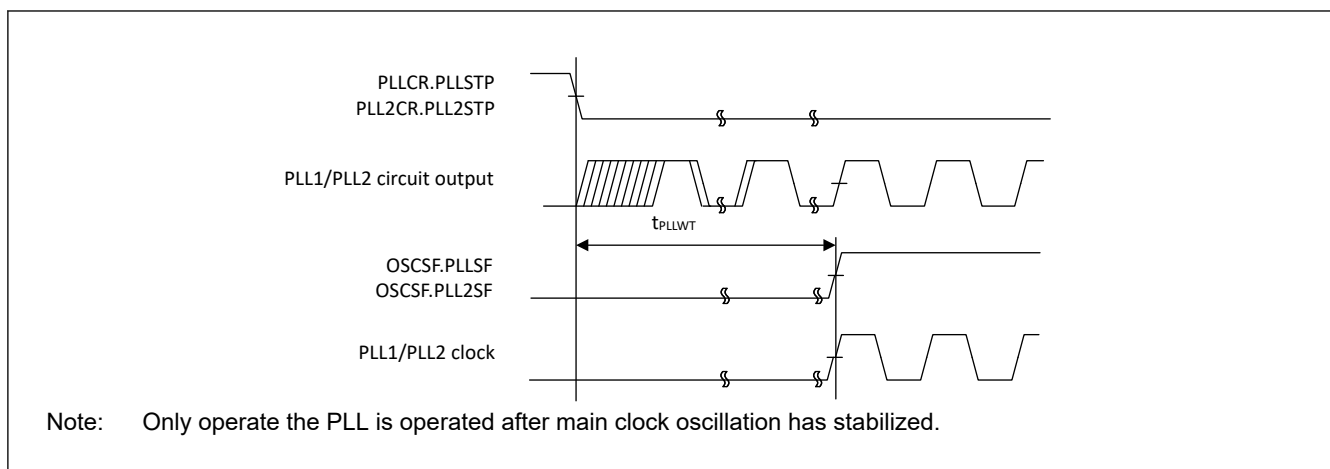


Figure 60.29 PLL1/PLL2 clock oscillation start timing

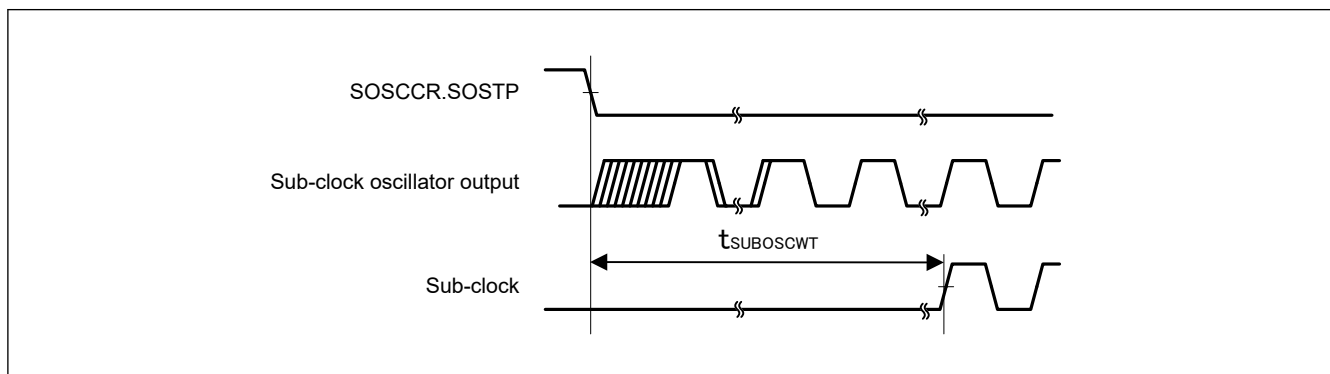
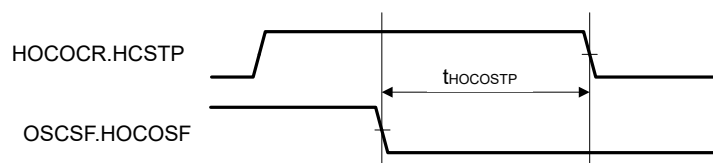


Figure 60.30 Sub-clock oscillation start timing

**Figure 60.31 HOCO stop width time**

60.3.3 Reset Timing

Table 60.50 Reset timing

Parameter			Symbol	Min	Typ	Max	Unit	Test conditions
RES pulse width	Power-on	—	t _{RESWP}	2.4	—	—	ms	Figure 60.32
	Deep Software Standby Mode 1	DPSBYCR.DCSSMODE = 01	t _{RESWD}	0.51	—	—	ms	Figure 60.33
		DPSBYCR.DCSSMODE = 10		0.67	—	—		
		DPSBYCR.DCSSMODE = 11		1.00	—	—		
	Deep Software Standby Mode 2	DPSBYCR.DCSSMODE = 01		0.51	—	—		
		DPSBYCR.DCSSMODE = 10		0.67	—	—		
		DPSBYCR.DCSSMODE = 11		1.00	—	—		
	Deep Software Standby Mode 3	DPSBYCR.DCSSMODE = 01		0.68	—	—		
		DPSBYCR.DCSSMODE = 10		0.84	—	—		
		DPSBYCR.DCSSMODE = 11		1.20	—	—		
	Software Standby Mode		t _{RESWS}	0.55	—	—	ms	
	CPU Deep Sleep mode (Subosc operation)		t _{RESWSODS}	0.16	—	—	ms	
	CPU Deep Sleep mode (Other than SOSC operation)		t _{RESWDS}	0.04	—	—	ms	
	SOSC operation	PGSCR.PGS = 1	t _{RESWSO}	0.27	—	—	ms	
		PGSCR.PGS = 0		0.30	—	—		
	Other than above	PGSCR.PGS = 1	t _{RESW}	0.15	—	—	ms	
		PGSCR.PGS = 0		0.18	—	—		
Wait time after RES cancellation			t _{RESWT}	—	78.7	79.1	μs	Figure 60.32
Wait time after internal reset cancellation (IWDT reset, WDT0/1 reset, CPU0/1 Lockup reset, Bus Error reset, Common Memory Error reset, Software reset, Local Memory 0/1 error reset, Temperature monitor reset)			t _{RESW2}	—	78.7	79.1	μs	—

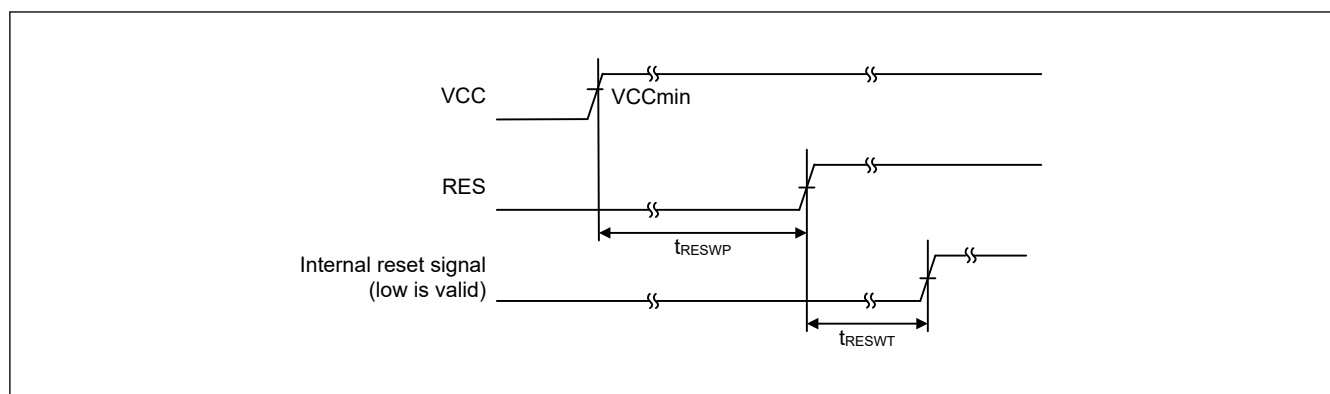


Figure 60.32 RES pin input timing under the condition that VCC exceeds V_{POR} voltage threshold

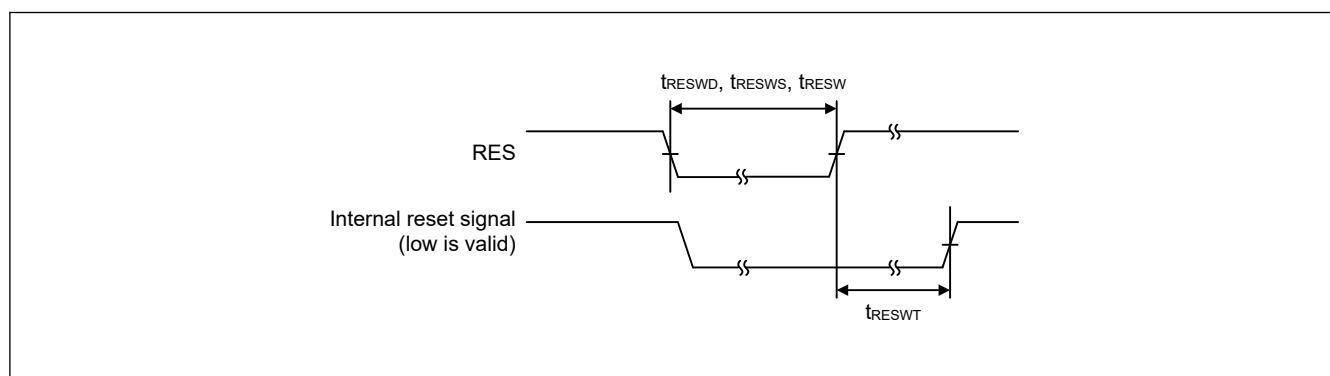


Figure 60.33 Reset input timing

60.3.4 Wakeup Timing

Table 60.51 Timing of recovery from low power modes (1 of 3)

Parameter		Fast return function ^{*9}	Symbol	Min	Typ	Max	Unit	Test conditions
Recovery time from CPU Deep Sleep mode	CPU0 Deep Sleep mode	—	$t_{DSL P}^{*11}$	—	6.14	9.45	μs	—
	CPU1 Deep Sleep mode	—		—	7.71	15.66	μs	

Table 60.51 Timing of recovery from low power modes (2 of 3)

Parameter			Fast return function ^{*9}	Symbol	Min	Typ	Max	Unit	Test conditions
Recovery time from Software Standby mode ^{*12}	Crystal resonator connected to main clock oscillator	System clock source is main clock oscillator ^{*1} MOSCSCR.MO SCSOKP = 0	Enabled	t_{SBYMC}^{*10}	—	2.09	2.14	ms	Figure 60.34 The division ratio of all oscillators is 1.
		System clock source is main clock oscillator ^{*1} MOSCSCR.MO SCSOKP = 1	Enabled		—	44.9	94.6	μs	
		System clock source is PLL1P with main clock oscillator ^{*2} MOSCSCR.MO SCSOKP = 0	Enabled	t_{SBYPC}^{*10}	—	2.21	2.27	ms	
		System clock source is PLL1P with main clock oscillator ^{*2} MOSCSCR.MO SCSOKP = 1	Enabled		—	135	197	μs	
	External clock input to main clock oscillator	System clock source is main clock oscillator ^{*3}	Enabled	t_{SBYEX}^{*10}	—	44.9	94.6	μs	
		System clock source is PLL1P with main clock oscillator ^{*4}	Enabled	t_{SBYPE}^{*10}	—	135	197		
	System clock source is sub-clock oscillator ^{*5}		Enabled	t_{SBYSC}^{*10}	—	480	481	μs	
	System clock source is HOCO clock oscillator ^{*6}		Enabled	t_{SBYHO}^{*10}	—	46.3	96.0	μs	
	System clock source is PLL1P with HOCO ^{*7}		Enabled	t_{SBYPH}^{*10}	—	146	208	μs	
	System clock source is MOCO clock oscillator ^{*8}		Enabled	t_{SBYMO}^{*10}	—	44.6	87.5	μs	

Table 60.51 Timing of recovery from low power modes (3 of 3)

Parameter			Fast return function* ⁹	Symbol	Min	Typ	Max	Unit	Test conditions
Recovery time from Deep Software Standby mode	Deep Software Standby mode 1	DPSBYCR.DC SSMODE[1:0] = 01	—	t_{DSBY}	—	296	346	μs	Figure 60.35
		DPSBYCR.DC SSMODE[1:0] = 10	—		—	456	506	μs	
		DPSBYCR.DC SSMODE[1:0] = 11	—		—	776	826	μs	
	Deep Software Standby mode 2	DPSBYCR.DC SSMODE[1:0] = 01	—		—	296	346	μs	
		DPSBYCR.DC SSMODE[1:0] = 10	—		—	456	506	μs	
		DPSBYCR.DC SSMODE[1:0] = 11	—		—	776	826	μs	
	Deep Software Standby mode 3	DPSBYCR.DC SSMODE[1:0] = 01	—		—	483	604	μs	
		DPSBYCR.DC SSMODE[1:0] = 10	—		—	643	764	μs	
		DPSBYCR.DC SSMODE[1:0] = 11	—		—	963	1084	μs	
Wait time after cancellation of Deep Software Standby mode			—	t_{DSBYWT}	22.2	—	33.6	μs	

Note 1. When the frequency of the crystal is 48 MHz (Main Clock Oscillator Wait Control Register (MOSCWTCR) is set to 0x05) and the greatest value of the internal clock division setting is 1.

Note 2. When the frequency of PLL1P is 1 GHz (Main Clock Oscillator Wait Control Register (MOSCWTCR) is set to 0x05) and the greatest value of the internal clock division setting is 16.

Note 3. When the frequency of the external clock is 48 MHz (Main Clock Oscillator Wait Control Register (MOSCWTCR) is set to 0x00) and the greatest value of the internal clock division setting is 1.

Note 4. When the frequency of PLL1P is 1 GHz (Main Clock Oscillator Wait Control Register (MOSCWTCR) is set to 0x00) and the greatest value of the internal clock division setting is 16.

Note 5. The Sub-clock oscillator frequency is 32.768 kHz and the greatest value of the internal clock division setting is 1.

Note 6. The HOCO frequency is 20 MHz and the greatest value of the internal clock division setting is 1.

Note 7. The PLL frequency is 1 GHz and the greatest value of the internal clock division setting is 16.

Note 8. The MOCO frequency is 8 MHz and the greatest value of the internal clock division setting is 1.

Note 9. For details, see SSCR1.SS2FR bit in the [section 11, Low Power Mode](#).

Note 10. The recovery time can be calculated with the equation of $t_{Common} + \max(t_{OSCSTB}, t_{PG1}, t_{PGCK}) + \max(t_{PG2}, t_{LPW})$. And they can be determined with the following values and equations. For n, the greatest value is selected from among the internal clock(CPUCLK0, CPUCLK1, NPUCLK, ICLK, MRCLK, MRPCLK, PCLKm, FCLK, BCLK and EBCLK) division settings (m = A to E). t_{OSCSTB} in the table below means the time when each oscillator is active. When multiple oscillators are active, t_{OSCSTB} is determined by the longest t_{OSCSTB} among the active oscillators.

Note 11. The ICLK frequency is 250 MHz. This recovery time corresponds to t_{PG2} .

Note 12. When Ccyc is 27. See [Table 60.53](#).

Table 60.52 Each element of recovery time

Mode	Wakeu p time	Oscillat ion keep	Fast return functio n	Typ						Max						Unit
				t _{Commo n}	t _{OSCSTB *1}	t _{PG1}	t _{PGCK}	t _{PG2}	t _{LPW}	t _{Commo n}	t _{OSCSTB *1}	t _{PG1}	t _{PGCK}	t _{PG2}	t _{LPW}	
Softwa re Standb y mode	t _{SBYMC}	MOSC disabled	Enabled	C _{cyc} ^{*2} / f _{MOCO} + 2/f _{ICLK}	t _{MAINOS} CWT	t _{OSC} STB/ f _{MOC} o + 208/ f _{MOC} o + 11.6	(10.5 + 2.5n)/ f _{MOCO} + 2.5/ f _{SRCC} L K + 2/ f _{ICLK}	18/ f _{MOCO} + 9/ f _{ICLK}	2/f _{ICLK} + 2n/ f _{MOSC} + 2/ f _{ICLK}	C _{cyc} ^{*2} / f _{MOCO} + 2/f _{ICLK}	t _{MAINOS} CWT + 11/0.23 6	t _{OSC} STB/ f _{MOC} o + 208/ f _{MOC} o + 51.0	(10.5 + 2.5n)/ f _{MOCO} + 2.5/ f _{SRCC} L K + 2/ f _{ICLK}	18/ f _{MOCO} + 9/ f _{ICLK}	2/f _{ICLK} + 2n/ f _{MOSC} + 2/ f _{ICLK}	μs
		MOSC enabled	Enabled		3/0.262						14/0.23 6				μs	
	t _{SBYPC}	MOSC disabled	Enabled		t _{MAINOS} CWT +31/0.2 62				2/f _{ICLK} + 2n/ f _{PLL} + 2/f _{ICLK}		t _{MAINOS} CWT + 42/0.23 6				2/f _{ICLK} + 2n/ f _{PLL} + 2/f _{ICLK}	μs
		MOSC enabled	Enabled		34/0.26 2						(14 + 31)/ 0.236					μs
	t _{SBYEX}	—	Enabled		3/0.262				2/f _{ICLK} + 2n/ f _{MOSC} + 2/ f _{ICLK}		14/0.23 6				2/f _{ICLK} + 2n/ f _{MOSC} + 2/ f _{ICLK}	μs
	t _{SBYPE}	—	Enabled		34/0.26 2				2/f _{ICLK} + 2n/ f _{PLL} + 2/f _{ICLK}		45/0.23 6				2/f _{ICLK} + 2n/ f _{PLL} + 2/f _{ICLK}	μs
	t _{SBYSC}	—	Enabled		0				2/f _{ICLK} + 2n/ f _{SOSC} + 2/ f _{ICLK}		0				2/f _{ICLK} + 2n/ f _{SOSC} + 2/ f _{ICLK}	μs
	t _{SBYHO}	—	Enabled		20				2/f _{ICLK} + 2n/ f _{HOCO} + 2/ f _{ICLK}		67				2/f _{ICLK} + 2n/ f _{HOCO} + 2/ f _{ICLK}	μs
	t _{SBYPH}	—	Enabled		140				2/f _{ICLK} + 2n/ f _{PLL} + 2/f _{ICLK}		202				2/f _{ICLK} + 2n/ f _{PLL} + 2/f _{ICLK}	μs
	t _{SBYMO}	—	Enabled		0				2/f _{ICLK} + 2n/ f _{MOCO} + 2/ f _{ICLK}		0				2/f _{ICLK} + 2n/ f _{MOCO} + 2/ f _{ICLK}	μs

Note: The unit of frequency is MHz.

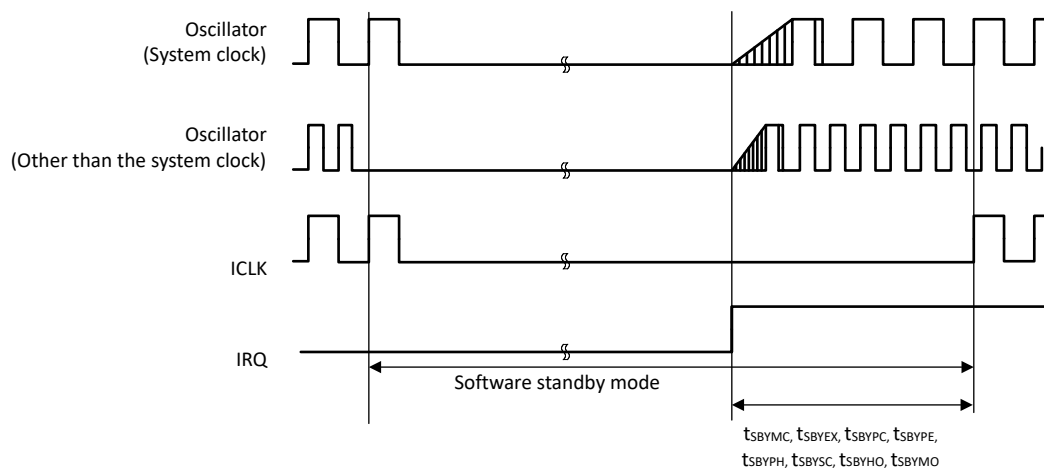
Note 1. If more than one oscillator is operating, the largest value of the operating oscillator in this column is applied.

Note 2. For C_{cyc}, see Table 60.53.

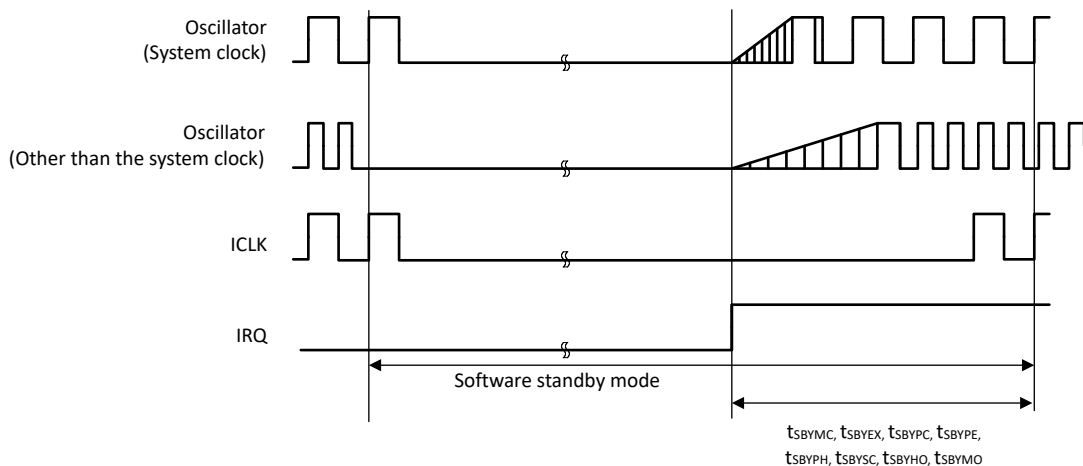
Table 60.53 Ccyc value

SSCR1.SS2LP [1:0]	VSCR.VSCM[2:0]	SVSCR.SVSCM [2:0]	{PLL1LDOCR.LD OSTP, PLL2LDOCR.LD OSTP, PLL1LDOCR.SK EEP, PLL2LDOCR.SK EEP}	{HOCOLDOCR.L DOSTP, HOCOLDOCR.S KEEP}	Ccyc	Unit
00: SS2LP_0	001 : VSCR_1	001 : SVSCR_1	{1, 1, x, x} or {x, x, 1, 1}	{0, 0}	56	cycle
				Other than above	27	cycle
			Other than above	Don't care	237	cycle
		010 : SVSCR_2	Don't care	Don't care	379	cycle
		011 : SVSCR_3	Don't care	Don't care	591	cycle
		100 : SVSCR_4	Don't care	Don't care	696	cycle
		101 : SVSCR_5	Don't care	Don't care	802	cycle
	010 : VSCR_2	001 : SVSCR_1	Don't care	Don't care	379	cycle
		010 : SVSCR_2	{1, 1, x, x} or {x, x, 1, 1}	{0, 0}	56	cycle
				Other than above	27	cycle
			Other than above	Don't care	237	cycle
		011 : SVSCR_3	Don't care	Don't care	538	cycle
		100 : SVSCR_4	Don't care	Don't care	643	cycle
		101 : SVSCR_5	Don't care	Don't care	749	cycle
01: SS2LP_1	001 : VSCR_1	010 : SVSCR_2	Don't care	Don't care	514	cycle
		011 : SVSCR_3	Don't care	Don't care	726	cycle
		100 : SVSCR_4	Don't care	Don't care	831	cycle
		101 : SVSCR_5	Don't care	Don't care	937	cycle
	010 : VSCR_2	010 : SVSCR_2	{1, 1, x, x} or {x, x, 1, 1}	Don't care	162	cycle
			Other than above	Don't care	327	cycle
		011 : SVSCR_3	Don't care	Don't care	673	cycle
		100 : SVSCR_4	Don't care	Don't care	778	cycle
		101 : SVSCR_5	Don't care	Don't care	884	cycle

Note: x: Don't care



When stabilization of the system clock oscillator is slower



When stabilization of an oscillator other than the system clock is slower

Figure 60.34 Software Standby mode cancellation timing

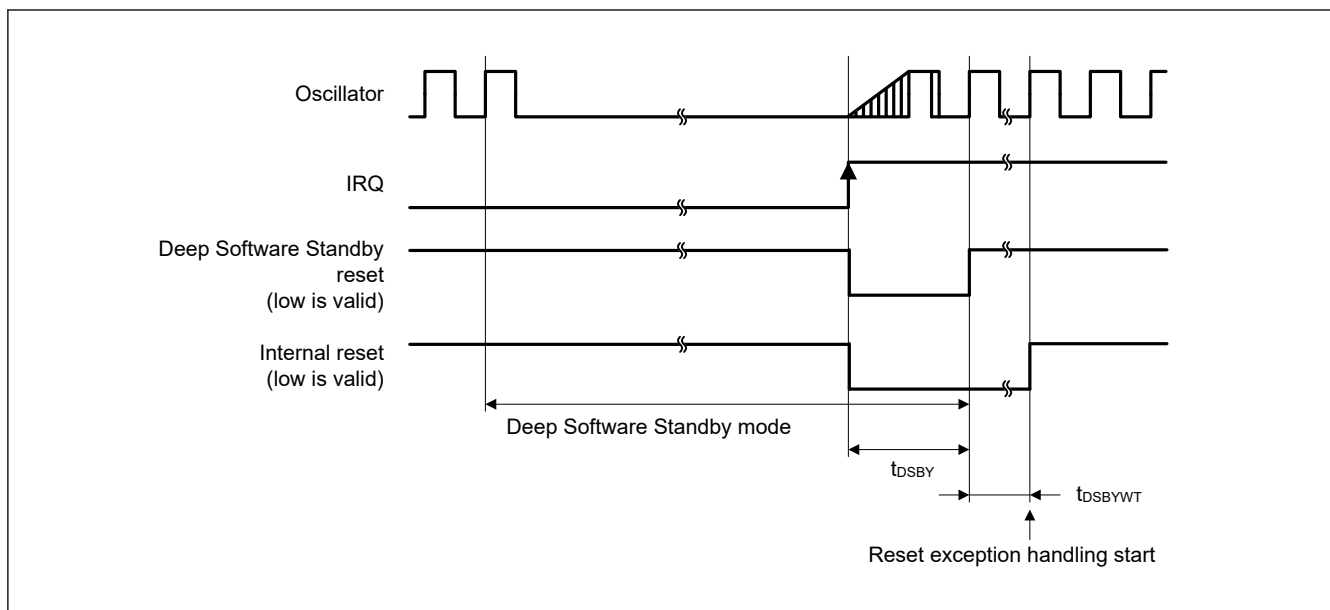


Figure 60.35 Deep Software Standby mode cancellation timing

60.3.5 NMI and IRQ Noise Filter

Table 60.54 NMI and IRQ noise filter

Parameter	Symbol	Min	Typ	Max	Unit	Test conditions	
NMI pulse width	t_{NMIW}	200	—	—	ns	NMI digital filter disabled	$t_{Pcyc} \times 2 \leq 200$ ns
		$t_{Pcyc} \times 2^{*1}$	—	—			$t_{Pcyc} \times 2 > 200$ ns
		200	—	—		NMI digital filter enabled	$t_{NMICK} \times 3 \leq 200$ ns
		$t_{NMICK} \times 3.5^{*2}$	—	—			$t_{NMICK} \times 3 > 200$ ns
IRQ pulse width	t_{IRQW}	200	—	—	ns	IRQ digital filter disabled	$t_{Pcyc} \times 2 \leq 200$ ns
		$t_{Pcyc} \times 2^{*1}$	—	—			$t_{Pcyc} \times 2 > 200$ ns
		200	—	—		IRQ digital filter enabled	$t_{IRQCK} \times 3 \leq 200$ ns
		$t_{IRQCK} \times 3.5^{*3}$	—	—			$t_{IRQCK} \times 3 > 200$ ns

Note: 200 ns minimum in Software Standby mode.

Note: If the system clock source is switched, add 4 clock cycles of the switched source.

Note 1. t_{Pcyc} indicates the PCLKB cycle.

Note 2. t_{NMICK} indicates the cycle of the NMI digital filter sampling clock.

Note 3. t_{IRQCK} indicates the cycle of the IRQi digital filter sampling clock.

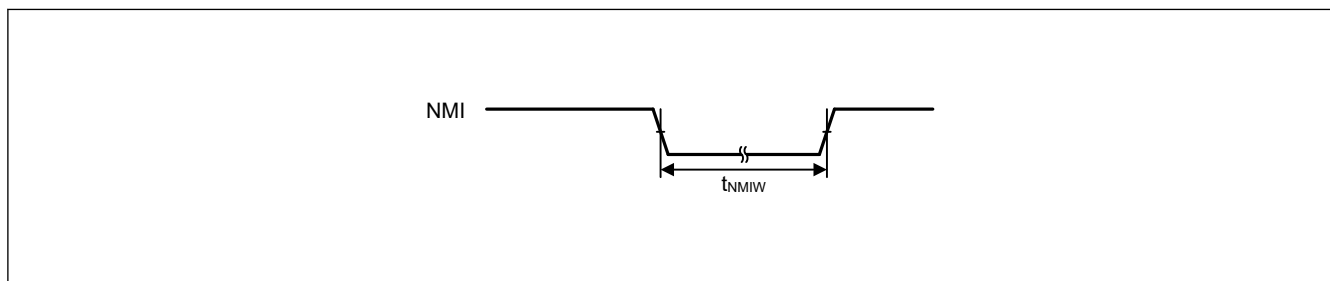


Figure 60.36 NMI interrupt input timing

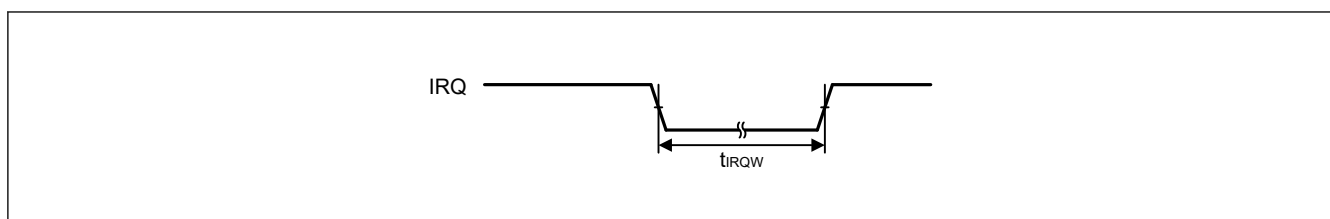


Figure 60.37 IRQ interrupt input timing

60.3.6 Bus Timing

Table 60.55 Bus timing (1 of 3)

Condition 1: When using the CS area controller (CSC).

VCC = VCC_DCDC = VBATT = 1.62 V to 3.6 V, VCC2 = 1.62 V to 3.63 V

BCLK = 8 to 120 MHz, BCLKA = 8 to 120 MHz, EBCLK = 8 to 60 MHz (When VCC = VCC_USB = VBATT = 2.70 to 3.63 V)

BCLK = BCLKA = 8 to 60 MHz, EBCLK = 8 to 30 MHz (When VCC = VCC_USB = VBATT = 1.62 to 3.63 V)

Output load conditions: VOH = VCC × 0.5, VOL = VCC × 0.5, C = 30 pF

EBCLK: High drive output is selected in the port drive capability bit in the PmnPFS register.

Others: Middle drive output is selected in the port drive capability bit in the PmnPFS register.

Condition 2: When using the SDRAM area controller (SDRAMC).

BCLK = SDCLK = 8 to 125 MHz, BCLKA = SDCLK = 8 to 133 MHz

VCC = VCC_DCDC = VBATT = 3.0 to 3.63 V, VCC2 = 1.62 V to 3.63 V

Output load conditions: VOH = VCC × 0.5, VOL = VCC × 0.5, C = 15 pF

SDCLK: High-speed high drive output is selected in the port drive capability bit in the PmnPFS register.

Others: High drive output is selected in the port drive capability bit in the PmnPFS register.

Condition 3: When using the SDRAM area controller (SDRAMC) and CS area controller (CSC) simultaneously.

BCLK = SDCLK = 8 to 66 MHz, BCLKA = SDCLK = 8 to 66 MHz

VCC = VCC_DCDC = VBATT = 3.0 to 3.63 V, VCC2 = 1.62 V to 3.63 V

Output load conditions: VOH = VCC × 0.5, VOL = VCC × 0.5, C = 15 pF

EBCLK/SDCLK: High drive output is selected in the port drive capability bit in the PmnPFS register.

Others: Middle drive output is selected in the port drive capability bit in the PmnPFS register.

Parameter	Condition	VCC/VCC2	Symbol	Min	Max	Unit	Test conditions
Address delay	Condition1	2.70 V or above	t_{AD}	1.0	12.5	ns	Figure 60.38 to Figure 60.44
		1.62 V or above		1.0	12.5	ns	
	Condition3	3.0 V or above		1.0	10.8	ns	
Byte control delay	Condition1	2.70 V or above	t_{BCD}	1.0	12.5	ns	
		1.62 V or above		1.0	12.5	ns	
	Condition3	3.0 V or above		1.0	10.8	ns	
CS delay	Condition1	2.70 V or above	t_{CSD}	1.0	12.5	ns	
		1.62 V or above		1.0	12.5	ns	
	Condition3	3.0 V or above		1.0	10.8	ns	
ALE delay time	Condition1	2.70 V or above	t_{ALED}	1.0	12.5	ns	
		1.62 V or above		1.0	12.5	ns	
	Condition3	3.0 V or above		1.0	10.8	ns	
RD delay	Condition1	2.70 V or above	t_{RSD}	1.0	12.5	ns	
		1.62 V or above		1.0	12.5	ns	
	Condition3	3.0 V or above		1.0	10.8	ns	
Read data setup time	Condition1	2.70 V or above	t_{RDS}	12.5	—	ns	
		1.62 V or above		20.5	—	ns	
	Condition3	3.0 V or above		10.8	—	ns	
Read data hold time	Condition1	2.70 V or above	t_{RDH}	0	—	ns	
		1.62 V or above		0	—	ns	
	Condition3	3.0 V or above		0	—	ns	
WR/WRn delay	Condition1	2.70 V or above	t_{WRD}	1.0	12.5	ns	
		1.62 V or above		1.0	12.5	ns	
	Condition3	3.0 V or above		1.0	10.8	ns	
Write data delay	Condition1	2.70 V or above	t_{WDD}	—	12.5	ns	
		1.62 V or above		—	12.5	ns	
	Condition3	3.0 V or above		1.0	10.8	ns	

Table 60.55 Bus timing (2 of 3)

Condition 1: When using the CS area controller (CSC).

VCC = VCC_DCDC = VBATT = 1.62 V to 3.6 V, VCC2 = 1.62 V to 3.63 V

BCLK = 8 to 120 MHz, BCLKA = 8 to 120 MHz, EBCLK = 8 to 60 MHz (When VCC = VCC_USB = VBATT = 2.70 to 3.63 V)

BCLK = BCLKA = 8 to 60 MHz, EBCLK = 8 to 30 MHz (When VCC = VCC_USB = VBATT = 1.62 to 3.63 V)

Output load conditions: VOH = VCC × 0.5, VOL = VCC × 0.5, C = 30 pF

EBCLK: High drive output is selected in the port drive capability bit in the PmnPFS register.

Others: Middle drive output is selected in the port drive capability bit in the PmnPFS register.

Condition 2: When using the SDRAM area controller (SDRAMC).

BCLK = SDCLK = 8 to 125 MHz, BCLKA = SDCLK = 8 to 133 MHz

VCC = VCC_DCDC = VBATT = 3.0 to 3.63 V, VCC2 = 1.62 V to 3.63 V

Output load conditions: VOH = VCC × 0.5, VOL = VCC × 0.5, C = 15 pF

SDCLK: High-speed high drive output is selected in the port drive capability bit in the PmnPFS register.

Others: High drive output is selected in the port drive capability bit in the PmnPFS register.

Condition 3: When using the SDRAM area controller (SDRAMC) and CS area controller (CSC) simultaneously.

BCLK = SDCLK = 8 to 66 MHz, BCLKA = SDCLK = 8 to 66 MHz

VCC = VCC_DCDC = VBATT = 3.0 to 3.63 V, VCC2 = 1.62 V to 3.63 V

Output load conditions: VOH = VCC × 0.5, VOL = VCC × 0.5, C = 15 pF

EBCLK/SDCLK: High drive output is selected in the port drive capability bit in the PmnPFS register.

Others: Middle drive output is selected in the port drive capability bit in the PmnPFS register.

Parameter	Condition	VCC/VCC2	Symbol	Min	Max	Unit	Test conditions
Write data hold time	Condition1	2.70 V or above	t_{WDH}	1.0	—	ns	Figure 60.38 to Figure 60.44
		1.62 V or above		1.0	—	ns	
	Condition3	3.0 V or above		1.0	10.8	ns	
WAIT setup time	Condition1	2.70 V or above	t_{WTS}	12.5	—	ns	
		1.62 V or above		20.5	—	ns	
	Condition3	3.0 V or above		10.8	—	ns	
WAIT hold time	Condition1	2.70 V or above	t_{WTH}	0	—	ns	
		1.62 V or above		0	—	ns	
	Condition3	3.0 V or above		0	—	ns	

Table 60.55 Bus timing (3 of 3)

Condition 1: When using the CS area controller (CSC).

VCC = VCC_DCDC = VBATT = 1.62 V to 3.6 V, VCC2 = 1.62 V to 3.63 V

BCLK = 8 to 120 MHz, BCLKA = 8 to 120 MHz, EBCLK = 8 to 60 MHz (When VCC = VCC_USB = VBATT = 2.70 to 3.63 V)

BCLK = BCLKA = 8 to 60 MHz, EBCLK = 8 to 30 MHz (When VCC = VCC_USB = VBATT = 1.62 to 3.63 V)

Output load conditions: VOH = VCC × 0.5, VOL = VCC × 0.5, C = 30 pF

EBCLK: High drive output is selected in the port drive capability bit in the PmnPFS register.

Others: Middle drive output is selected in the port drive capability bit in the PmnPFS register.

Condition 2: When using the SDRAM area controller (SDRAMC).

BCLK = SDCLK = 8 to 125 MHz, BCLKA = SDCLK = 8 to 133 MHz

VCC = VCC_DCDC = VBATT = 3.0 to 3.63 V, VCC2 = 1.62 V to 3.63 V

Output load conditions: VOH = VCC × 0.5, VOL = VCC × 0.5, C = 15 pF

SDCLK: High-speed high drive output is selected in the port drive capability bit in the PmnPFS register.

Others: High drive output is selected in the port drive capability bit in the PmnPFS register.

Condition 3: When using the SDRAM area controller (SDRAMC) and CS area controller (CSC) simultaneously.

BCLK = SDCLK = 8 to 66 MHz, BCLKA = SDCLK = 8 to 66 MHz

VCC = VCC_DCDC = VBATT = 3.0 to 3.63 V, VCC2 = 1.62 V to 3.63 V

Output load conditions: VOH = VCC × 0.5, VOL = VCC × 0.5, C = 15 pF

EBCLK/SDCLK: High drive output is selected in the port drive capability bit in the PmnPFS register.

Others: Middle drive output is selected in the port drive capability bit in the PmnPFS register.

Parameter	Condition	VCC/VCC2	Symbol	Min	Max	Unit	Test conditions
Address delay 2 (SDRAM)	Condition 2	3.0 V or above	t_{AD2}	0.8	6.0	ns	Figure 60.45 to Figure 60.51
	Condition 3	3.0 V or above		0.8	10		
CS delay 2 (SDRAM)	Condition 2	3.0 V or above	t_{CSD2}	0.8	6.0	ns	
	Condition 3	3.0 V or above		0.8	10		
DQM delay (SDRAM)	Condition 2	3.0 V or above	t_{DQMD}	0.8	6.0	ns	
	Condition 3	3.0 V or above		0.8	10		
CKE delay (SDRAM)	Condition 2	3.0 V or above	t_{CKED}	0.8	6.0	ns	
	Condition 3	3.0 V or above		0.8	10		
Read data setup time 2 (SDRAM)	Condition 2	3.0 V or above	t_{RDS2}	2.1	—	ns	
	Condition 3	3.0 V or above		6.1	—		
Read data hold time 2 (SDRAM)	Condition 2	3.0 V or above	t_{RDH2}	1.5	—	ns	
	Condition 3	3.0 V or above		1.5	—		
Write data delay 2 (SDRAM)	Condition 2	3.0 V or above	t_{WDD2}	—	6.0	ns	
	Condition 3	3.0 V or above		—	10		
Write data hold time 2 (SDRAM)	Condition 2	3.0 V or above	t_{WDH2}	0.8	—	ns	
	Condition 3	3.0 V or above		0.8	—		
WE delay (SDRAM)	Condition 2	3.0 V or above	t_{WED}	0.8	6.0	ns	
	Condition 3	3.0 V or above		0.8	10		
RAS delay (SDRAM)	Condition 2	3.0 V or above	t_{RASD}	0.8	6.0	ns	
	Condition 3	3.0 V or above		0.8	10		
CAS delay (SDRAM)	Condition 2	3.0 V or above	t_{CASD}	0.8	6.0	ns	
	Condition 3	3.0 V or above		0.8	10		

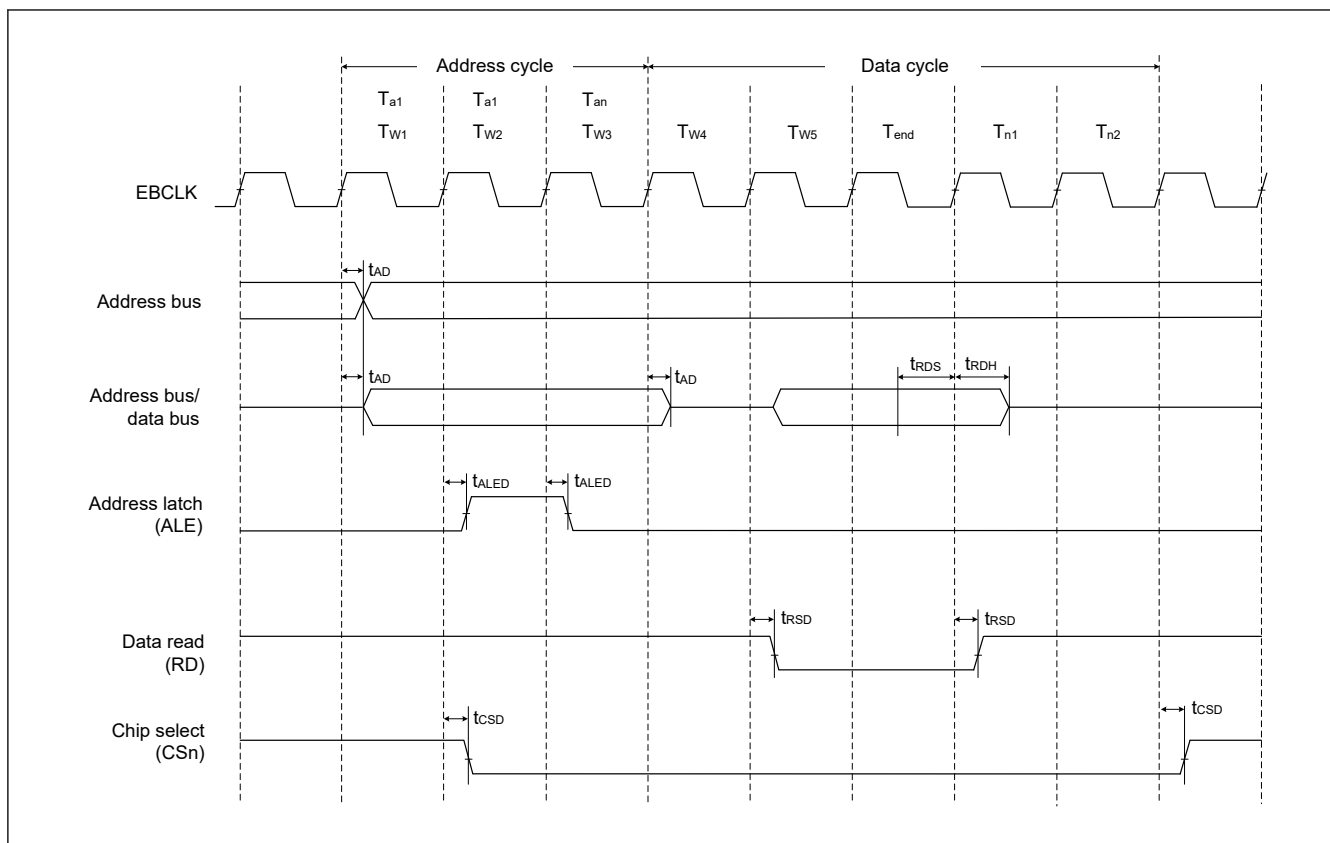


Figure 60.38 Address/data multiplexed bus read access timing

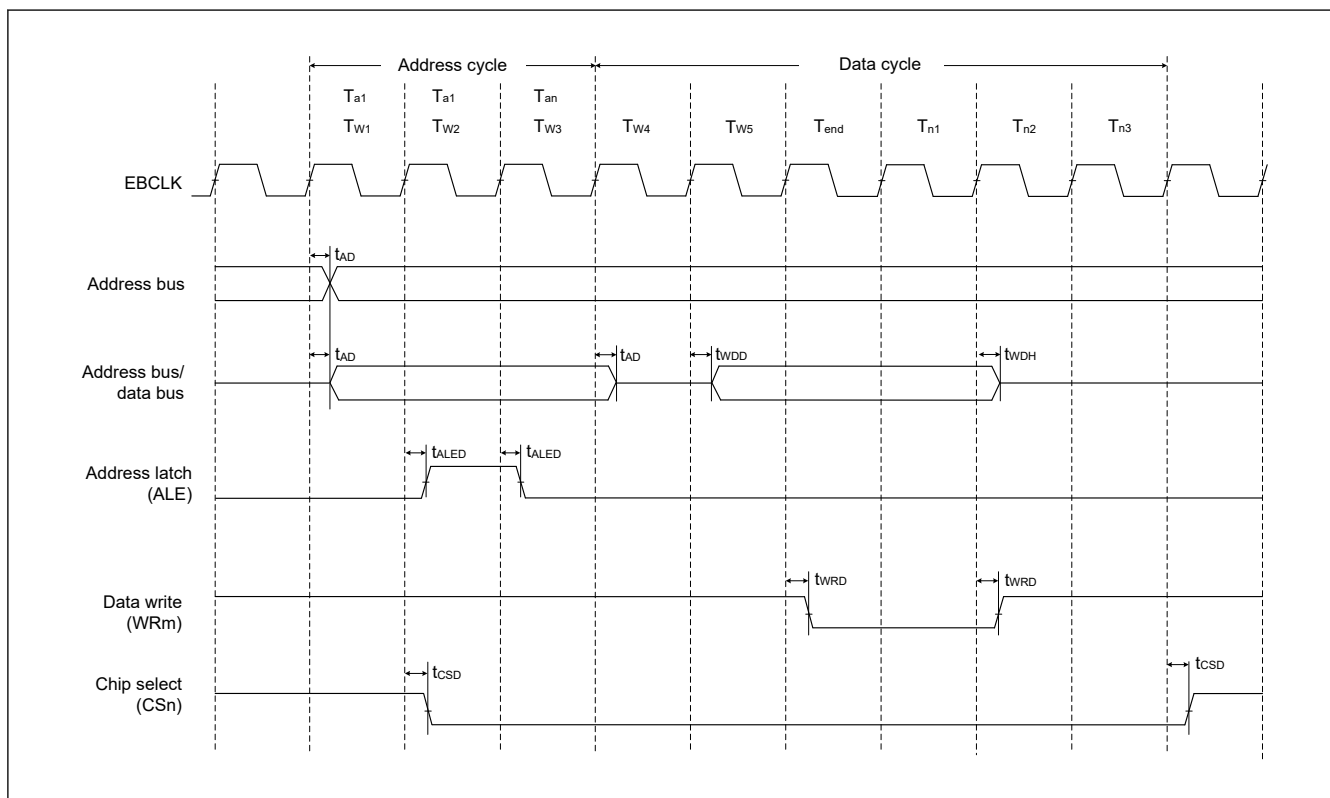


Figure 60.39 Address/data multiplexed bus write access timing

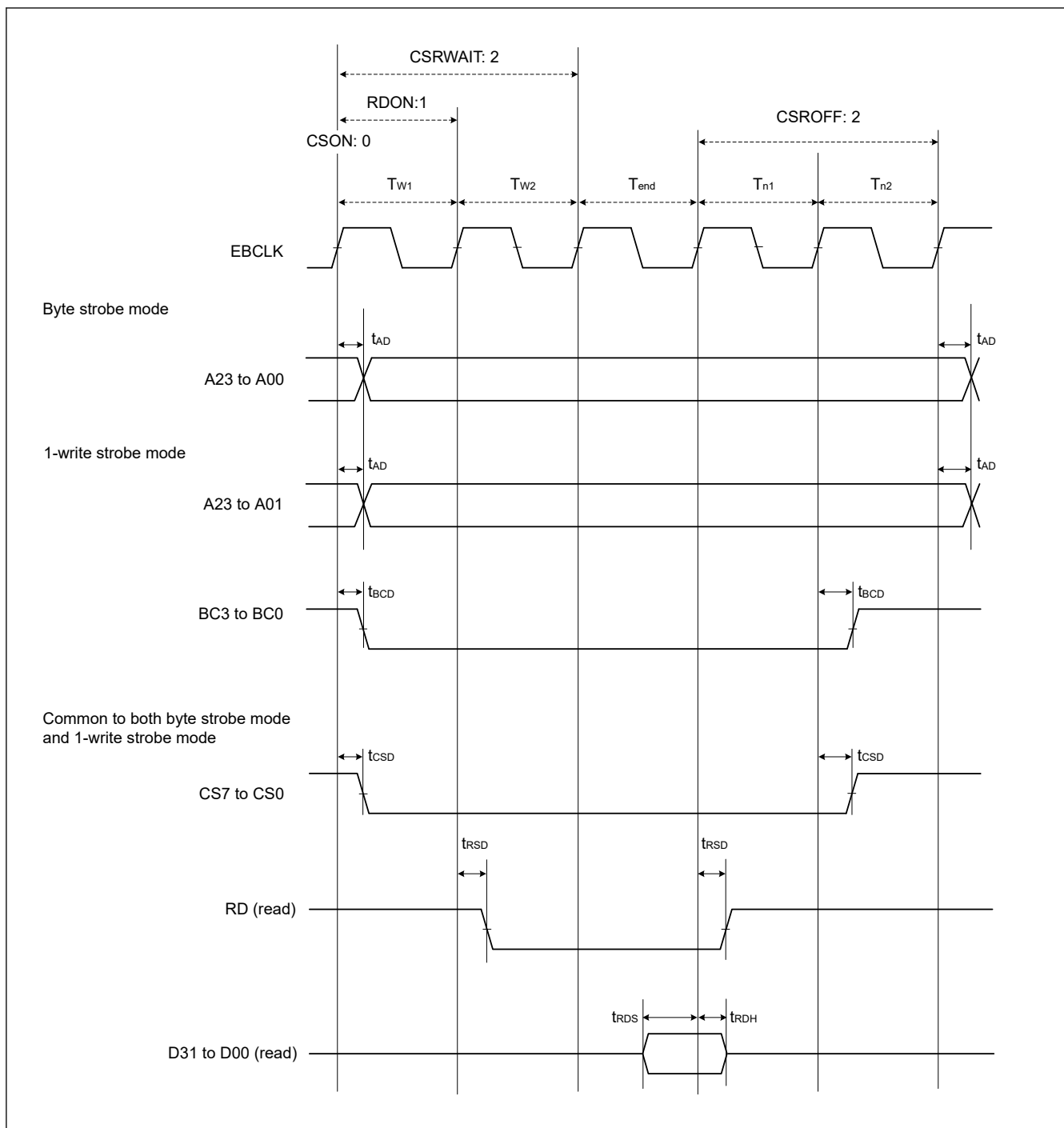


Figure 60.40 External bus timing for normal read cycle with bus clock synchronized

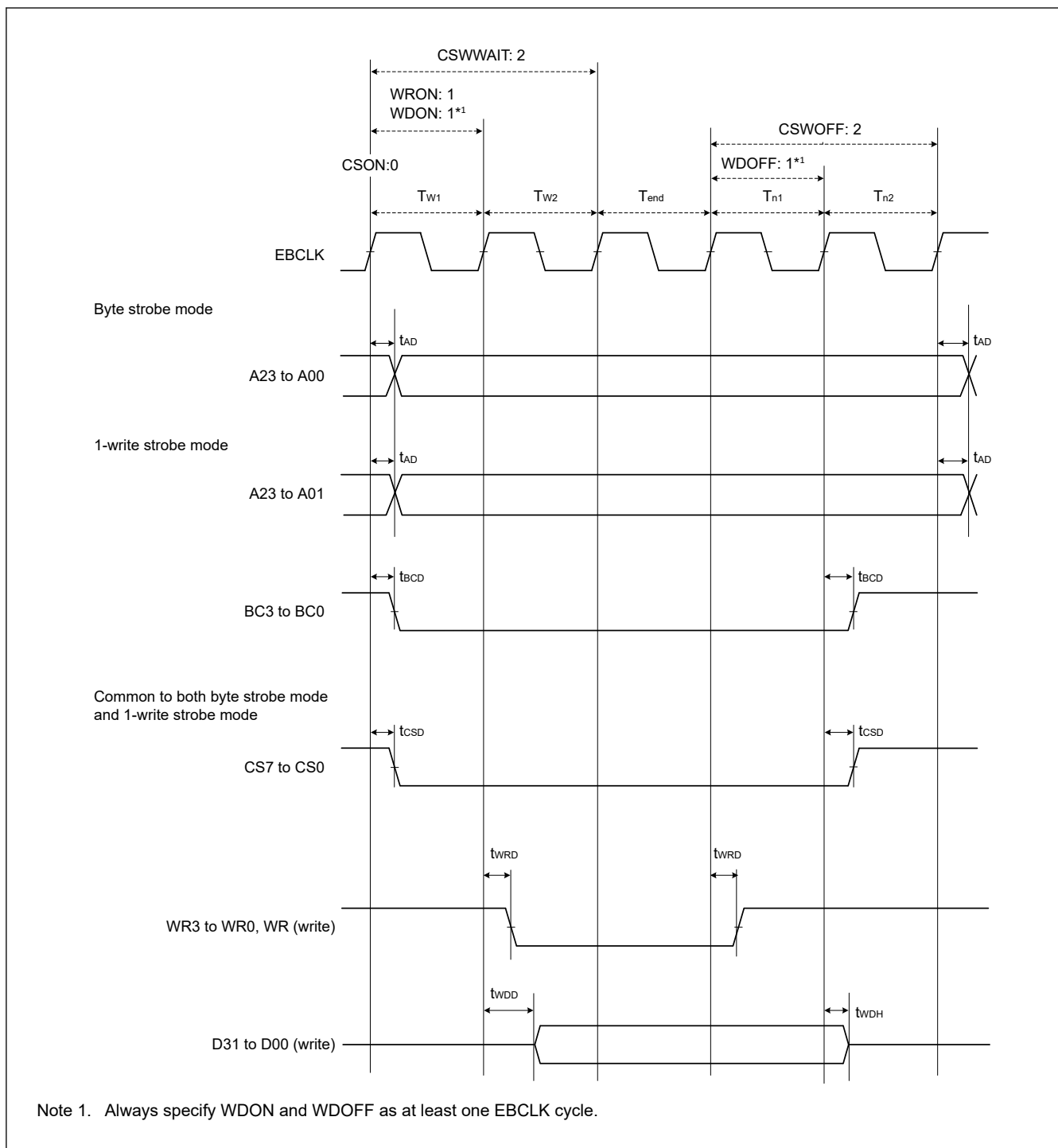


Figure 60.41 External bus timing for normal write cycle with bus clock synchronized

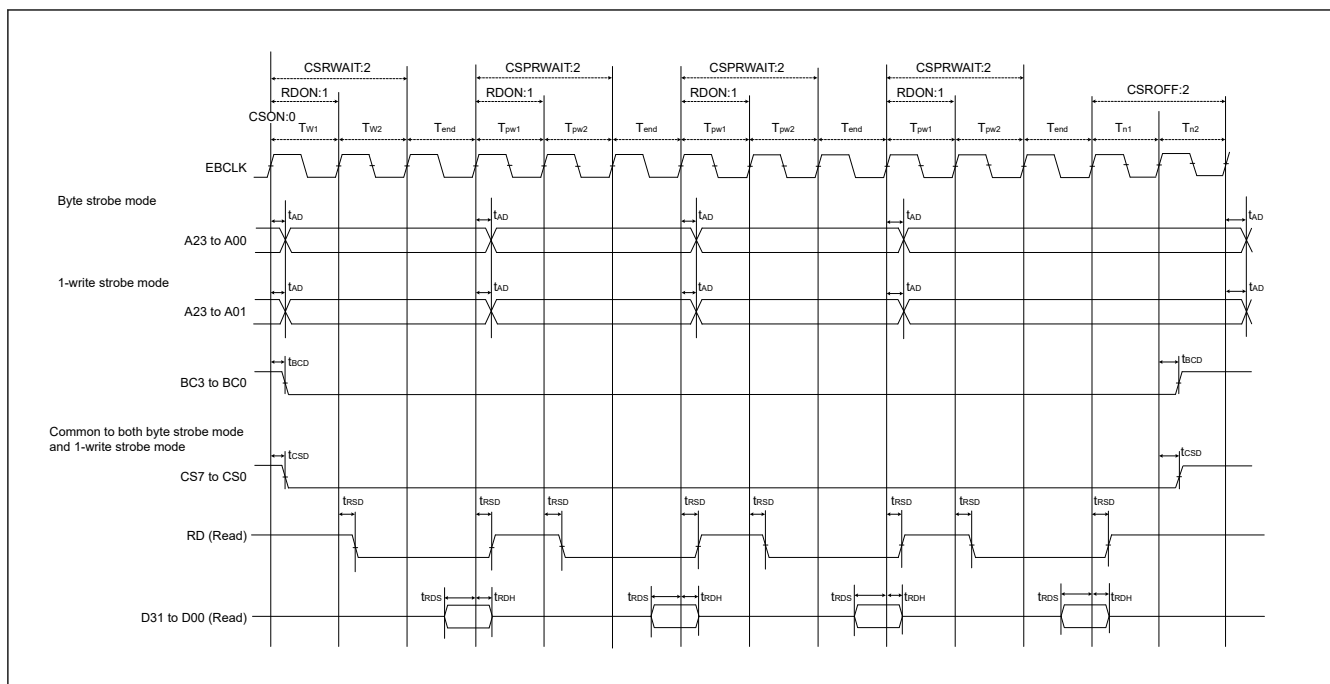


Figure 60.42 External bus timing for page read cycle with bus clock synchronized

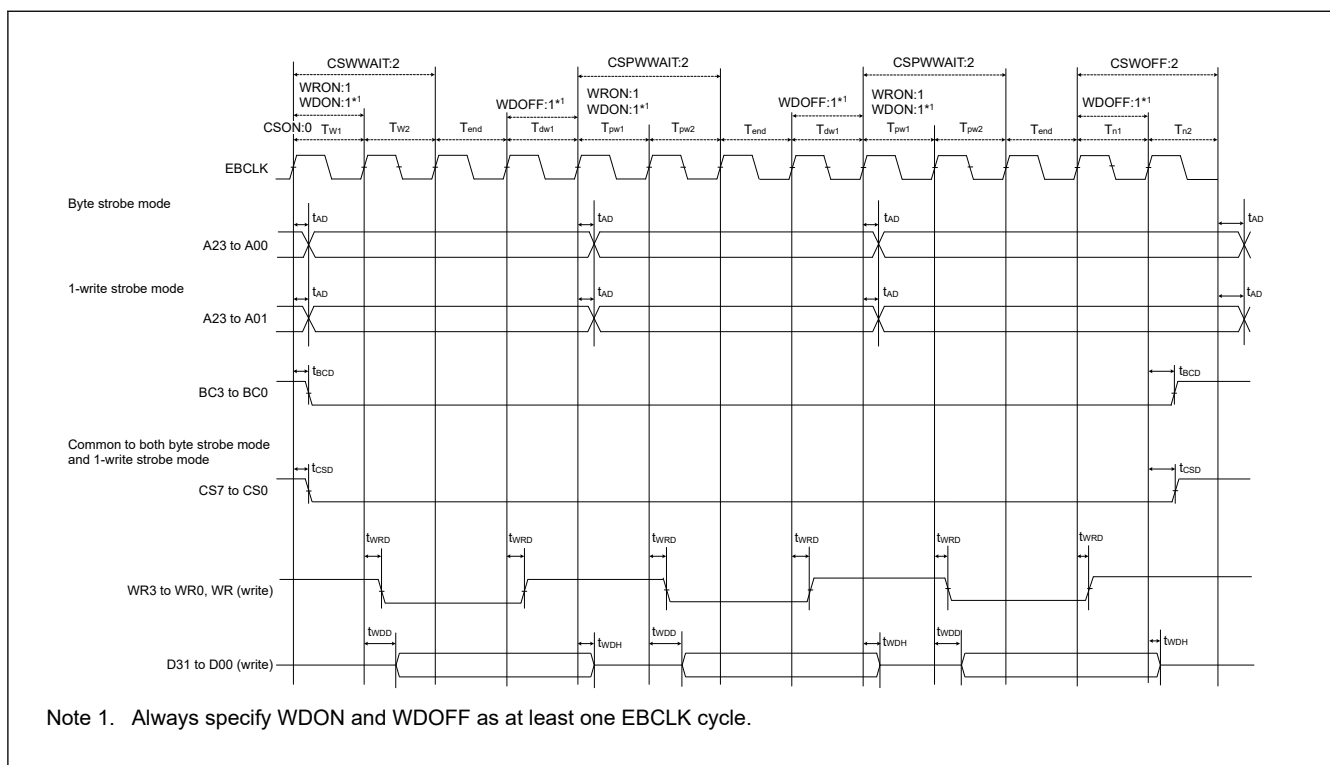


Figure 60.43 External bus timing for page write cycle with bus clock synchronized

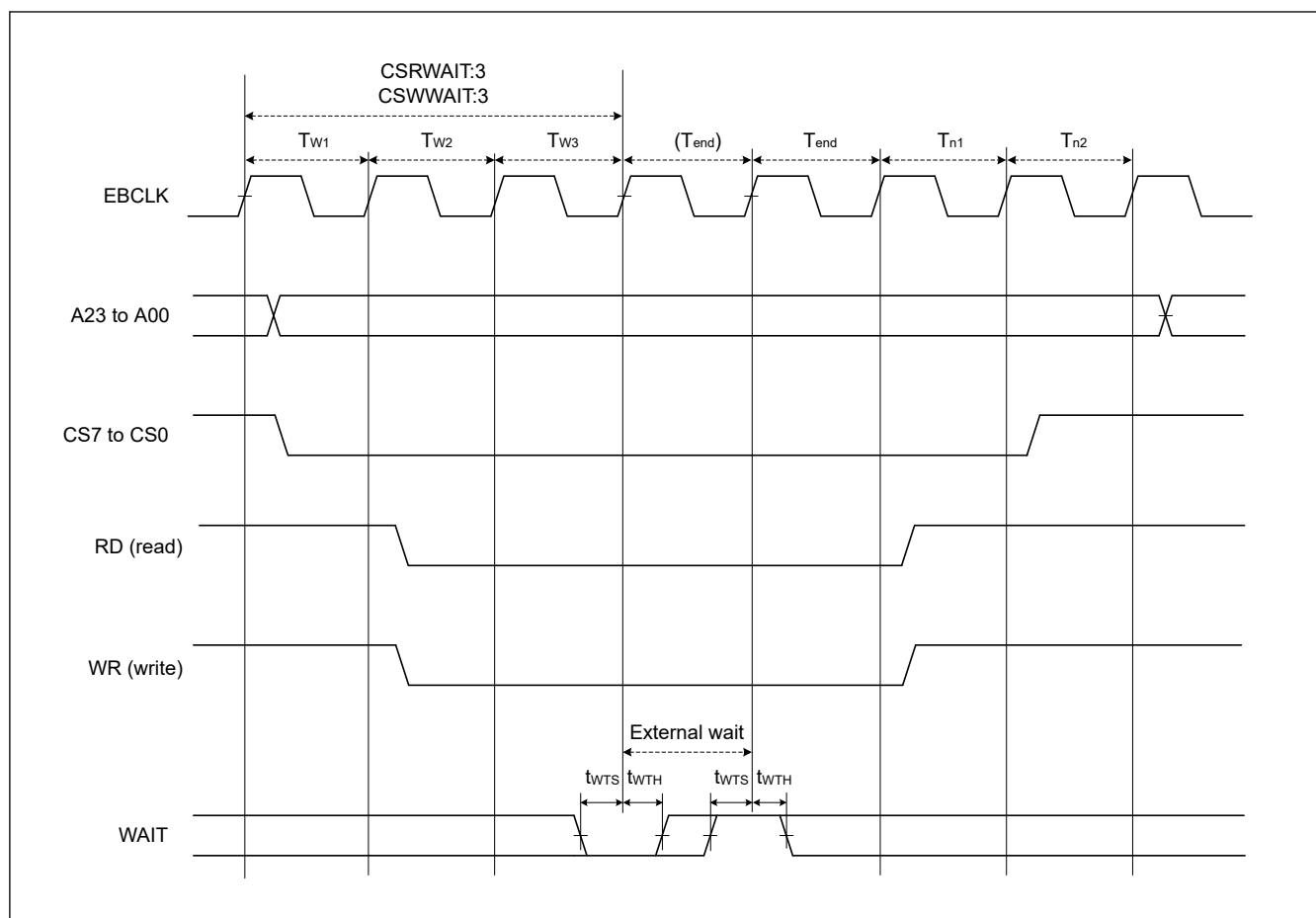


Figure 60.44 External bus timing for external wait control

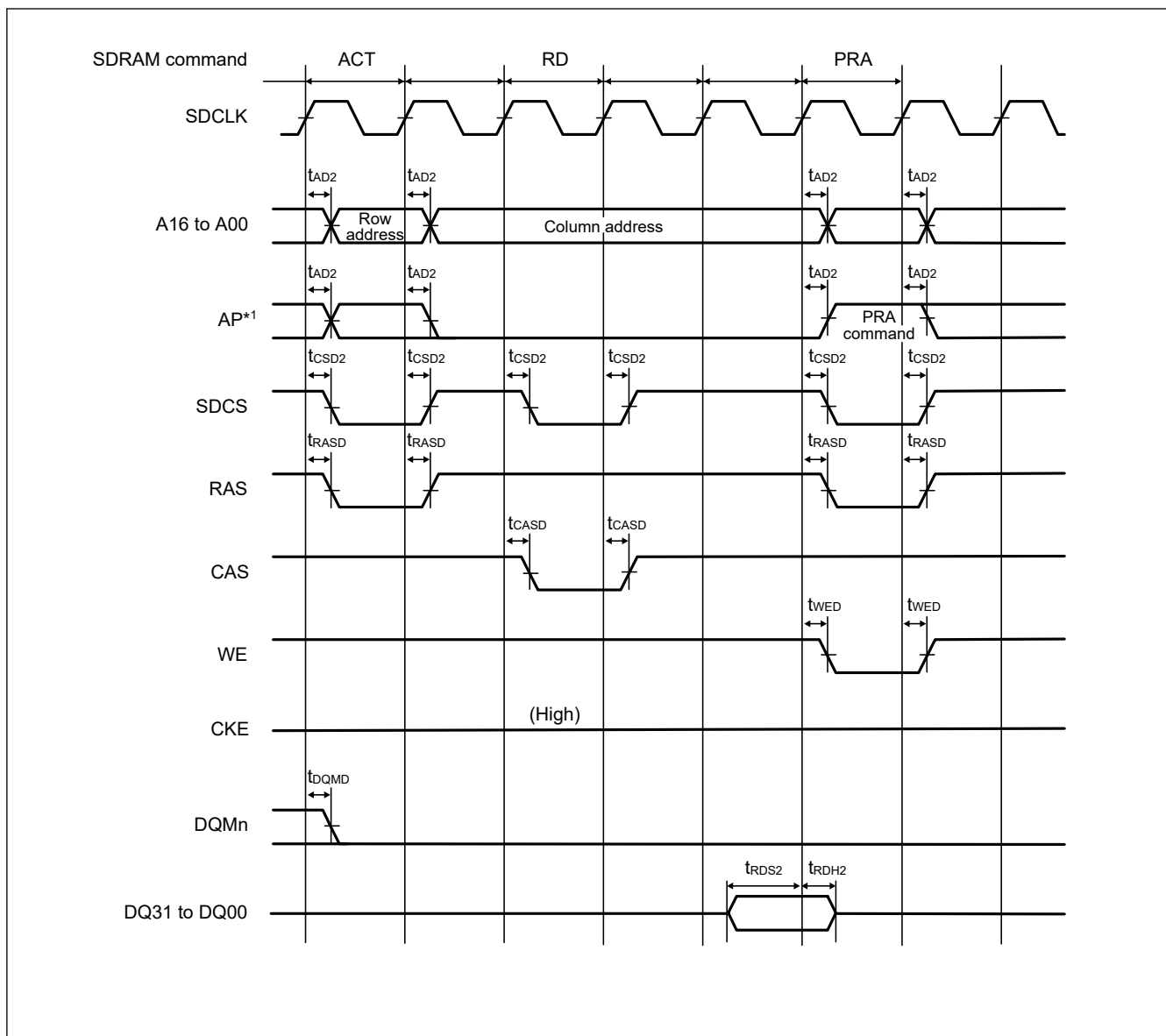


Figure 60.45 SDRAM single read timing

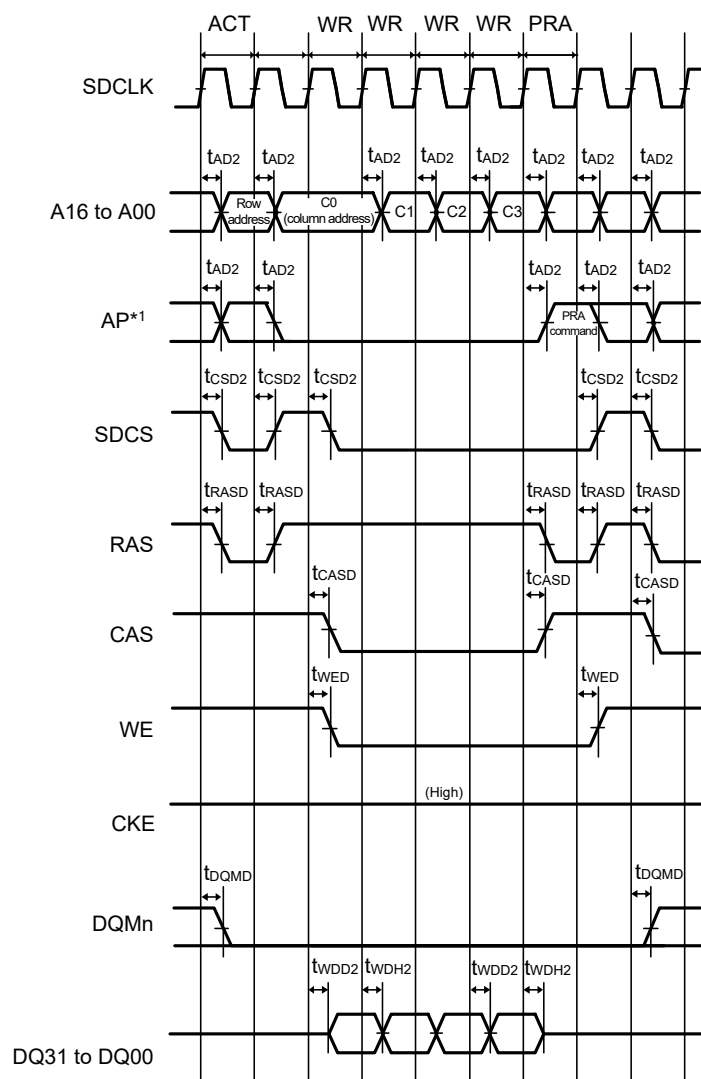


Figure 60.46 SDRAM single write timing

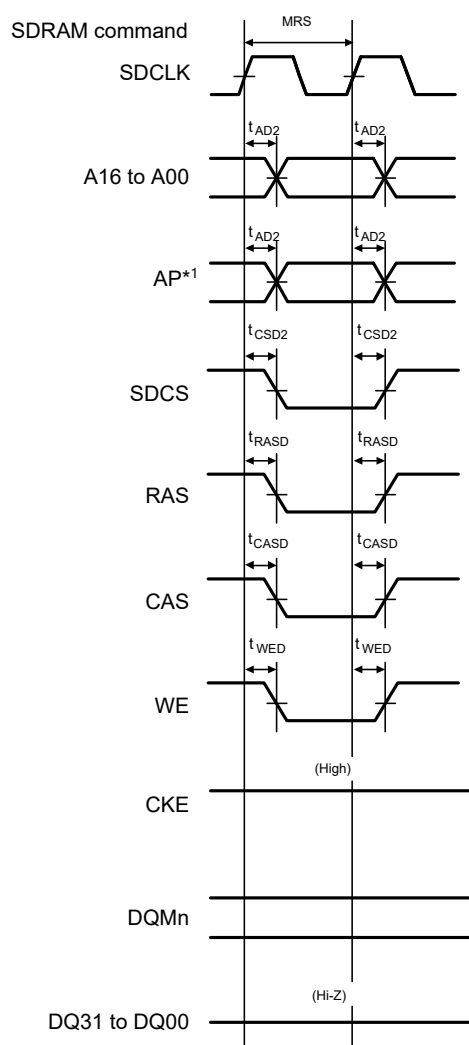


Figure 60.47 SDRAM multiple read timing

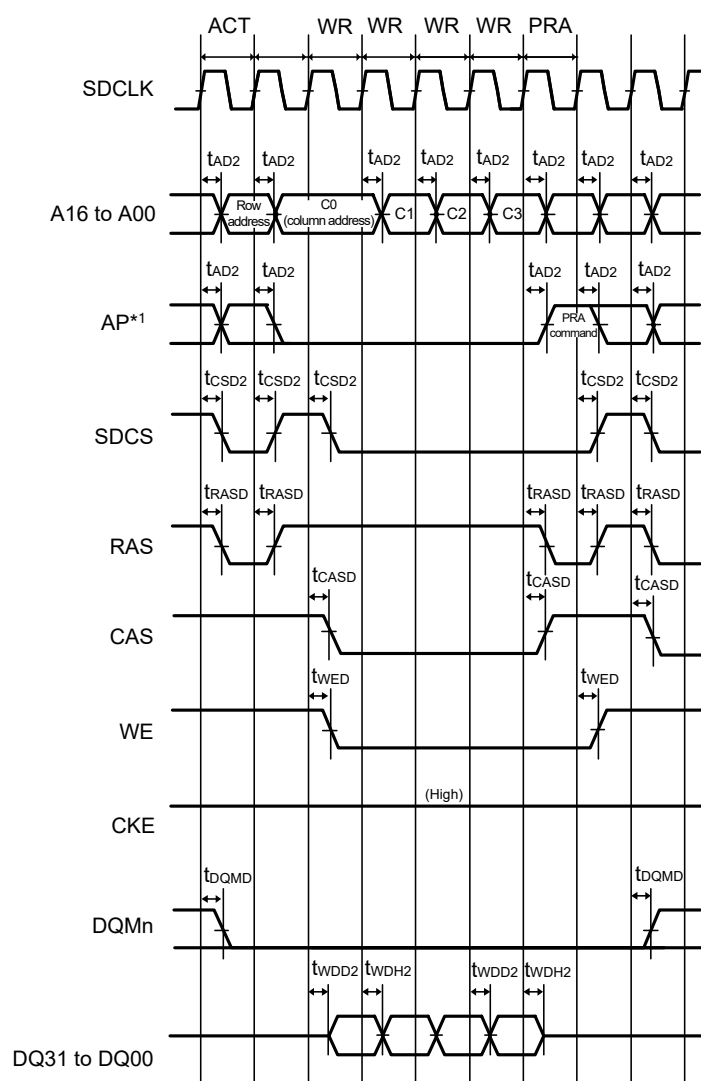


Figure 60.48 SDRAM multiple write timing

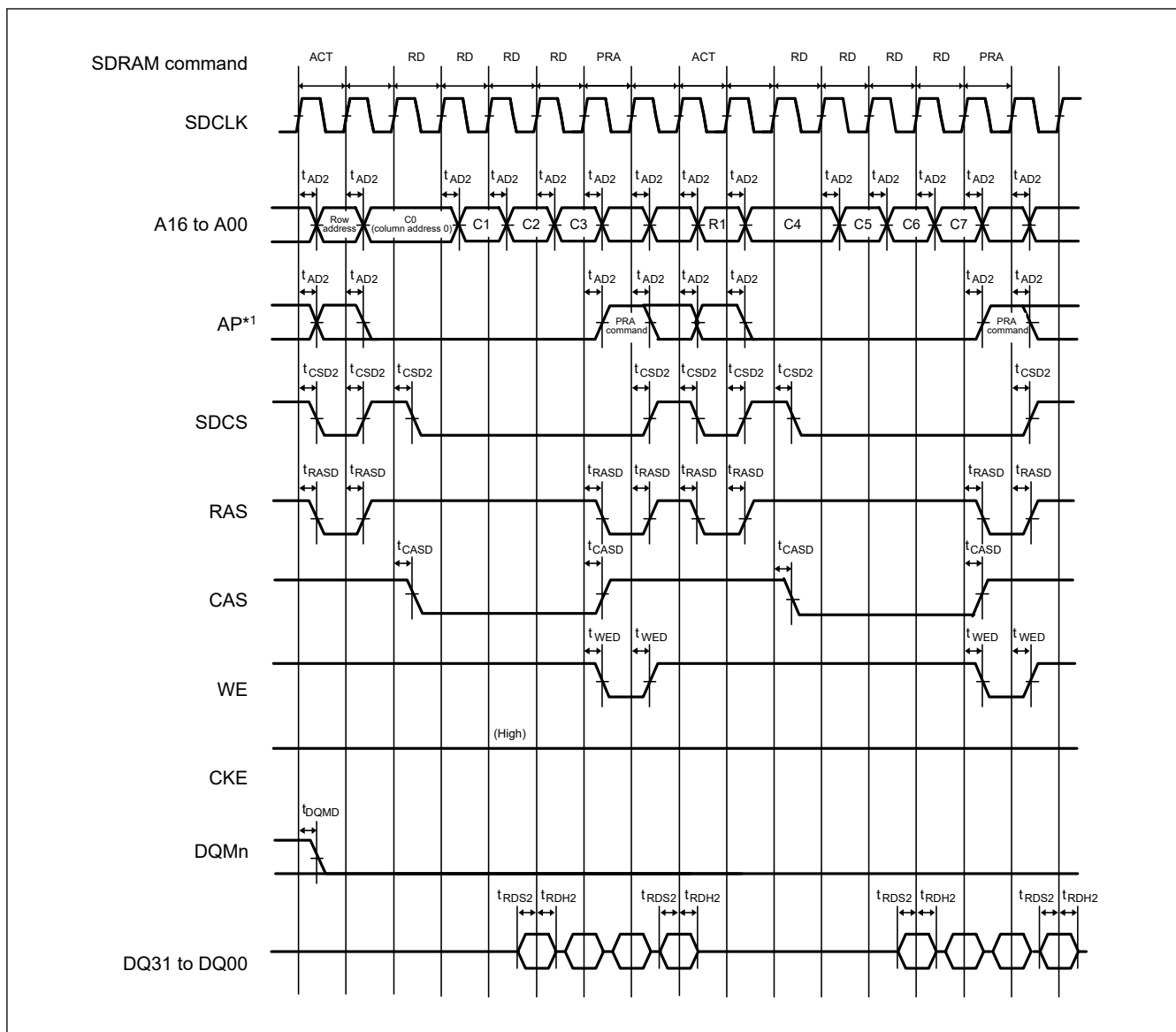


Figure 60.49 SDRAM multiple read line stride timing

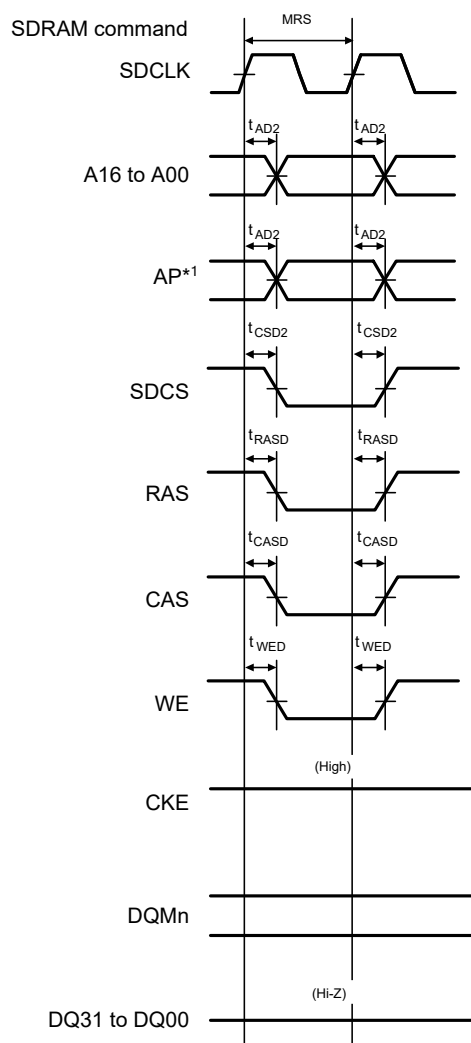


Figure 60.50 SDRAM mode register set timing

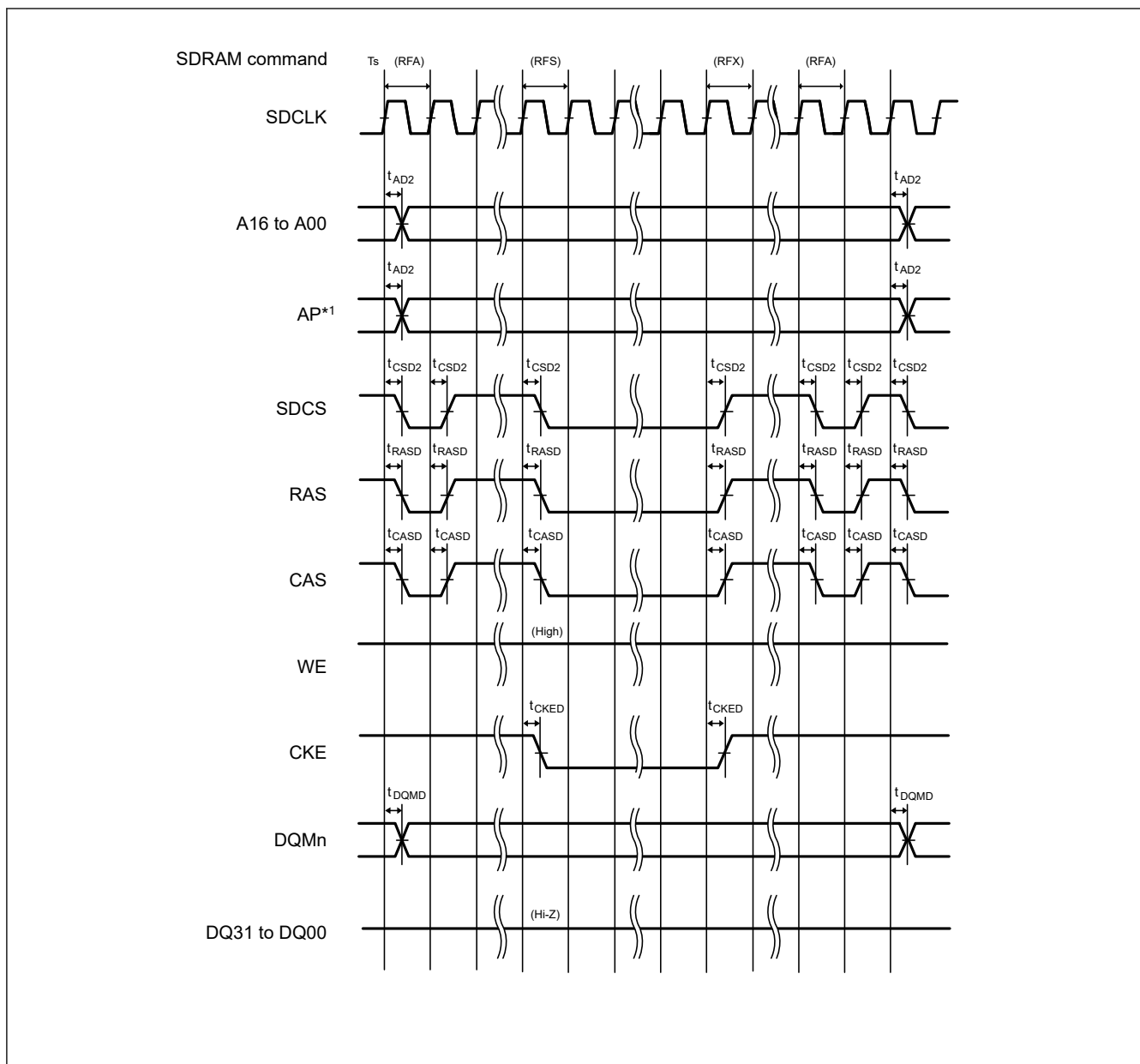


Figure 60.51 SDRAM self-refresh timing

60.3.7 I/O Ports, POEG, GPT, AGT, ULPT and ADC Trigger Timing

Table 60.56 I/O ports, POEG, GPT, AGT, ULPT and ADC trigger timing (1 of 4)

GPT32 Conditions:

Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If GPT pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC = VCC2.

AGT Conditions:

Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register.

Parameter		Symbol	Min	Max	Unit	Test conditions
I/O ports	Input data pulse width	t_{PRW}	5.5	—	$t_{l_{cyc}}$	Figure 60.52
	EXCIN input frequency	t_{EXCIN}	—	36	kHz	
	RTCICn (n = 0 to 2) input pulse width	t_{RTCICW}	13.89	—	μs	Figure 60.53

Table 60.56 I/O ports, POEG, GPT, AGT, ULPT and ADC trigger timing (2 of 4)

GPT32 Conditions:

Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If GPT pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC = VCC2.

AGT Conditions:

Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register.

Parameter			Symbol	Min	Max	Unit	Test conditions
POEG	POEG input trigger pulse width		t _{POEW}	3	—	t _{pcyc}	Figure 60.54
	Output disable time	Input level detection of the GTETRGN pin (via flag)	t _{POEGDI}	—	2 PCLK B + 0.34	μs	Figure 60.55 When the digital noise filter is not in use (POEGGn.NFEN = 0 (n = A to D))
		Detection of the output stopping signal from GPT (deadtime error, simultaneous high output, or simultaneous low output)	t _{POEGDE}	—	0.5	μs	Figure 60.56
		Edge detection signal from a comparator	t _{POEGDC}	—	3 PCLK B + 0.5	μs	Figure 60.57 The time is that when the noise filter for ACMPHS is not in use (CMPCTL.CDFS [1:0] = 00b) and excludes the time for detection by ACMPHS.
		Register setting	t _{POEGDS}	—	0.3	μs	Figure 60.58 Time for access to the register is not included.
		Oscillation stop detection	t _{POEGDOS}	—	1.3	μs	Figure 60.59
		Level detection signal from a comparator	t _{POEGDDC}	—	0.5	μs	Figure 60.60 The time is that when the noise filter for ACMPHS is not in use (CMPCTL.CDFS [1:0] = 00b) and excludes the time for detection by ACMPHS.
		Overcurrent Detection Window Notification detection from DSMIF	t _{POEGDER}	—	0.5	μs	Figure 60.61 The time is that when the noise filter for ACMPHS is not in use (CMPCTL.CDFS [1:0] = 00b) and excludes the time for detection by ACMPHS.

Table 60.56 I/O ports, POEG, GPT, AGT, ULPT and ADC trigger timing (3 of 4)

GPT32 Conditions:

Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If GPT pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC = VCC2.

AGT Conditions:

Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register.

Parameter				Symbol	Min	Max	Unit	Test conditions
GPT	Input capture pulse width (Cycle)		Single edge	t_{GTICW}^{*1}	1.5	—	t_{pDcyc}	Figure 60.62
			Dual edge		2.5	—		
	Input capture pulse width (Times)		2.70 V or above	t_{GTICW}^{*1}	8.3	—	ns	
			1.62 V or above		10.0	—		
	GTIOCxY output skew (x = 0 to 3, Y = A or B)	Middle drive output	2.70 V or above	t_{GTISK}	—	4	ns	Figure 60.63
			1.62 V or above		—	6		
		High drive output	2.70 V or above			3.5		
			1.62 V or above			4.5		
	GTIOCxY output skew (x = 4 to 13, Y = A or B)	Middle drive output	2.70 V or above		—	4		
			1.62 V or above		—	6		
		High drive output	2.70 V or above		—	3.5		
			1.62 V or above		—	4.5		
	GTIOCxY output skew (x = 0 to 13, Y = A or B)	Middle drive output	2.70 V or above	—	6			
			1.62 V or above	—	7			
		High drive output	2.70 V or above	—	3.5			
			1.62 V or above	—	5			
OPS output skew GTOUUP, GTOULO, GTOVUP, GTOVLO, GTOWUP, GTOWLO	Middle drive output	2.70 V or above	t_{GTOSK}	—	5	ns	Figure 60.64	
		1.62 V or above		—	6			
GPT (PWM Delay Generation Circuit)	GTIOCxY output skew (x = 0 to 3, Y = A or B)	Middle drive output	t_{HRSK}	2.70 V or above	—	4	ns	Figure 60.65
				1.62 V or above	—	6		
		High drive output		2.70 V or above	—	3.5		
				1.62 V or above	—	5		

Table 60.56 I/O ports, POEG, GPT, AGT, ULPT and ADC trigger timing (4 of 4)

GPT32 Conditions:

Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If GPT pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC = VCC2.

AGT Conditions:

Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register.

Parameter			Symbol	Min	Max	Unit	Test conditions
AGT	AGTIO, AGTEE input cycle	2.70 V or above	t_{ACYC}^{*2}	100	—	ns	Figure 60.66
		1.62 V or above		100	—		
	AGTIO, AGTEE input high width, low width	2.70 V or above	t_{ACKWH}, t_{ACKWL}	40	—	ns	
		1.62 V or above		40	—		
	AGTIO, AGTO, AGTOA, AGTOB output cycle	2.70 V or above	t_{ACYC2}	62.5	—	ns	
		1.62 V or above		62.5	—		
ULPT	ULPTEE, ULPTEVI input cycle	2.70 V or above	t_{ULCYC}^{*3}	32	—	μs	Figure 60.67
		1.62 V or above		32	—		
	ULPTEE, ULPTVI input high width, low width	2.70 V or above	t_{ULCKWH}, t_{ULCKWL}	12	—	μs	
		1.62 V or above		12	—		
	ULPTO, ULPTOA, ULPTOB output cycle	2.70 V or above	t_{ULCYC2}	64	—	μs	
		1.62 V or above		64	—		
ADC	ADC trigger input pulse width	2.70 V or above	t_{TRGW}	1.5	—	t_{ADcyc}	Figure 60.68
		1.62 V or above		3.0	—		

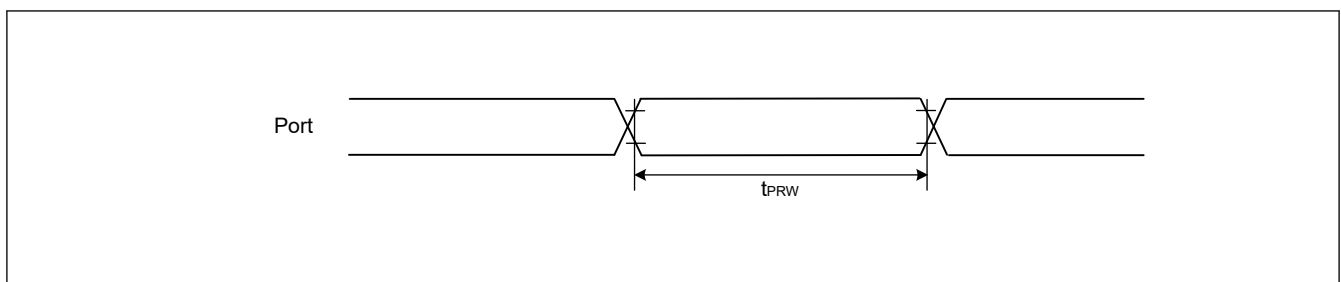
Note: t_{cyc} : ICLK cycle, t_{Pcyc} : PCLKB cycle, t_{PDcyc} : GTCLK cycle, $t_{ULPTCLK}$: ULPTCLK cycle, t_{ADcyc} : ADCLK cycle.

Note 1. For Cycle and Time, the longer time characteristics are applied.

Note 2. Constraints on input cycle:

When not switching the source clock: $t_{Pcyc} \times 2 < t_{ACYC}$ should be satisfied.When switching the source clock: $t_{Pcyc} \times 6 < t_{ACYC}$ should be satisfied.

Note 3. Constraints on input cycle:

ULPTEVI: $t_{Pcyc} \times 2 < t_{ULCYC}$ should be satisfied.ULPTEE: $t_{ULPTCLK} \times 2 < t_{ULCYC}$ should be satisfied.**Figure 60.52 I/O ports input timing**

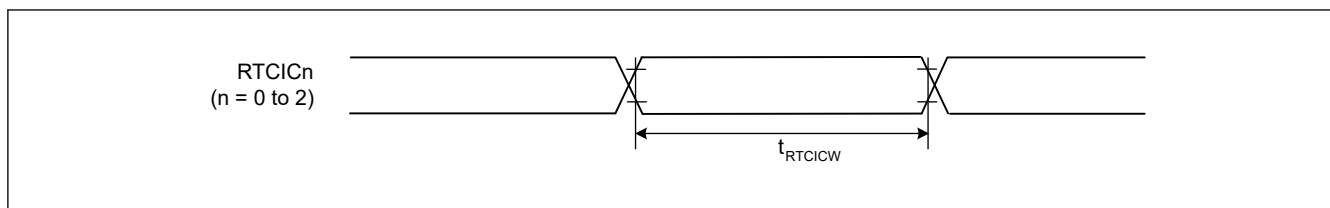


Figure 60.53 RTCICn input timing

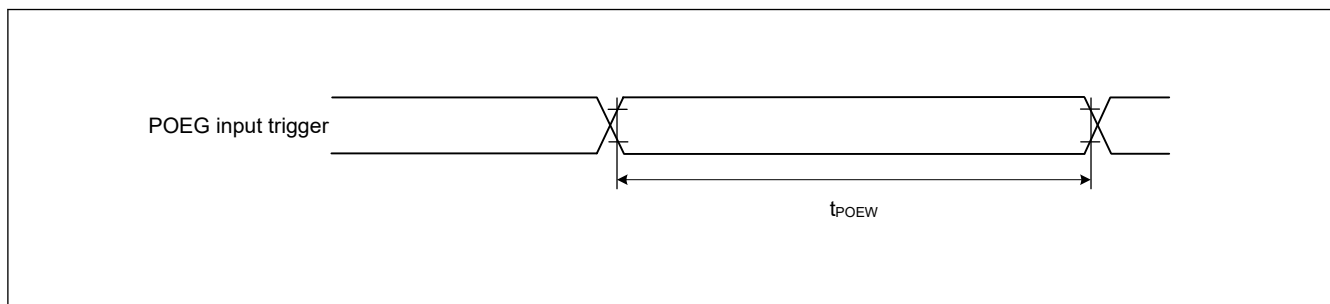


Figure 60.54 POEG input trigger timing

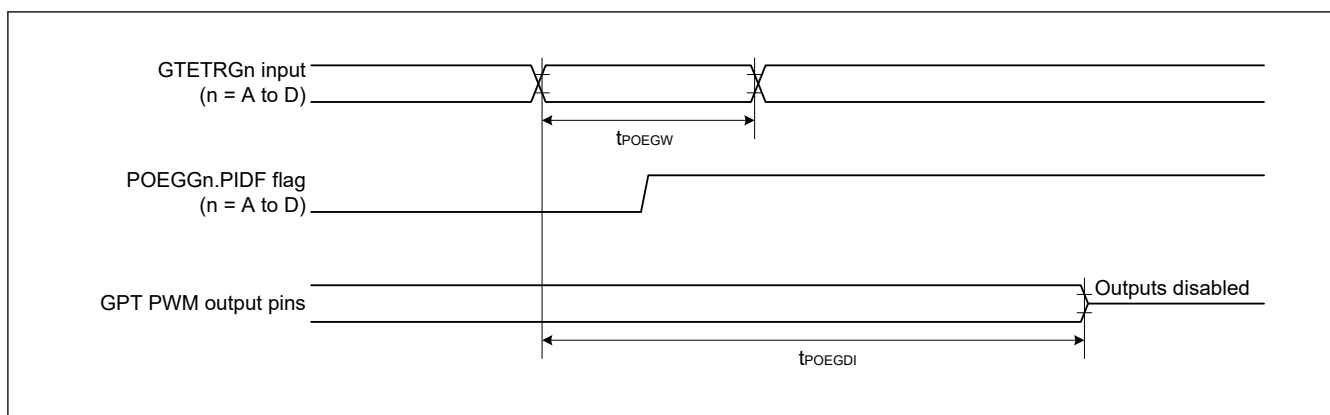


Figure 60.55 Output Disable Time for POEG via Detection Flag in Response to the Input Level Detection of the GTETRn pin

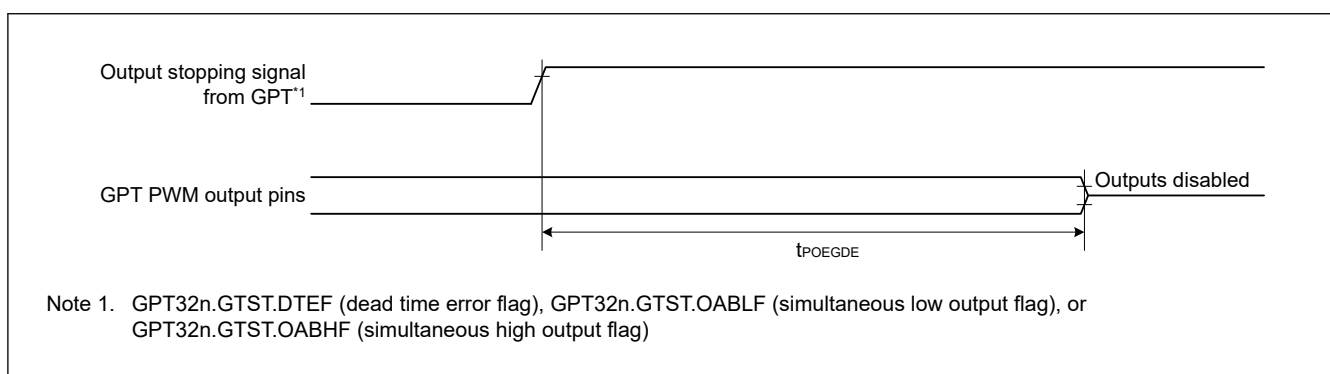


Figure 60.56 Output Disable Time for POEG in Response to Detection of the Output Stopping Signal from GPT

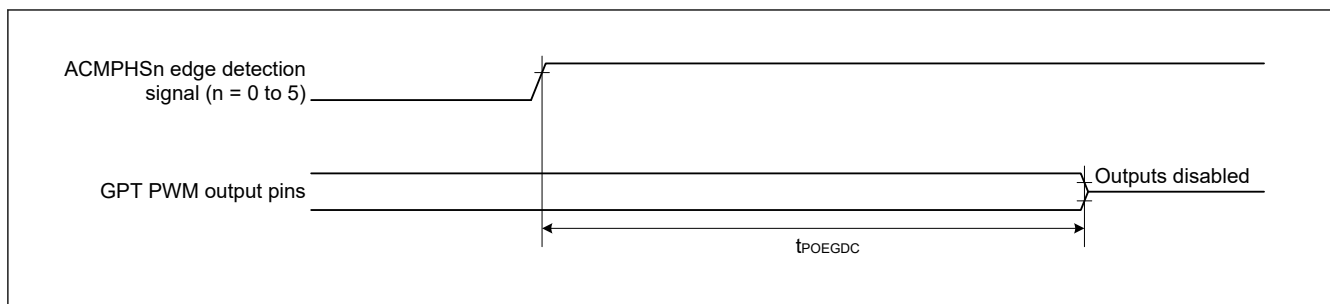


Figure 60.57 Output Disable Time for POEG in Response to Edge Detection Signal from a Comparator

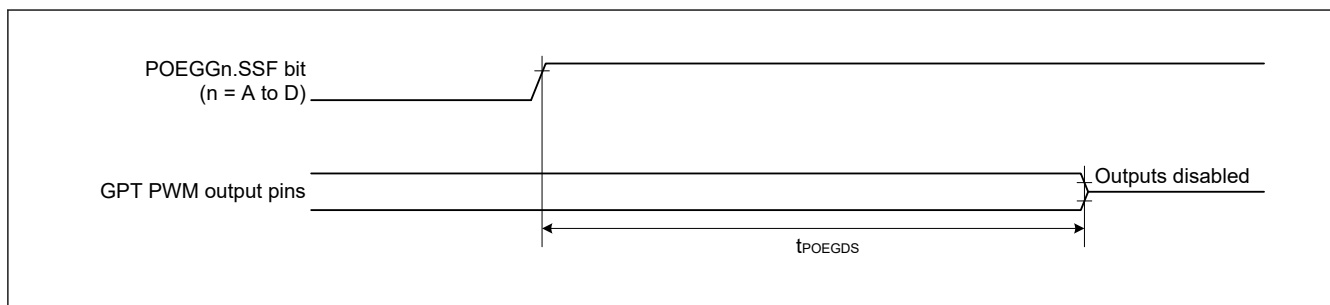


Figure 60.58 Output Disable Time for POEG in Response to the Register Setting

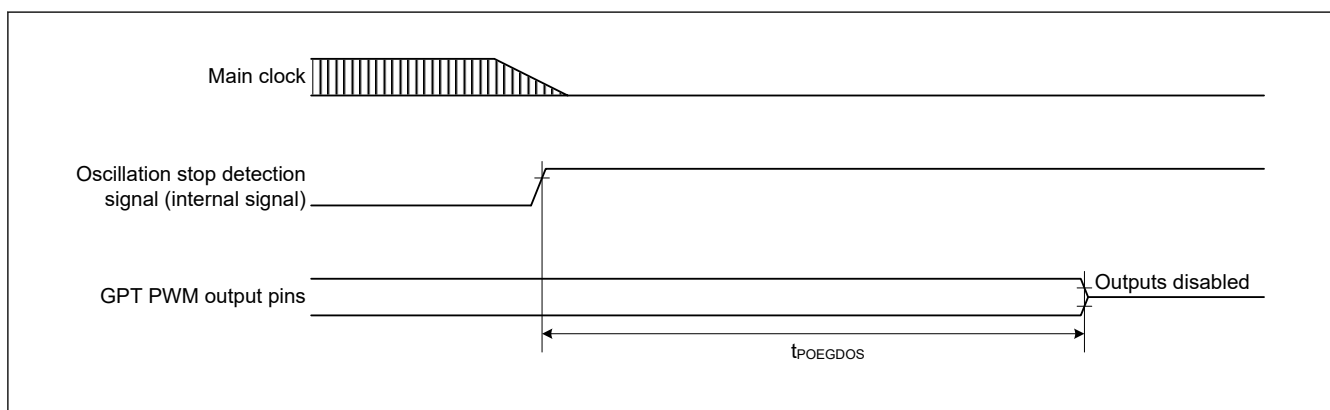


Figure 60.59 Output Disable Time of POEG in Response to the Oscillation Stop Detection

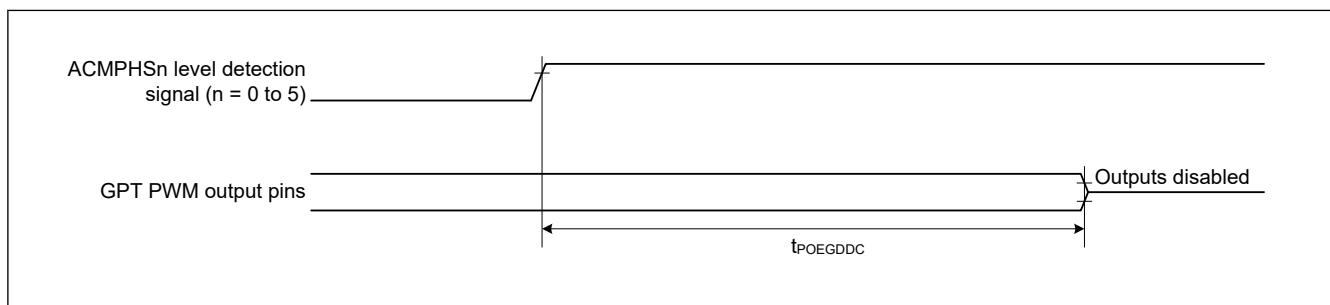


Figure 60.60 Output Disable Time for POEG in Response to Level Detection Signal from a Comparator

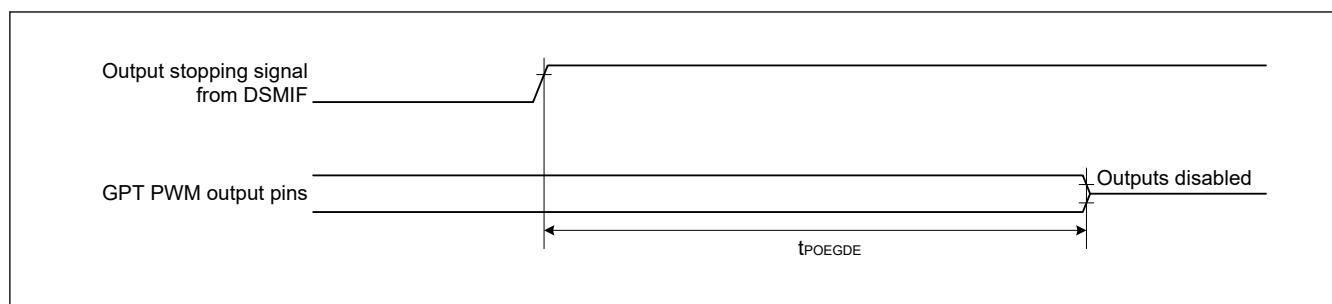


Figure 60.61 Output Disable Time for POEG in Response to Detection of the Output Stopping Signal from DSMIF

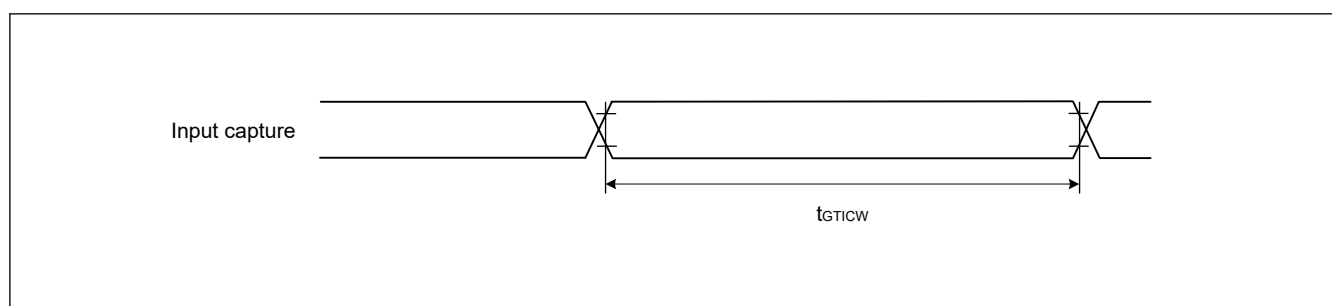


Figure 60.62 GPT input capture timing

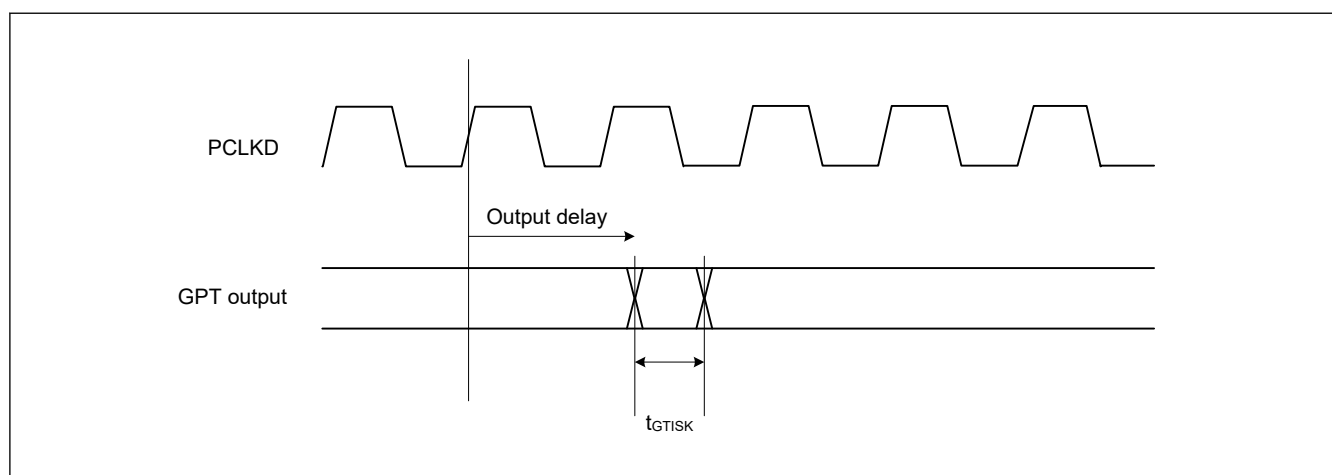


Figure 60.63 GPT output delay skew

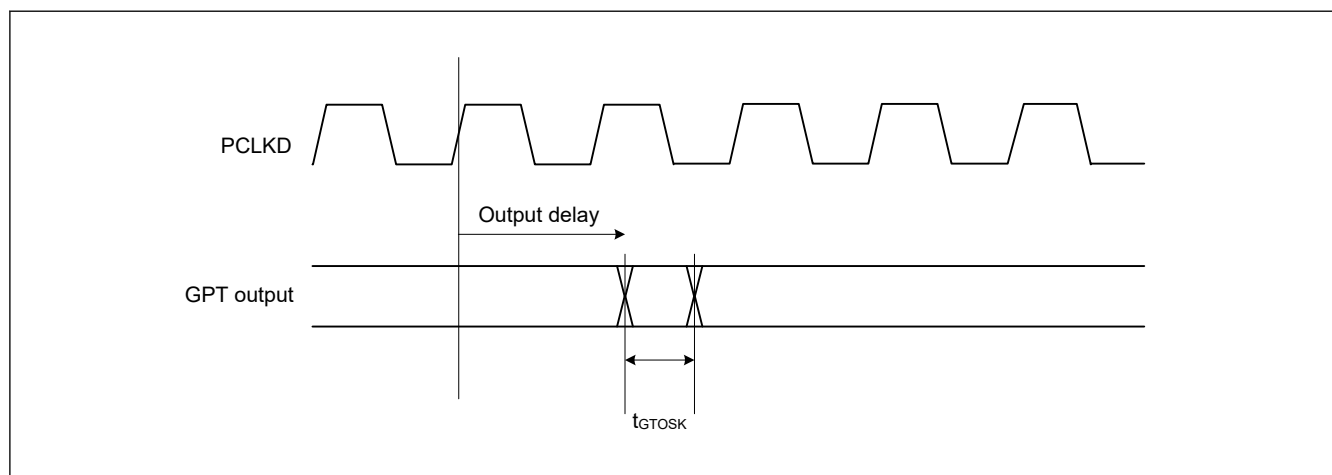


Figure 60.64 GPT output delay skew for OPS

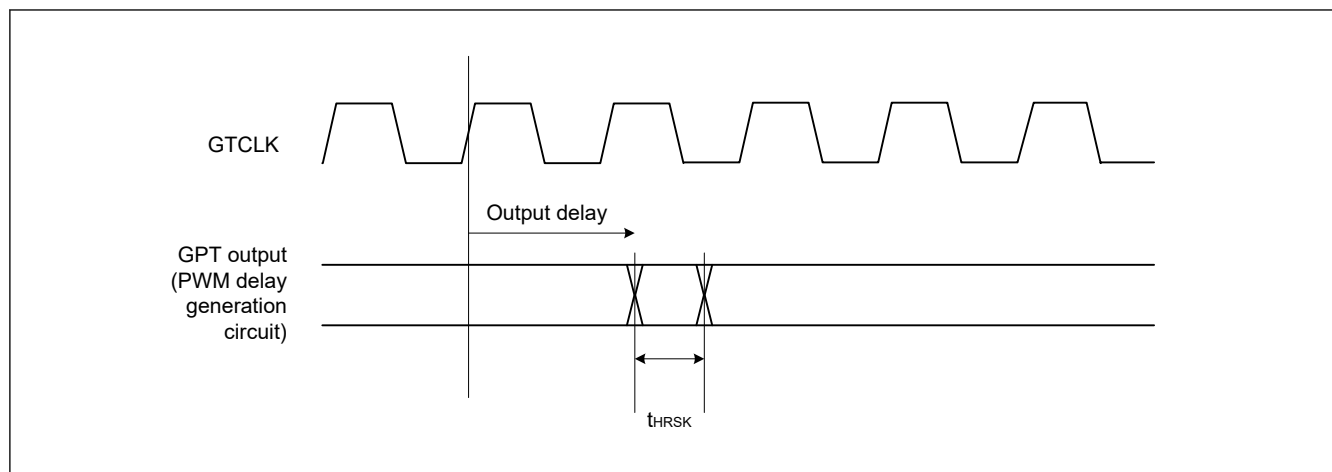


Figure 60.65 GPT (PDG) output delay skew

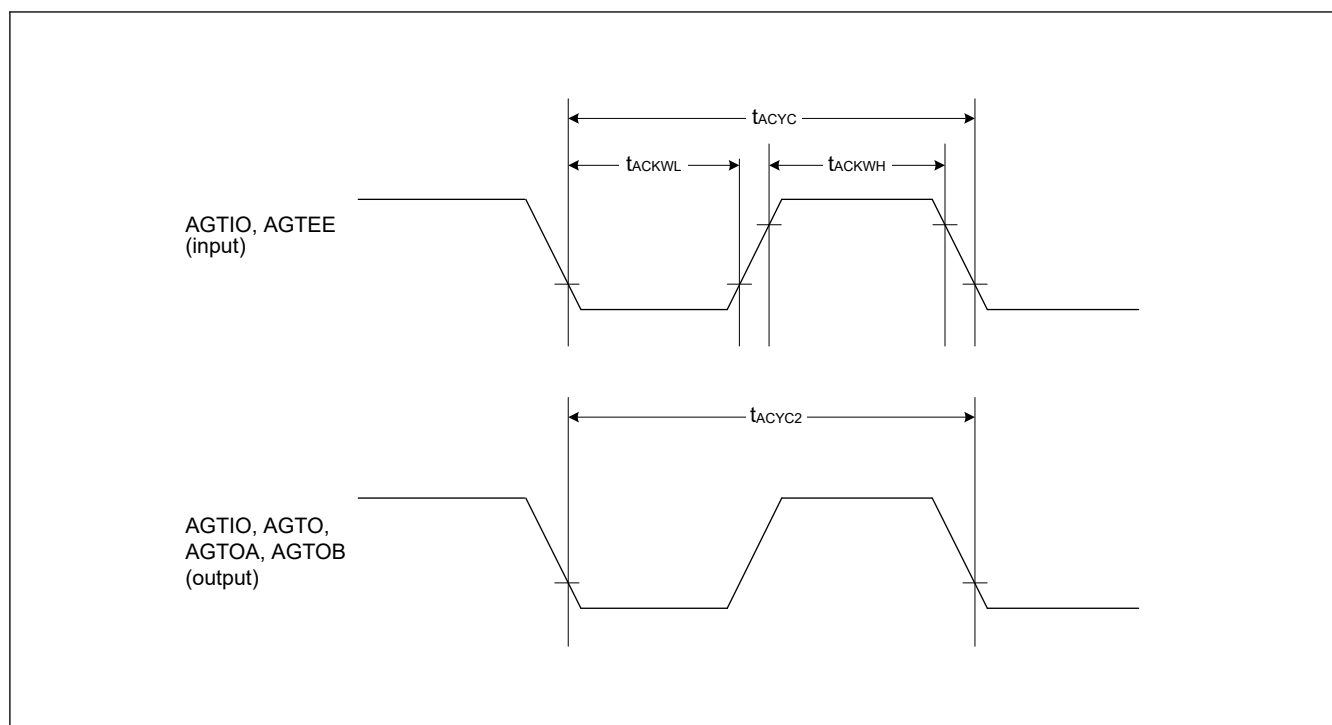


Figure 60.66 AGT input/output timing

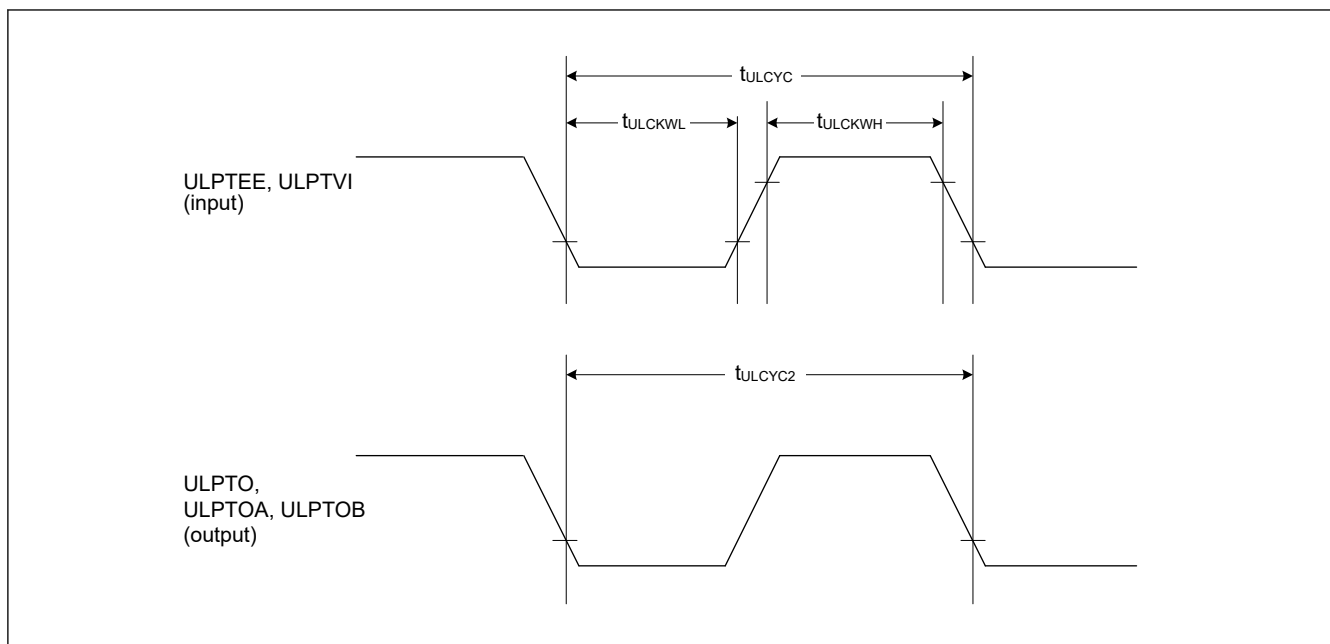


Figure 60.67 ULPT input/output timing

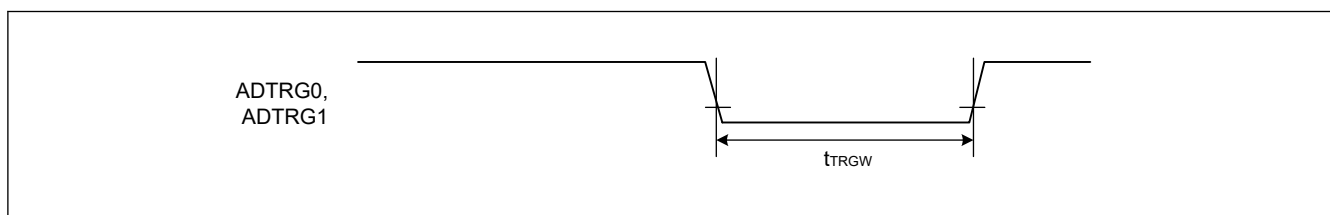


Figure 60.68 ADC trigger input timing

60.3.8 CAC Timing

Table 60.57 CAC timing

Parameter			Symbol	Min	Typ	Max	Unit	Test conditions
CAC	CACREF input pulse width	$t_{PBcyc} \leq t_{cac}^{*1}$	t_{CACREF}	$4.5 \times t_{cac} + 3 \times t_{PBcyc}$	—	—	ns	—
		$t_{PBcyc} > t_{cac}^{*1}$		$5 \times t_{cac} + 6.5 \times t_{PBcyc}$	—	—	ns	

Note: t_{PBcyc} : PCLKB cycle.

Note 1. t_{cac} : CAC count clock source cycle.

60.3.9 SCI Timing

Table 60.58 SCI timing (Asynchronous mode)

Conditions:

High drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If SCI pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC = VCC2.

Parameter	VCC/VCC2	Symbol	Min	Max	Unit	Note
Input clock cycle	1.62 V or above	t _{Scyc}	4.0	—	t _{Tcyc}	Figure 60.69
Input clock pulse width	1.62 V or above	t _{SCKW}	0.4	0.6	t _{Scyc}	
Input clock rise time	1.62 V or above	t _{SCKr}	—	0.1*1	t _{Scyc}	
Input clock fall time	1.62 V or above	t _{SCKf}	—	0.1*1	t _{Scyc}	
Output clock cycle	1.62 V or above	t _{Scyc}	6.0	—	t _{Tcyc}	
Output clock pulse width	1.62 V or above	t _{SCKW}	0.4	0.6	t _{Scyc}	
Output clock rise time	2.70 V or above	t _{SCKr}	—	3.3	ns	
	1.62 V or above		—	6.6		
Output clock fall time	2.70 V or above	t _{SCKf}	—	3.3	ns	
	1.62 V or above		—	6.6		

Note: t_{Tcyc} : TCLK cycle.

Note 1. 1 μ s at the longest

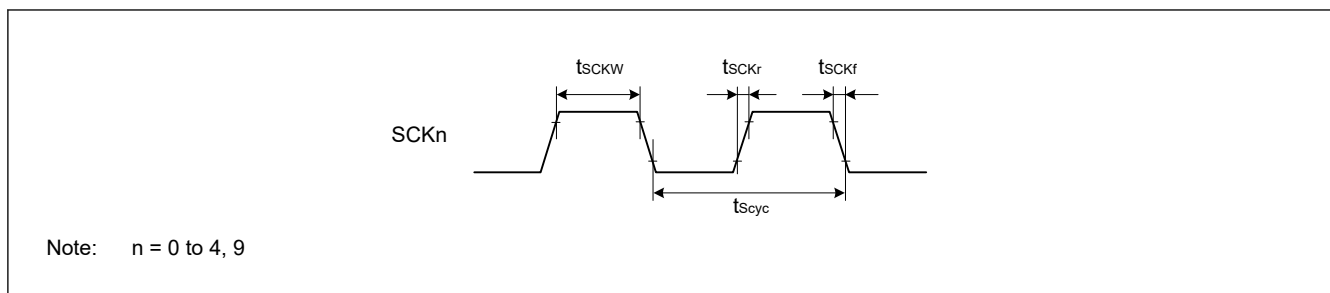


Figure 60.69 SCK clock input/output timing

Table 60.59 SCI timing (Simple SPI) (1 of 3)

Condition 1: VCC/VCC2 = 2.70 V or above

High drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If SCI pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC = VCC2.

Condition 2: VCC/VCC2 = 1.62 V or above

Following pins have high-speed high-drive output selected in the Port Drive Capability bit in the PmnPFS register: SCK1_A, SCK1_C, SCK3_A, SCK4_A, SCK4_B, SCK4_C and SCK6_B

Other pins have high drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If SCI pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC=VCC2.

Parameter		High Speed/ Default	VCC/VCC2	Symbol	Min	Max	Unit	Note	
SCK clock cycle output	Master	—	2.70 V or above	t _{SPcyc}	2 (TCLK ≤ 120MHz) 4 (TCLK > 120MHz)	65536	t _{Tcyc}	Figure 60.70	
			1.62 V or above		2 (TCLK ≤ 60MHz) 4 (TCLK ≤ 120MHz) 8 (TCLK > 120MHz)	65536			
SCK clock cycle input	Slave	—	2.70 V or above		2	—			
			1.62 V or above		2 (TCLK ≤ 100MHz) 4 (TCLK > 100MHz)	—			
SCK clock high pulse width	Master	—	1.62 V or above		t _{SPCKWH}	0.4	—		t _{SPcyc}
	Slave	—							
SCK clock low pulse width	Master	—	1.62 V or above	t _{SPCKWL}	0.4	—	t _{SPcyc}		
	Slave	—							
SCK clock rise and fall time	Output	—	2.70 V or above	t _{SPCKr} , t _{SPCKf}	—	3.3	ns		
			1.62 V or above		—	6.6			
	Input	—	2.70 V or above	—	0.1*3	t _{SPcyc}			
			1.62 V or above	—	0.1*3				

Table 60.59 SCI timing (Simple SPI) (2 of 3)

Condition 1: VCC/VCC2 = 2.70 V or above

High drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If SCI pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC = VCC2.

Condition 2: VCC/VCC2 = 1.62 V or above

Following pins have high-speed high-drive output selected in the Port Drive Capability bit in the PmnPFS register: SCK1_A, SCK1_C, SCK3_A, SCK4_A, SCK4_B, SCK4_C and SCK6_B

Other pins have high drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If SCI pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC=VCC2.

Parameter		High Speed/ Default	VCC/VCC2	Symbol	Min	Max	Unit	Note
Data input setup time	Master	High Speed*1	2.70 V or above	t_{SU}	-1.5	—	ns	Figure 60.71, Figure 60.72
			1.62 V or above		-1.5	—		
		Default*2	2.70 V or above		2.0	—		
			1.62 V or above		2.0	—		
	Slave	Default*2	2.70 V or above		2.5	—		
			1.62 V or above		4.5	—		
Data input hold time	Master	High Speed*1	2.70 V or above	t_H	7.5	—	ns	
			1.62 V or above		9.5	—		
		Default*2	2.70 V or above		7.5	—		
			1.62 V or above		9.5	—		
	Slave	Default*2	2.70 V or above		2.5	—		
			1.62 V or above		4.5	—		
Data output delay	Master	High Speed*1	2.70 V or above	t_{OD}	—	3.0	ns	
			1.62 V or above		—	4.5		
		Default*2	2.70 V or above		—	3.5		
			1.62V or above		—	5.5		
	Slave	High Speed*1	2.70 V or above		—	12.5		
			1.62 V or above		—	20.5		
		Default*2	2.70 V or above		—	18.5		
			1.62 V or above		—	26.5		
Data output hold time	Master	High Speed*1	2.70 V or above	t_{OH}	-3.0	—	ns	
			1.62 V or above		-4.5	—		
		Default*2	2.70 V or above		-3.5	—		
			1.62 V or above		-5.5	—		
	Slave	Default*2	2.70 V or above		0.0	—		
			1.62 V or above		0.0	—		
Data rise and fall time	Output	—	2.70 V or above	t_{Dr}, t_{Df}	—	3.3	ns	
			1.62 V or above		—	6.6		
	Input	—	2.70 V or above		—	1		
			1.62 V or above		—	1		
SS input setup time	—	—	1.62 V or above	t_{LEAD}	1.0	—	t_{SPcyc}	Figure 60.73, Figure 60.74
SS input hold time	—	—	1.62 V or above	t_{LAG}	1.0	—	t_{SPcyc}	
SS input rise and fall time	—	—	1.62 V or above	t_{SSLr}, t_{SSLf}	—	1	μs	—

Table 60.59 SCI timing (Simple SPI) (3 of 3)

Condition 1: VCC/VCC2 = 2.70 V or above

High drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If SCI pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC = VCC2.

Condition 2: VCC/VCC2 = 1.62 V or above

Following pins have high-speed high-drive output selected in the Port Drive Capability bit in the PmnPFS register: SCK1_A, SCK1_C, SCK3_A, SCK4_A, SCK4_B, SCK4_C and SCK6_B

Other pins have high drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If SCI pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC=VCC2.

Parameter	High Speed/ Default	VCC/VCC2	Symbol	Min	Max	Unit	Note
Slave access time	—	2.70 V or above	t _{SA}	—	3 × t _{TCYC} + 25	ns	Figure 60.73, Figure 60.74
		1.62 V or above		—	3 × t _{TCYC} + 32		
Slave output release time	—	2.70 V or above	t _{REL}	—	3 × t _{TCYC} + 25	ns	
		1.62 V or above		—	3 × t _{TCYC} + 32		

Note: t_{Tcyc} : TCLK cycle.

Note 1. Must use pins that have a letter appended to their name, for instance _A, _B, _C, to indicate group membership. SCI0, SCI1, SCI2, SCI3 and SCI9 are instance _A, SCI4 and SCI5 are instance _B, SCI6, SCI7 and SCI8 are instance _C.

Note 2. All pins of group membership can be used.

Note 3. 1 μ s at the longest**Table 60.60 SCI timing (Clock synchronous mode) (1 of 2)**

Condition 1: VCC/VCC2 = 2.70 V or above

High drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If SCI pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC = VCC2.

Condition 2: VCC/VCC2 = 1.62 V or above

Following pins have high-speed high-drive output selected in the Port Drive Capability bit in the PmnPFS register: SCK1_A, SCK1_C, SCK3_A, SCK4_A, SCK4_B, SCK4_C and SCK6_B

Other pins have high drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If SCI pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC=VCC2.

Parameter		High Speed/ Default	VCC/VCC2	Symbol	Min	Max	Unit	Note
SCK clock cycle output	Master	—	2.70 V or above	t_{Scyc}	2 (TCLK $\leq$ 120MHz) 4 (TCLK > 120MHz)	—	t_{Tcyc}	
			1.62 V or above		2 (TCLK $\leq$ 60MHz) 4 (TCLK $\leq$ 120MHz) 8 (TCLK > 120MHz)	—		
SCK clock cycle input	Slave	—	2.70 V or above		2	—		
			1.62 V or above		2 (TCLK $\leq$ 100MHz) 4 (TCLK > 100MHz)	—		
SCK clock high pulse width	Master	—	1.62 V or above	t_{SCKWH}	0.4	0.6	t_{Scyc}	
	Slave		1.62 V or above					
SCK clock low pulse width	Master	—	1.62 V or above	t_{SCKWL}	0.4	0.6	t_{Scyc}	
	Slave		1.62 V or above					

Table 60.60 SCI timing (Clock synchronous mode) (2 of 2)

Condition 1: VCC/VCC2 = 2.70 V or above

High drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If SCI pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC = VCC2.

Condition 2: VCC/VCC2 = 1.62 V or above

Following pins have high-speed high-drive output selected in the Port Drive Capability bit in the PmnPFS register: SCK1_A, SCK1_C, SCK3_A, SCK4_A, SCK4_B, SCK4_C and SCK6_B

Other pins have high drive output is selected in the Port Drive Capability bit in the PmnPFS register.

If SCI pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC=VCC2.

Parameter		High Speed/ Default	VCC/VCC2	Symbol	Min	Max	Unit	Note
SCK clock rise and fall time	Output	—	2.70 V or above	t_{SCKr}, t_{SCKf}	—	3.3	ns	
			1.62 V or above		—	6.6		
	Input	—	1.62 V or above		—	0.1*3	t_{Scyc}	
Data input setup time	Master	High Speed*1	2.70 V or above	t_{SU}	2.6	—	ns	
			1.62 V or above		2.6	—		
		Default*2	2.70 V or above		2.8	—		
			1.62 V or above		2.8	—		
	Slave	Default*2	2.70 V or above		3.3	—		
			1.62 V or above		5.3	—		
Data input hold time	Master	High Speed*1	2.70 V or above	t_H	7.5	—	ns	
			1.62 V or above		9.5	—		
		Default*2	2.70 V or above		7.5	—		
			1.62 V or above		9.5	—		
	Slave	Default*2	2.70 V or above		3.0	—		
			1.62 V or above		5.0	—		
Data output delay	Master	High Speed*1	2.70 V or above	t_{OD}	—	5	ns	
			1.62 V or above		—	5		
		Default*2	2.70 V or above		—	7.3		
			1.62 V or above		—	7.3		
	Slave	High Speed*1	2.70 V or above		—	12.5		
			1.62 V or above		—	20.5		
		Default*2	2.70 V or above		—	18.5		
			1.62 V or above		—	26.5		
Data rise and fall time	Output	—	2.70 V or above	t_{Dr}, t_{Df}	—	3.3	ns	
			1.62 V or above		—	6.6		
	Input	—	1.62 V or above		—	1	μs	

Note: t_{Tcyc} : TCLK cycle.

Note 1. Must use pins that have a letter appended to their name, for instance _A, _B, _C, to indicate group membership. SCI0, SCI1, SCI2, SCI3 and SCI9 are instance _A, SCI4 and SCI5 are instance _B, SCI6, SCI7 and SCI8 are instance _C.

Note 2. All pins of group membership can be used.

Note 3. 1 μs at the longest

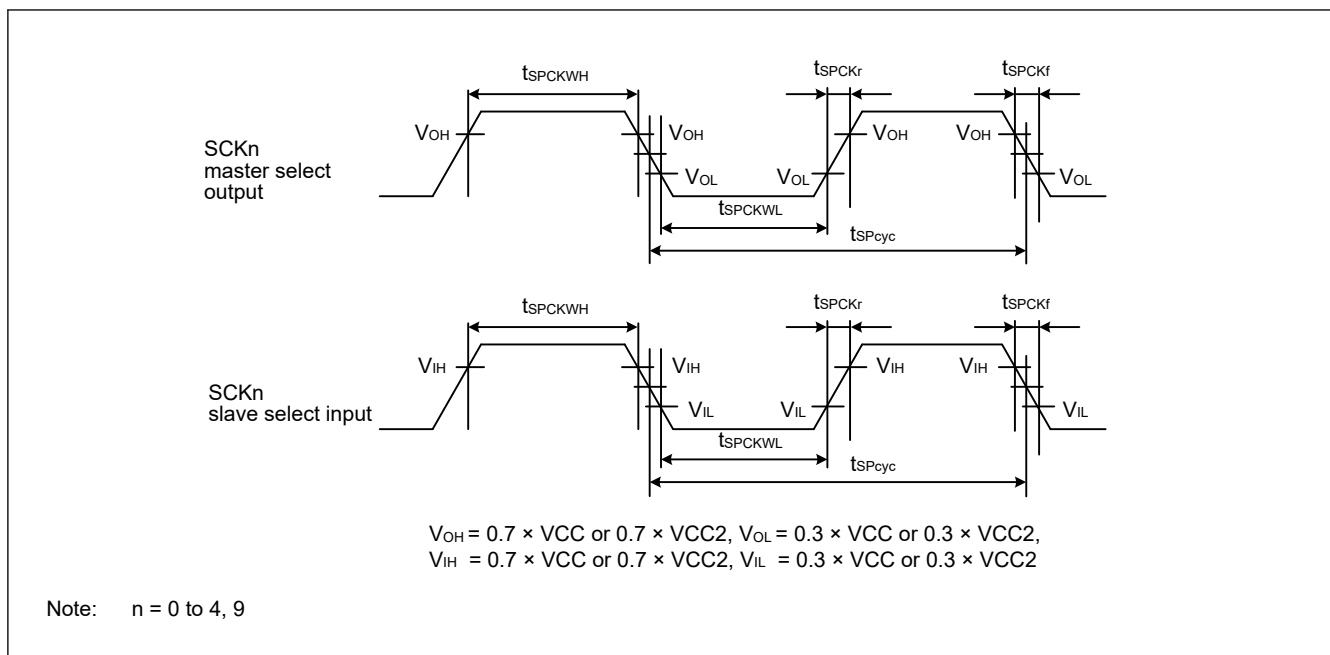


Figure 60.70 SCI simple SPI mode clock timing

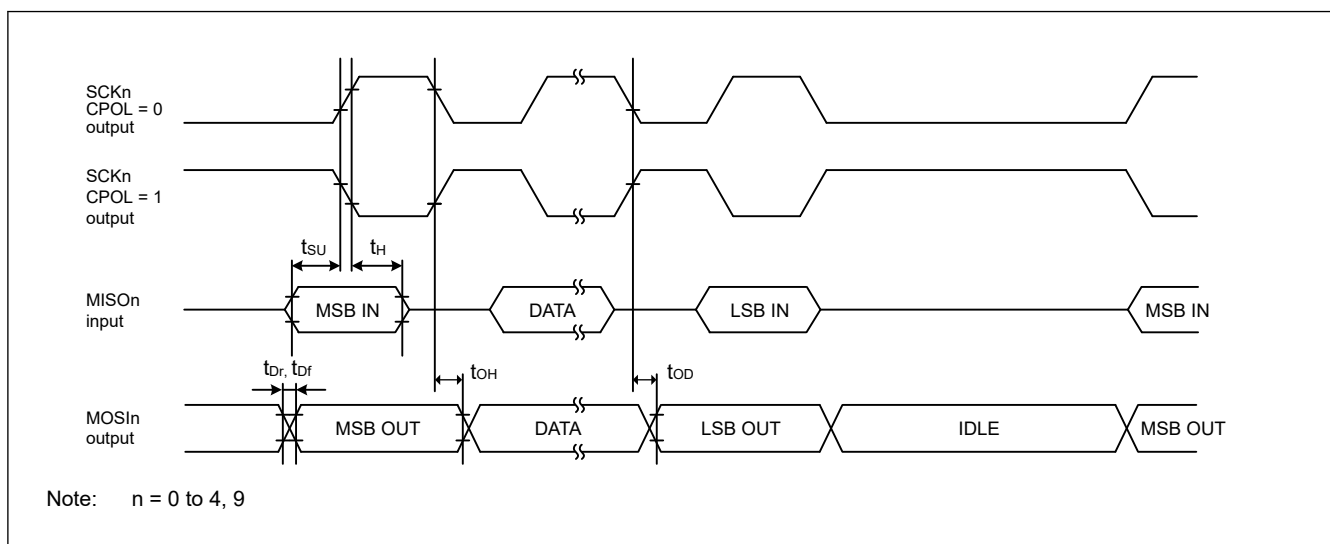


Figure 60.71 SCI simple SPI mode timing for master when CPHA = 0

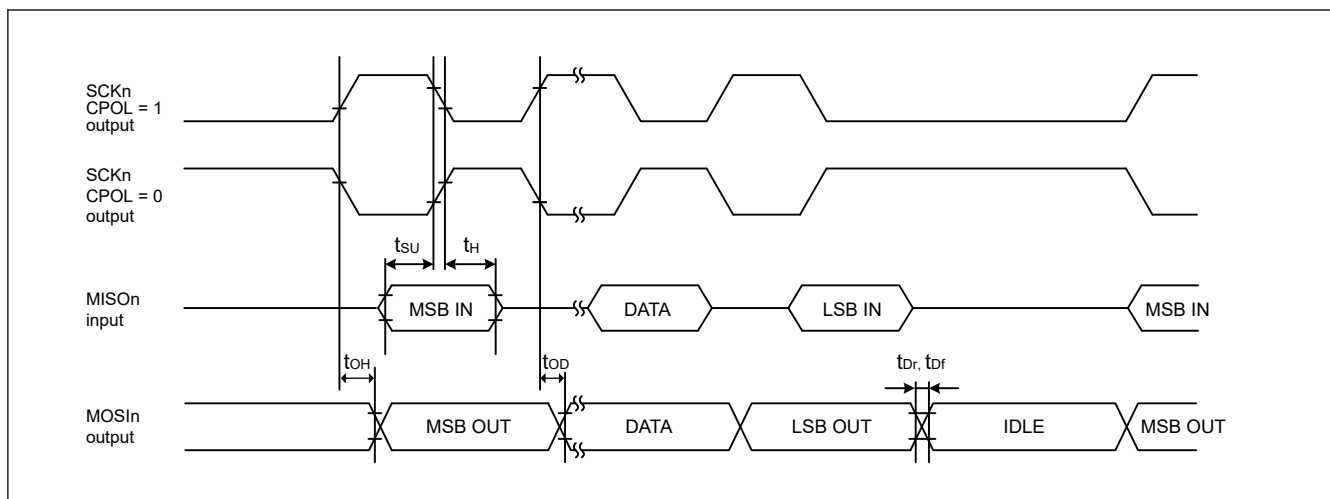


Figure 60.72 SCI simple SPI mode timing for master when $CPHA = 1$

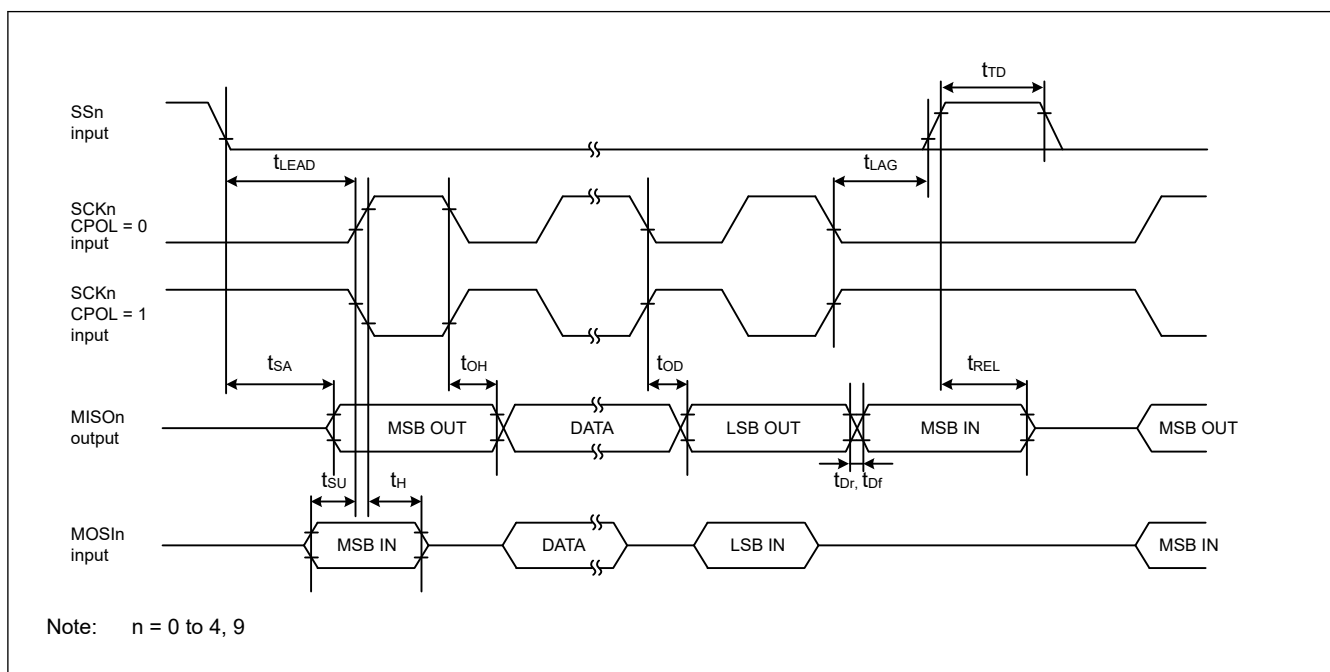


Figure 60.73 SCI simple SPI mode timing for slave when $CPHA = 0$

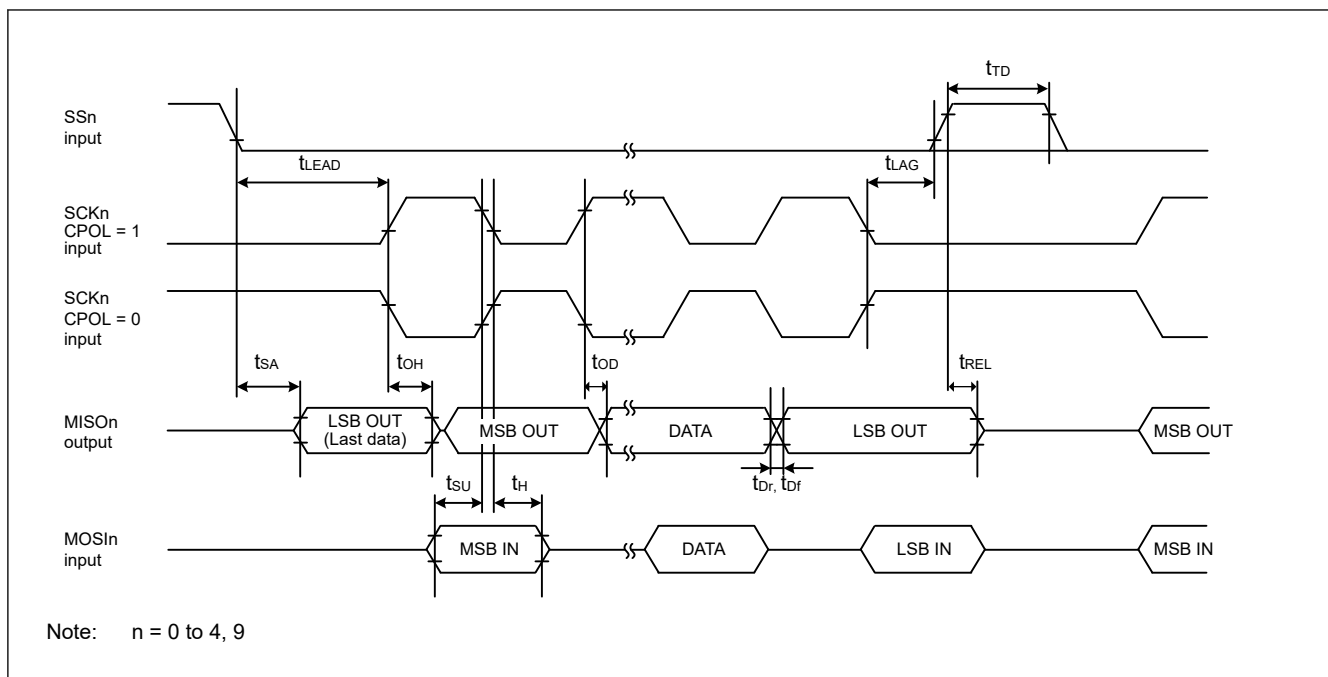


Figure 60.74 SCI simple SPI mode timing for slave when CPHA = 1

Table 60.61 SCI timing (Simple IIC mode)

Conditions:

Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register.

VCC/VCC2 : 1.62 V or above

If SCI pins are specified across the VCC I/O and VCC2 I/O, characteristics below is guaranteed only when VCC = VCC2.

Parameter	Symbol	Min	Max	Unit	Note
Simple IIC (Standard mode)	SCL, SDA input rise time	t_{Sr}	—	1000	ns
	SCL, SDA input fall time	t_{Sf}	—	300	ns
	SCL, SDA input spike pulse removal time	t_{SP}	0	$4 \times t_{Tcyc}$	ns
	Data input setup time	t_{SDAS}	250	—	ns
	Data input hold time	t_{SDAH}	0	—	ns
	SCL, SDA capacitive load	C_b^{*1}	—	400	pF
Simple IIC (Fast mode)	SCL, SDA input rise time	t_{Sr}	—	300	ns
	SCL, SDA input fall time	t_{Sf}	—	300	ns
	SCL, SDA input spike pulse removal time	t_{SP}	0	$4 \times t_{Tcyc}$	ns
	Data input setup time	t_{SDAS}	100	—	ns
	Data input hold time	t_{SDAH}	0	—	ns
	SCL, SDA capacitive load	C_b^{*1}	—	400	pF

Note: t_{Tcyc} : TCLK cycle.

Note 1. C_b indicates the total capacity of the bus line.

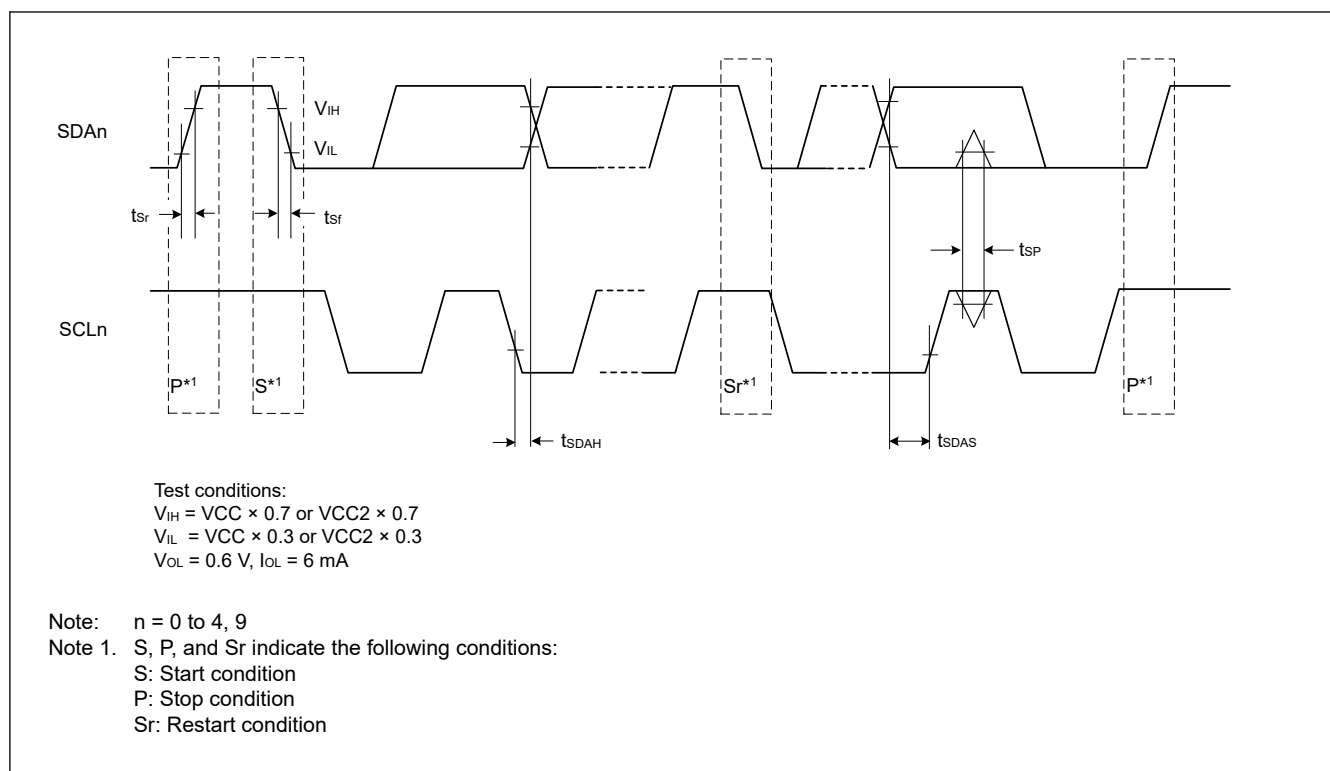


Figure 60.75 SCI simple IIC mode timing

60.3.10 SPI Timing

Table 60.62 SPI timing (1 of 5)

Conditions:

- High-speed high drive output is selected in the Port Drive Capability bit in the PmnPFS register for the following pins:
RSPCLKA_B, RSPCLKB_B.
For other pins, high drive output is selected in the Port Drive Capability bit in the PmnPFS register.
- Use pins that have a letter appended to their names, for instance “_A” or “_B” to indicate group membership.
- Load capacitance C = 15pF is applied to the VCC/VCC2 condition “3.00 V or above”.

Parameter		High Speed/ Default	VCC/VCC2	Symbol	Min	Max	Unit	Note
RSPCK clock cycle	Master	High Speed* ¹	3.00 V or above	t_{SPcyc}	2	4096	t_{Tcyc}	Figure 60.76
		High Speed* ¹	2.70 V or above		2 (TCLK ≤ 166.6MHz) 4 (TCLK > 166.6 MHz)	4096		
		High Speed* ¹	1.62 V or above		2 (TCLK ≤ 83.3MHz) 4 (TCLK ≤ 166.6 MHz) 8 (TCLK > 166.6 MHz)	4096		
		Default* ²	3.00 V or above		2 (TCLK ≤ 166.6 MHz) 4 (TCLK > 166.6 MHz)	4096		
		Default* ²	2.70 V or above		2 (TCLK ≤ 120MHz) 4 (TCLK ≤ 240 MHz) 8 (TCLK > 240 MHz)	4096		
		Default* ²	1.62 V or above		2 (TCLK ≤ 60 MHz) 4 (TCLK ≤ 120 MHz) 8 (TCLK ≤ 240 MHz) 16 (TCLK > 240 MHz)	4096		
	Slave	High Speed* ¹	3.00 V or above		2 (TCLK ≤ 266 MHz) 4 (TCLK > 266 MHz)	—		
		High Speed* ¹	2.70 V or above		2 (TCLK ≤ 166.6 MHz) 4 (TCLK > 166.6 MHz)	—		
		High Speed* ¹	1.62 V or above		2 (TCLK ≤ 83.3MHz) 4 (TCLK ≤ 166.6 MHz) 8 (TCLK > 166.6 MHz)	—		
		Default* ²	3.00 V or above		2 (TCLK ≤ 166.6 MHz) 4 (TCLK > 166.6 MHz)	—		
		Default* ²	2.70 V or above		2 (TCLK ≤ 120 MHz) 4 (TCLK ≤ 240 MHz) 8 (TCLK > 240 MHz)	—		
		Default* ²	1.62 V or above		2 (TCLK ≤ 60 MHz) 4 (TCLK ≤ 120 MHz) 8 (TCLK ≤ 240 MHz) 16 (TCLK > 240 MHz)	—		

Table 60.62 SPI timing (2 of 5)

Conditions:

- High-speed high drive output is selected in the Port Drive Capability bit in the PmnPFS register for the following pins:
RSPCLKA_B, RSPCLKB_B.
For other pins, high drive output is selected in the Port Drive Capability bit in the PmnPFS register.
- Use pins that have a letter appended to their names, for instance “_A” or “_B” to indicate group membership.
- Load capacitance C = 15pF is applied to the VCC/VCC2 condition “3.00 V or above”.

Parameter		High Speed/ Default	VCC/VCC2	Symbol	Min	Max	Unit	Note
RSPCK clock high pulse width	Master	—	3.00 V or above	t _{SPCKWH}	(t _{SPcyc} – t _{SPCKr} – t _{SPCKf}) / 2 – 1	—	ns	Figure 60.76
		—	2.70 V or above		(t _{SPcyc} – t _{SPCKr} – t _{SPCKf}) / 2 – 2	—		
		—	1.62 V or above		(t _{SPcyc} – t _{SPCKr} – t _{SPCKf}) / 2 – 3	—		
	Slave	—	1.62 V or above	0.4	—	t _{SPcyc}		
RSPCK clock low pulse width	Master	—	3.00 V or above	t _{SPCKWL}	(t _{SPcyc} – t _{SPCKr} – t _{SPCKf}) / 2 – 1	—	ns	
		—	2.70 V or above		(t _{SPcyc} – t _{SPCKr} – t _{SPCKf}) / 2 – 2	—		
		—	1.62 V or above		(t _{SPcyc} – t _{SPCKr} – t _{SPCKf}) / 2 – 3	—		
	Slave	—	1.62 V or above	0.4	—	t _{SPcyc}		
RSPCK clock rise and fall time	Output	High Speed*1	3.00 V or above	t _{SPCKr} , t _{SPCKf}	—	0.80	ns	
		High Speed*1	2.70 V or above		—	1.40		
		High Speed*1	1.62 V or above		—	2.50		
		Default*2	3.00 V or above		—	1.66		
		Default*2	2.70 V or above		—	3.30		
		Default*2	1.62 V or above		—	6.60		
	Input	—	3.00 V or above	—	0.1*3	μs		
		—	2.70 V or above	—	0.1*3			
		—	1.62 V or above	—	0.1*3			

Table 60.62 SPI timing (3 of 5)

Conditions:

- High-speed high drive output is selected in the Port Drive Capability bit in the PmnPFS register for the following pins:
RSPCLKA_B, RSPCLKB_B.
For other pins, high drive output is selected in the Port Drive Capability bit in the PmnPFS register.
- Use pins that have a letter appended to their names, for instance "_A" or "_B" to indicate group membership.
- Load capacitance C = 15pF is applied to the VCC/VCC2 condition "3.00 V or above".

Parameter		High Speed/ Default	VCC/VCC2	Symbol	Min	Max	Unit	Note
Data input setup time	Master	—	3.00 V or above	t _{SU}	−2.5	—	ns	Figure 60.77, Figure 60.78
		—	2.70 V or above		0.0	—		
		—	1.62 V or above		0.0	—		
	Slave	High Speed*1	3.00 V or above		1.5	—		
		High Speed*1	2.70 V or above		1.5	—		
		High Speed*1	1.62 V or above		1.5	—		
		Default*2	3.00 V or above		2.5	—		
		Default*2	2.70 V or above		2.5	—		
		Default*2	1.62 V or above		2.5	—		
Data input hold time	Master	—	3.00 V or above	t _H	7.5	—	ns	
		—	2.70 V or above		7.5	—		
		—	1.62 V or above		9.5	—		
	Slave	High Speed*1	3.00 V or above		1.5	—		
		High Speed*1	2.70 V or above		1.5	—		
		High Speed*1	1.62 V or above		1.5	—		
		Default*2	3.00 V or above		2.5	—		
		Default*2	2.70 V or above		2.5	—		
		Default*2	1.62 V or above		5.5	—		
SSL setup time	Master	—	3.00 V or above	t _{LEAD}	1 × t _{SPcyc} − 10	8 × t _{SPcyc} + 10	ns	Figure 60.77, Figure 60.78
		—	2.70 V or above		1 × t _{SPcyc} − 10	8 × t _{SPcyc} + 10		
		—	1.62 V or above		1 × t _{SPcyc} − 10	8 × t _{SPcyc} + 10		
	Slave	—	3.00 V or above		5.0	—	t _{TCyc}	
		—	2.70 V or above		5.0	—		
		—	1.62 V or above		5.0	—		
	SSL hold time	Master	—		3.00 V or above	t _{LAG}	1 × t _{SPcyc} − 10	
—			2.70 V or above	1 × t _{SPcyc} − 10	8 × t _{SPcyc} + 10			
—			1.62 V or above	1 × t _{SPcyc} − 10	8 × t _{SPcyc} + 10			
Slave		—	3.00 V or above	5.0	—		t _{TCyc}	
		—	2.70 V or above	5.0	—			
		—	1.62 V or above	5.0	—			

Table 60.62 SPI timing (4 of 5)

Conditions:

- High-speed high drive output is selected in the Port Drive Capability bit in the PmnPFS register for the following pins:
RSPCLKA_B, RSPCLKB_B.
For other pins, high drive output is selected in the Port Drive Capability bit in the PmnPFS register.
- Use pins that have a letter appended to their names, for instance "_A" or "_B" to indicate group membership.
- Load capacitance C = 15pF is applied to the VCC/VCC2 condition "3.00 V or above".

Parameter		High Speed/ Default	VCC/VCC2	Symbol	Min	Max	Unit	Note
TI SSP SS input setup time	Slave	—	3.00 V or above	t _{TISS}	2.5	—	ns	Figure 60.82
		—	2.70 V or above		2.5	—		
		—	1.62 V or above		2.5	—		
TI SSP SS input hold time	Slave	—	3.00 V or above	t _{TISH}	2.5	—	ns	
		—	2.70 V or above		2.5	—		
		—	1.62 V or above		5.5	—		
TI SSP next- access time	Slave	—	3.00 V or above	t _{TIND}	2 × t _{TCyc} + SLNDL × t _{TCyc}	—	ns	
		—	2.70 V or above		2 × t _{TCyc} + SLNDL × t _{TCyc}	—		
		—	1.62 V or above		2 × t _{TCyc} + SLNDL × t _{TCyc}	—		
TI SSP master SS output delay	Master	—	3.00 V or above	t _{TISSOD}	—	4.0	ns	Figure 60.79
		—	2.70 V or above		—	8.0		
		—	1.62 V or above		—	8.0		
Data output delay time	Master	—	3.00 V or above	t _{OD1}	—	2.0	ns	Figure 60.77, Figure 60.78
		—	2.70 V or above		—	3.0		
		—	1.62 V or above		—	6.0		
		High Speed*1	3.00 V or above	t _{OD2}	—	1.5		
		High Speed*1	2.70 V or above		—	2.5		
		High Speed*1	1.62 V or above		—	4.5		
		Default*2	3.00 V or above		—	2.5		
		Default*2	2.70 V or above		—	2.5		
		Default*2	1.62 V or above		—	4.5		
	Slave	—	3.00 V or above	t _{OD}	—	10.0		
		—	2.70 V or above		—	13.5		
		—	1.62 V or above		—	21.5		
Data output hold time	Master	High Speed*1	3.00 V or above	t _{OH}	–1.5	—	ns	
		High Speed*1	2.70 V or above		–2.5	—		
		High Speed*1	1.62 V or above		–4.5	—		
		Default*2	3.00 V or above		–2.5	—		
		Default*2	2.70 V or above		–2.5	—		
		Default*2	1.62 V or above		–4.5	—		
	Slave	—	3.00 V or above		0.0	—		
		—	2.70 V or above		0.0	—		
		—	1.62 V or above		0.0	—		

Table 60.62 SPI timing (5 of 5)

Conditions:

- High-speed high drive output is selected in the Port Drive Capability bit in the PmnPFS register for the following pins:
RSPCLKA_B, RSPCLKB_B.
For other pins, high drive output is selected in the Port Drive Capability bit in the PmnPFS register.
- Use pins that have a letter appended to their names, for instance "_A" or "_B" to indicate group membership.
- Load capacitance C = 15pF is applied to the VCC/VCC2 condition "3.00 V or above".

Parameter		High Speed/ Default	VCC/VCC2	Symbol	Min	Max	Unit	Note
Successive transmission delay time	Master	—	3.00 V or above	t_{TD}	$t_{SPcyc} + 2 \times t_{TCyc}$	$8 \times t_{SPcyc} + 2 \times t_{TCyc}$	ns	Figure 60.77, Figure 60.78
		—	2.70 V or above		$t_{SPcyc} + 2 \times t_{TCyc}$	$8 \times t_{SPcyc} + 2 \times t_{TCyc}$		
		—	1.62 V or above		$t_{SPcyc} + 2 \times t_{TCyc}$	$8 \times t_{SPcyc} + 2 \times t_{TCyc}$		
	Slave	—	3.00 V or above		t_{TCyc}	—	ns	
		—	2.70 V or above		t_{TCyc}	—		
		—	1.62 V or above		t_{TCyc}	—		
MOSI and MISO rise and fall time	Output	—	3.00 V or above	t_{Dr}, t_{Df}	—	1.66	ns	
		—	2.70 V or above		—	3.30		
		—	1.62V or above		—	6.60		
	Input	—	3.00 V or above		—	1.0	μs	
		—	2.70 V or above		—	1.0		
		—	1.62 V or above		—	1.0		
SSL rise and fall time	Output	—	3.00- V or above	t_{SSLr}, t_{SSLf}	—	1.66	ns	
		—	2.70 V or above		—	3.30		
		—	1.62 V or above		—	6.60		
	Input	—	3.00 V or above		—	1.0	μs	
		—	2.70 V or above		—	1.0		
		—	1.62 V or above		—	1.0		
Slave access time	Slave	—	3.00 V or above	t_{SA}	—	20.0	ns	Figure 60.80, Figure 60.81
		—	2.70 V or above		—	20.0		
		—	1.62 V or above		—	25.0		
Slave output release time	Slave	—	3.00 V or above	t_{REL}	—	20.0	ns	
		—	2.70 V or above		—	20.0		
		—	1.62 V or above		—	25.0		

Note: t_{TCyc} : TCLK cycle.

Note 1. Must use pins that have a letter appended to their name, for instance _A, _B, to indicate group membership. SPI0 and SPI1 are instance _B.

Note 2. All pins of group membership can be used.

Note 3. 1 μs at the longest

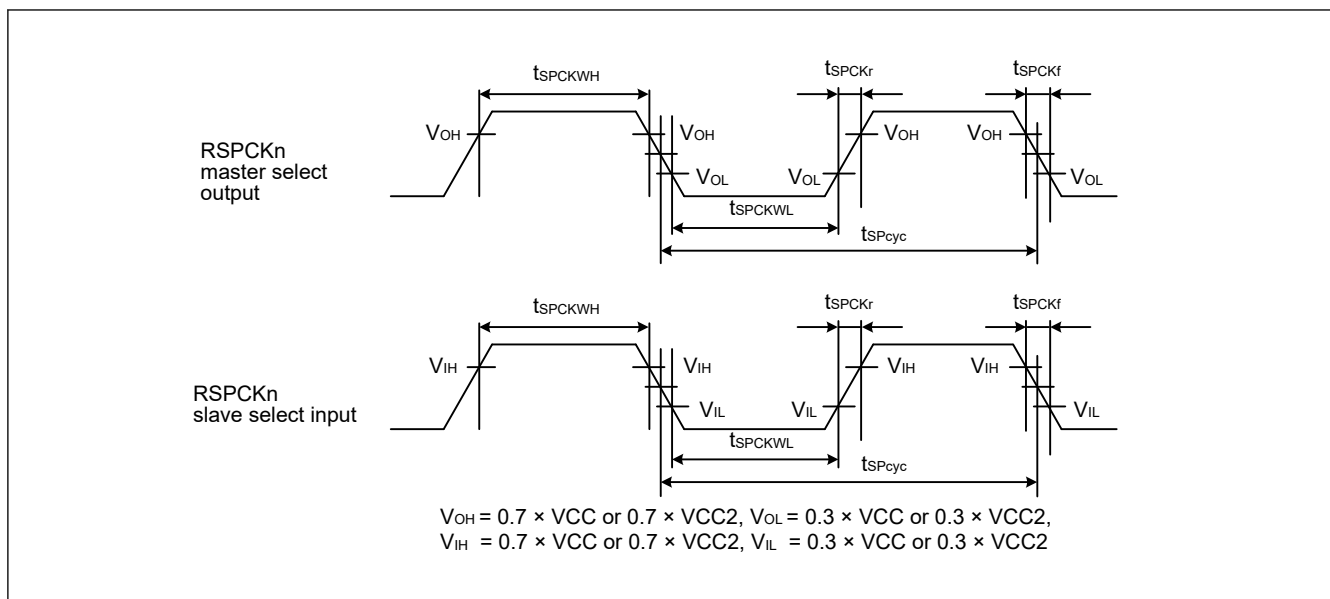


Figure 60.76 SPI clock timing

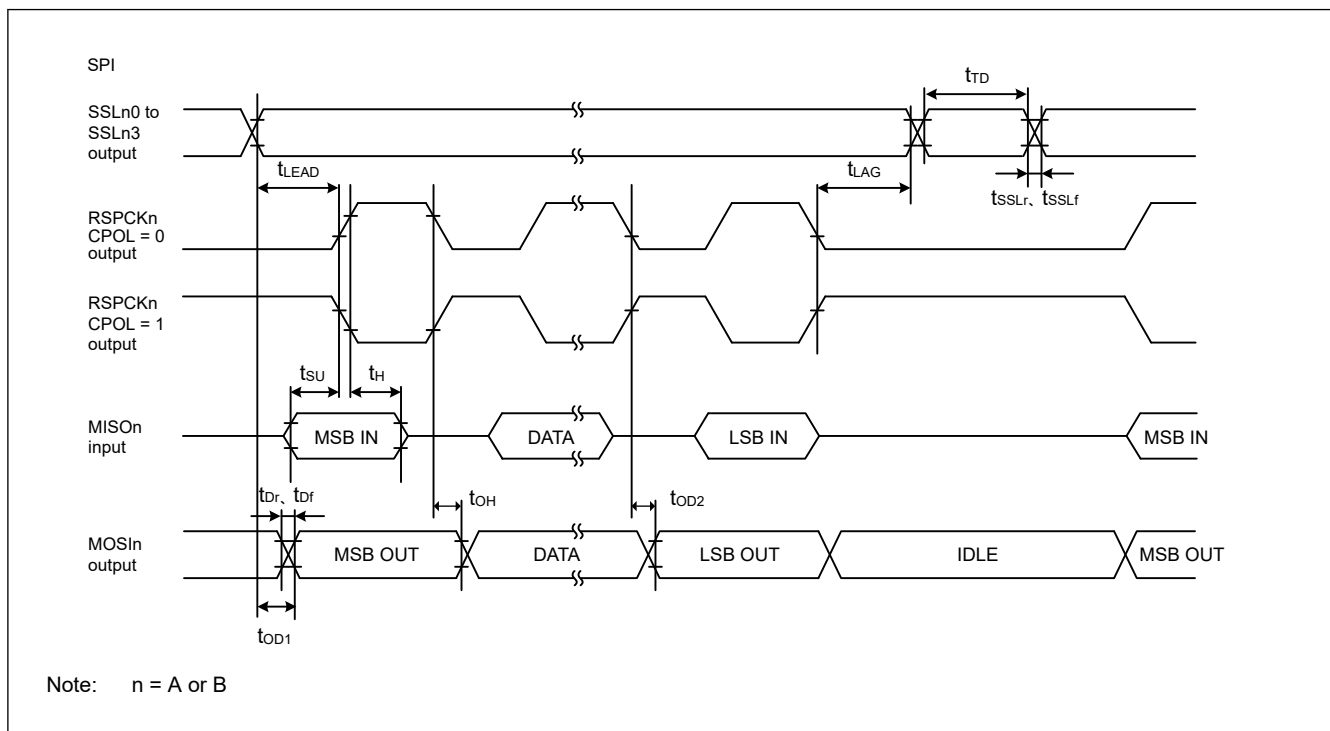


Figure 60.77 SPI timing for Motorola SPI master when CPHA = 0

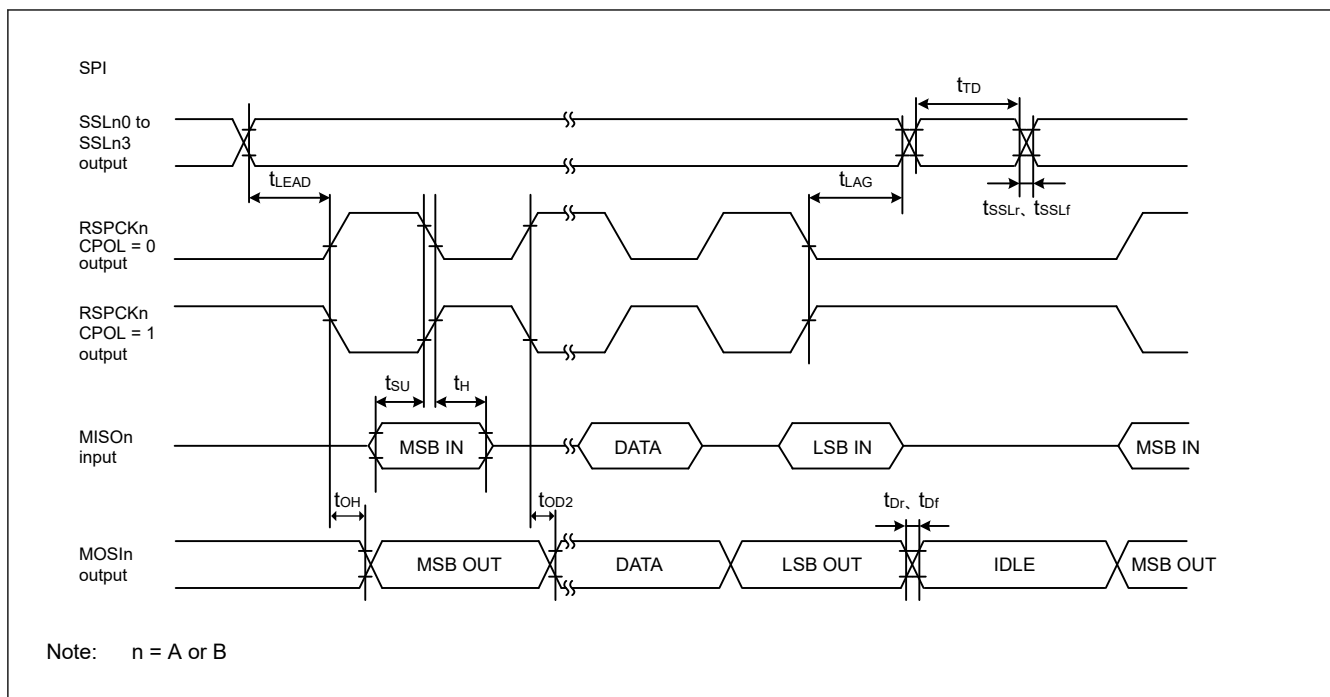


Figure 60.78 SPI timing for Motorola SPI master when CPHA = 1

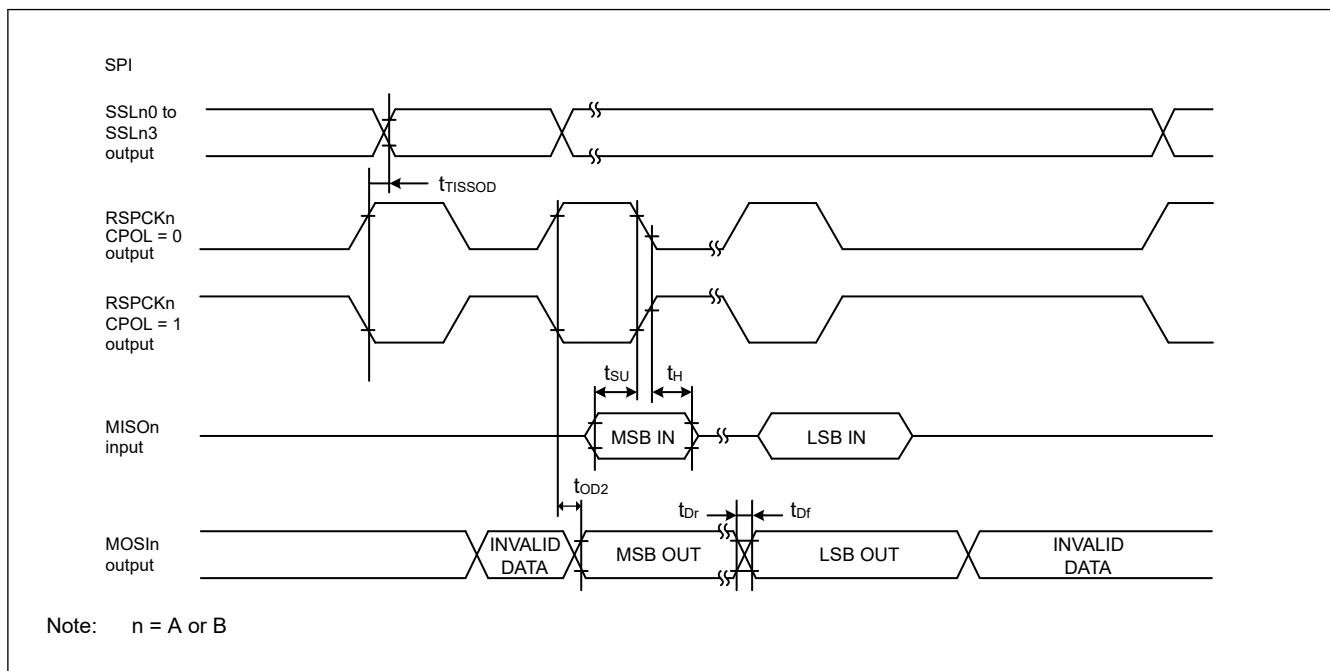


Figure 60.79 SPI timing for TI SSP master

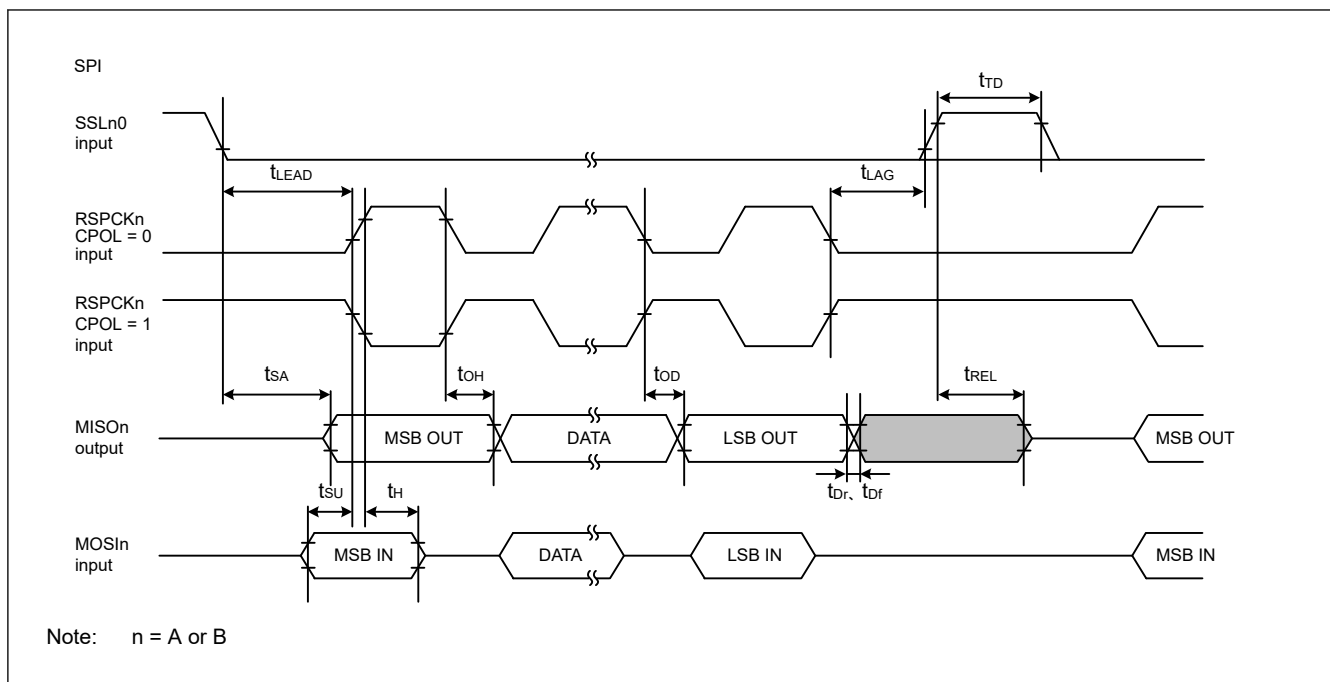


Figure 60.80 SPI timing for Motorola SPI slave when CPHA = 0

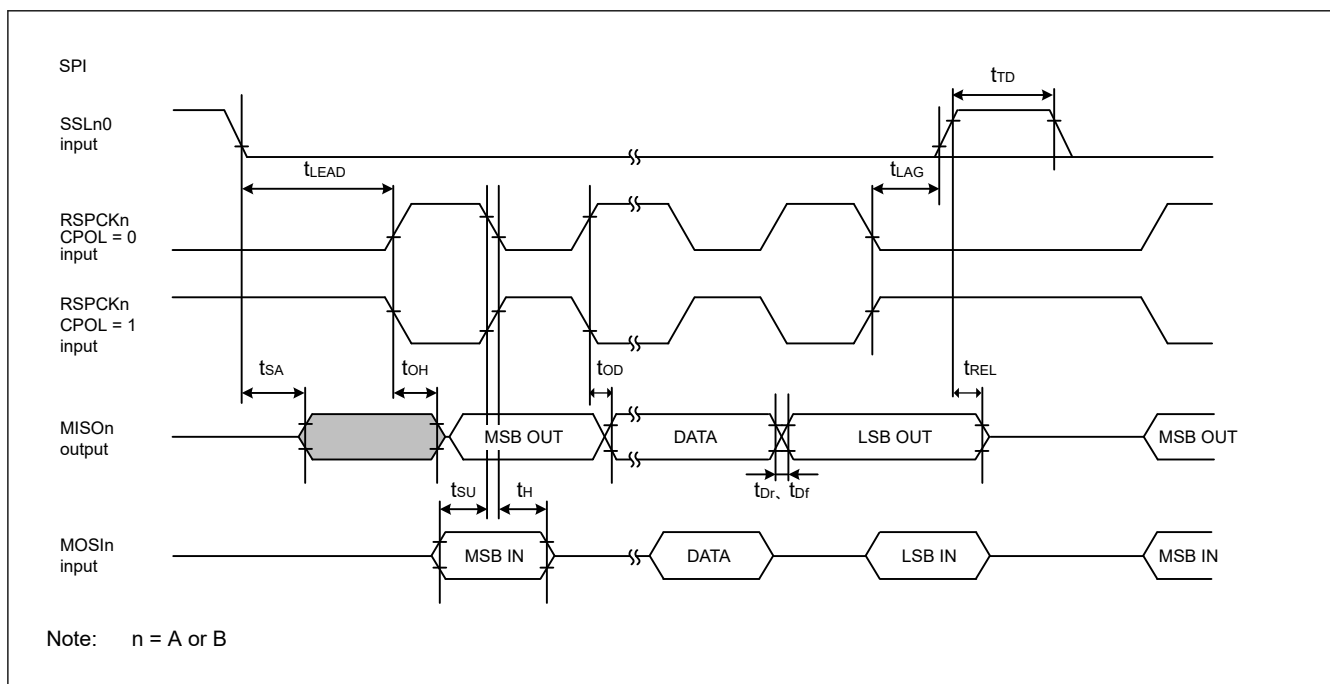


Figure 60.81 SPI timing for Motorola SPI slave when CPHA = 1

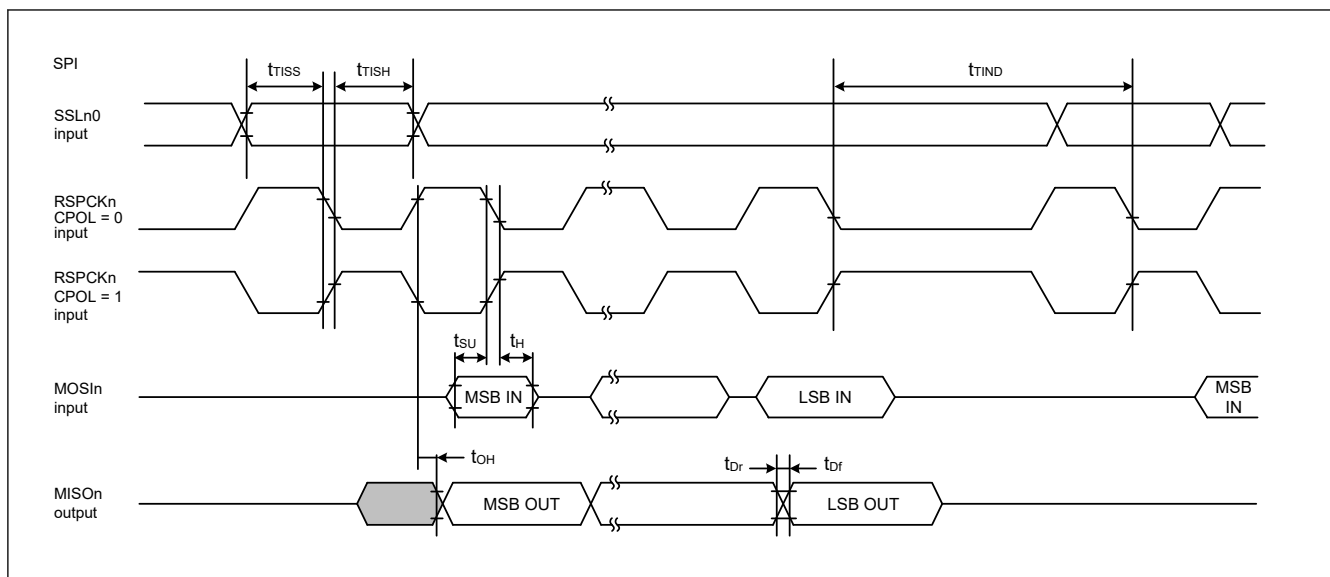


Figure 60.82 SPI timing for TI SSP slave when transmit with delay between frames

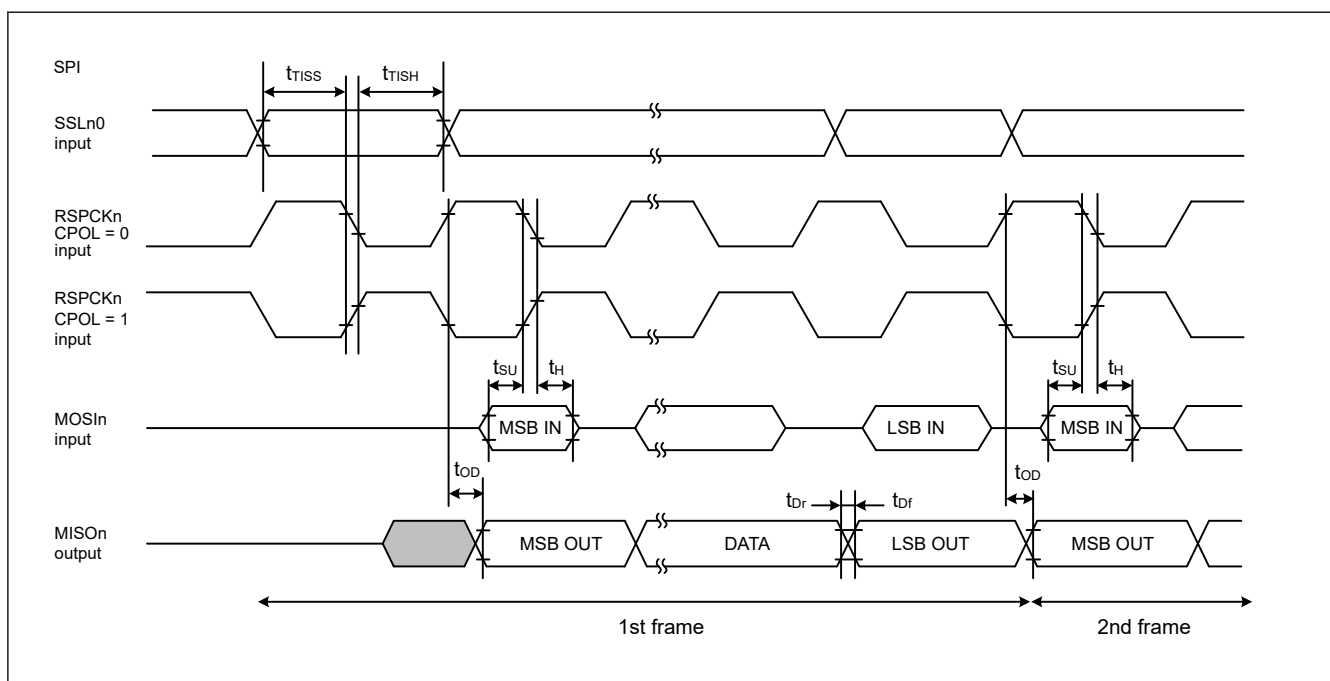


Figure 60.83 SPI timing for TI SSP slave when transmit with no delay between frames

60.3.11 OSPI Timing

Table 60.63 OSPI timing (1 of 9)

Conditions:

High-speed high drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_SCLK, OM_n_SCLKN, OM_n_SIO7-0, OM_n_DQS.

Middle drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_CS0, OM_n_CS1.

Load capacitance C = 15pF

Item		Symbol	VCC/VCC2	VDD	Min	Max	Unit	Note
Cycle time	SDR without OM_DQS	t _{PERIOD}	2.70 V or above	VSCR_1, voltage range 1	16.67	—	ns	Figure 60.84
				VSCR_2, voltage range 2				
			1.62 V to 2.00 V	VSCR_1, voltage range 1	20.00	—		
				VSCR_2, voltage range 2				
	SDR with OM_DQS/DDR	t _{PERIOD}	2.70 V or above	VSCR_1, voltage range 1	6.00	—	ns	
				VSCR_2, voltage range 2	7.50	—		
			1.62 V to 2.00 V	VSCR_1, voltage range 1				
				VSCR_2, voltage range 2				
Clock output slew rate		t _{SRck}	2.70 V or above	VSCR_1, voltage range 1	0.94	—	V/ns	Figure 60.84
				VSCR_2, voltage range 2	0.75	—		
			1.62 V to 2.00 V	VSCR_1, voltage range 1				
				VSCR_2, voltage range 2				
Clock Duty cycle-distortion		t _{CKDCD}	2.70 V or above	VSCR_1, voltage range 1	0	0.3	ns	Figure 60.84
				VSCR_2, voltage range 2	0	0.375		
			1.62 V to 2.00 V	VSCR_1, voltage range 1				
				VSCR_2, voltage range 2				

Table 60.63 OSPI timing (2 of 9)

Conditions:

High-speed high drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_SCLK, OM_n_SCLKN, OM_n_SIO7-0, OM_n_DQS.

Middle drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_CS0, OM_n_CS1.

Load capacitance C = 15pF

Item	Symbol	VCC/VCC2	VDD	Min	Max	Unit	Note
Clock Minimum Pulse width	t_{CKMPW}	2.70 V or above	VSCR_1, voltage range 1	2.7	—	ns	Figure 60.84
			VSCR_2, voltage range 2	3.375	—		
		1.62 V to 2.00 V	VSCR_1, voltage range 1				
			VSCR_2, voltage range 2				
Differential clock crossing voltage	$V_{ox(AC)}$	2.70 V or above	VSCR_1, voltage range 1	$0.2 \times VCC2$	$0.6 \times VCC2$	V	Figure 60.84
			VSCR_2, voltage range 2				
		1.62 V to 2.00 V	VSCR_1, voltage range 1				
			VSCR_2, voltage range 2				
DS Duty cycle distortion	t_{DSDCD}	2.70 V or above	VSCR_1, voltage range 1	0	0.24	ns	Figure 60.84
			VSCR_2, voltage range 2	0	0.3		
		1.62 V to 2.00 V	VSCR_1, voltage range 1				
			VSCR_2, voltage range 2				
DS Minimum Pulse width	t_{DSMPW}	2.70 V or above	VSCR_1, voltage range 1	2.46	—	ns	Figure 60.84
			VSCR_2, voltage range 2	3.075	—		
		1.62 V to 2.00 V	VSCR_1, voltage range 1				
			VSCR_2, voltage range 2				

Table 60.63 OSPI timing (3 of 9)

Conditions:

High-speed high drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_SCLK, OM_n_SCLKN, OM_n_SIO7-0, OM_n_DQS.

Middle drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_CS0, OM_n_CS1.

Load capacitance C = 15pF

Item		Symbol	VCC/VCC2	VDD	Min	Max	Unit	Note	
Data input/output slew rate		t _{SR}	2.70 V or above	VSCR_1, voltage range 1	1.72	—	ns	Figure 60.84	
				VSCR_2, voltage range 2	1.37	—			
			1.62 V to 2.00 V	VSCR_1, voltage range 1	0.75	—			
				VSCR_2, voltage range 2					
Data input setup time (to OM_SCLK/OM_SCLKN)	SDR without OM_DQS	t _{SU}	2.70 V or above	VSCR_1, voltage range 1	8.17	—	ns	Figure 60.85	
				VSCR_2, voltage range 2					
			1.62 V to 2.00 V	VSCR_1, voltage range 1	13.0	—			
				VSCR_2, voltage range 2					
Data input hold time (to OM_SCLK/OM_SCLKN)		t _H	2.70 V or above	VSCR_1, voltage range 1	0.5	—	ns		
				VSCR_2, voltage range 2					
			1.62 V to 2.00 V	VSCR_1, voltage range 1	0.5	—			
				VSCR_2, voltage range 2					

Table 60.63 OSPI timing (4 of 9)

Conditions:

High-speed high drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_SCLK, OM_n_SCLKN, OM_n_SIO7-0, OM_n_DQS.

Middle drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_CS0, OM_n_CS1.

Load capacitance C = 15pF

Item		Symbol	VCC/VCC2	VDD	Min	Max	Unit	Note
Data output valid time	SDR without OM_DQS	t_{OV}^{*2}	2.70 V or above	VSCR_1, voltage range 1	—	5.4	ns	Figure 60.85
				VSCR_2, voltage range 2				
			1.62 V to 2.00 V	VSCR_1, voltage range 1	—	6.9		
				VSCR_2, voltage range 2				
Data output hold time		t_{OH}	2.70 V or above	VSCR_1, voltage range 1	-5.4	—	ns	
				VSCR_2, voltage range 2				
			1.62 V to 2.00 V	VSCR_1, voltage range 1	-6.9	—		
				VSCR_2, voltage range 2				
Data output buffer off time		t_{BOFF}	2.70 V or above	VSCR_1, voltage range 1	-5.4	—	ns	
				VSCR_2, voltage range 2				
			1.62 V to 2.00 V	VSCR_1, voltage range 1	-6.9	—		
				VSCR_2, voltage range 2				

Table 60.63 OSPI timing (5 of 9)

Conditions:

High-speed high drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_SCLK, OM_n_SCLKN, OM_n_SIO7-0, OM_n_DQS.

Middle drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_CS0, OM_n_CS1.

Load capacitance C = 15pF

Item		Symbol	VCC/VCC2	VDD	Min	Max	Unit	Note
Data input setup time (to OM_DQS)	SDR with OM_DQS/DDR	t _{su}	2.70 V or above	VSCR_1, voltage range 1	−0.58	—	ns	Figure 60.86 , Figure 60.87
				VSCR_2, voltage range 2	−0.7	—		
			1.62 V to 2.00 V	VSCR_1, voltage range 1				
				VSCR_2, voltage range 2				
Data input hold time (to OM_DQS)		t _h	2.70 V or above	VSCR_1, voltage range 1	1.88	—	ns	
				VSCR_2, voltage range 2	2.375	—		
			1.62 V to 2.00 V	VSCR_1, voltage range 1				
				VSCR_2, voltage range 2				

Table 60.63 OSPI timing (6 of 9)

Conditions:

High-speed high drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_SCLK, OM_n_SCLKN, OM_n_SIO7-0, OM_n_DQS.

Middle drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_CS0, OM_n_CS1.

Load capacitance C = 15pF

Item		Symbol	VCC/VCC2	VDD	Min	Max	Unit	Note				
Data output valid time	SDR with OM_DQS/DDR	t_{OV}^{*2}	2.70 V or above	VSCR_1, voltage range 1	—	$t_{PERIOD}/4 + 0.5$	ns	Figure 60.86 , Figure 60.87				
				VSCR_2, voltage range 2	—	$t_{PERIOD}/4 + 0.6$						
			1.62 V to 2.00 V	VSCR_1, voltage range 1								
				VSCR_2, voltage range 2								
Data output hold time		t_{OH}	2.70 V or above	VSCR_1, voltage range 1	0.7	—	ns					
				VSCR_2, voltage range 2	0.9	—						
			1.62 V to 2.00 V	VSCR_1, voltage range 1								
				VSCR_2, voltage range 2								
Data output buffer off time			t_{BOFF}	2.70 V or above	VSCR_1, voltage range 1	0.7			—	ns		
					VSCR_2, voltage range 2	0.9			—			
				1.62 V to 2.00 V	VSCR_1, voltage range 1							
					VSCR_2, voltage range 2							

Table 60.63 OSPI timing (7 of 9)

Conditions:

High-speed high drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_SCLK, OM_n_SCLKN, OM_n_SIO7-0, OM_n_DQS.

Middle drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_CS0, OM_n_CS1.

Load capacitance C = 15pF

Item	Symbol	VCC/VCC2	VDD	Min	Max	Unit	Note
Clock Low to CS Low	t_{CKLCSL}	2.70 V or above	VSCR_1, voltage range 1	4.8	—	ns	Figure 60.85 , Figure 60.86 , Figure 60.87
			VSCR_2, voltage range 2	6	—		
		1.62 V to 2.00 V	VSCR_1, voltage range 1				
			VSCR_2, voltage range 2				
CS Low to Clock High	t_{CSLCKH}^{*3}	2.70 V or above	VSCR_1, voltage range 1	4.8	—	ns	
			VSCR_2, voltage range 2	6	—		
		1.62 V to 2.00 V	VSCR_1, voltage range 1				
			VSCR_2, voltage range 2				
Clock Low to CS High	t_{CKLCSH}	2.70 V or above	VSCR_1, voltage range 1	4.8	—	ns	
			VSCR_2, voltage range 2	6	—		
		1.62 V to 2.00 V	VSCR_1, voltage range 1				
			VSCR_2, voltage range 2				
CS High to Clock High	t_{CSHCKH}	2.70 V or above	VSCR_1, voltage range 1	4.8	—	ns	
			VSCR_2, voltage range 2	6	—		
		1.62 V to 2.00 V	VSCR_1, voltage range 1				
			VSCR_2, voltage range 2				

Table 60.63 OSPI timing (8 of 9)

Conditions:

High-speed high drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_SCLK, OM_n_SCLKN, OM_n_SIO7-0, OM_n_DQS.

Middle drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_CS0, OM_n_CS1.

Load capacitance C = 15pF

Item	Symbol	VCC/VCC2	VDD	Min	Max	Unit	Note
DS Low output to CS High	t_{CSCHKH}	2.70 V or above	VSCR_1, voltage range 1	$0.8 \times t_{PERIOD}$	—	ns	Figure 60.88
			VSCR_2, voltage range 2	$0.8 \times t_{PERIOD}$	—		
		1.62 V to 2.00 V	VSCR_1, voltage range 1				
			VSCR_2, voltage range 2				
CS High to DS Tri-State	t_{CSHDST}	2.70 V or above	VSCR_1, voltage range 1	—	t_{PERIOD}	ns	
			VSCR_2, voltage range 2	—	t_{PERIOD}		
		1.62 V to 2.00V	VSCR_1, voltage range 1				
			VSCR_2, voltage range 2				
CS Low to DS Low input*1 *3	t_{CSLDSL}	2.70 V or above	VSCR_1, voltage range 1	0	12.5	ns	
			VSCR_2, voltage range 2	0	20		
		1.62 V to 2.00 V	VSCR_1, voltage range 1	0	12.5		
			VSCR_2, voltage range 2				
DS Tri-State to CS Low	t_{DSTCSL}	2.70 V or above	VSCR_1, voltage range 1	0	—	ns	
			VSCR_2, voltage range 2	0	—		
		1.62 V to 2.00 V	VSCR_1, voltage range 1				
			VSCR_2, voltage range 2				

Table 60.63 OSPI timing (9 of 9)

Conditions:

High-speed high drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_SCLK, OM_n_SCLKN, OM_n_SIO7-0, OM_n_DQS.

Middle drive output is selected in the port drive capability bit in the PmnPFS register for the following pins: OM_n_CS0, OM_n_CS1.

Load capacitance C = 15pF

Item	Symbol	VCC/VCC2	VDD	Min	Max	Unit	Note
Clock High to DQS input* ⁴	t _{CKHDSH}	2.70 V or above	VSCR_1, voltage range 1	—	t _{PERIOD} × (1 + DDRSMPEX [3:0]) – 8.5	ns	Figure 60.86
			VSCR_2, voltage range 2				
		1.62 V to 2.00 V	VSCR_1, voltage range 1	—	t _{PERIOD} × (1 + DDRSMPEX [3:0]) – 12.5		
			VSCR_2, voltage range 2				

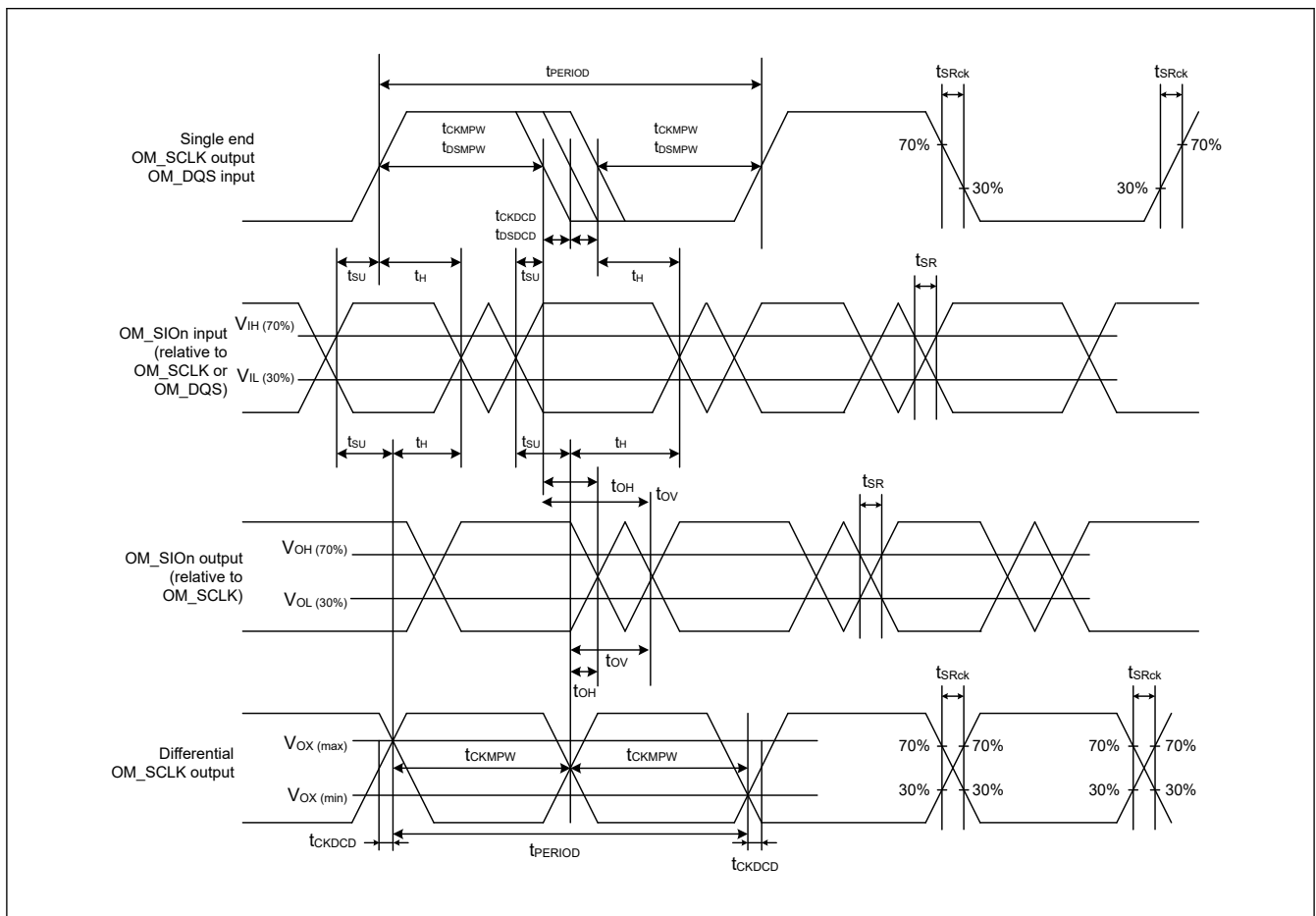
Note: n = 0, 1

Note 1. This restriction does not need to be met when using the JESD251 Profile 1.0 memory with an external pull-down attached to the OM_DQS pin.

Note 2. Condition: COMCFG.OEASTEX = 1

Note 3. Condition: LIOCFGCSx.CSASTEX = 1

Note 4. See the datasheet of memory and set DDRSMPEX[3:0] bits to satisfy this value.

**Figure 60.84 OSPI clock / DS timing**

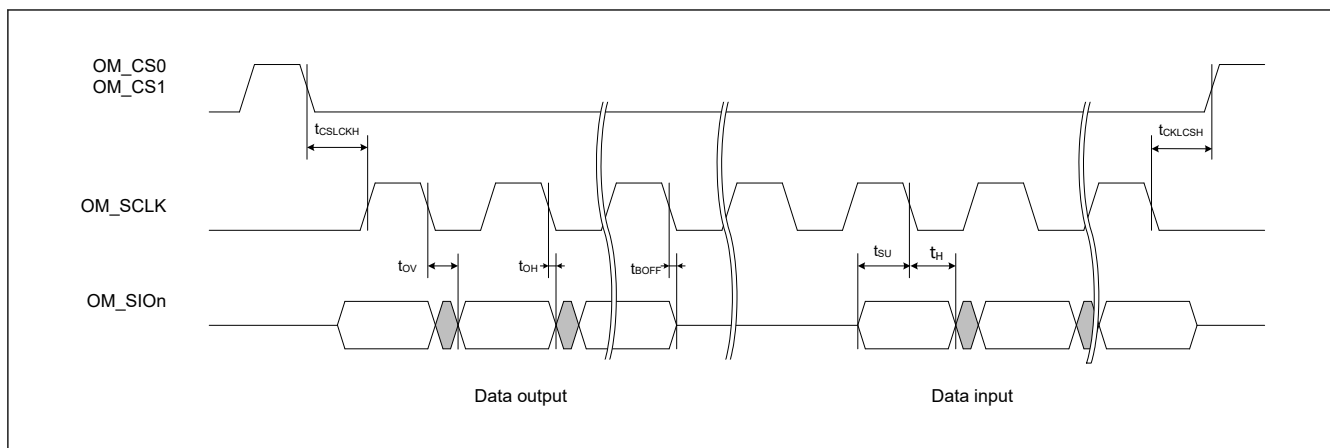


Figure 60.85 SDR transmit/receive timing (1S-1S-1S, 1S-2S-2S, 2S-2S-2S, 1S-4S-4S, 4S-4S-4S)

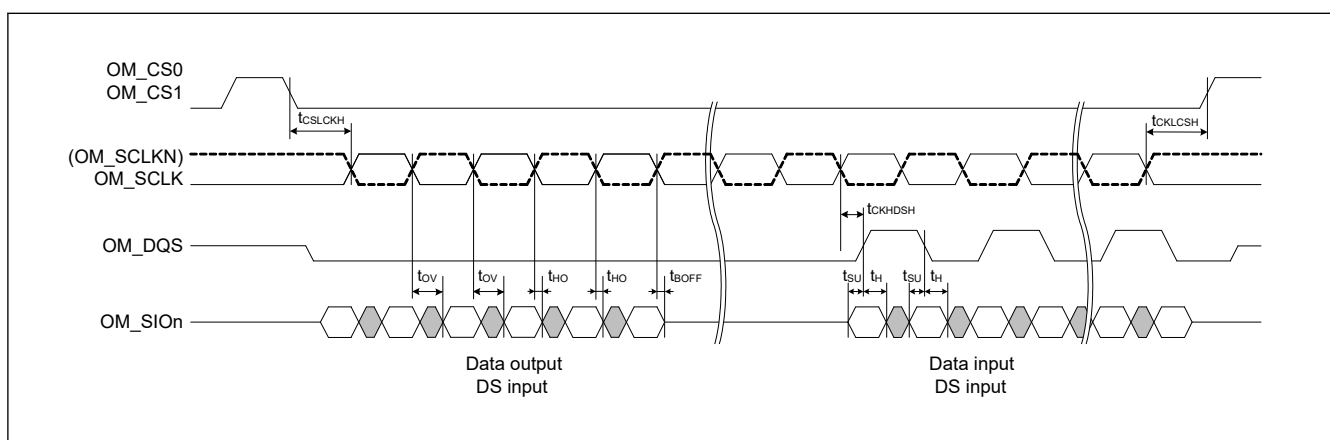


Figure 60.86 DDR transmit/receive timing (4S-4D-4D, 8D-8D-8D)

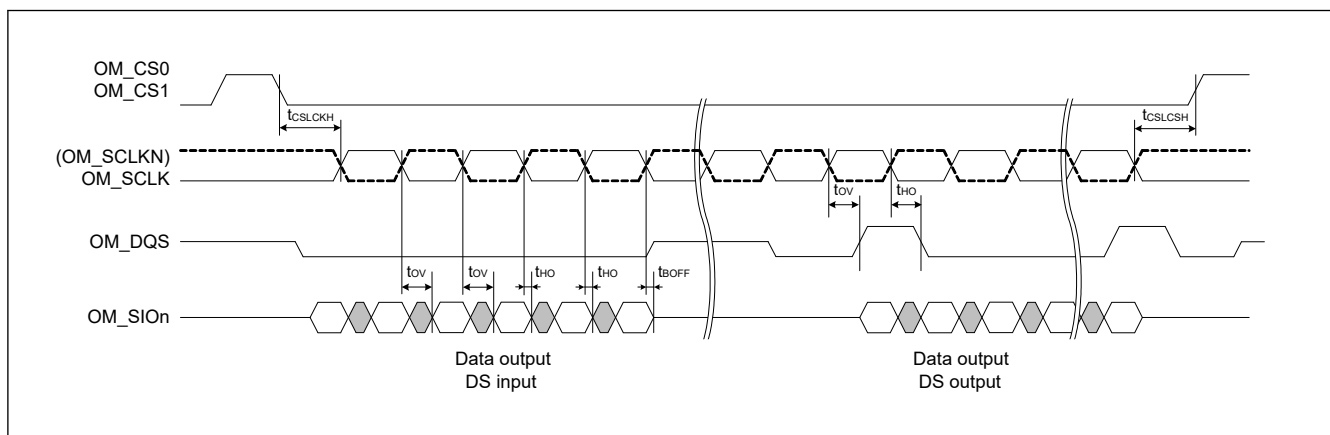
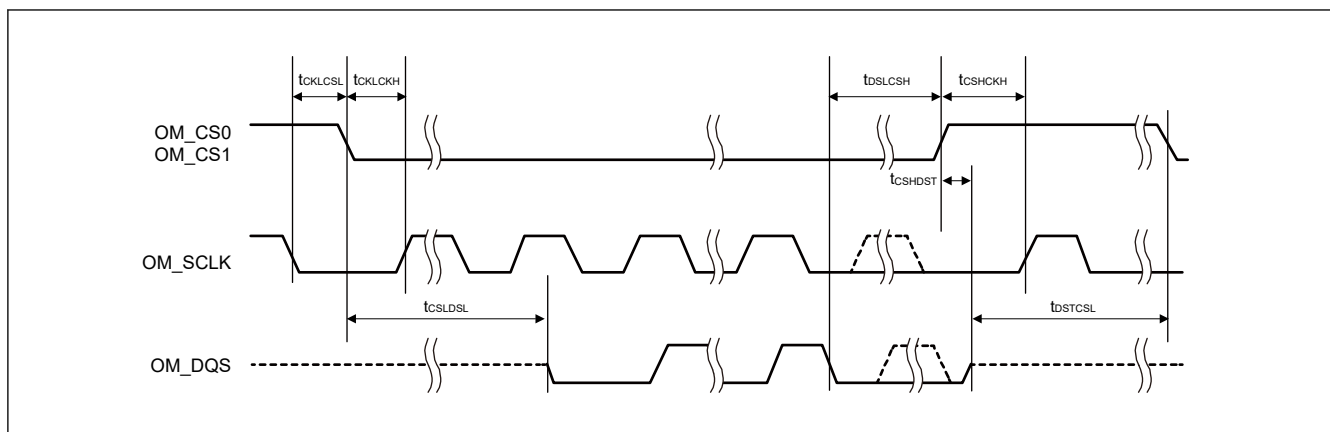


Figure 60.87 DDR transmit/receive timing (HyperRAM write)

**Figure 60.88 DS to CS signal timing**

60.3.12 IIC Timing

Table 60.64 IIC timing (1) (1 of 2)

Conditions:

1. Middle drive output is selected when VCC is 2.70 V or above, High drive output is selected when VCC is 1.62 to 1.95 V in the port drive capability bit in the PmnPFS register for the following pins: SDA0_B, SCL0_B, SDA1_B, SCL1_B, SDA2_B, SCL2_B
2. The following pins do not require setting: SCL0_A, SDA0_A, SCL1_A, SDA1_A, SCL2_A, SDA2_A
3. Use pins that have a letter appended to their names, for instance “_A” or “_B”, to indicate group membership.

For the IIC interface, the AC portion of the electrical characteristics is measured for each group.

Parameter		Symbol	VCC	Min	Max	Unit	Test conditions
IIC (Standard mode, SMBus) ICFER.FMPE = 0	SCL input cycle time	t_{SCL}	2.70 V or above	$6 (12) \times t_{IICcyc} + 1300$	—	ns	Figure 60.89
			1.62 to 1.95 V				
	SCL input high pulse width	t_{SCLH}	2.70 V or above	$3 (6) \times t_{IICcyc} + 300$	—	ns	
			1.62 to 1.95 V				
	SCL input low pulse width	t_{SCLL}	2.70 V or above	$3 (6) \times t_{IICcyc} + 300$	—	ns	
			1.62 to 1.95 V				
	SCL, SDA rise time	t_{Sr}	2.70 V or above	—	1000	ns	
			1.62 to 1.95 V				
	SCL, SDA fall time	t_{Sf}	2.70 V or above	—	300	ns	
			1.62 to 1.95 V				
	SCL, SDA input spike pulse removal time	t_{SP}	2.70 V or above	0	$1 (4) \times t_{IICcyc}$	ns	
			1.62 to 1.95 V				
	SDA input bus free time when wakeup function is disabled	t_{BUF}	2.70 V or above	$3 (6) \times t_{IICcyc} + 300$	—	ns	
			1.62 to 1.95 V				
	SDA input bus free time when wakeup function is enabled	t_{BUF}	2.70 V or above	$3 (6) \times t_{IICcyc} + 4 \times t_{Pcyc} + 300$	—	ns	
			1.62 to 1.95 V				
	START condition input hold time when wakeup function is disabled	t_{STAH}	2.70 V or above	$t_{IICcyc} + 300$	—	ns	
			1.62 to 1.95 V				
	START condition input hold time when wakeup function is enabled	t_{STAH}	2.70 V or above	$1 (5) \times t_{IICcyc} + t_{Pcyc} + 300$	—	ns	
			1.62 to 1.95 V				
	Repeated START condition input setup time	t_{STAS}	2.70 V or above	1000	—	ns	
			1.62 to 1.95 V				
	STOP condition input setup time	t_{STOS}	2.70 V or above	1000	—	ns	
			1.62 to 1.95 V				
	Data input setup time	t_{SDAS}	2.70 V or above	$t_{IICcyc} + 50$	—	ns	
			1.62 to 1.95 V				
	Data input hold time	t_{SDAH}	2.70 V or above	0	—	ns	
			1.62 to 1.95 V				
	SCL, SDA capacitive load	C_b	2.70 V or above	—	400	pF	
			1.62 to 1.95 V				

Table 60.64 IIC timing (1) (2 of 2)

Conditions:

1. Middle drive output is selected when VCC is 2.70 V or above, High drive output is selected when VCC is 1.62 to 1.95 V in the port drive capability bit in the PmnPFS register for the following pins: SDA0_B, SCL0_B, SDA1_B, SCL1_B, SDA2_B, SCL2_B
2. The following pins do not require setting: SCL0_A, SDA0_A, SCL1_A, SDA1_A, SCL2_A, SDA2_A
3. Use pins that have a letter appended to their names, for instance “_A” or “_B”, to indicate group membership.

For the IIC interface, the AC portion of the electrical characteristics is measured for each group.

Parameter		Symbol	VCC	Min	Max	Unit	Test conditions
IIC (Fast mode) ICFER.FMPE = 0	SCL input cycle time	t _{SCL}	2.70 V or above	6 (12) × t _{IICcyc} + 600	—	ns	Figure 60.89
			1.62 to 1.95 V				
	SCL input high pulse width	t _{SCLH}	2.70 V or above	3 (6) × t _{IICcyc} + 300	—	ns	
			1.62 to 1.95 V				
	SCL input low pulse width	t _{SCLL}	2.70 V or above	3 (6) × t _{IICcyc} + 300	—	ns	
			1.62 to 1.95 V				
	SCL, SDA rise time	t _{Sr}	2.70 V or above	20	300	ns	
			1.62 to 1.95 V				
	SCL, SDA fall time	t _{Sf}	2.70 V or above	20 × (external pullup voltage/5.5 V)*1	300	ns	
			1.62 to 1.95 V				
	SCL, SDA input spike pulse removal time	t _{SP}	2.70 V or above	0	1 (4) × t _{IICcyc}	ns	
			1.62 to 1.95 V				
	SDA input bus free time when wakeup function is disabled	t _{BUF}	2.70 V or above	3 (6) × t _{IICcyc} + 300	—	ns	
			1.62 to 1.95 V				
	SDA input bus free time when wakeup function is enabled	t _{BUF}	2.70 V or above	3 (6) × t _{IICcyc} + 4 × t _{Pcyc} + 300	—	ns	
			1.62 to 1.95 V				
	START condition input hold time when wakeup function is disabled	t _{STAH}	2.70 V or above	t _{IICcyc} + 300	—	ns	
			1.62 to 1.95 V				
	START condition input hold time when wakeup function is enabled	t _{STAH}	2.70 V or above	1 (5) × t _{IICcyc} + t _{Pcyc} + 300	—	ns	
			1.62 to 1.95 V				
Repeated START condition input setup time	t _{STAS}	2.70 V or above	300	—	ns		
		1.62 to 1.95 V					
STOP condition input setup time	t _{STOS}	2.70 V or above	300	—	ns		
		1.62 to 1.95 V					
Data input setup time	t _{SDAS}	2.70 V or above	t _{IICcyc} + 50	—	ns		
		1.62 to 1.95 V					
Data input hold time	t _{SDAH}	2.70 V or above	0	—	ns		
		1.62 to 1.95 V					
SCL, SDA capacitive load	C _b	2.70 V or above	—	400	pF		
		1.62 to 1.95 V					

Note: t_{IICcyc} : IIC internal reference clock (IIC ϕ) cycle, t_{Pcyc} : PCLKB cycle.

Note: Values in parentheses apply when ICMR3.NF[1:0] is set to 11b while the digital filter is enabled with ICFER.NFE set to 1.

Note: Must use pins that have a letter appended to their name, for instance “_A”, “_B”, to indicate group membership. For the IIC interface, the AC portion of the electrical characteristics is measured for each group.

Note 1. Only supported for SCL0_A, SDA0_A, SCL1_A, SDA1_A, SCL2_A, and SDA2_A.

Table 60.65 IIC timing (2)

Setting of the SCL0_A, SDA0_A, SCL1_A, SDA1_A, SCL2_A, SDA2_A pins are not required with the port drive capability bit in the PmnPFS register.

Parameter		Symbol	VCC	Min	Max	Unit	Test conditions
IIC (Fast-mode+) ICFER.FMPE = 1	SCL input cycle time	t _{SCL}	2.70 V or above	6 (12) × t _{IICcyc} + 240	—	ns	Figure 60.89
			1.62 to 1.95V				
	SCL input high pulse width	t _{SCLH}	2.70 V or above	3 (6) × t _{IICcyc} + 120	—	ns	
			1.62 to 1.95V				
	SCL input low pulse width	t _{SCLL}	2.70 V or above	3 (6) × t _{IICcyc} + 120	—	ns	
			1.62 to 1.95V				
	SCL, SDA rise time	t _{Sr}	2.70 V or above	—	120	ns	
			1.62 to 1.95V				
	SCL, SDA fall time	t _{Sf}	2.70 V or above	20 × (external pullup voltage/ 5.5V)	120	ns	
			1.62 to 1.95V				
	SCL, SDA input spike pulse removal time	t _{SP}	2.70 V or above	0	1 (4) × t _{IICcyc}	ns	
			1.62 to 1.95V				
	SDA input bus free time when wakeup function is disabled	t _{BUF}	2.70 V or above	3 (6) × t _{IICcyc} + 120	—	ns	
			1.62 to 1.95V				
	SDA input bus free time when wakeup function is enabled	t _{BUF}	2.70 V or above	3 (6) × t _{IICcyc} + 4 × t _{Pcyc} + 120	—	ns	
			1.62 to 1.95V				
	Start condition input hold time when wakeup function is disabled	t _{STAH}	2.70 V or above	t _{IICcyc} + 120	—	ns	
			1.62 to 1.95V				
	START condition input hold time when wakeup function is enabled	t _{STAH}	2.70 V or above	1 (5) × t _{IICcyc} + t _{Pcyc} + 120	—	ns	
			1.62 to 1.95V				
	Restart condition input setup time	t _{STAS}	2.70 V or above	120	—	ns	
			1.62 to 1.95V				
	Stop condition input setup time	t _{STOS}	2.70 V or above	120	—	ns	
			1.62 to 1.95V				
	Data input setup time	t _{SDAS}	2.70 V or above	t _{IICcyc} + 30	—	ns	
			1.62 to 1.95V				
	Data input hold time	t _{SDAH}	2.70 V or above	0	—	ns	
			1.62 to 1.95V				
SCL, SDA capacitive load	C _b *1	2.70 V or above	—	550	pF		
		1.62 to 1.95V					

Note: t_{IICcyc} : IIC internal reference clock (IIC ϕ) cycle, t_{Pcyc} : PCLKB cycle.

Note: Values in parentheses apply when ICMR3.NF[1:0] is set to 11b while the digital filter is enabled with ICFER.NFE set to 1.

Note 1. Cb indicates the total capacity of the bus line.

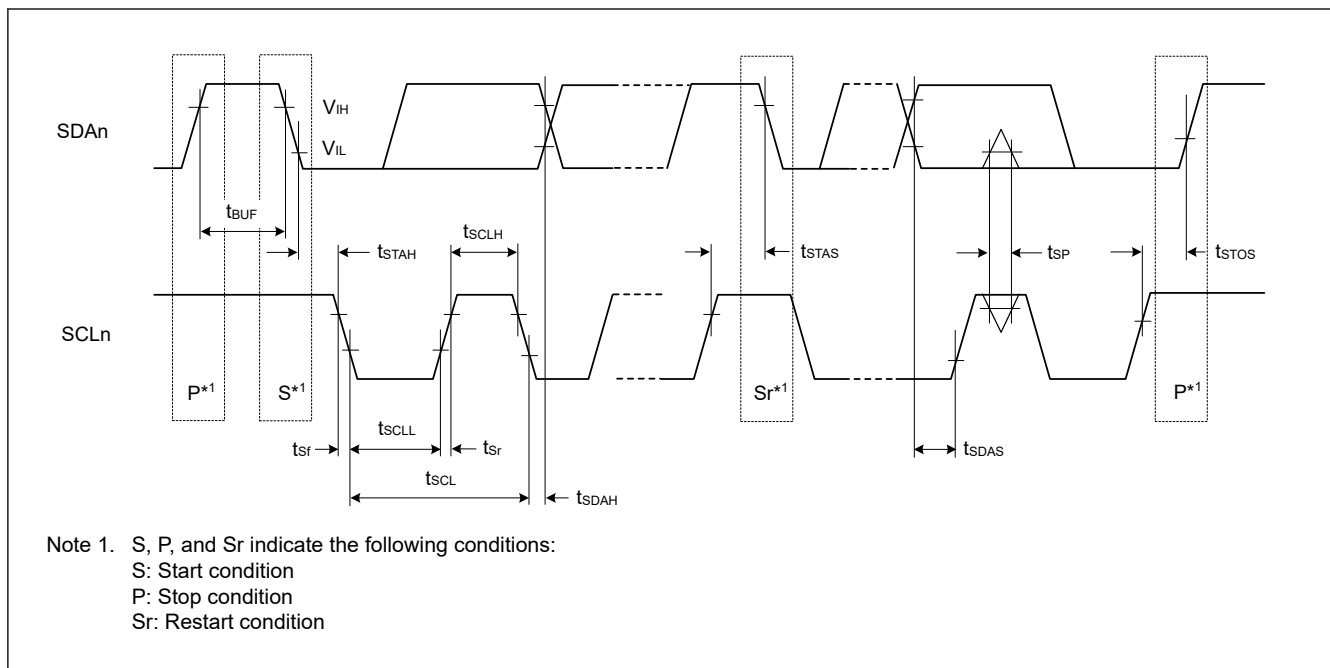


Figure 60.89 I²C bus interface input/output timing

60.3.13 I3C Timing

Table 60.66 IIC timing (1)-1

Setting of the I3C_SCL0, I3C_SDA0 pins are not required with the Port Drive Capability bit in the PmnPFS register.

Parameter		Symbol	VCC	Min	Max	Unit
IIC (Standard mode, SMBus) BFCTL.FMPE = 0	SCL input cycle time	t _{SCL}	2.70 V or above, 1.65 to 1.95 V	10 (18) × t _{I3C_{Cyc}} + 1300	—	ns
	SCL input high pulse width	t _{SCLH}	2.70 V or above, 1.65 to 1.95 V	5 (9) × t _{I3C_{Cyc}} + 300	—	ns
	SCL input low pulse width	t _{SCLL}	2.70 V or above, 1.65 to 1.95 V	5 (9) × t _{I3C_{Cyc}} + 300	—	ns
	SCL, SDA rise time	t _{Sr}	2.70 V or above, 1.65 to 1.95 V	—	1000	ns
	SCL, SDA fall time	t _{Sf}	2.70 V or above, 1.65 to 1.95 V	—	300	ns
	SCL, SDA input spike pulse removal time	t _{SP}	2.70 V or above, 1.65 to 1.95 V	0	1 (4) × t _{I3C_{Cyc}}	ns
	SDA input bus free time when wakeup function is disabled	t _{BUF}	2.70 V or above, 1.65 to 1.95 V	5(9) × t _{I3C_{Cyc}} + 300	—	ns
	SDA input bus free time when wakeup function is enabled	t _{BUF}	2.70 V or above, 1.65 to 1.95 V	5(9) × t _{I3C_{Cyc}} + 4 × t _{TCyc} + 300	—	ns
	START condition input hold time when wakeup function is disabled	t _{STAH}	2.70 V or above, 1.65 to 1.95 V	t _{I3C_{Cyc}} + 300	—	ns
	START condition input hold time when wakeup function is enabled	t _{STAH}	2.70 V or above, 1.65 to 1.95 V	1(5) × t _{I3C_{Cyc}} + t _{TCyc} + 300	—	ns
	Repeated START condition input setup time	t _{STAS}	2.70 V or above, 1.65 to 1.95 V	1000	—	ns
	STOP condition input setup time	t _{STOS}	2.70 V or above, 1.65 to 1.95 V	1000	—	ns
	Data input setup time	t _{SDAS}	2.70 V or above, 1.65 to 1.95 V	t _{I3C_{Cyc}} + 50	—	ns
	Data input hold time	t _{SDAH}	2.70 V or above, 1.65 to 1.95 V	0	—	ns
	SCL, SDA capacitive load	C _b ^{*1}	2.70 V or above, 1.65 to 1.95 V	—	400	pF

Note: $t_{I3C_{Cyc}}$: I3C internal reference clock (I3C ϕ) cycle, t_{TCyc} : TCLK cycle.

Note: Values in parentheses apply when INCTL.DNFS[3:0] is set to 0011b while the digital filter is enabled with INCTL.DNFE set to 1.

Note 1. C_b indicates the total capacity of the bus line.

Table 60.67 IIC timing (1)-2

Setting of the I3C_SCL0, I3C_SDA0 pins are not required with the Port Drive Capability bit in the PmnPFS register.

Parameter		Symbol	VCC	Min	Max	Unit
IIC (Fast-mode)	SCL input cycle time	t_{SCL}	2.70 V or above, 1.65 to 1.95 V	$10 (18) \times t_{I3C_{Cyc}} + 600$	—	ns
	SCL input high pulse width	t_{SCLH}	2.70 V or above, 1.65 to 1.95 V	$5 (9) \times t_{I3C_{Cyc}} + 300$	—	ns
	SCL input low pulse width	t_{SCLL}	2.70 V or above, 1.65 to 1.95 V	$5 (9) \times t_{I3C_{Cyc}} + 300$	—	ns
	SCL, SDA rise time	t_{Sr}	2.70 V or above, 1.65 to 1.95 V	20	300	ns
	SCL, SDA fall time	t_{Sf}	2.70 V or above, 1.65 to 1.95 V	$20 \times (\text{external pull-up voltage}/3.6 \text{ V})$	300	ns
	SCL, SDA input spike pulse removal time	t_{SP}	2.70 V or above, 1.65 to 1.95 V	0	$1 (4) \times t_{I3C_{Cyc}}$	ns
	SDA input bus free time when wakeup function is disabled	t_{BUF}	2.70 V or above, 1.65 to 1.95 V	$5 (9) \times t_{I3C_{Cyc}} + 300$	—	ns
	SDA input bus free time when wakeup function is enabled		2.70 V or above, 1.65 to 1.95 V	$5(9) \times t_{I3C_{Cyc}} + 4 \times t_{T_{Cyc}} + 300$	—	ns
	START condition input hold time when wakeup function is disabled	t_{STAH}	2.70 V or above, 1.65 to 1.95 V	$t_{I3C_{Cyc}} + 300$	—	ns
	START condition input hold time when wakeup function is enabled		2.70 V or above, 1.65 to 1.95 V	$1(5) \times t_{I3C_{Cyc}} + t_{T_{Cyc}} + 300$	—	ns
	Repeated START condition input setup time	t_{STAS}	2.70 V or above, 1.65 to 1.95 V	300	—	ns
	STOP condition input setup time	t_{STOS}	2.70 V or above, 1.65 to 1.95 V	300	—	ns
	Data input setup time	t_{SDAS}	2.70 V or above, 1.65 to 1.95 V	$t_{I3C_{Cyc}} + 50$	—	ns
	Data input hold time	t_{SDAH}	2.70 V or above, 1.65 to 1.95 V	0	—	ns
	SCL, SDA capacitive load	C_b^{*1}	2.70 V or above, 1.65 to 1.95 V	—	400	pF

Note: $t_{I3C_{Cyc}}$: I3C internal reference clock (I3C ϕ) cycle, $t_{T_{Cyc}}$: TCLK cycle.

Note: Values in parentheses apply when INCTL.DNFS[3:0] is set to 0011b while the digital filter is enabled with INCTL.DNFE set to 1.

Note 1. C_b indicates the total capacity of the bus line.

Table 60.68 IIC timing (1)-3

Setting of the I3C_SCL0, I3C_SDA0 pins are not required with the Port Drive Capability bit in the PmnPFS register.

Parameter		Symbol	VCC	Min	Max	Unit
IIC (Fast-mode +) BFCTL.FMPE = 1	SCL input cycle time	t _{SCL}	2.70 V or above, 1.65 to 1.95 V	10 (18) × t _{I3CCyc} + 240	—	ns
	SCL input high pulse width	t _{SCLH}	2.70 V or above, 1.65 to 1.95 V	5 (9) × t _{I3CCyc} + 120	—	ns
	SCL input low pulse width	t _{SCLL}	2.70 V or above, 1.65 to 1.95 V	5 (9) × t _{I3CCyc} + 120	—	ns
	SCL, SDA rise time	t _{Sr}	2.70 V or above, 1.65 to 1.95 V	—	120	ns
	SCL, SDA fall time	t _{Sf}	2.70 V or above, 1.65 to 1.95 V	20 × (external pull-up voltage/3.3V)	120	ns
	SCL, SDA input spike pulse removal time	t _{SP}	2.70 V or above, 1.65 to 1.95 V	0	1 (4) × t _{I3CCyc}	ns
	SDA input bus free time when wakeup function is disabled	t _{BUF}	2.70 V or above, 1.65 to 1.95 V	5 (9) × t _{I3CCyc} + 120	—	ns
	SDA input bus free time when wakeup function is enabled			5(9) × t _{I3CCyc} + 4 × t _{TCyc} + 120		ns
	START condition input hold time when wakeup function is disabled	t _{STAH}	2.70 V or above, 1.65 to 1.95 V	t _{I3CCyc} + 120	—	ns
	START condition input hold time when wakeup function is enabled			1(5) × t _{I3CCyc} + t _{TCyc} + 120		ns
	Restart condition input setup time	t _{STAS}	2.70 V or above, 1.65 to 1.95 V	120	—	ns
	Stop condition input setup time	t _{STOS}	2.70 V or above, 1.65 to 1.95 V	120	—	ns
	Data input setup time	t _{SDAS}	2.70 V or above, 1.65 to 1.95 V	t _{I3CCyc} + 30	—	ns
	Data input hold time	t _{SDAH}	2.70 V or above, 1.65 to 1.95 V	0	—	ns
	SCL, SDA capacitive load	C _b ^{*1}	2.70 V or above, 1.65 to 1.95 V	—	550	pF

Note: $t_{I3C_{Cyc}}$: I3C internal reference clock (I3C ϕ) cycle, t_{Tcyc} : TCLK cycle.

Note: Values in parentheses apply when INCTL.DNFS[3:0] is set to 0011b while the digital filter is enabled with INCTL.DNFE set to 1.

Note 1. C_b indicates the total capacity of the bus line.

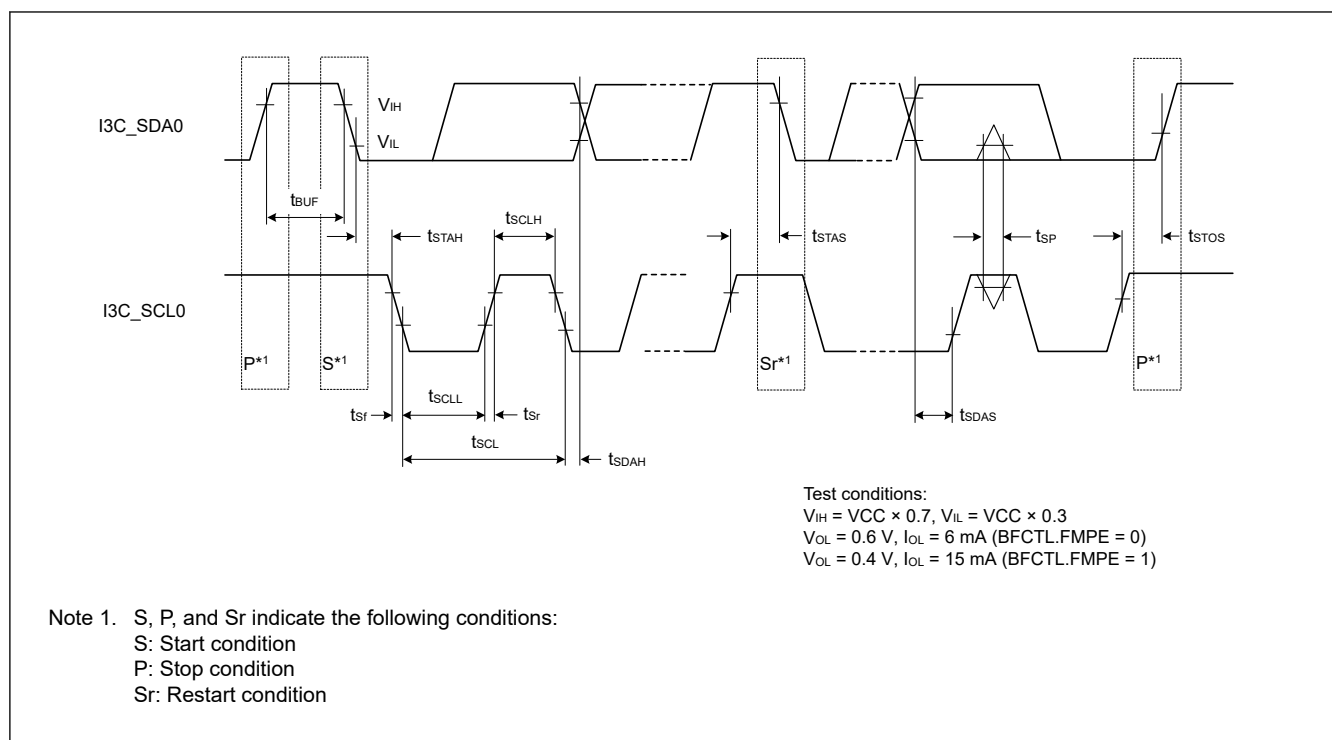
Figure 60.90 I²C bus interface input/output timing

Table 60.69 IIC timing (2)

Setting of the I3C_SCL0, I3C_SDA0 pins are not required with the Port Drive Capability bit in the PmnPFS register.

Parameter			Symbol	VCC	Min	Max	Unit
IIC (Hs-mode) BFCTL.HS ME = 1	SCL input cycle time		t _{SCL}	3.00 V or above	46 (48) × t _{I3CCyc}	—	ns
				1.65 to 1.95 V	46 (48) × t _{I3CCyc}	—	
	SCL input high pulse width	Cb = 400 pF	t _{SCLH}	3.00 V or above	29 (30) × t _{I3CCyc}	—	ns
				1.65 to 1.95 V	29 (30) × t _{I3CCyc}	—	
		Cb = 100 pF		3.00 V or above	13 (14) × t _{I3CCyc}	—	
				1.65 to 1.95 V	13 (14) × t _{I3CCyc}	—	
	SCL input low pulse width	Cb = 400 pF	t _{SCLL}	3.00 V or above	69 (70) × t _{I3CCyc}	—	ns
				1.65 to 1.95 V	69 (70) × t _{I3CCyc}	—	
		Cb = 100 pF		3.00 V or above	33 (34) × t _{I3CCyc}	—	
				1.65 to 1.95 V	33 (34) × t _{I3CCyc}	—	
	SCL rise time	Cb = 400 pF	t _{SrCL}	3.00 V or above	—	80	ns
				1.65 to 1.95 V	—	80	
		Cb = 100 pF		3.00 V or above	—	40	
				1.65 to 1.95 V	—	40	
	SDA rise time	Cb = 400 pF	t _{SrDA}	3.00 V or above	—	160	ns
				1.65 to 1.95 V	—	160	
		Cb = 100 pF		3.00 V or above	—	80	
				1.65 to 1.95 V	—	80	
	SCL fall time	Cb = 400 pF	t _{SfCL}	3.00 V or above	—	80	ns
				1.65 to 1.95 V	—	80	
		Cb = 100 pF		3.00 V or above	—	40	
				1.65 to 1.95 V	—	40	
	SDA fall time	Cb = 400 pF	t _{SfDA}	3.00 V or above	—	160	ns
				1.65 to 1.95 V	—	160	
		Cb = 100 pF		3.00 V or above	—	80	
				1.65 to 1.95 V	—	80	
	SCL, SDA input spike pulse removal time		t _{SP}	3.00 V or above	0	1 (1) × t _{I3CCyc}	ns
				1.65 to 1.95 V	0	1 (1) × t _{I3CCyc}	
	Repeated START condition input setup time		t _{STAS}	3.00 V or above	40	—	ns
				1.65 to 1.95 V	40	—	
STOP condition input setup time		t _{STOS}	3.00 V or above	40	—	ns	
			1.65 to 1.95 V	40	—		
Data input setup time		t _{SDAS}	3.00 V or above	10	—	ns	
			1.65 to 1.95 V	10	—		
Data input hold time	Cb = 400 pF	t _{SDAH}	3.00 V or above	0	150	ns	
			1.65 to 1.95 V	0	150		
	Cb = 100 pF		3.00 V or above	0	70		
			1.65 to 1.95 V	0	70		
SCL, SDA capacitive load		C _b ^{*1}	3.00 V or above	—	400	pF	
			1.65 to 1.95 V	—	400		

Table 60.70 I3C timing (Open Drain Timing Parameters)

Setting of the I3C_SCL0, I3C_SDA0 pins are not required with the Port Drive Capability bit in the PmnPFS register.

Parameter	Symbol	VCC	Min	Max	Unit	Test conditions
I3C Open Drain Timing Parameters	SCL Clock Low Period	$t_{\text{LOW_OD}}^{*1 *2}$	3.00 V or above	—	ns	Figure 60.94
			1.65 to 1.95 V	200	—	
		$t_{\text{DIG_OD_L}}$	3.00 V or above	$t_{\text{LOW_ODmin}} + t_{\text{fDA_ODmin}}$	ns	Figure 60.94
			1.65 to 1.95 V	$t_{\text{LOW_ODmin}} + t_{\text{fDA_ODmin}}$	—	
	SCL Clock High Period	$t_{\text{HIGH}}^{*3 *4}$	3.00 V or above	—	ns	Figure 60.92
			1.65 to 1.95 V	41	—	
		$t_{\text{DIG_H}}$	3.00 V or above	$t_{\text{HIGH}} + t_{\text{CF}}$	ns	Figure 60.92
			1.65 to 1.95 V	$t_{\text{HIGH}} + t_{\text{CF}}$	—	
	SDA Signal Fall Time	$t_{\text{fDA_OD}}$	3.00 V or above	t_{CF}	ns	Figure 60.94
			1.65 to 1.95 V	t_{CF}	12	
	SDA Data Setup Time Open Drain Mode	$t_{\text{SU_OD}}^{*1}$	3.00 V or above	—	ns	Figure 60.93
			1.65 to 1.95 V	18	—	
	Clock After START (S) Condition	$t_{\text{CAS}}^{*5 *6}$	3.00 V or above	38.4 nano	seconds	Figure 60.94
				For ENAS0: 1 μ		
				For ENAS1: 100 μ		
				For ENAS2: 2 milli		
				For ENAS3: 50 milli		
			1.65 to 1.95 V	38.4 nano		
				For ENAS0: 1 μ		
				For ENAS1: 100 μ		
				For ENAS2: 2 milli		
				For ENAS3: 50 milli		
	Clock Before STOP (P) Condition	t_{CBP}	3.00 V or above	$t_{\text{CASmin}} / 2$	seconds	Figure 60.95
			1.65 to 1.95 V	$t_{\text{CASmin}} / 2$	—	
	Current Master to Secondary Master Overlap time during handoff	$t_{\text{MMOverlap}}$	3.00 V or above	$t_{\text{DIG_OD_Lmin}}$	ns	Figure 60.101
			1.65 to 1.95 V	$t_{\text{DIG_OD_Lmin}}$	—	
	Bus Available Condition	t_{AVAL}^{*7}	3.00 V or above	1	μ s	—
			1.65 to 1.95 V	1	—	
	Bus Idle Condition	t_{IDLE}	3.00 V or above	1	ms	—
			1.65 to 1.95 V	1	—	
	Time Internal Where New Master Not Driving SDA Low	t_{MMLock}	3.00 V or above	t_{AVALmin}	μ s	Figure 60.101
			1.65 to 1.95 V	t_{AVALmin}	—	

Note 1. This is approximately equal to $t_{\text{LOWmin}} + t_{\text{DS_ODmin}} + t_{\text{rDA_ODtyp}} + t_{\text{SU_ODmin}}$.

Note 2. The Master may use a shorter Low period if it knows that this is safe, i.e., that SDA is already above VIH.

Note 3. Based on t_{SPIKE} , rise and fall times, and interconnect.Note 4. This maximum High period may be exceeded when the signals can be safely seen by Legacy I²C Devices, and/or in consideration of the interconnect (e.g., a short Bus).

As a product specification, if this Max value cannot be guaranteed, change this Max value and specify that it cannot be used in the Mixed Bus.

Note 5. On a Legacy Bus where I²C Devices need to see Start

Note 6. Slaves that do not support the optional ENTASx CCCs shall use the t_{CAS} Max value shown for ENTAS3

Note 7. On a Mixed Bus with Fm Legacy I²C Devices, t_{AVAIL} is 300 ns shorter than the Fm Bus Free Condition time (t_{BUF})

Table 60.71 I3C timing (Push-Pull Timing Parameters for SDR and HDR-DDR Modes)

Setting of the I3C_SCL0, I3C_SDA0 pins are not required with the Port Drive Capability bit in the PmnPFS register.

Parameter		Symbol	VCC	Min	Max	Unit	Test conditions
I3C Push-Pull Timing Parameters for SDR and HDR-DDR Modes	SCL Clock Frequency	f _{SCL} ^{*1}	3.00 V or above	0.01	12.5	MHz	—
			1.65 to 1.95 V	0.01	12.5		
	SCL Clock Low Period	t _{LOW}	3.00 V or above	27	—	ns	Figure 60.92
			1.65 to 1.95 V	32	—		
		t _{DIG_L} ^{*2 *4}	3.00 V or above	35	—	ns	Figure 60.92
			1.65 to 1.95 V	40	—		
	SCL Clock High Period for Mixed Bus	t _{HIGH_MIXED}	3.00 V or above	24	—	ns	Figure 60.92
			1.65 to 1.95 V	27	—		
		t _{DIG_H_MIXED} ^{*2 *3}	3.00 V or above	32	45	ns	Figure 60.92
			1.65 to 1.95 V	35	45		
	SCL Clock High Period	t _{HIGH}	3.00 V or above	24	—	ns	Figure 60.92
			1.65 to 1.95 V	27	—		
		t _{DIG_H} ^{*2}	3.00 V or above	32	—	ns	Figure 60.92
			1.65 to 1.95 V	35	—		
	Clock in to Data Out for Slave	t _{SCO}	3.00 V or above	—	12	ns	Figure 60.97
			1.65 to 1.95 V	—	12		
	SCL Clock Rise Time	t _{CR}	3.00 V or above	—	150 × 1 / f _{SCL} (capped at 60)	ns	Figure 60.92
			1.65 to 1.95 V	—	150 × 1 / f _{SCL} (capped at 60)		
	SCL Clock Fall Time	t _{CF}	3.00 V or above	—	150 × 1 / f _{SCL} (capped at 60)	ns	Figure 60.92
			1.65 to 1.95 V	—	150 × 1 / f _{SCL} (capped at 60)		
	SDA Signal Data Hold in Push-Pull Mode	Master	t _{HD_PP} ^{*4*5}	3.00 V or above	t _{CR} + 3 and t _{CF} + 3	—	Figure 60.96
			1.65 to 1.95 V	t _{CR} + 3 and t _{CF} + 3	—		
		Slave	t _{HD_PP} ^{*5}	3.00 V or above	0	—	Figure 60.96
			1.65 to 1.95 V	0	—		
	SDA Signal Data Setup in Push-Pull Mode	t _{SU_PP}	3.00 V or above	12	N/A	ns	Figure 60.98
			1.65 to 1.95 V	18	N/A		
	Clock After Repeated START (Sr)	t _{CASr}	3.00 V or above	t _{CASmin}	N/A	ns	Figure 60.100
			1.65 to 1.95 V	t _{CASmin}	N/A		
	Clock Before Repeated START (Sr)	t _{CBSr}	3.00 V or above	t _{CASmin} / 2	N/A	ns	Figure 60.100
			1.65 to 1.95 V	t _{CASmin} / 2	N/A		
	Capacitive Load per Bus Line (SDA/SCL)	C _b	3.00 V or above	—	50	pF	—
			1.65 to 1.95 V	—	50		

Note 1. $f_{SCL} = 1 / (t_{DIG_L} + t_{DIG_H})$

Note 2. t_{DIG_L} and t_{DIG_H} are the clock Low and High periods as seen at the receiver end of the I3C Bus using V_{IL} and V_{IH} .

Note 3. When communicating with an I3C Device on a mixed Bus, the $t_{DIG_H_MIXED}$ period must be constrained in order to make sure that I²C Devices do not interpret I3C signaling as valid I²C signaling.

Note 4. As both edges are used, the hold time needs to be satisfied for the respective edges; i.e., $t_{CF} + 3$ for falling edge clocks, and $t_{CR} + 3$ for rising edge clocks.

Note 5. In SDR Mode the Hold time parameter is referred to as t_{HD_SDR} , and in DDR Mode it is referred to as t_{HD_DDR} .

Table 60.72 I3C timing (Push-Pull Timing Parameters for HDR-TSP and HDR-TSL Modes)

Setting of the I3C_SCL0, I3C_SDA0 pins are not required with the Port Drive Capability bit in the PmnPFS register.

Parameter	Symbol	VCC	Min	Max	Unit	Test conditions
I3C Push-Pull Timing Parameters for HDR-TSP and HDR-TSL Modes	Edge-to-Edge Period	$t_{EDGE}^{*1 *2}$	3.00 V or above	t_{DIG_H}	—	Figure 60.102
			1.65 to 1.95 V	t_{DIG_H}	—	
	Allowed Difference Between Signals for 'Simultaneous' Change	t_{SKEW}	3.00 V or above	—	11	
			1.65 to 1.95 V	—	11	
	Stable Condition Between Symbols	t_{EYE}	3.00 V or above	12	—	
			1.65 to 1.95 V	12	—	
	Time Between Successive Symbols	t_{SYMBOL}	3.00 V or above	$t_{EDGE \text{ Min}}$	—	
			1.65 to 1.95 V	$t_{EDGE \text{ Min}}$	—	
	Symbol Clock	t_{CLOCK}	3.00 V or above	$1 / f_{SCL} \text{ (Max)}$	—	
			1.65 to 1.95 V	$1 / f_{SCL} \text{ (Max)}$	—	

Note 1. Edges occur at the rate of $1 / (t_{EDGE} \times 2)$

Note 2. In a Mixed Bus, HDR-TSL shall respect the maximum $t_{DIG_H_MIXED}$ shown in Figure 60.95.

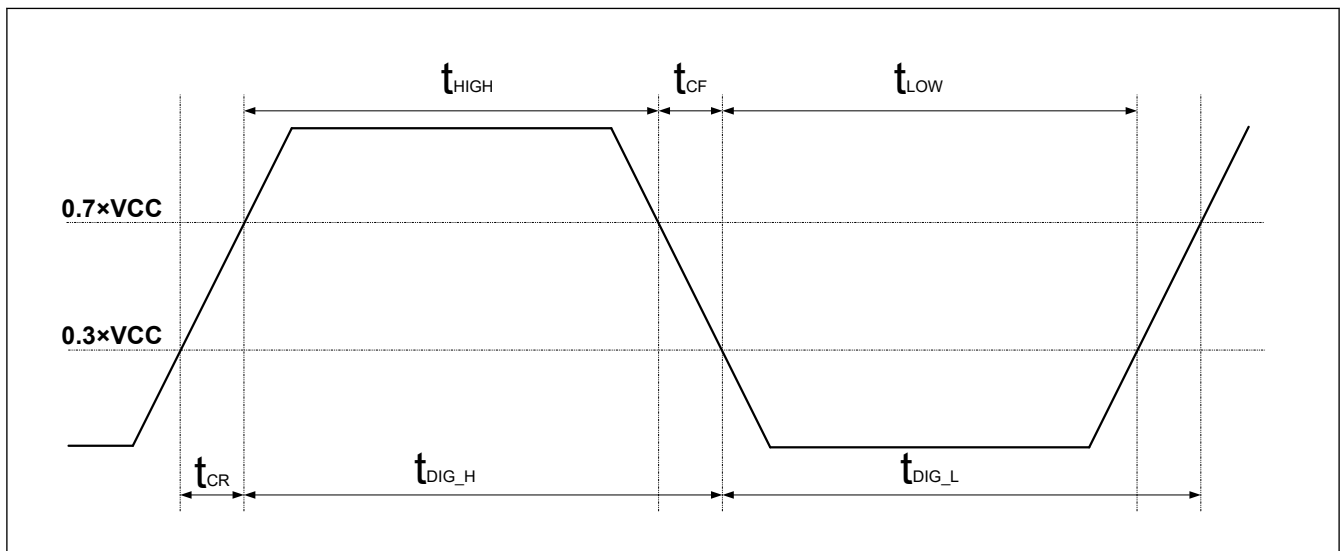


Figure 60.92 t_{DIG_H} and t_{DIG_L}

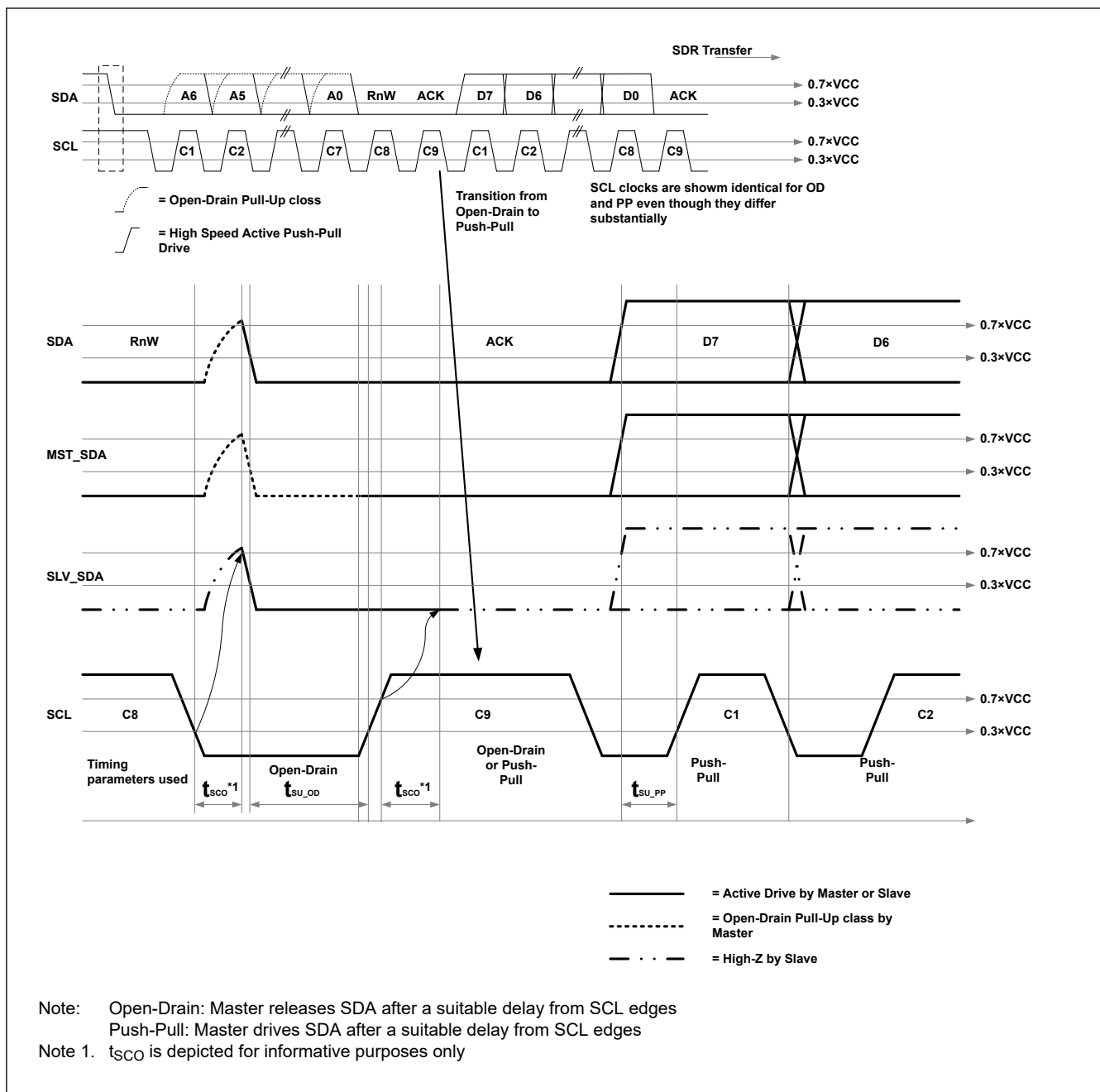


Figure 60.93 I3C Data Transfer – ACK by Slave

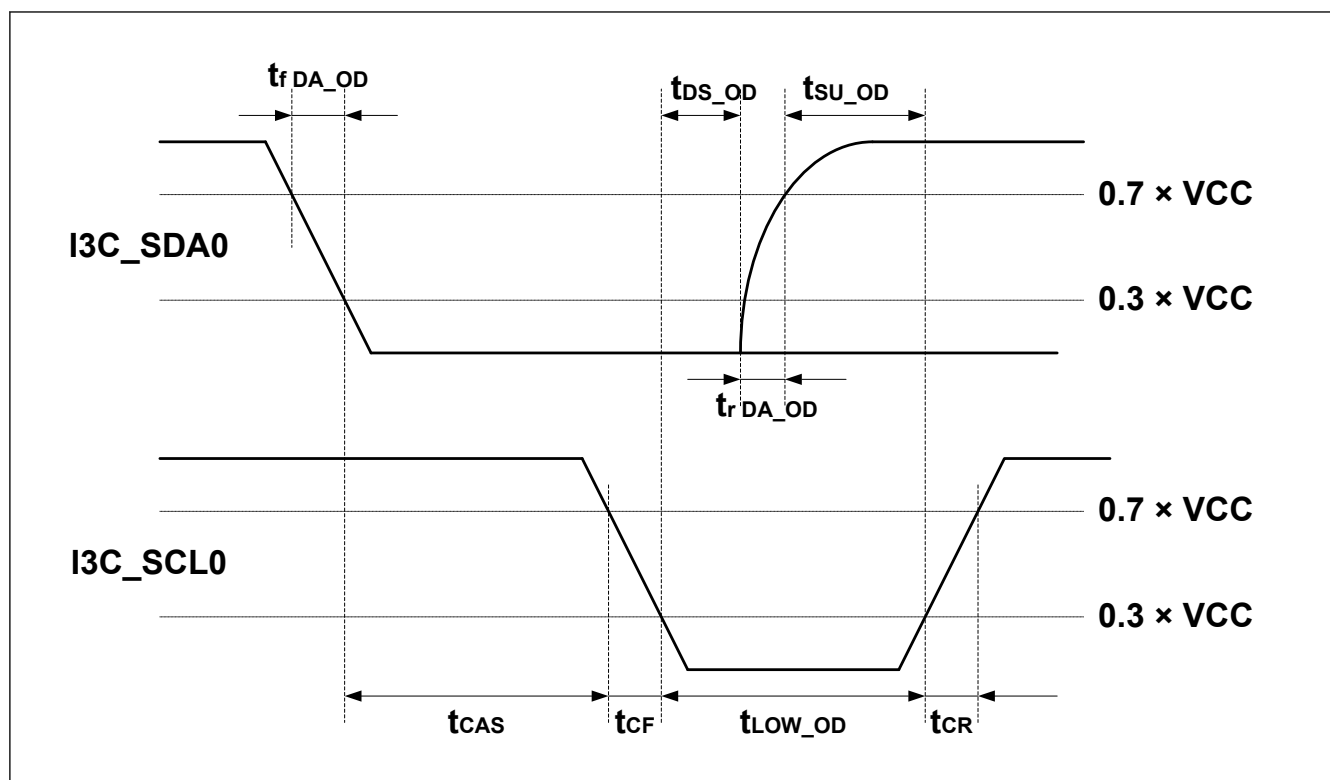


Figure 60.94 I3C START condition Timing

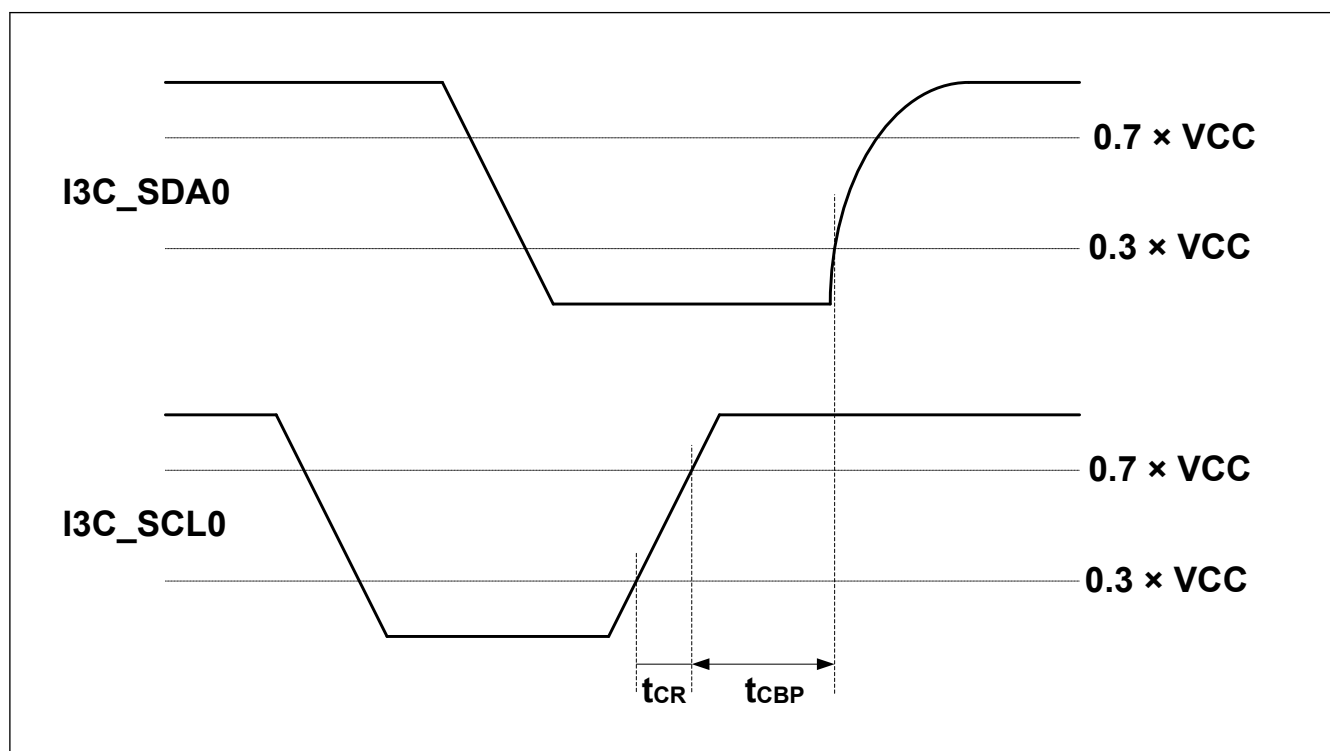


Figure 60.95 I3C STOP condition Timing

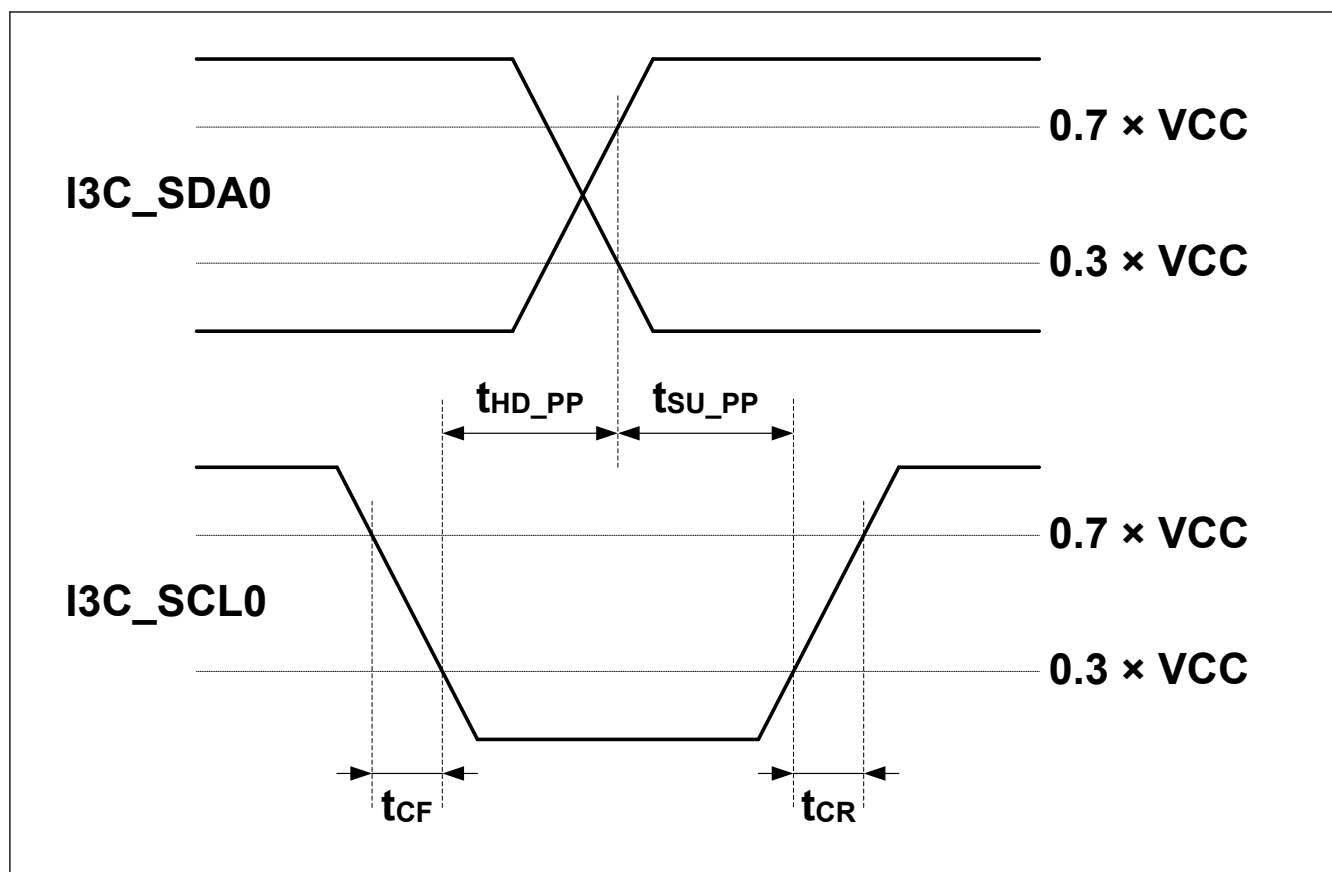


Figure 60.96 I3C Master Out Timing

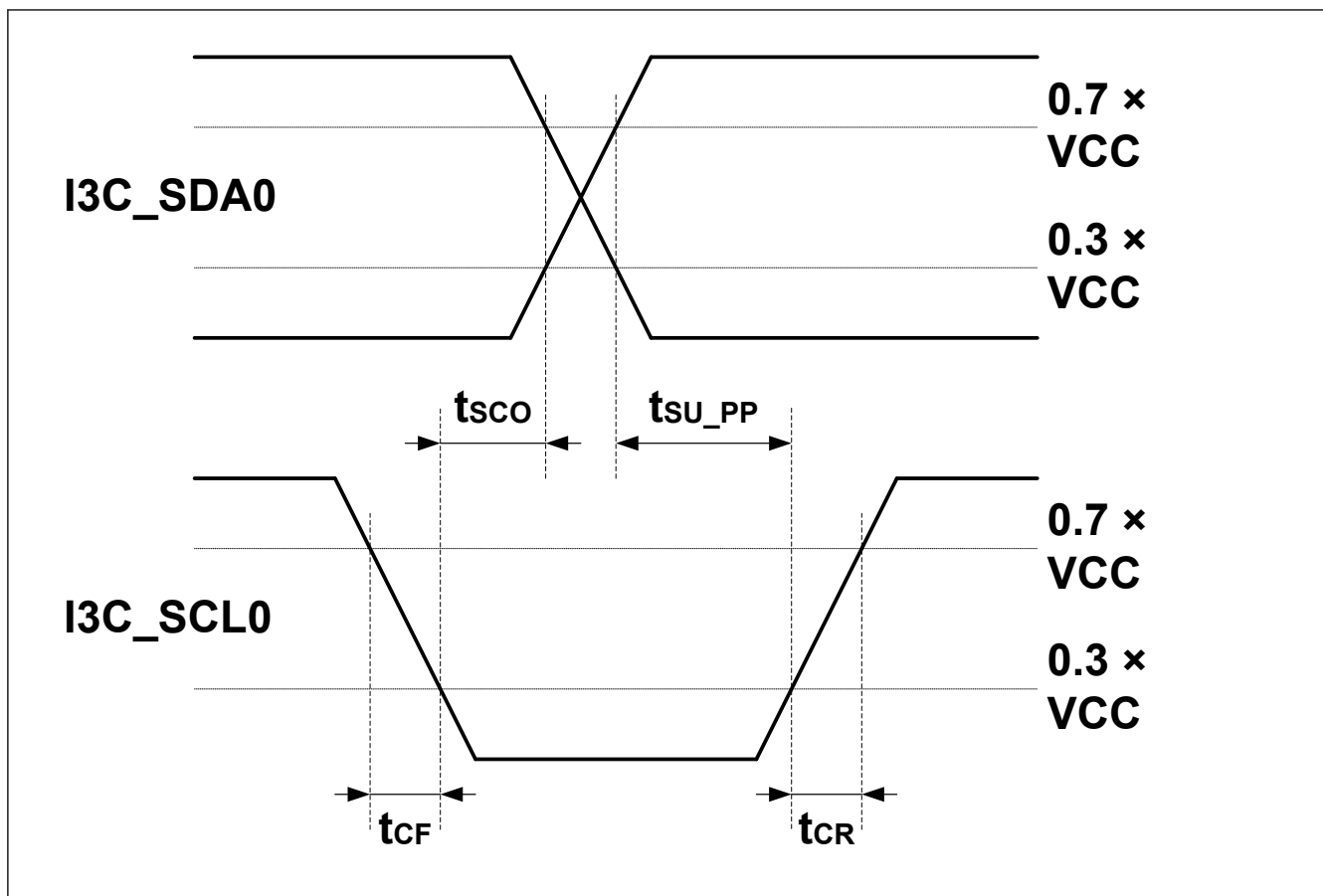


Figure 60.97 I3C Slave Out Timing

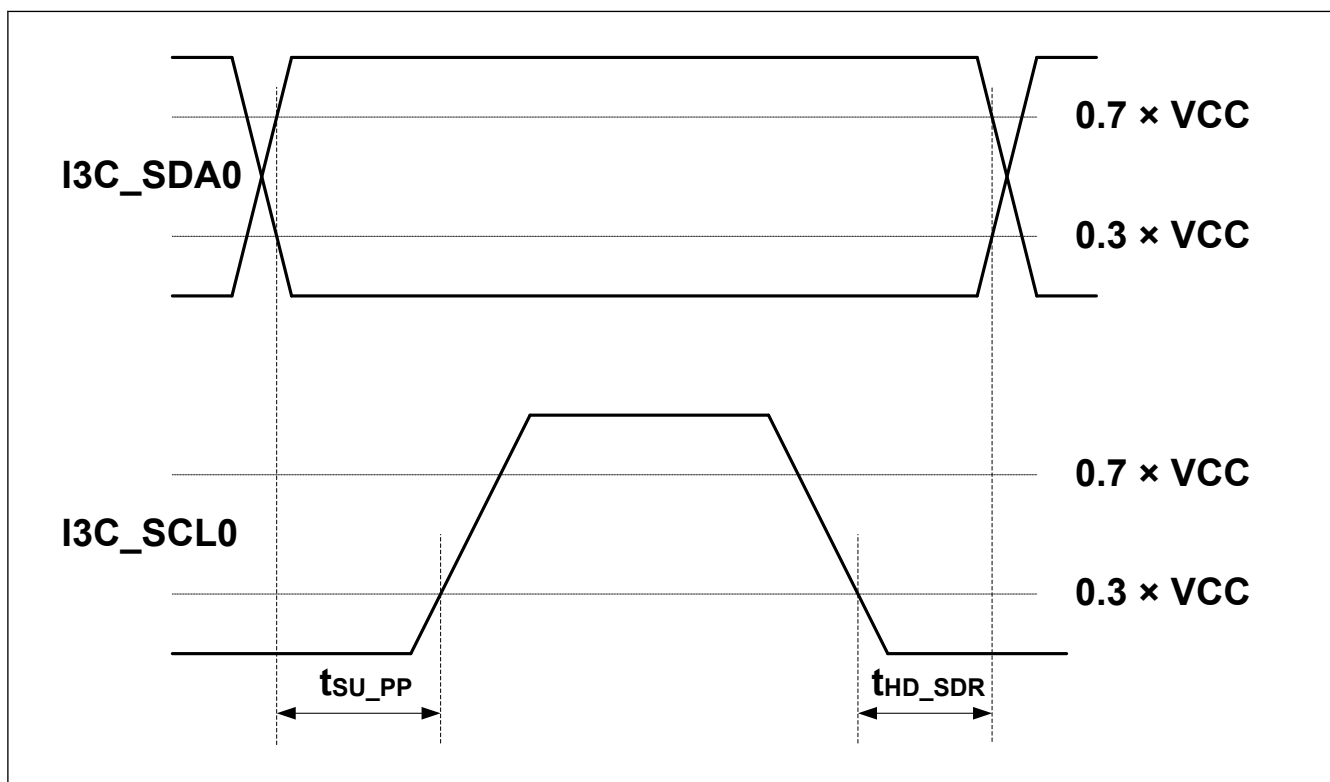


Figure 60.98 Master SDR Timing

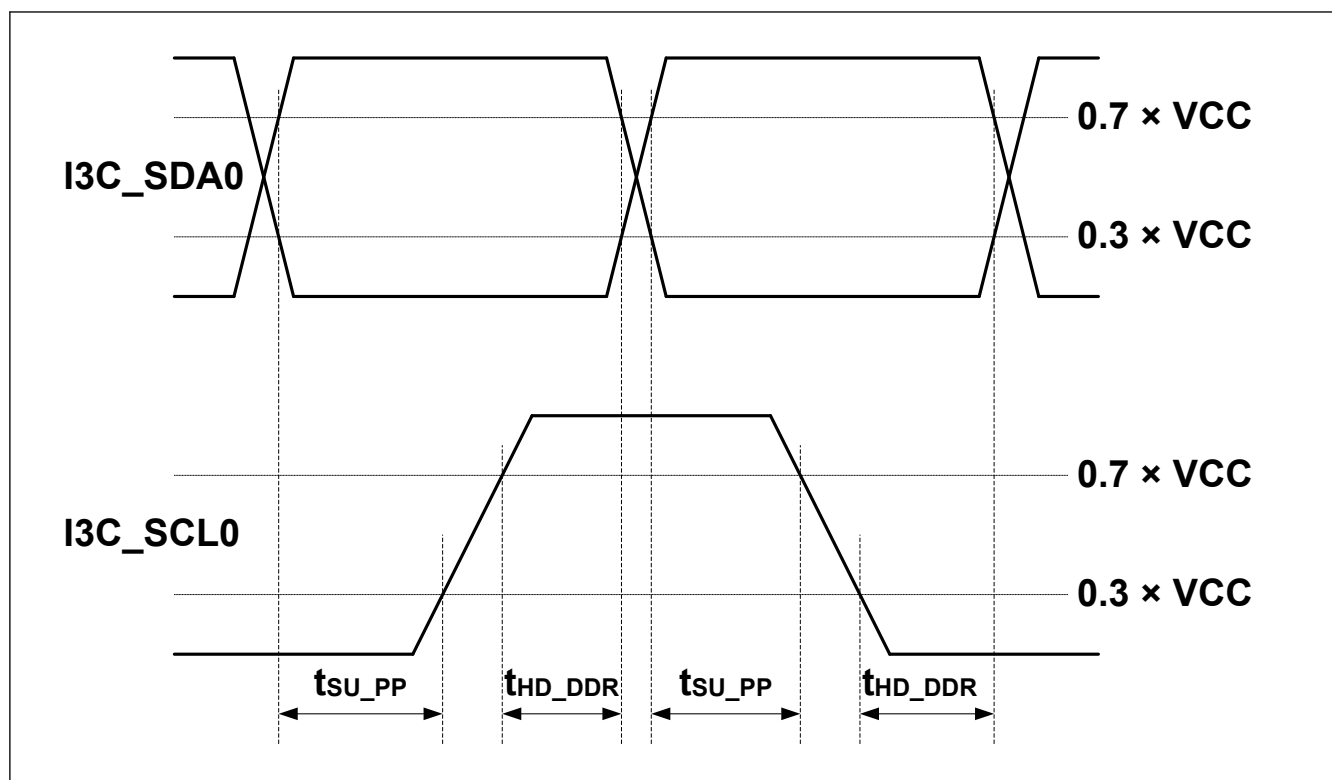


Figure 60.99 Master DDR Timing

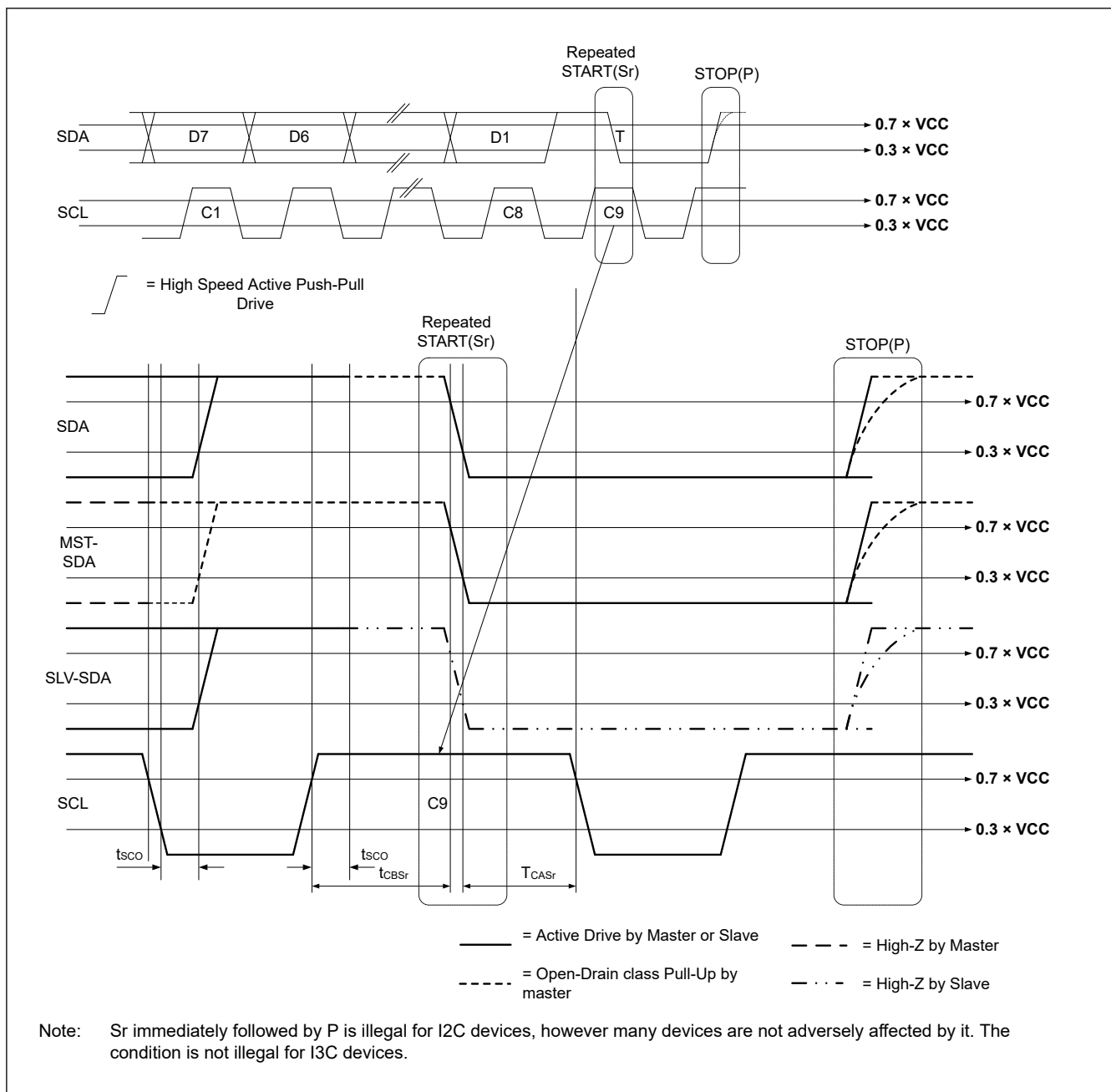


Figure 60.100 T-Bit When Master Ends Read with Repeated START and STOP

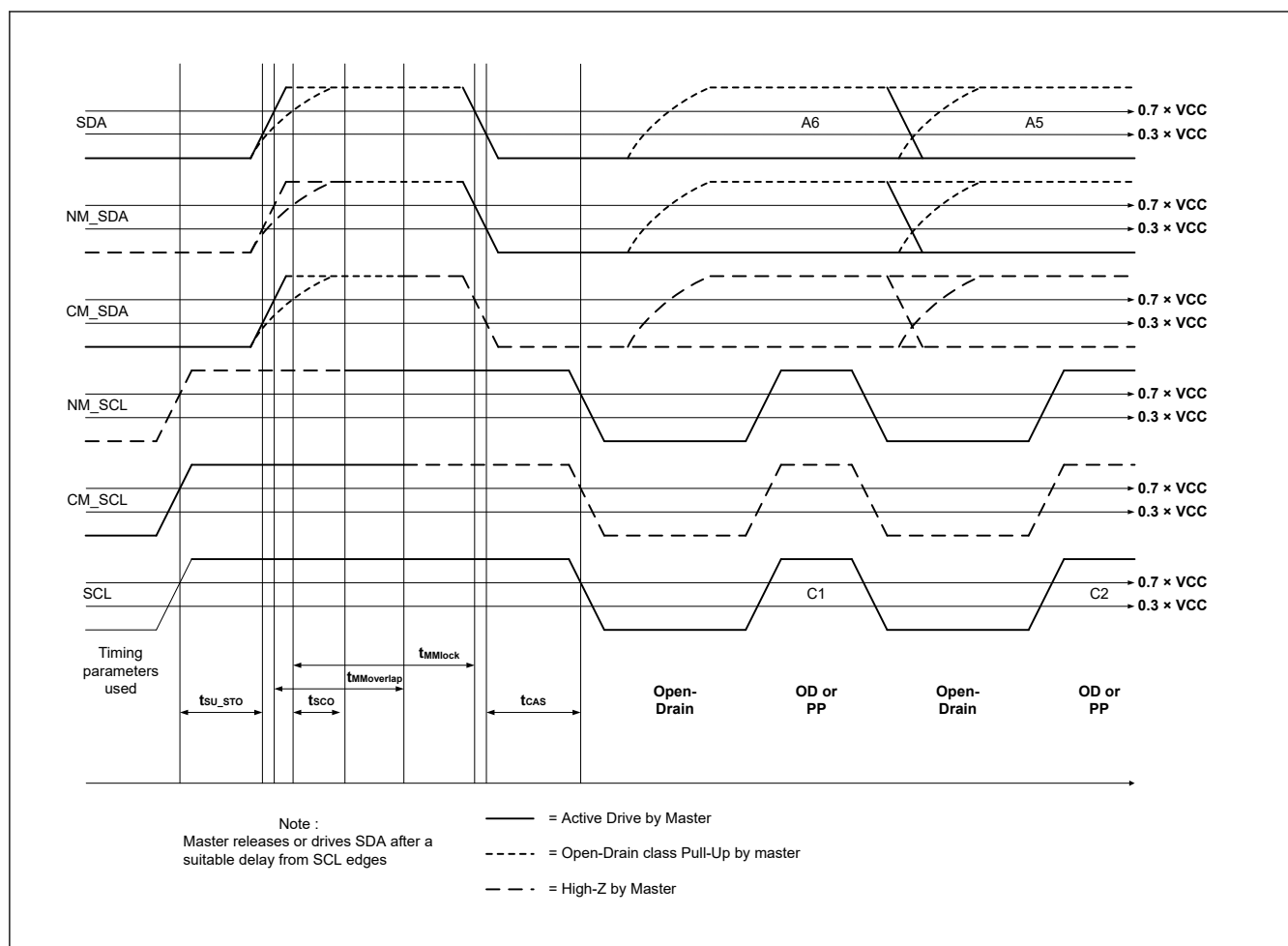
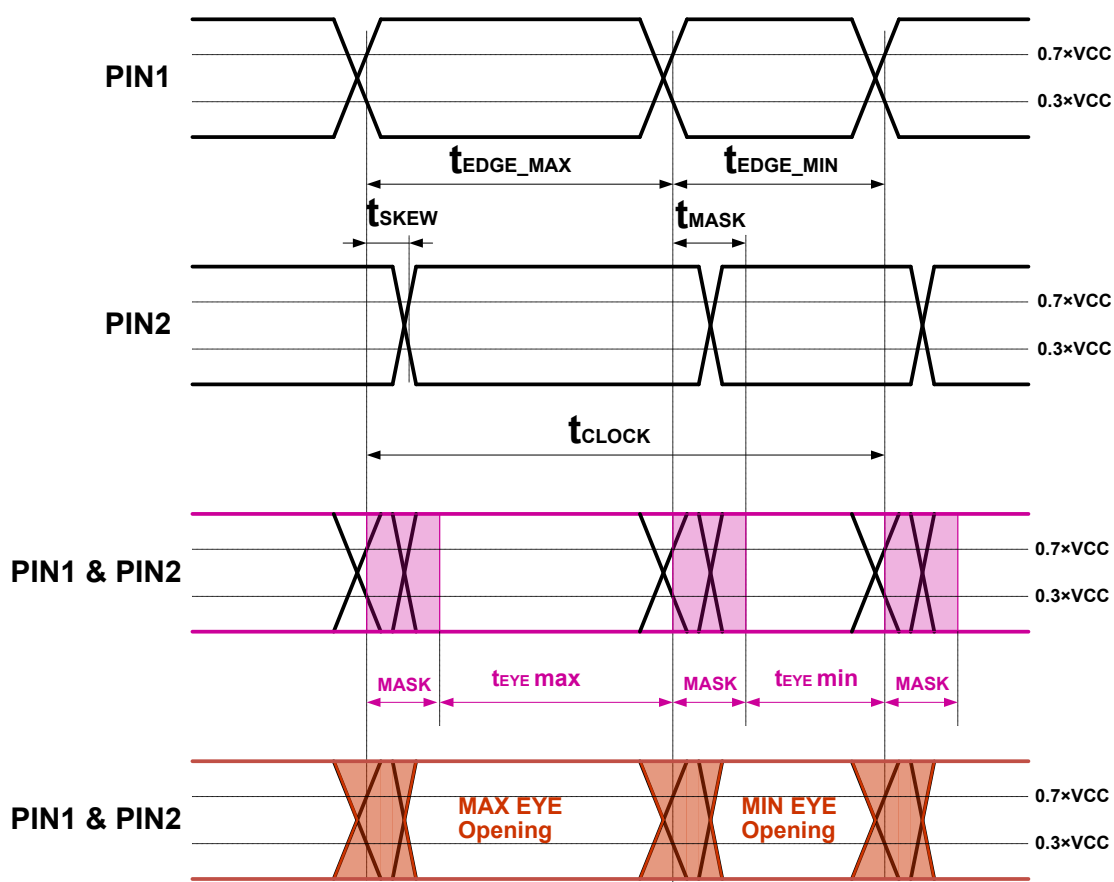


Figure 60.101 Master to Master Bus Handoff



Design Conditions

1. The differences of load magnitude and drive strength between the SDA and SCL lines result in different slopes, as shown
2. MASK period $\geq$ MAX t_{SKEW} + MAX rise/fall time between $0.3 \times V_{DD}$ and $0.7 \times V_{DD}$, respectively, decreased by the time crossing the hysteresis window.

Simplifying Assumptions

1. t_{SKEW} is the same for both edges
2. Base CLOCK (e.g. 12.5MHz) is asymmetrical
3. Falling edges are equal rising edges
4. The MASK is shown as starting, from the last possible moment, whereas in reality it could be shifted left (towards the start point) delayed by the hysteresis window

Figure 60.102 Ternary Protocol Timing

60.3.14 SD/MMC Host Interface Timing

Table 60.73 SD/MMC Host Interface signal timing

Conditions:

High-speed high drive output is selected in the Port Drive Capability bit in the PmnPFS register for the following pins:

SD0CLK_A, SD0CLK_B, SD0CLK_C, SD1CLK_A, SD1CLK_B

For other pins, high drive output is selected in the Port Drive Capability bit in the PmnPFS register.

Clock duty ratio is 50%.

Parameter	Symbol	VCC/VCC2	Min	Max	Unit	Test conditions
SDCLK clock cycle	t _{SDCYC}	2.70 V or above	20	—	ns	Figure 60.103
		1.70 to 1.95 V*1	20	—		
		1.70 to 1.95 V	40	—		
SDCLK clock high pulse width	t _{SDWH}	2.70 V or above	6.5	—	ns	
		1.70 to 1.95 V*1	6.5	—		
		1.70 to 1.95 V	13.0	—		
SDCLK clock low pulse width	t _{SDWL}	2.70 V or above	6.5	—	ns	
		1.70 to 1.95 V*1	6.5	—		
		1.70 to 1.95 V	13.0	—		
SDCLK clock rise time	t _{SDLH}	2.70 V or above	—	3	ns	
		1.70 to 1.95 V*1	—	4		
		1.70 to 1.95 V	—	8		
SDCLK clock fall time	t _{SDHL}	2.70 V or above	—	3	ns	
		1.70 to 1.95 V*1	—	4		
		1.70 to 1.95 V	—	8		
SDCMD/SDDAT output data delay	t _{SDODLY}	2.70 V or above	−7.0	4.0	ns	
		1.70 to 1.95 V*1	−7.0	7.0		
		1.70 to 1.95 V	−15.0	15.0		
SDCMD/SDDAT input data setup	t _{SDIS}	2.70 V or above	4.5	—	ns	
		1.70 to 1.95 V*1	4.5	—		
		1.70 to 1.95 V	20.0	—		
SDCMD/SDDAT input data hold	t _{SDIH}	2.70 V or above	1.5	—	ns	
		1.70 to 1.95 V	1.5	—		

Note: Must use pins that have a letter appended to their name, for instance “_A”, “_B”, to indicate group membership. For the SD/MMC Host interface, the AC portion of the electrical characteristics is measured for each group.

Note: If SD1DAT4_A to SD1DAT7_A are used, characteristics above is guaranteed only when VCC = VCC2.

Note 1. Only supported for Ch0 group B (“SD0*_B”) and Ch1 group A (“SD1*_A”)

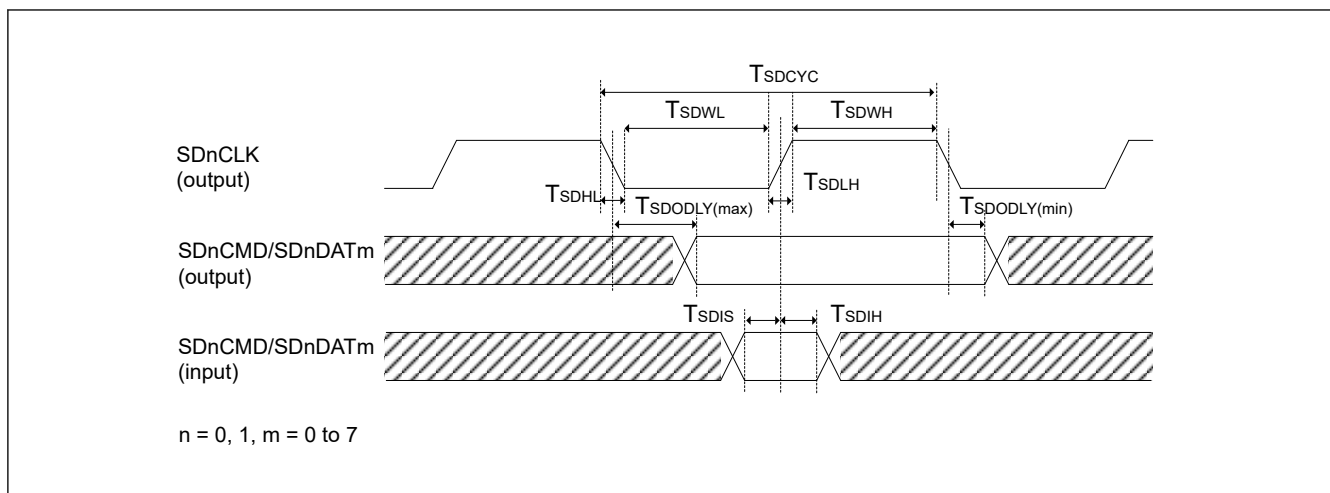


Figure 60.103 SD/MMC Host Interface signal timing

60.3.15 CANFD Timing

Table 60.74 CANFD interface timing

Conditions: Low drive output is selected when VCC is 2.70V or above. Middle drive output is selected when VCC is 1.62V or above.

Conditions: Low drive outputs selected when VCC is 2.70 V or above; Middle drive outputs selected when VCC is 1.62 V or above.						
Parameter	Symbol	VCC/VCC2	Min	Max	Unit	Test conditions
Internal delay time	t _{node}	2.70 V or above	—	50	ns	Figure 60.104
		1.62 V or above	—	50		
Transmission rate		2.70 V or above	—	8	Mbps	
		1.62 V or above	—	8		

Note: Internal delay time (t_{node}) = Internal transfer delay time (t_{output}) + Internal receive delay time (t_{input})

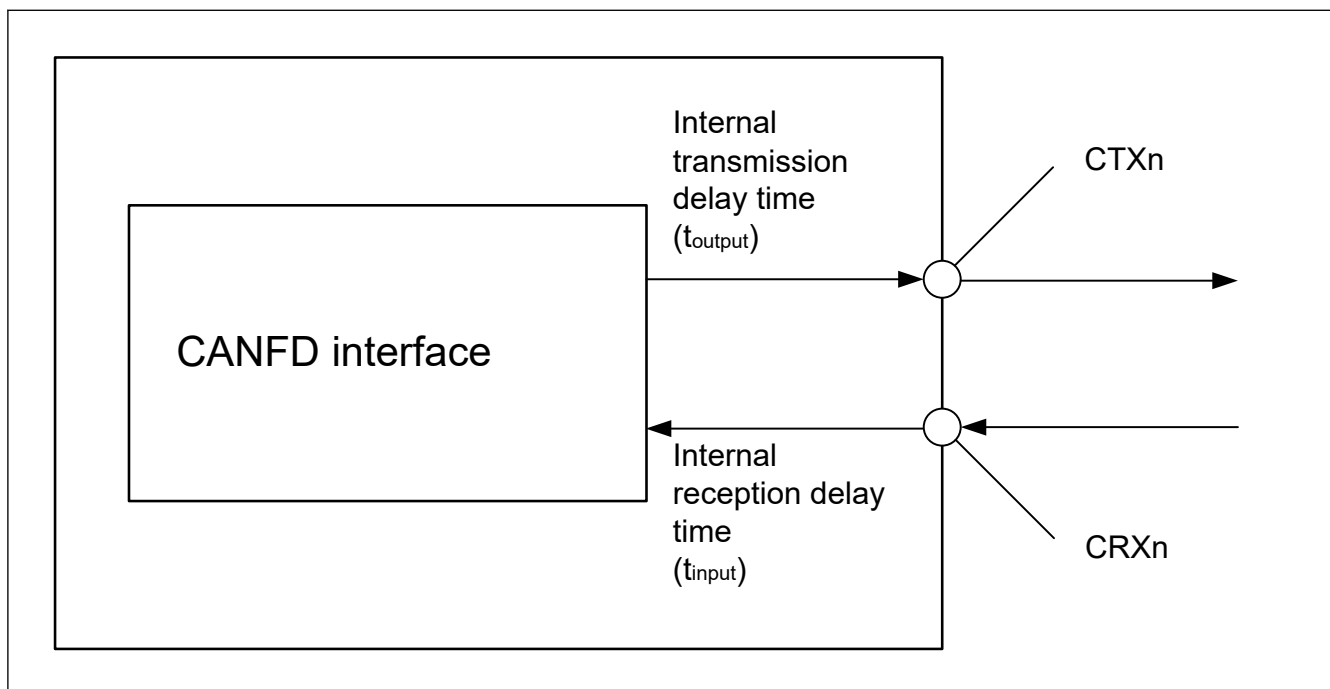


Figure 60.104 CANFD interface condition

60.3.16 PDG Timing

Table 60.75 PDG Timing

Parameter	Min	Typ	Max	Unit	Test conditions
Operation frequency	80	—	300	MHz	—
Resolution	—	48.8	—	ps	GTDLYCR.FRANGE [1:0] = 00 and GPTCLK = 160 MHz
	—	52.1	—	ps	GTDLYCR.FRANGE [1:0] = 01 and GPTCLK = 300 MHz
DNL ^{*1}	—	±2.0	—	LSB	—

Note 1. This value normalizes the differences between lines in 1-LSB resolution.

60.3.17 ESWM Timing

Table 60.76 ESWM timing (RMII)

Conditions:

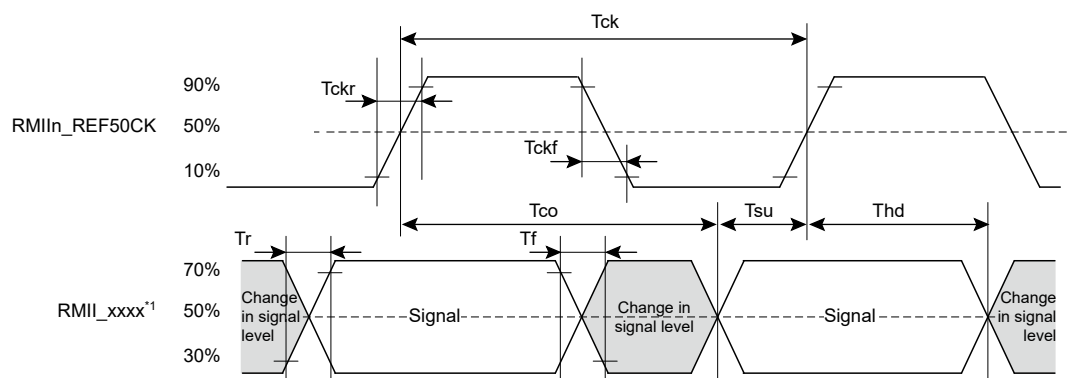
- Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register for the following pins: ETn_MDC, ETn_MDIO.
- The drive capacity selection and voltage range for pins other than ETn_MDC, ETn_MDIO.

- RMII use: High drive select, VCC = 2.7 to 3.6

Parameter	Symbol	VCC	Min	Max	Unit	Test conditions
ESWM (RMII)	RMIIIn_REF50CK cycle time	2.70V or above	20	—	ns	Figure 60.106
	RMIIIn_REF50CK frequency, typical 50 MHz		—	50 + 100 ppm	MHz	
	RMIIIn_REF50CK duty		35	65	%	
	RMIIIn_REF50CK rise/fall time		0.5	3.5	ns	
	RMIIIn_xxxx ^{*1} output delay		2.5	12	ns	
	RMIIIn_xxxx ^{*2} setup time		3	—	ns	
	RMIIIn_xxxx ^{*2} hold time		1	—	ns	
	RMIIIn_xxxx ^{*1 *2} rise/fall time		0.5	4	ns	

Note 1. RMIIIn_TX_EN, RMIIIn_TXD1, RMIIIn_TXD0.

Note 2. RMIIIn_CRS_DV, RMIIIn_RXD1, RMIIIn_RXD0, RMIIIn_RX_ER.



Note 1. RMII_n_TX_EN, RMII_n_TXD1, RMII_n_TXD0, RMII_n_CRS_DV, RMII_n_RXD1, RMII_n_RXD0, RMII_n_RX_ER

Figure 60.106 RMII_n_REF50CK and RMII signal timing

Table 60.77 ESWM timing (MII)

Conditions:

- Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register for the following pins: ET_n_MDC, ET_n_MDIO.
- The drive capacity selection and voltage range for pins other than ET_n_MDC, ET_n_MDIO.

- MII use only: Middle drive select, VCC = 2.7 to 3.6
- GMII and MII use: RGMII 2.50 V drive select, VCC = 2.3 to 2.7
- GMII and MII use: RGMII 3.30 V drive select, VCC = 3.0 to 3.6

Parameter	Symbol	VCC	Min	Max	Unit	Test conditions
ESWM (MII) ET _n _TX_CLK cycle time	t _{Tcyc}	2.30 V or above	40	—	ns	—
ET _n _TX_EN output delay	t _{TEND}		1	20	ns	Figure 60.107
ET _n _ETXD0 to ET _n _ETXD3 output delay	t _{MTDd}		1	20	ns	
ET _n _RX_CLK cycle time	t _{TRcyc}		40	—	ns	—
ET _n _RX_DV setup time	t _{RDVs}		10	—	ns	Figure 60.108
ET _n _RX_DV hold time	t _{RDVh}		10	—	ns	
ET _n _ERXD0 to ET _n _ERXD3 setup time	t _{MRDs}		10	—	ns	
ET _n _ERXD0 to ET _n _ERXD3 hold time	t _{MRDh}		10	—	ns	
ET _n _RX_ER setup time	t _{RERs}		10	—	ns	
ET _n _RX_ER hold time	t _{RESh}		10	—	ns	

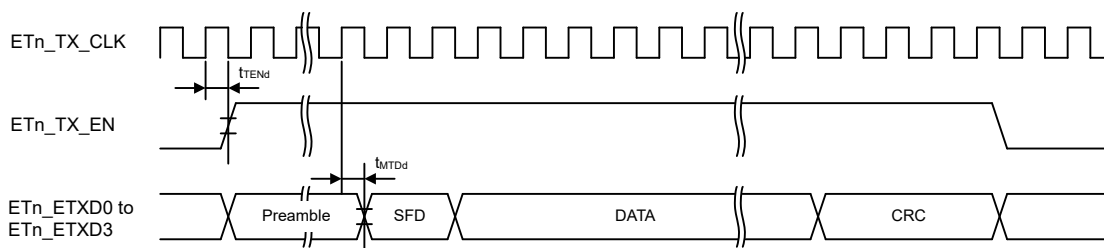


Figure 60.107 MII transmission timing in normal operation

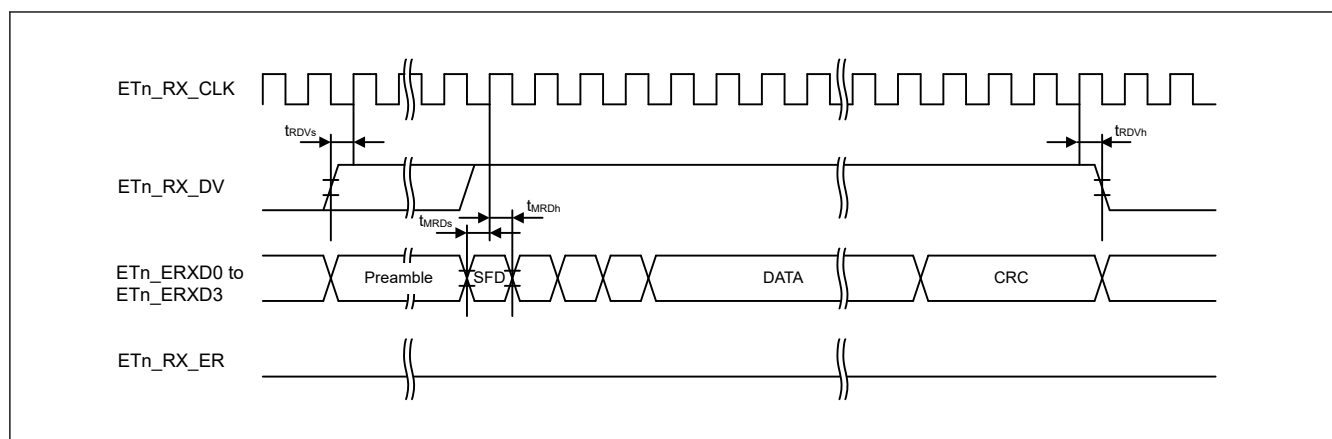


Figure 60.108 MII reception timing in normal operation

Table 60.78 ESWM timing (GMII)

Conditions:

- Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register for the following pins: ETn_MDC, ETn_MDIO.
- The drive capacity selection and voltage range for pins other than ETn_MDC, ETn_MDIO.
 - GMII use at 2.5 V: RGMII 2.50V drive select, VCC = 2.3 to 2.7
 - GMII use at 3.3 V: RGMII 3.30V drive select, VCC = 3.0 to 3.6

Parameter	Symbol	VCC	Min	Max	Unit	Test conditions
ESWM (GMII)	ETn_GTX_CLK Frequency	2.30 V to 3.60 V	125 - 100 ppm	125 + 100 ppm	MHz	Figure 60.109 Figure 60.110
	ETn_GTX_CLK Period		7.5	8.5	ns	
	ETn_RX_CLK Period		7.5	—	ns	
	ETn_GTX_CLK, ETn_RX_CLK Time High		2.5	—	ns	
	ETn_GTX_CLK, ETn_RX_CLK Time Low		2.5	—	ns	
	ETn_GTX_CLK, ETn_RX_CLK Rise Time		—	1	ns	
	ETn_GTX_CLK, ETn_RX_CLK Fall Time		—	1	ns	
	Magnitude of ETn_GTX_CLK, ETn_RX_CLK Slew Rate (rising)*1		0.6	—	V/ns	
	Magnitude of ETn_GTX_CLK, ETn_RX_CLK Slew Rate (falling)*1		0.6	—	V/ns	
	ETn_TXD, ETn_TX_EN, ETn_TX_ER Setup to ↑ETn_GTX_CLK and ETn_RXD, ETn_RX_DV, ETn_RX_ER Setup to ↑ETn_RX_CLK		2.5	—	ns	
	ETn_TXD, ETn_TX_EN, ETn_TX_ER Hold from ↑ETn_GTX_CLK and ETn_RXD, ETn_RX_DV, ETn_RX_ER Hold from ↑ETn_RX_CLK		0.5	—	ns	
	ETn_TXD, ETn_TX_EN, ETn_TX_ER Setup to ↑ETn_GTX_CLK and ETn_RXD, ETn_RX_DV, ETn_RX_ER Setup to ↑ETn_RX_CLK		2	—	ns	
	TXD, TX_EN, TX_ER, ETn_TX_ER Hold from ↑GTX_CLK and RXD, RX_DV, RX_ER Hold from ↑RX_CLK		0	—	ns	

Note 1. Clock Skew rate is the instantaneous rate of change of the clock potential with respect to time (dV/dt), not an average value over the entire rise or fall time interval. Conformance with this specification guarantees that the clock signals will rise and fall monotonically through the switching region.

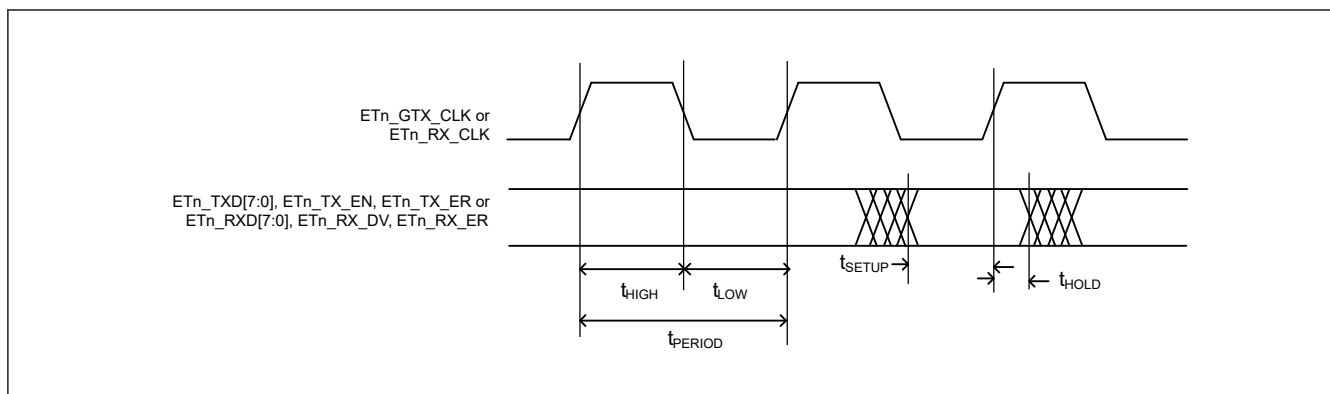


Figure 60.109 GMII timing

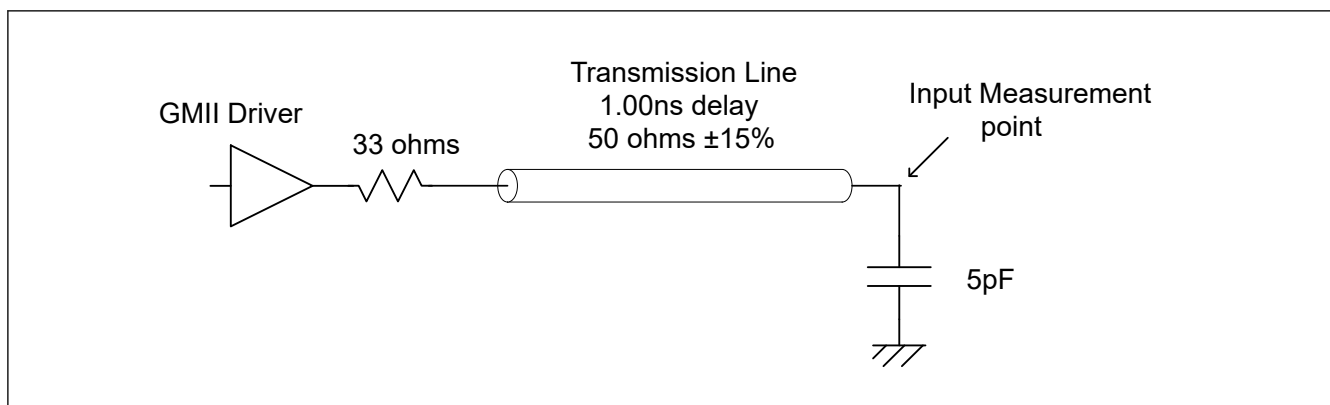


Figure 60.110 GMII output timing measurement conditions

Table 60.79 ESWM timing (RGMII)

Conditions:

- Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register for the following pins: ETn_MDC, ETn_MDIO.
- The drive capacity selection and voltage range for pins other than ETn_MDC, ETn_MDIO.
 - RGMII use at 2.5 V: RGMII 2.50V drive select, VCC = 2.3 to 2.7
 - RGMII use at 3.3 V: RGMII 3.30V drive select, VCC = 3.0 to 3.6

Parameter	Symbol	VCC	Min	Max	Unit	Test conditions
ESWM (RGMII)	Data to Clock output Skew (at Transmitter)*1	2.30 V to 3.60 V	-500	500	ps	Figure 60.111 Figure 60.112 Figure 60.113
	Data to Clock input Skew (at Receiver)*1		1	2.6	ns	
	Data to Clock output Setup (at Transmitter-integrated delay)		1.2	—	ns	
	Clock to Data output Hold (at Transmitter-integrated delay)		1.2	—	ns	
	Data to Clock input Setup (at Receiver-integrated delay)		1	—	ns	
	Data to Clock input Hold (at Receiver-integrated delay)		1	—	ns	
	Clock Cycle Duration*2		7.2	8.8	ns	
	Duty Cycle for Gigabit*3		45	55	%	
	Duty Cycle for 10/100T*3		40	60	%	
	Rise / Fall Time (20-80%)		—	0.75	ns	

- Note 1. This implies that PC board design will require clocks to be routed such that an additional trace delay of greater than 1.5ns and less than 2.0ns will be added to the associated clock signal.
- Note 2. For 10Mbps and 100Mbps, T_{cy} will scale to 400ns+40ns and 40ns+4ns respectively.
- Note 3. Duty cycle may be stretched/shrunk during speed changes or while transitioning to a received packet's clock domain as long as minimum duty cycle is not violated and stretching occurs for no more than three T_{cy} of the lowest speed transitioned between.

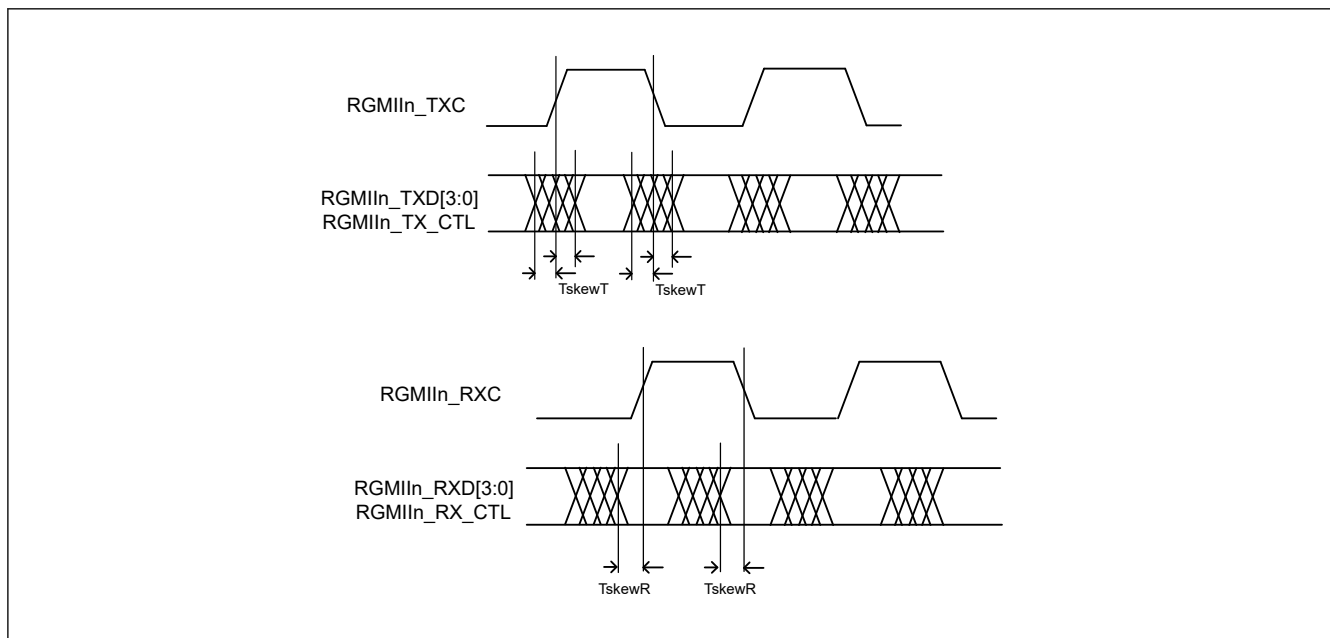


Figure 60.111 RGMII timing

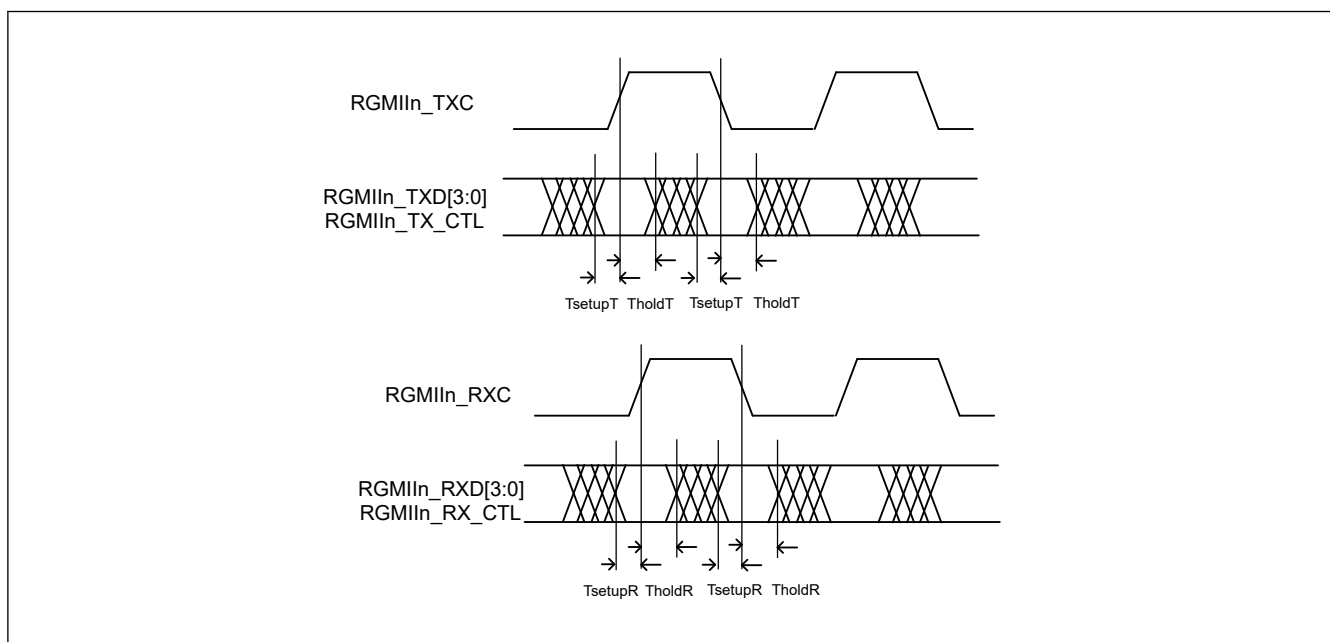


Figure 60.112 RGMII timing (RGMII-ID)

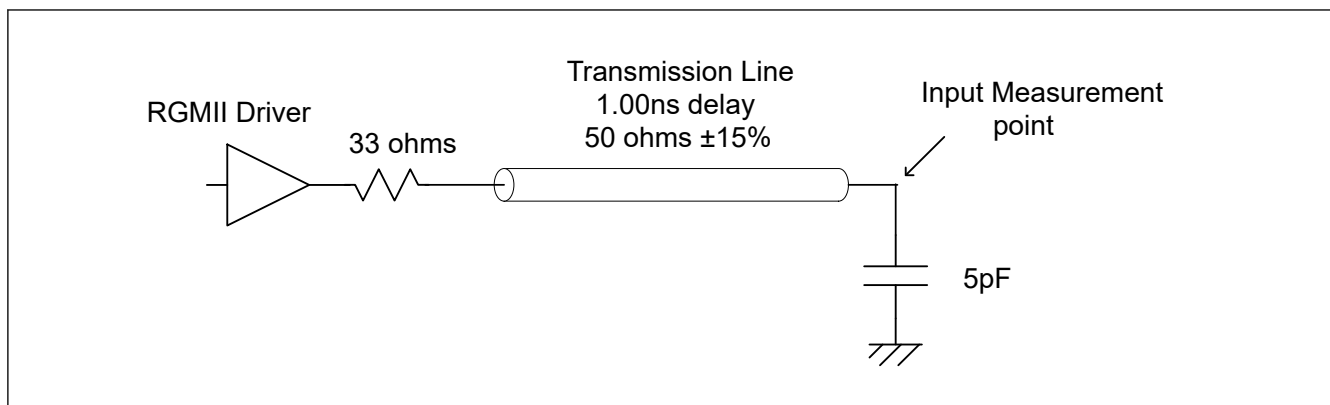


Figure 60.113 RGMII output timing measurement conditions

Table 60.80 ESWM timing (ETn_MDIO, ETn_MDC)

Conditions:

1. Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register.

Parameter		Symbol	VCC	Min	Max	Unit	Test conditions
ESWM (ETn_MDIO, ETn_MDC)	ETHn_MDC output cycle	t_{MDC}	2.70 V or above	80	—	ns	Figure 60.114
			2.30 V or above	160	—	ns	
	ETHn_MDIO setup time (relative to ETHn_MDC↑)	t_{SMDIO}	2.70 V or above	20	—	ns	
			2.30 V or above	40	—	ns	
	ETHn_MDIO hold time (relative to ETHn_MDC↑)	t_{HMDIO}	2.70 V or above	0	—	ns	
			2.30 V or above	0	—	ns	
	ETHn_MDIO output delay time (relative to ETHn_MDC↑)	t_{DMDIO}	2.70 V or above	0	20	ns	
			2.30 V or above	0	40	ns	

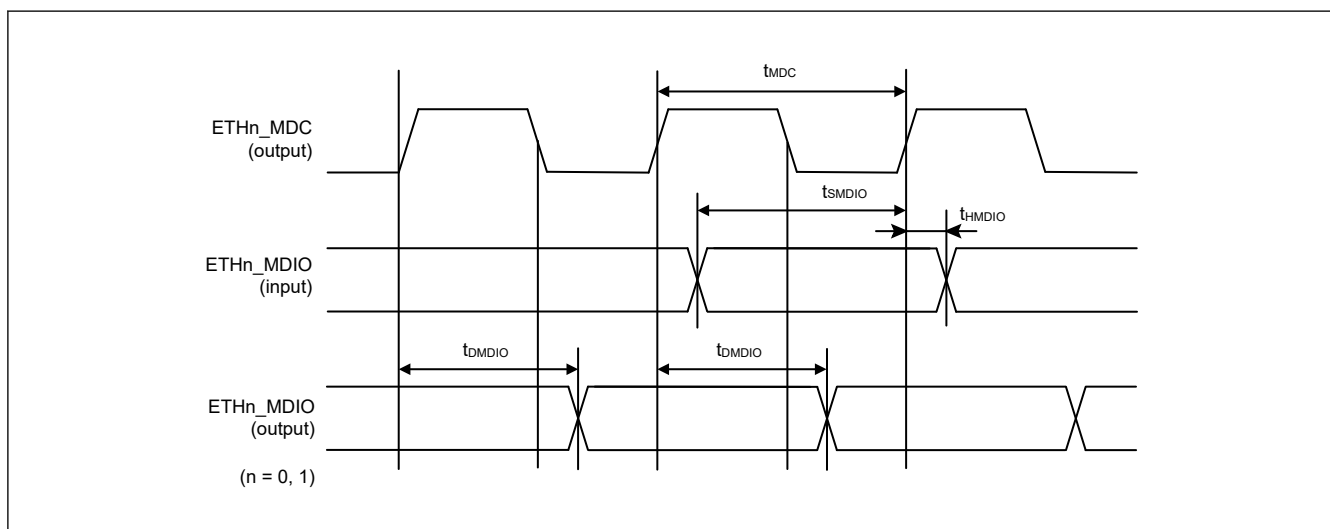


Figure 60.114 ETn_MDIO, ETn_MDC timing

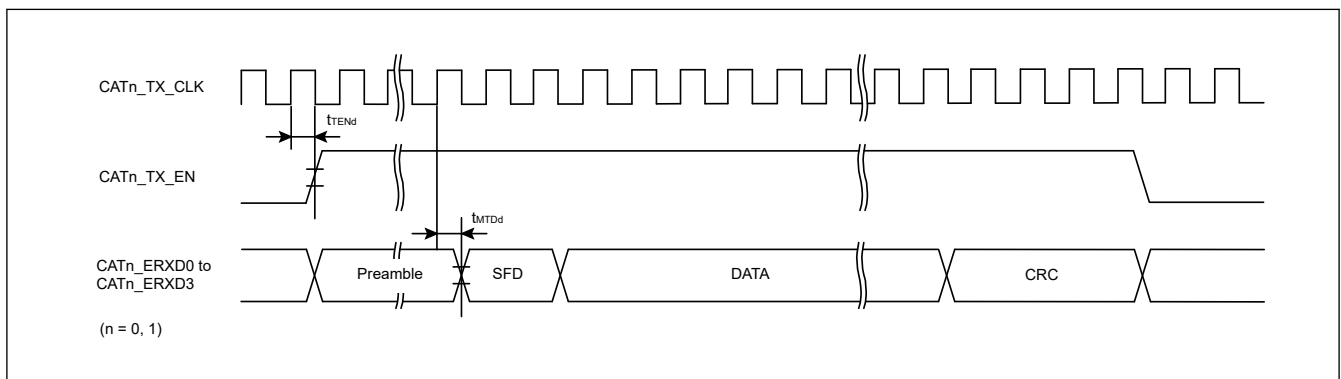
60.3.18 ESC Timing

Table 60.81 ESC timing

Conditions:

1. Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register for the following pins: CAT0_MDC, CAT0_MDIO.
2. High drive output is selected in the Port Drive Capability bit in the PmnPFS register for pins other than CAT0_MDC, CAT0_MDIO.

Parameter		Symbol	VCC	Min	Max	Unit	Test conditions
ESC (MII)	CATn_TX_CLK cycle time	t_{Tcyc}	2.70V or above	40	—	ns	Figure 60.114
	CATn_TX_EN output delay time	t_{TEND}		1	25	ns	
	CATn_ETXD0 to CATn_ETXD3 output delay time	t_{MTDd}		1	25	ns	
	CATn_RX_CLK cycle time	t_{TRcyc}		40	—	ns	Figure 60.115
	CATn_RX_DV setup time	t_{RDVs}		10	—	ns	
	CATn_RX_DV hold time	t_{RDVh}		10	—	ns	
	CATn_ERXD0 to CATn_ERXD3 setup time	t_{MRDs}		10	—	ns	
	CATn_ERXD0 to CATn_ERXD3 hold time	t_{MRDh}		10	—	ns	Figure 60.116
	CATn_RX_ER setup time	t_{RERs}		10	—	ns	
	CATn_RX_ER hold time	t_{RERh}		10	—	ns	
ESC (MDIO)	CAT0_MDIO setup time (CAT0_MDC↑)	t_{SMDIO}		60	—	ns	Figure 60.117
	CAT0_MDIO hold time (CAT0_MDC↑)	t_{HMDIO}		0	—	ns	
	CAT0_MDIO output delay time (CAT0_MDC↓)	t_{DMDIO}		0	30	ns	

**Figure 60.114 MII Transmission Timing (Normal Operation)**

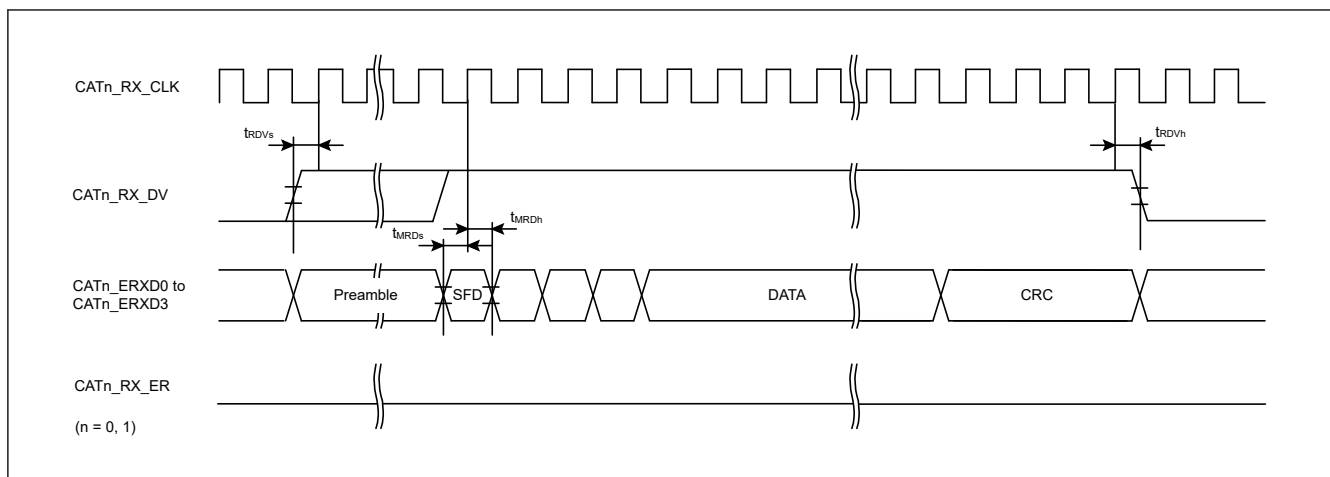


Figure 60.115 MII Reception Timing (Normal Operation)

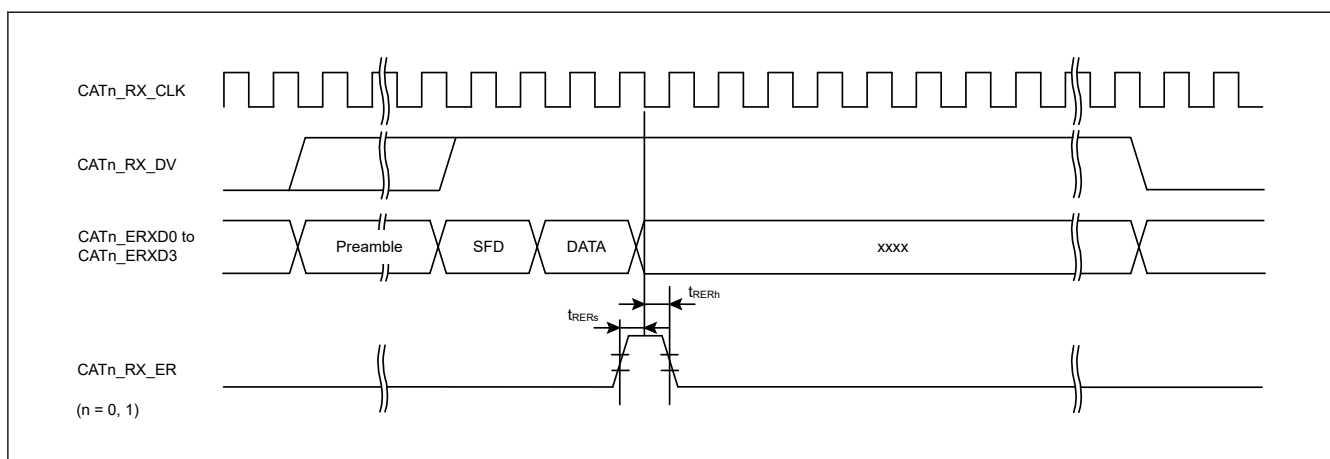


Figure 60.116 MII Reception Timing (Error Occurrence)

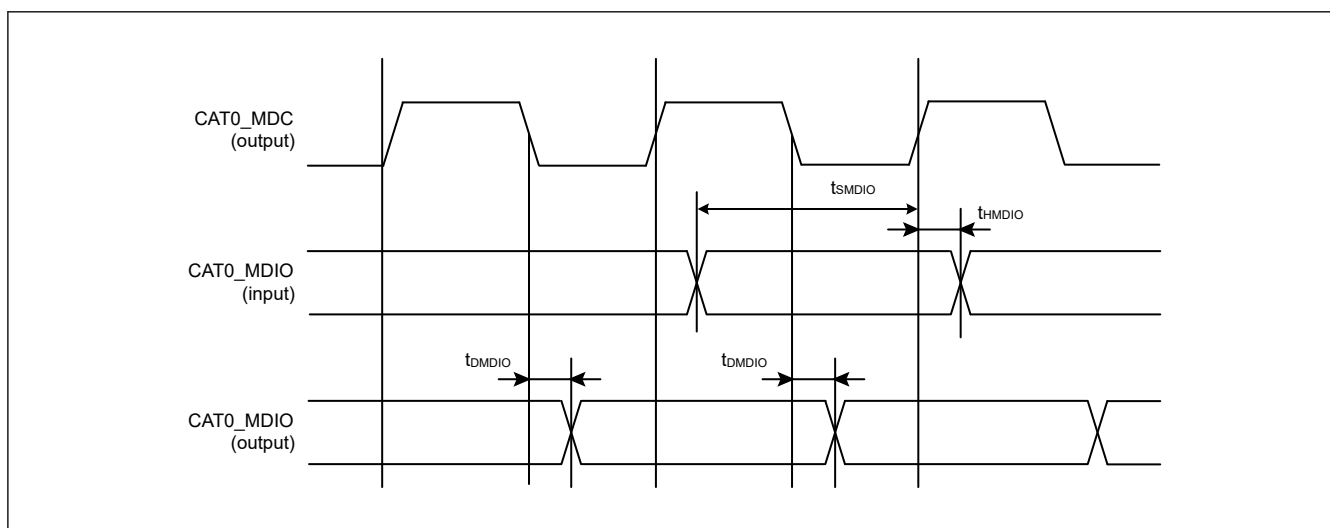


Figure 60.117 Timing of Serial Management Access

60.3.19 DSMIF Timing

Table 60.82 DSMIF timing

Conditions: Middle drive output is selected in the Port Drive Capability bit in the PmnPFS register.

Parameter		Symbol	VCC	Min	Max	Unit	Test conditions
Clock period	Master	t_{DScyc}	2.70 V or above	40	200	ns	Figure 60.118
			1.62 V or above	80	200		
	Slave		2.70 V or above	40	200		
			1.62 V or above	40	200		
Clock high level pulse width	Master	t_{DSCKWH}	2.70 V or above	16	—	ns	
			1.62 V or above	32	—		
	Slave		2.70 V or above	16	—		
			1.62 V or above	16	—		
Clock low level pulse width	Master	t_{DSCKWL}	2.70 V or above	16	—	ns	
			1.62 V or above	32	—		
	Slave		2.70 V or above	16	—		
			1.62 V or above	16	—		
Setup time	Master	t_{SU}	2.70 V or above	15	—	ns	Figure 60.119 Figure 60.120
			1.62 V or above	30	—		
	Slave		2.70 V or above	10	—		
			1.62 V or above	10	—		
Hold time	Master	t_H	2.70 V or above	0	—	ns	
			1.62 V or above	0	—		
	Slave		2.70 V or above	10	—		
			1.62 V or above	10	—		

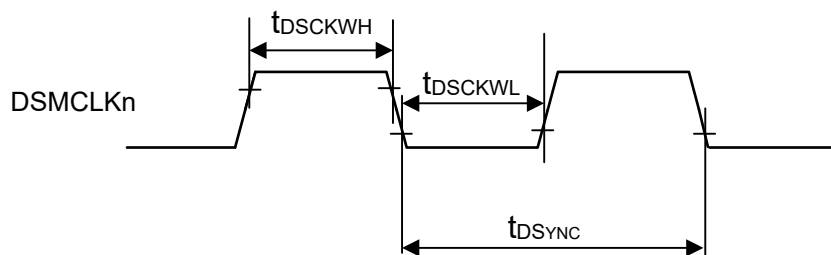


Figure 60.118 Timing of clock input / output

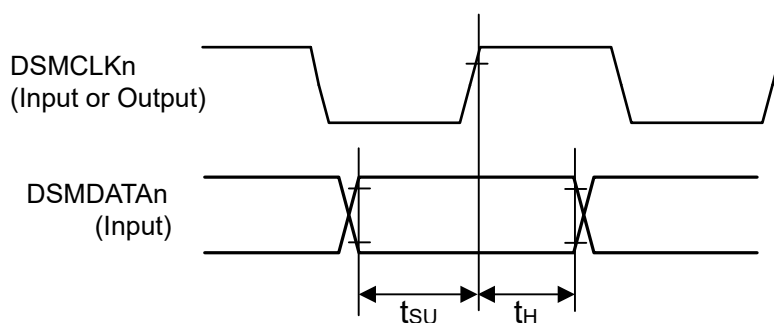


Figure 60.119 Receive Timing (Synchronized with the rise of DSMCLKn)

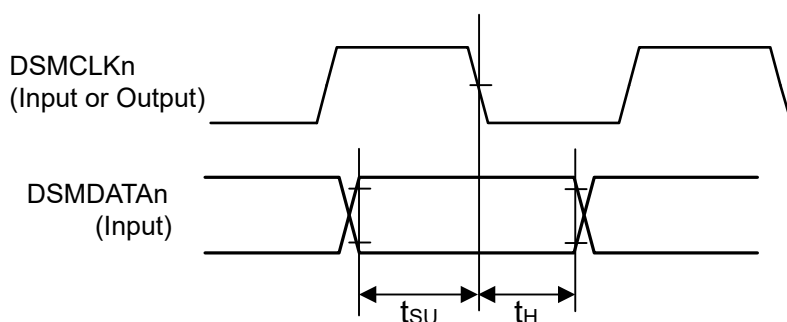


Figure 60.120 Receive Timing (Synchronized with the fall of DSMCLKn)

60.4 USB Characteristics

60.4.1 USBFS Timing

Table 60.83 USBFS low-speed characteristics for host only (USB_DP and USB_DM pin characteristics) (1 of 2)

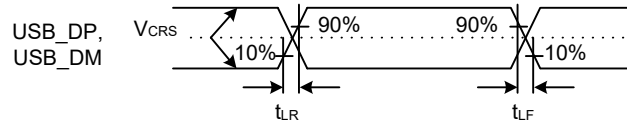
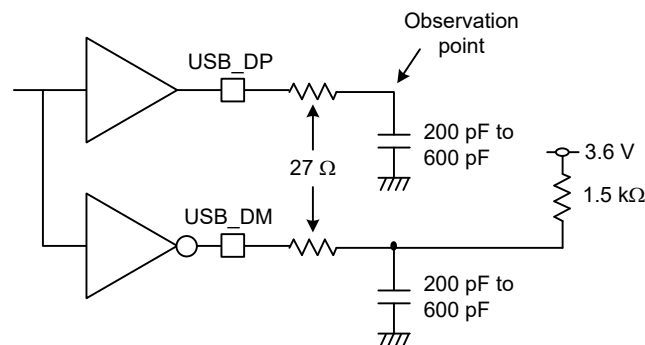
Conditions: VCC = VCC_USB = 3.0 to 3.6 V, USBCLK = 48 MHz

Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
Input characteristics	Input high voltage	V _{IH}	2.0	—	V	—
	Input low voltage	V _{IL}	—	0.8	V	—
	Differential input sensitivity	V _{DI}	0.2	—	V	USB_DP - USB_DM
	Differential common-mode range	V _{CM}	0.8	2.5	V	—

Table 60.83 USBFS low-speed characteristics for host only (USB_DP and USB_DM pin characteristics) (2 of 2)

Conditions: VCC = VCC_USB = 3.0 to 3.6 V, USBCLK = 48 MHz

Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
Output characteristics	Output high voltage	V _{OH}	2.8	—	3.6	V I _{OH} = -200 μA
	Output low voltage	V _{OL}	0.0	—	0.3	V I _{OL} = 2 mA
	Cross-over voltage	V _{CRS}	1.3	—	2.0	V Figure 60.121
	Rise time	t _{LR}	75	—	300	ns
	Fall time	t _{LF}	75	—	300	ns
	Rise/fall time ratio	t _{LR} / t _{LF}	80	—	125	% t _{LR} / t _{LF}
Pull-up and pull-down characteristics	USB_DP and USB_DM pull-down resistance in host controller mode	R _{pd}	14.25	—	24.80	kΩ —

**Figure 60.121 USB_DP and USB_DM output timing in low-speed mode****Figure 60.122 Test circuit in low-speed mode****Table 60.84 USBFS full-speed characteristics (USB_DP and USB_DM pin characteristics) (1 of 2)**

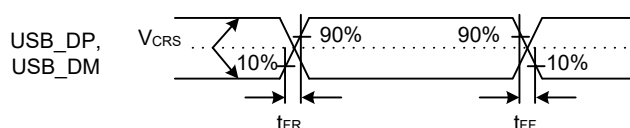
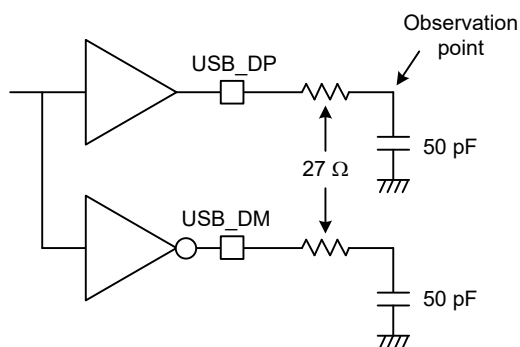
Conditions: VCC = VCC_USB = 3.0 to 3.6 V, USBCLK = 48 MHz

Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
Input characteristics	Input high voltage	V _{IH}	2.0	—	—	V
	Input low voltage	V _{IL}	—	—	0.8	V
	Differential input sensitivity	V _{DI}	0.2	—	—	V USB_DP - USB_DM
	Differential common-mode range	V _{CM}	0.8	—	2.5	V

Table 60.84 USBFS full-speed characteristics (USB_DP and USB_DM pin characteristics) (2 of 2)

Conditions: VCC = VCC_USB = 3.0 to 3.6 V, USBCLK = 48 MHz

Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
Output characteristics	Output high voltage	V _{OH}	2.8	—	3.6	V I _{OH} = -200 μA
	Output low voltage	V _{OL}	0.0	—	0.3	V I _{OL} = 2 mA
	Cross-over voltage	V _{CRS}	1.3	—	2.0	V Figure 60.123
	Rise time	t _{LR}	4	—	20	ns
	Fall time	t _{LF}	4	—	20	ns
	Rise/fall time ratio	t _{LR} / t _{LF}	90	—	111.11	% t _{FR} / t _{FF}
	Output resistance	Z _{DRV}	28	—	44	Ω USBFS: R _s = 27 Ω included
Pull-up and pull-down characteristics	DM pull-up resistance in device controller mode	R _{pu}	0.900	—	1.575	kΩ During idle state
			1.425	—	3.090	kΩ During transmission and reception
	USB_DP and USB_DM pull-down resistance in host controller mode	R _{pd}	14.25	—	24.80	kΩ —

**Figure 60.123 USB_DP and USB_DM output timing in full-speed mode****Figure 60.124 Test circuit in full-speed mode**

60.5 ADC Characteristics

Table 60.85 A/D conversion characteristics (common) (1 of 3)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter	Min	Typ	Max	Unit	Test conditions
A/D conversion clock frequency (ADCLK)	25	50	60	MHz	AVCC: 2.7 to 3.63 V VCC: 2.7 to 3.63 V
Successive approximation time	100	—	200	ns	VREFH0/VREFH: 2.7 V to AVCC

Table 60.85 A/D conversion characteristics (common) (2 of 3)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter					Min	Typ	Max	Unit	Test conditions	
A/D sampling time (Normal mode)	Self-calibration				SAR mode	$1 \times t_{ADcyc} + 40$	—	—	ns	AVCC: 2.7 to 3.63 V VCC: 2.7 to 3.63 V VREFH0/VREFH: 2.7 V to AVCC tCmp=100ns
					Oversampling mode	$1 \times t_{ADcyc} + 40$	—	—	ns	
	Self-diagnosis				SAR mode	$1 \times t_{ADcyc} + 40$	—	—	ns	
					Oversampling mode	$1 \times t_{ADcyc} + 40$	—	—	ns	
	A/D conversion	High-speed channels	Without channel dedicated sample and hold circuits	(AN000 to AN005) (AN006 to AN011)	SAR mode	$1 \times t_{ADcyc} + 40$	—	—	ns	
					Oversampling mode (One-channel continuous scan mode)	40	—	—	ns	
					Oversampling mode (Single/Continuous scan mode)	$1 \times t_{ADcyc} + 40$	—	—	ns	
			With channel-dedicated sample-and-hold circuits	(AN000 to AN005) (AN006 to AN011)	SAR mode	$1 \times t_{ADcyc} + 160$	—	—	ns	
					Hybrid mode	$1 \times t_{ADcyc} + 160$	—	—	ns	
				Middle-speed channels		(AN012 to AN015)	SAR mode	180	—	
		Oversampling mode	200				—	—	ns	
		Low-speed channels		(AN016 to AN022)	SAR mode	400	—	—	ns	
					Oversampling mode	400	—	—	ns	
A/D sampling time (High accuracy mode)		Self-calibration				SAR mode	$1 \times t_{ADcyc} + 140^{*1}$	—	—	ns
	Hybrid mode					$1 \times t_{ADcyc} + 140^{*1}$	—	—	ns	
	Self-diagnosis				SAR mode	$1 \times t_{ADcyc} + 140^{*1}$	—	—	ns	
					Hybrid mode	$1 \times t_{ADcyc} + 140^{*1}$	—	—	ns	
	A/D conversion	High-speed channels	Without channel dedicated sample and hold circuits	(AN000 to AN005) (AN006 to AN011)	SAR mode	$1 \times t_{ADcyc} + 140^{*1}$	—	—	ns	
					Hybrid mode	$1 \times t_{ADcyc} + 140^{*1}$	—	—	ns	
			With channel-dedicated sample-and-hold circuits	(AN000 to AN005) (AN006 to AN011)	SAR mode	$1 \times t_{ADcyc} + 320$	—	—	ns	
					Hybrid mode	$1 \times t_{ADcyc} + 320$	—	—	ns	
		Middle-speed channels		(AN012 to AN015)	SAR mode	400	—	—	ns	
					Hybrid mode	440	—	—	ns	
		Low-speed channels		(AN016 to AN022)	SAR mode	840	—	—	ns	
					Hybrid mode	840	—	—	ns	

Table 60.85 A/D conversion characteristics (common) (3 of 3)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter				Min	Typ	Max	Unit	Test conditions
Channel dedicated sample and hold circuits	Sampling time	Self-calibration		400	—	—	ns	AVCC: 2.7 to 3.63 V VCC: 2.7 to 3.63 V VREFH0/VREFH: 2.7 V to AVCC
		A/D conversion		400	—	—	ns	
	Hold mode switching time		40	—	—	ns		
	Hold time		—	—	5	μs		
Operation stabilization time	A/D start-up time		2	—	—	μs		
	Channel-dedicated sample-and-hold circuits start-up time		2	—	—	μs		
	A/D shut-down time		1	—	—	μs		
Analog input voltage range	Single-ended input voltage	Unit 0	AN000 to AN005, AN012, AN014	VREFL0	—	VREFH0	V	—
			AN016 to AN018	VREFL0	—	VREFH0	V	VCC ≥ VREFH0
				VREFL0	—	VCC	V	VCC < VREFH0
			AN019 to AN022	VREFL0	—	VREFH0	V	VCC2 ≥ VREFH0
		VREFL0		—	VCC2	V	VCC2 < VREFH0	
		Unit 1	AN006 to AN011, AN013, AN015	VREFL	—	VREFH	V	—
			AN016 to AN018	VREFL	—	VREFH	V	VCC ≥ VREFH
				VREFL	—	VCC	V	VCC < VREFH
			AN019 to AN022	VREFL	—	VREFH	V	VCC2 ≥ VREFH
		VREFL		—	VCC2	V	VCC2 < VREFH	
	Differential input voltage ^{*2}	Unit 0	AN000 to AN005	-VREFH0	—	+VREFH0	V	—
		Unit 1	AN006 to AN011	-VREFH	—	+VREFH	V	—

Note: t_{ADcyc} : ADCLK cycleNote: t_{Cmp} : Successive approximation timeNote 1. If t_{Cmp} is greater than 100ns, the A/D sampling time should be greater than the following equation.

$$1 \times t_{ADcyc} + 1.6 \times t_{Cmp}$$

Note 2. Differential input voltage is ($A_{INP} - A_{INN}$).

A/D Converter Unit 0:

- A_{INP} is input voltage of A_{Nx} , and $VREFL0 \leq A_{INP} \leq VREFH0$.
- A_{INN} is input voltage of A_{Ny} , and $VREFL0 \leq A_{INN} \leq VREFH0$.

A/D Converter Unit 1:

- A_{INP} is input voltage of A_{Nx} , and $VREFL \leq A_{INP} \leq VREFH$.
- A_{INN} is input voltage of A_{Ny} , and $VREFL \leq A_{INN} \leq VREFH$.

(x = 2i, y = 2i + 1, i = 0, 1, 2... (any integer))

Table 60.86 A/D conversion characteristics (common) (1 of 2)

Conditions: AVCC: 1.62 to 2.7 V, VCC: 1.62 to 2.7 V, VREFH0/VREFH: 1.62 V to AVCC

Parameter					Min	Typ	Max	Unit	Test conditions	
A/D conversion clock frequency (ADCLK)					25	50	60	MHz	AVCC: 1.62 to 2.7 V	
Successive approximation time					200	—	200	ns	VCC: 1.62 to 2.7 V VREFH0/VREFH: 1.62 V to AVCC	
A/D sampling time (Normal mode)	Self-calibration				SAR mode	1 × t _{ADcyc} + 420	—	—	ns	AVCC: 1.62 to 2.7 V VCC: 1.62 to 2.7 V VREFH0/VREFH: 1.62 V to AVCC tCmp=200ns
					Oversampling mode	1 × t _{ADcyc} + 420	—	—	ns	
	Self-diagnosis				SAR mode	1 × t _{ADcyc} + 420	—	—	ns	
					Oversampling mode	1 × t _{ADcyc} + 420	—	—	ns	
	A/D conversion	High-speed channels	Without channel dedicated sample and hold circuits	(AN000 to AN005) (AN006 to AN011)	SAR mode	1 × t _{ADcyc} + 420	—	—	ns	
					Oversampling mode (One-channel continuous scan mode)	440	—	—	ns	
					Oversampling mode (Single/Continuous scan mode)	1 × t _{ADcyc} + 420	—	—	ns	
		Middle-speed channels		(AN012 to AN015)	SAR mode	560	—	—	ns	
					Oversampling mode	560	—	—	ns	
		Low-speed channels		(AN016 to AN022)	SAR mode	800	—	—	ns	
					Oversampling mode	800	—	—	ns	
		A/D sampling time (High accuracy mode)	Self-calibration				SAR mode	1 × t _{ADcyc} + 780	—	—
Hybrid mode	1 × t _{ADcyc} + 780						—	—	ns	
Self-diagnosis				SAR mode	1 × t _{ADcyc} + 780	—	—	ns		
				Hybrid mode	1 × t _{ADcyc} + 780	—	—	ns		
A/D conversion	High-speed channels		Without channel dedicated sample and hold circuits	(AN000 to AN005) (AN006 to AN011)	SAR mode	1 × t _{ADcyc} + 780	—	—	ns	
					Hybrid mode	1 × t _{ADcyc} + 780	—	—	ns	
	Middle-speed channels		(AN012 to AN015)	SAR mode	1200	—	—	ns		
				Hybrid mode	1200	—	—	ns		
	Low-speed channels		(AN016 to AN022)	SAR mode	1680	—	—	ns		
				Hybrid mode	1680	—	—	ns		
Operation	A/D start-up time				2	—	—	ns	AVCC: 1.62 to 2.7 V VCC: 1.62 to 2.7 V VREFH0/VREFH: 1.62 V to AVCC	
	A/D shut-down time				1	—	—	ns		

Table 60.86 A/D conversion characteristics (common) (2 of 2)

Conditions: AVCC: 1.62 to 2.7 V, VCC: 1.62 to 2.7 V, VREFH0/VREFH: 1.62 V to AVCC

Parameter				Min	Typ	Max	Unit	Test conditions
Analog input voltage range	Single-ended input voltage	Unit 0	AN000 to AN005, AN012, AN014	VREFL0	—	VREFH0	V	—
			AN016 to AN018	VREFL0	—	VREFH0	V	VCC ≥ VREFH0
				VREFL0	—	VCC	V	VCC < VREFH0
			AN019 to AN022	VREFL0	—	VREFH0	V	VCC2 ≥ VREFH0
				VREFL0	—	VCC2	V	VCC2 < VREFH0
		Unit 1	AN006 to AN011, AN013, AN015	VREFL	—	VREFH	V	—
			AN016 to AN018	VREFL	—	VREFH	V	VCC ≥ VREFH
				VREFL	—	VCC	V	VCC < VREFH
			AN019 to AN022	VREFL	—	VREFH	V	VCC2 ≥ VREFH
				VREFL	—	VCC2	V	VCC2 < VREFH
	Differential input voltage*1	Unit 0	AN000 to AN005	-VREFH0	—	+VREFH0	V	—
		Unit 1	AN006 to AN011	-VREFH	—	+VREFH	V	—

Note: t_{ADcyc} : ADCLK cycleNote: t_{Cmp} : Successive approximation timeNote 1. Differential input voltage is ($A_{INP} - A_{INN}$).

A/D Converter Unit 0:

- A_{INP} is input voltage of A_{Nx} , and $VREFL0 \leq A_{INP} \leq VREFH0$.
- A_{INN} is input voltage of A_{Ny} , and $VREFL0 \leq A_{INN} \leq VREFH0$.

A/D Converter Unit 1:

- A_{INP} is input voltage of A_{Nx} , and $VREFL \leq A_{INP} \leq VREFH$.
- A_{INN} is input voltage of A_{Ny} , and $VREFL \leq A_{INN} \leq VREFH$.

(x = 2i, y = 2i + 1, i = 0, 1, 2... (any integer))

Table 60.87 A/D conversion characteristics (SAR mode : DCDC mode) (1 of 6)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter						Min	Typ	Max	Unit	Test conditions
SAR mode	Resolution					—	—	12	bit	—
SAR mode	Single-ended input	Normal mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time ^{*1}	0.16	—	—	μs	- ADCLK: 50 MHz - Sampling time: 3 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±3	±6.5	LSB	BGA package
						—	±3	±7.5	LSB	LQFP package
					Full-scale error	—	±3	±6.5	LSB	BGA package
						—	±3	±7.5	LSB	LQFP package
					Absolute accuracy	—	±4	±11	LSB	BGA package
						—	±4	±14	LSB	LQFP package
					DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	BGA package
						—	±1	-1 to +2.0	LSB	LQFP package
					INL integral nonlinearity error	—	±2	±3	LSB	BGA package
						—	±2	±5	LSB	LQFP package
		High-speed channels (AN000 to AN005) (AN006 to AN011)	With channel dedicated sample and hold circuits	Conversion time ^{*2}		1.00	—	—	μs	- ADCLK: 50 MHz - Sampling time of channel-dedicated sample-and-hold circuits: 35 ADCLK - Hold mode switching time of channel-dedicated sample-and-hold circuits: 2 ADCLK - Sampling time: 8 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1.5	±6.75	LSB	BGA package
						—	±1.5	±7.5	LSB	LQFP package
					Full-scale error	—	±1.5	±6.75	LSB	BGA package
						—	±1.5	±7.5	LSB	LQFP package
					Absolute accuracy	—	±5	±10.5	LSB	BGA package
						—	±5	±11.5	LSB	LQFP package
					DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	BGA package
						—	±1	-1 to +2.0	LSB	LQFP package
					INL integral nonlinearity error	—	±2.5	±3.5	LSB	BGA package
						—	±2.5	±5.5	LSB	LQFP package

Table 60.87 A/D conversion characteristics (SAR mode : DCDC mode) (2 of 6)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter						Min	Typ	Max	Unit	Test conditions
SAR mode	Single-ended input	High accuracy mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time ^{*1}	0.26	—	—	μs	- ADCLK: 50 MHz - Sampling time: 8 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1.5	±4.5	LSB	BGA package
						—	±1.5	±5.5	LSB	LQFP package
					Full-scale error	—	±1.5	±4.5	LSB	BGA package
						—	±1.5	±5.5	LSB	LQFP package
					Absolute accuracy	—	±4	±7	LSB	BGA package
						—	±4	±10	LSB	LQFP package
					DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	BGA package
						—	±1	-1 to +2.0	LSB	LQFP package
					INL integral nonlinearity error	—	±2	±3	LSB	BGA package
						—	±2	±5	LSB	LQFP package
			High-speed channels (AN000 to AN005) (AN006 to AN011)	With channel dedicated sample and hold circuits	Conversion time ^{*2}	1.72	—	—	μs	- ADCLK: 50 MHz - Sampling time of channel-dedicated sample-and-hold circuits: 63 ADCLK - Hold mode switching time of channel-dedicated sample-and-hold circuits: 2 ADCLK - Sampling time: 16 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1.5	±6.75	LSB	BGA package
						—	±1.5	±7.5	LSB	LQFP package
					Full-scale error	—	±1.5	±6.75	LSB	BGA package
						—	±1.5	±7.5	LSB	LQFP package
					Absolute accuracy	—	±4.5	±9	LSB	BGA package
						—	±4.5	±9.5	LSB	LQFP package
					DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	BGA package
						—	±1	-1 to +2.0	LSB	LQFP package
					INL integral nonlinearity error	—	±2.5	±3.5	LSB	BGA package
						—	±2.5	±4.5	LSB	LQFP package

Table 60.87 A/D conversion characteristics (SAR mode : DCDC mode) (3 of 6)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter					Min	Typ	Max	Unit	Test conditions
SAR mode	Single-ended input	Normal mode	Middle-speed channels (AN012 to AN015)	Conversion time ^{*1}	0.28	—	—	μs	- ADCLK: 50 MHz - Sampling time: 9 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	±1.5	±6.5	LSB	BGA package
					—	±1.5	±6.5	LSB	LQFP package
				Full-scale error	—	±1.5	±6.5	LSB	BGA package
					—	±1.5	±6.5	LSB	LQFP package
				Absolute accuracy	—	±4	±11	LSB	BGA package
					—	±4	±15	LSB	LQFP package
				DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	BGA package
					—	±1	-1 to +2.0	LSB	LQFP package
				INL integral nonlinearity error	—	±2	±3	LSB	BGA package
					—	±2	±3	LSB	LQFP package
			Low-speed channels (AN016 to AN022)	Conversion time ^{*1}	0.5	—	—	μs	- ADCLK: 50 MHz - Sampling time: 20 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	±1.5	±6.5	LSB	—
				Full-scale error	—	±1.5	±6.5	LSB	—
				Absolute accuracy	—	±5.5	±11	LSB	—
				DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	—
				INL integral nonlinearity error	—	±2	±4	LSB	—

Table 60.87 A/D conversion characteristics (SAR mode : DCDC mode) (4 of 6)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter					Min	Typ	Max	Unit	Test conditions
SAR mode	Single-ended input	High accuracy mode	Middle-speed channels (AN012 to AN015)	Conversion time ^{*1}	0.5	—	—	μs	- ADCLK: 50 MHz - Sampling time: 20 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	±1.5	±4.5	LSB	BGA package
					—	±1.5	±4.5	LSB	LQFP package
				Full-scale error	—	±1.5	±4.5	LSB	BGA package
					—	±1.5	±4.5	LSB	LQFP package
				Absolute accuracy	—	±4	±7	LSB	BGA package
					—	±4	±11	LSB	LQFP package
				DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	BGA package
					—	±1	-1 to +2.0	LSB	LQFP package
				INL integral nonlinearity error	—	±2	±3	LSB	BGA package
					—	±2	±4.5	LSB	LQFP package
			Low-speed channels (AN016 to AN022)	Conversion time ^{*1}	0.94	—	—	μs	- ADCLK: 50 MHz - Sampling time: 42 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	±1.5	±4.5	LSB	—
				Full-scale error	—	±1.5	±4.5	LSB	—
				Absolute accuracy	—	±5.5	±8	LSB	—
				DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	—
				INL integral nonlinearity error	—	±2	±4	LSB	—

Table 60.87 A/D conversion characteristics (SAR mode : DCDC mode) (5 of 6)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter						Min	Typ	Max	Unit	Test conditions
SAR mode	Differential input	Normal mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time ^{*1}	0.16	—	—	μs	- ADCLK: 50 MHz - Sampling time: 3 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±2	±3.5	LSB	—
					Full-scale error	—	±2	±3.5	LSB	—
					Absolute accuracy	—	±3	±6	LSB	—
					DNL differential nonlinearity error ^{*3}	—	±0.75	±1	LSB	—
					INL integral nonlinearity error	—	±1.5	±2	LSB	—
			High-speed channels (AN000 to AN005) (AN006 to AN011)	With channel dedicated sample and hold circuits	Conversion time ^{*2}	1.00	—	—	μs	- ADCLK: 50 MHz - Sampling time of channel-dedicated sample-and-hold circuits: 35 ADCLK - Hold mode switching time of channel-dedicated sample-and-hold circuits: 2 ADCLK - Sampling time: 8 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1.5	±6.75	LSB	—
					Full-scale error	—	±1.5	±6.75	LSB	—
					Absolute accuracy	—	±3.5	±10.5	LSB	—
					DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	—
					INL integral nonlinearity error	—	±2.5	±3.5	LSB	—

Table 60.87 A/D conversion characteristics (SAR mode : DCDC mode) (6 of 6)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter						Min	Typ	Max	Unit	Test conditions
SAR mode	Differential input	High accuracy mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time ^{*1}	0.26	—	—	μs	- ADCLK: 50 MHz - Sampling time: 8 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1	±2.5	LSB	—
					Full-scale error	—	±1	±2.5	LSB	—
					Absolute accuracy	—	±2	±4	LSB	—
					DNL differential nonlinearity error ^{*3}	—	±0.75	±1	LSB	—
					INL integral nonlinearity error	—	±1.5	±2	LSB	—
			High-speed channels (AN000 to AN005) (AN006 to AN011)	With channel dedicated sample and hold circuits	Conversion time ^{*2}	1.72	—	—	μs	- ADCLK: 50 MHz - Sampling time of channel-dedicated sample-and-hold circuits: 63 ADCLK - Hold mode switching time of channel-dedicated sample-and-hold circuits: 2 ADCLK - Sampling time: 16 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1.5	±6.75	LSB	—
					Full-scale error	—	±1.5	±6.75	LSB	—
					Absolute accuracy	—	±3.5	±9	LSB	—
					DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	—
					INL integral nonlinearity error	—	±2.5	±3.5	LSB	—

Note: These specification values are applicable when only one ADC16H is operating, DAC12 and ACMPHS are not operating, and there is no access to the external bus during A/D conversion.

If other ADC unit, DAC12, or ACMPHS is operating or bus access occurs during A/D conversion, values might not fall within the indicated ranges.

The use of ports 0 as digital outputs is not allowed when the ADC16H is used.

The characteristics apply when AVCC0, AVSS0, VREFH0, VREFH, VREFL0, VREFL, and ADC16H input voltage is stable.

Note 1. Without channel dedicated sample and hold circuits; The conversion time is the sum of the sampling time and the successive approximation time. Each of the above state is indicated for the test conditions.

Note 2. With channel-dedicated sample-and-hold circuits; The conversion time is the sum of the sampling time of channel-dedicated sample-and-hold circuits, the hold mode switching time, the sampling time and the successive approximation time. Each of the above state is indicated for the test conditions.

Note 3. DNL is measured using the Histogram Method, so the lower limit value is -1.

Table 60.88 A/D conversion characteristics (SAR mode : External VDD mode) (1 of 6)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter						Min	Typ	Max	Unit	Test conditions
SAR mode	Resolution					—	—	12	bit	—
SAR mode	Single-ended input	Normal mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time ^{*1}	0.16	—	—	μs	- ADCLK: 50 MHz - Sampling time: 3 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±3	±6.5	LSB	—
					Full-scale error	—	±3	±6.5	LSB	—
					Absolute accuracy	—	±4	±11	LSB	—
					DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	—
					INL integral nonlinearity error	—	±2	±3	LSB	—
			High-speed channels (AN000 to AN005) (AN006 to AN011)	With channel dedicated sample and hold circuits	Conversion time ^{*2}	1.00	—	—	μs	- ADCLK: 50 MHz - Sampling time of channel-dedicated sample-and-hold circuits: 35 ADCLK - Hold mode switching time of channel-dedicated sample-and-hold circuits: 2 ADCLK - Sampling time: 8 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1.5	±6.75	LSB	—
					Full-scale error	—	±1.5	±6.75	LSB	—
					Absolute accuracy	—	±5	±10.5	LSB	—
					DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	—
					INL integral nonlinearity error	—	±2.5	±3.5	LSB	—

Table 60.88 A/D conversion characteristics (SAR mode : External VDD mode) (2 of 6)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter						Min	Typ	Max	Unit	Test conditions
SAR mode	Single-ended input	High accuracy mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time ^{*1}	0.26	—	—	μs	- ADCLK: 50 MHz - Sampling time: 8 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1.5	±4.5	LSB	—
					Full-scale error	—	±1.5	±4.5	LSB	—
					Absolute accuracy	—	±4	±7	LSB	—
					DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	—
					INL integral nonlinearity error	—	±2	±3	LSB	—
			High-speed channels (AN000 to AN005) (AN006 to AN011)	With channel dedicated sample and hold circuits	Conversion time ^{*2}	1.72	—	—	μs	- ADCLK: 50 MHz - Sampling time of channel-dedicated sample-and-hold circuits: 63 ADCLK - Hold mode switching time of channel-dedicated sample-and-hold circuits: 2 ADCLK - Sampling time: 16 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1.5	±6.75	LSB	—
					Full-scale error	—	±1.5	±6.75	LSB	—
					Absolute accuracy	—	±4.5	±9	LSB	—
					DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	—
					INL integral nonlinearity error	—	±2.5	±3.5	LSB	—

Table 60.88 A/D conversion characteristics (SAR mode : External VDD mode) (3 of 6)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter					Min	Typ	Max	Unit	Test conditions
SAR mode	Single-ended input	Normal mode	Middle-speed channels (AN012 to AN015)	Conversion time*1	0.28	—	—	μs	- ADCLK: 50 MHz - Sampling time: 9 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	±1.5	±6.5	LSB	—
				Full-scale error	—	±1.5	±6.5	LSB	—
				Absolute accuracy	—	±4	±11	LSB	—
				DNL differential nonlinearity error*3	—	±1	-1 to +1.5	LSB	—
				INL integral nonlinearity error	—	±2	±3	LSB	—
			Low-speed channels (AN016 to AN022)	Conversion time*1	0.5	—	—	μs	- ADCLK: 50 MHz - Sampling time: 20 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	±1.5	±6.5	LSB	—
				Full-scale error	—	±1.5	±6.5	LSB	—
				Absolute accuracy	—	±5.5	±11	LSB	—
				DNL differential nonlinearity error*3	—	±1	-1 to +1.5	LSB	—
				INL integral nonlinearity error	—	±2	±4	LSB	—

Table 60.88 A/D conversion characteristics (SAR mode : External VDD mode) (4 of 6)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter					Min	Typ	Max	Unit	Test conditions
SAR mode	Single-ended input	High accuracy mode	Middle-speed channels (AN012 to AN015)	Conversion time*1	0.5	—	—	μs	- ADCLK: 50 MHz - Sampling time: 20 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	±1.5	±4.5	LSB	—
				Full-scale error	—	±1.5	±4.5	LSB	—
				Absolute accuracy	—	±4	±7	LSB	—
				DNL differential nonlinearity error*3	—	±1	-1 to +1.5	LSB	—
				INL integral nonlinearity error	—	±2	±3	LSB	—
			Low-speed channels (AN016 to AN022)	Conversion time*1	0.94	—	—	μs	- ADCLK: 50 MHz - Sampling time: 42 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	±1.5	±4.5	LSB	—
				Full-scale error	—	±1.5	±4.5	LSB	—
				Absolute accuracy	—	±5.5	±8	LSB	—
				DNL differential nonlinearity error*3	—	±1	-1 to +1.5	LSB	—
				INL integral nonlinearity error	—	±2	±4	LSB	—

Table 60.88 A/D conversion characteristics (SAR mode : External VDD mode) (5 of 6)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter						Min	Typ	Max	Unit	Test conditions
SAR mode	Differential input	Normal mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time ^{*1}	0.16	—	—	μs	- ADCLK: 50 MHz - Sampling time: 3 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±2	±3.5	LSB	—
					Full-scale error	—	±2	±3.5	LSB	—
					Absolute accuracy	—	±3	±6	LSB	—
					DNL differential nonlinearity error ^{*3}	—	±0.75	±1	LSB	—
					INL integral nonlinearity error	—	±1.5	±2	LSB	—
			High-speed channels (AN000 to AN005) (AN006 to AN011)	With channel dedicated sample and hold circuits	Conversion time ^{*2}	1.00	—	—	μs	- ADCLK: 50 MHz - Sampling time of channel-dedicated sample-and-hold circuits: 35 ADCLK - Hold mode switching time of channel-dedicated sample-and-hold circuits: 2 ADCLK - Sampling time: 8 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1.5	±6.75	LSB	—
					Full-scale error	—	±1.5	±6.75	LSB	—
					Absolute accuracy	—	±3.5	±10.5	LSB	—
					DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	—
					INL integral nonlinearity error	—	±2.5	±3.5	LSB	—

Table 60.88 A/D conversion characteristics (SAR mode : External VDD mode) (6 of 6)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter						Min	Typ	Max	Unit	Test conditions
SAR mode	Differential input	High accuracy mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time ^{*1}	0.26	—	—	μs	- ADCLK: 50 MHz - Sampling time: 8 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1	±2.5	LSB	—
					Full-scale error	—	±1	±2.5	LSB	—
					Absolute accuracy	—	±2	±4	LSB	—
					DNL differential nonlinearity error ^{*3}	—	±0.75	±1	LSB	—
					INL integral nonlinearity error	—	±1.5	±2	LSB	—
			High-speed channels (AN000 to AN005) (AN006 to AN011)	With channel dedicated sample and hold circuits	Conversion time ^{*2}	1.72	—	—	μs	- ADCLK: 50 MHz - Sampling time of channel-dedicated sample-and-hold circuits: 63 ADCLK - Hold mode switching time of channel-dedicated sample-and-hold circuits: 2 ADCLK - Sampling time: 16 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1.5	±6.75	LSB	—
					Full-scale error	—	±1.5	±6.75	LSB	—
					Absolute accuracy	—	±3.5	±9	LSB	—
					DNL differential nonlinearity error ^{*3}	—	±1	-1 to +1.5	LSB	—
					INL integral nonlinearity error	—	±2.5	±3.5	LSB	—

Note: These specification values are applicable when only one ADC16H is operating, DAC12 and ACMPHS are not operating, and there is no access to the external bus during A/D conversion.

If other ADC unit, DAC12, or ACMPHS is operating or bus access occurs during A/D conversion, values might not fall within the indicated ranges.

The use of ports 0 as digital outputs is not allowed when the ADC16H is used.

The characteristics apply when AVCC0, AVSS0, VREFH0, VREFH, VREFL0, VREFL, and ADC16H input voltage is stable.

Note 1. Without channel dedicated sample and hold circuits; The conversion time is the sum of the sampling time and the successive approximation time. Each of the above state is indicated for the test conditions.

Note 2. With channel-dedicated sample-and-hold circuits; The conversion time is the sum of the sampling time of channel-dedicated sample-and-hold circuits, the hold mode switching time, the sampling time and the successive approximation time. Each of the above state is indicated for the test conditions.

Note 3. DNL is measured using the Histogram Method, so the lower limit value is -1.

Table 60.89 A/D conversion characteristics (SAR mode : DCDC mode) (1 of 4)

Conditions: AVCC: 1.62 to 2.7 V, VCC: 1.62 to 2.7 V, VREFH0/VREFH: 1.62 V to AVCC

Parameter						Min	Typ	Max	Unit	Test conditions
SAR mode	Resolution					—	—	12	bit	—
SAR mode	Single-ended input	Normal mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time ^{*1}	0.64	—	—	μs	- ADCLK: 50 MHz - Sampling time: 22 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±3	±6.5	LSB	BGA package
						—	±3	±8	LSB	LQFP package
					Full-scale error	—	±3	±6.5	LSB	BGA package
						—	±3	±8	LSB	LQFP package
					Absolute accuracy	—	±5.5	±11	LSB	BGA package
						—	±5.5	±15	LSB	LQFP package
					DNL differential nonlinearity error ^{*2}	—	±1	-1 to +1.5	LSB	BGA package
						—	±1	-1 to +2.0	LSB	LQFP package
					INL integral nonlinearity error	—	±2	±3	LSB	BGA package
						—	±2	±3.5	LSB	LQFP package
SAR mode	Single-ended input	High accuracy mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time ^{*1}	1	—	—	μs	- ADCLK: 50 MHz - Sampling time: 40 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1.5	±4.5	LSB	BGA package
						—	±1.5	±5	LSB	LQFP package
					Full-scale error	—	±1.5	±4.5	LSB	BGA package
						—	±1.5	±5	LSB	LQFP package
					Absolute accuracy	—	±5.0	±8	LSB	BGA package
						—	±5.0	±11	LSB	LQFP package
					DNL differential nonlinearity error ^{*2}	—	±1	-1 to +1.5	LSB	BGA package
						—	±1	-1 to +1.5	LSB	LQFP package
					INL integral nonlinearity error	—	±2	±3	LSB	BGA package
						—	±2	±3.5	LSB	LQFP package

Table 60.89 A/D conversion characteristics (SAR mode : DCDC mode) (2 of 4)

Conditions: AVCC: 1.62 to 2.7 V, VCC: 1.62 to 2.7 V, VREFH0/VREFH: 1.62 V to AVCC

Parameter					Min	Typ	Max	Unit	Test conditions
SAR mode	Single-ended input	Normal mode	Middle-speed channels (AN012 to AN015)	Conversion time*1	0.76	—	—	μs	- ADCLK: 50 MHz - Sampling time: 28 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	—	±6.5	LSB	BGA package
					—	±1.5	±7	LSB	LQFP package
				Full-scale error	—	±1.5	±6.5	LSB	BGA package
					—	±1.5	±7	LSB	LQFP package
				Absolute accuracy	—	±4	±11	LSB	BGA package
					—	±4	±15	LSB	LQFP package
				DNL differential nonlinearity error*2	—	±1	-1 to +1.5	LSB	BGA package
					—	±1	-1 to +2.0	LSB	LQFP package
			Low-speed channels (AN016 to AN022)	INL integral nonlinearity error	—	±2	±3	LSB	BGA package
					—	±2	±4	LSB	LQFP package
				Conversion time*1	1	—	—	μs	- ADCLK: 50 MHz - Sampling time: 40 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	±1.5	±6.5	LSB	—
				Full-scale error	—	±1.5	±6.5	LSB	—
				Absolute accuracy	—	±5.5	±11	LSB	—
				DNL differential nonlinearity error*2	—	±1	-1 to +1.5	LSB	—
				INL integral nonlinearity error	—	±2	±4	LSB	—

Table 60.89 A/D conversion characteristics (SAR mode : DCDC mode) (3 of 4)

Conditions: AVCC: 1.62 to 2.7 V, VCC: 1.62 to 2.7 V, VREFH0/VREFH: 1.62 V to AVCC

Parameter					Min	Typ	Max	Unit	Test conditions
SAR mode	Single-ended input	High accuracy mode	Middle-speed channels (AN012 to AN015)	Conversion time*1	1.4	—	—	μs	- ADCLK: 50 MHz - Sampling time: 60 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	±1.5	±4.5	LSB	BGA package
					—	±1.5	±5	LSB	LQFP package
				Full-scale error	—	±1.5	±4.5	LSB	BGA package
					—	±1.5	±5	LSB	LQFP package
				Absolute accuracy	—	±4	±8	LSB	BGA package
					—	±4	±13	LSB	LQFP package
				DNL differential nonlinearity error *2	—	±1	-1 to +1.5	LSB	BGA package
					—	±1	-1 to +2.0	LSB	LQFP package
			Low-speed channels (AN016 to AN022)	INL integral nonlinearity error	—	±2	±3	LSB	BGA package
					—	±2	±4	LSB	LQFP package
				Conversion time*1	1.88	—	—	μs	- ADCLK: 50 MHz - Sampling time: 84 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	±1.5	±4.5	LSB	—
				Full-scale error	—	±1.5	±4.5	LSB	—
				Absolute accuracy	—	±5.5	±8	LSB	—
				DNL differential nonlinearity error *2	—	±1	-1 to +1.5	LSB	—
				INL integral nonlinearity error	—	±2	±4	LSB	—
SAR mode	Differential input	Normal mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Conversion time*1	0.64	—	—	μs	- ADCLK: 50 MHz - Sampling time: 22 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	±2	±3.5	LSB	—
				Full-scale error	—	±2	±3.5	LSB	—
				Absolute accuracy	—	±4.5	±6	LSB	—
				DNL differential nonlinearity error *2	—	±0.75	±1	LSB	—
				INL integral nonlinearity error	—	±1.5	±2	LSB	—

Table 60.89 A/D conversion characteristics (SAR mode : DCDC mode) (4 of 4)

Conditions: AVCC: 1.62 to 2.7 V, VCC: 1.62 to 2.7 V, VREFH0/VREFH: 1.62 V to AVCC

Parameter						Min	Typ	Max	Unit	Test conditions
SAR mode	Differential input	High accuracy mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time ^{*1}	1	—	—	μs	- ADCLK: 50 MHz - Sampling time: 40 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1	±2.5	LSB	—
					Full-scale error	—	±1	±2.5	LSB	—
					Absolute accuracy	—	±3.5	±4.5	LSB	—
					DNL differential nonlinearity error ^{*2}	—	±0.75	±1	LSB	—
					INL integral nonlinearity error	—	±1.5	±2	LSB	—

Note: These specification values are applicable when only one ADC16H is operating, DAC12 and ACMPHS are not operating, and there is no access to the external bus during A/D conversion.

If other ADC unit, DAC12, or ACMPHS is operating or bus access occurs during A/D conversion, values might not fall within the indicated ranges.

The use of ports 0 as digital outputs is not allowed when the ADC16H is used.

The characteristics apply when AVCC0, AVSS0, VREFH0, VREFH, VREFL0, VREFL, and ADC16H input voltage is stable.

Note 1. Without channel dedicated sample and hold circuits; The conversion time is the sum of the sampling time and the successive approximation time. Each of the above state is indicated for the test conditions.

Note 2. DNL is measured using the Histogram Method, so the lower limit value is -1.

Table 60.90 A/D conversion characteristics (SAR mode : External VDD mode) (1 of 4)

Conditions: AVCC: 1.62 to 2.7 V, VCC: 1.62 to 2.7 V, VREFH0/VREFH: 1.62 V to AVCC

Parameter						Min	Typ	Max	Unit	Test conditions
SAR mode	Resolution					—	—	12	bit	—
SAR mode	Single-ended input	Normal mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time ^{*1}	0.64	—	—	μs	- ADCLK: 50 MHz - Sampling time: 22 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±3	±6.5	LSB	—
					Full-scale error	—	±3	±6.5	LSB	—
					Absolute accuracy	—	±5.5	±11	LSB	—
					DNL differential nonlinearity error ^{*2}	—	±1	-1 to +1.5	LSB	—
					INL integral nonlinearity error	—	±2	±3	LSB	—
SAR mode	Single-ended input	High accuracy mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time ^{*1}	1	—	—	μs	- ADCLK: 50 MHz - Sampling time: 40 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1.5	±4.5	LSB	—
					Full-scale error	—	±1.5	±4.5	LSB	—
					Absolute accuracy	—	±5.0	±8	LSB	—
					DNL differential nonlinearity error ^{*2}	—	±1	-1 to +1.5	LSB	—
					INL integral nonlinearity error	—	±2	±3	LSB	—

Table 60.90 A/D conversion characteristics (SAR mode : External VDD mode) (2 of 4)

Conditions: AVCC: 1.62 to 2.7 V, VCC: 1.62 to 2.7 V, VREFH0/VREFH: 1.62 V to AVCC

Parameter					Min	Typ	Max	Unit	Test conditions
SAR mode	Single-ended input	Normal mode	Middle-speed channels (AN012 to AN015)	Conversion time ^{*1}	0.76	—	—	μs	- ADCLK: 50 MHz - Sampling time: 28 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	±1.5	±6.5	LSB	—
				Full-scale error	—	±1.5	±6.5	LSB	—
				Absolute accuracy	—	±4	±11	LSB	—
				DNL differential nonlinearity error ^{*2}	—	±1	-1 to +1.5	LSB	—
				INL integral nonlinearity error	—	±2	±3	LSB	—
			Low-speed channels (AN016 to AN022)	Conversion time ^{*1}	1	—	—	μs	- ADCLK: 50 MHz - Sampling time: 40 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
				Offset error	—	±1.5	±6.5	LSB	—
				Full-scale error	—	±1.5	±6.5	LSB	—
				Absolute accuracy	—	±5.5	±11	LSB	—
				DNL differential nonlinearity error ^{*2}	—	±1	-1 to +1.5	LSB	—
				INL integral nonlinearity error	—	±2	±4	LSB	—

Table 60.90 A/D conversion characteristics (SAR mode : External VDD mode) (3 of 4)

Conditions: AVCC: 1.62 to 2.7 V, VCC: 1.62 to 2.7 V, VREFH0/VREFH: 1.62 V to AVCC

Parameter					Min	Typ	Max	Unit	Test conditions	
SAR mode	Single-ended input	High accuracy mode	Middle-speed channels (AN012 to AN015)		Conversion time*1	1.4	—	—	μs	- ADCLK: 50 MHz - Sampling time: 60 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1.5	±4.5	LSB	—
					Full-scale error	—	±1.5	±4.5	LSB	—
					Absolute accuracy	—	±4	±8	LSB	—
					DNL differential nonlinearity error *2	—	±1	-1 to +1.5	LSB	—
					INL integral nonlinearity error	—	±2	±3	LSB	—
			Low-speed channels (AN016 to AN022)		Conversion time*1	1.88	—	—	μs	- ADCLK: 50 MHz - Sampling time: 84 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1.5	±4.5	LSB	—
					Full-scale error	—	±1.5	±4.5	LSB	—
					Absolute accuracy	—	±5.5	±8	LSB	—
					DNL differential nonlinearity error *2	—	±1	-1 to +1.5	LSB	—
					INL integral nonlinearity error	—	±2	±4	LSB	—
SAR mode	Differential input	Normal mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time*1	0.64	—	—	μs	- ADCLK: 50 MHz - Sampling time: 22 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±2	±3.5	LSB	—
					Full-scale error	—	±2	±3.5	LSB	—
					Absolute accuracy	—	±4.5	±6	LSB	—
					DNL differential nonlinearity error *2	—	±0.75	±1	LSB	—
					INL integral nonlinearity error	—	±1.5	±2	LSB	—

Table 60.90 A/D conversion characteristics (SAR mode : External VDD mode) (4 of 4)

Conditions: AVCC: 1.62 to 2.7 V, VCC: 1.62 to 2.7 V, VREFH0/VREFH: 1.62 V to AVCC

Parameter						Min	Typ	Max	Unit	Test conditions
SAR mode	Differential input	High accuracy mode	High-speed channels (AN000 to AN005) (AN006 to AN011)	Without channel dedicated sample and hold circuits	Conversion time*1	1	—	—	μs	- ADCLK: 50 MHz - Sampling time: 40 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less
					Offset error	—	±1	±2.5	LSB	—
					Full-scale error	—	±1	±2.5	LSB	—
					Absolute accuracy	—	±3.5	±4.5	LSB	—
					DNL differential nonlinearity error*2	—	±0.75	±1	LSB	—
					INL integral nonlinearity error	—	±1.5	±2	LSB	—

Note: These specification values are applicable when only one ADC16H is operating, DAC12 and ACMPHS are not operating, and there is no access to the external bus during A/D conversion.

If other ADC unit, DAC12, or ACMPHS is operating or bus access occurs during A/D conversion, values might not fall within the indicated ranges.

The use of ports 0 as digital outputs is not allowed when the ADC16H is used.

The characteristics apply when AVCC0, AVSS0, VREFH0, VREFH, VREFL0, VREFL, and ADC16H input voltage is stable.

Note 1. Without channel dedicated sample and hold circuits; The conversion time is the sum of the sampling time and the successive approximation time. Each of the above state is indicated for the test conditions.

Note 2. DNL is measured using the Histogram Method, so the lower limit value is -1.

Table 60.91 A/D conversion characteristics (Oversampling mode and Hybrid mode) (1)

Parameter				Min	Typ	Max	Unit	Test conditions
Oversampling mode and Hybrid mode	Resolution			—	—	16	bit	—
	Oversampling period	Oversampling mode		0.16	—	—	μs	- ADCLK: 50 MHz - Sampling time: 3 ADCLK - Successive approximation time: 5 ADCLK - Without disconnection detection assist function - Signal source impedance: 50 Ω or less
		Hybrid mode*2		0.18	—	—	μs	- ADCLK: 50 MHz - Sampling time: 8 ADCLK - Successive approximation time: 5 ADCLK - Without disconnection detection assist function - Signal source impedance: 50 Ω or less
	Digital filter characteristics*1	Sinc filter	Initial delay	—	22	—	/Fos	—
			Group delay	—	11	—		—
			Normalized Cutoff Frequency	—	0.033	—	Fin/Fos	—

Note: Fos is oversampling frequency.

When in Hybrid mode, Fos is 1/ (the sum of the oversampling periods of each analog channel assigned to the scan group).

Note 1. See [Figure 60.125](#).

Note 2. Value per channel.

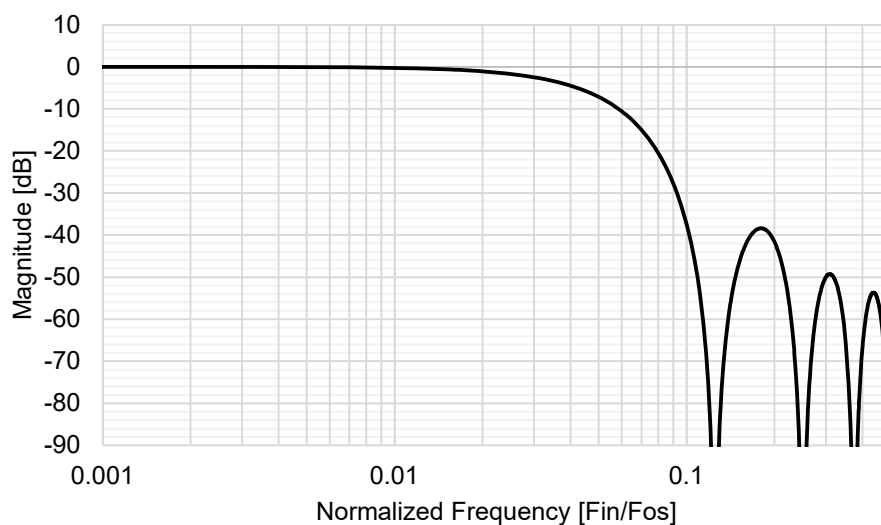


Figure 60.125 Digital filter characteristics (Sinc filter)

Table 60.92 A/D conversion characteristics (Oversampling mode and Hybrid mode) (2)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter				Min	Typ	Max	Unit	Test conditions
Oversampling mode and Hybrid mode	Sinc filter	Single-ended input (AN000 to AN005) (AN006 to AN011) (AN012 to AN015)	SNDR signal-to-noise and distortion ratio	—	80	—	dB	- ADCLK: 50 MHz - Sampling time: - High-speed channels (Oversampling mode): 3 ADCLK - High-speed channels (Hybrid mode): 8 ADCLK - Middle-speed channels (Oversampling mode): 10 ADCLK - Middle-speed channels (Hybrid mode): 22 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less - Input frequency: - Oversampling mode: 5 kHz - Hybrid mode: 5 kHz - Without channel dedicated sample and hold circuits
			ENOB effective number of bits	—	13	—	bit	
		Differential input (AN000 to AN005) (AN006 to AN011)	SNDR signal-to-noise and distortion ratio	—	86	—	dB	
			ENOB effective number of bits	—	14	—	bit	

Table 60.93 A/D conversion characteristics (Oversampling mode and Hybrid mode) (3)

Conditions: AVCC: 1.62 to 2.7 V, VCC: 1.62 to 2.7 V, VREFH0/VREFH: 1.62 V to AVCC

Parameter				Min	Typ	Max	Unit	Test conditions
Oversampling mode and Hybrid mode	Sinc filter	Single-ended input (AN000 to AN005) (AN006 to AN011) (AN012 to AN015)	SNDR signal-to-noise and distortion ratio	—	74	—	dB	- ADCLK: 50 MHz - Sampling time: - High-speed channels (Oversampling mode): 22 ADCLK - High-speed channels (Hybrid mode): 40 ADCLK - Middle-speed channels (Oversampling mode): 28 ADCLK - Middle-speed channels (Hybrid mode): 60 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less - Input frequency: - Oversampling mode: 5 kHz - Hybrid mode: 5 kHz - Without channel dedicated sample and hold circuits
			ENOB effective number of bits	—	12	—	bit	
		Differential input (AN000 to AN005) (AN006 to AN011)	SNDR signal-to-noise and distortion ratio	—	80	—	dB	
			ENOB effective number of bits	—	13	—	bit	

Table 60.94 A/D conversion characteristics (Oversampling mode)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter				Min	Typ	Max	Unit	Test conditions
Oversampling mode	Single-ended input	High-speed channels (AN000 to AN005) (AN006 to AN011)	Offset error ^{*3}	—	±0.5	±4	LSB	- ADCLK: 50 MHz - Sampling time: 3 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less - Digital filter: Sinc filter
			Gain error (Single/Continuous mode) ^{*3}	—	±1	±4	LSB	
			Gain error (One-channel continuous mode) ^{*3}	—	±1	±5	LSB	
			DNL differential nonlinearity error ^{*1 *2}	—	-1 to +1.5	-1 to +2.5	LSB	
			INL integral nonlinearity error ^{*1}	—	±4	±8	LSB	
		Middle-speed channels (AN012 to AN015)	Offset error ^{*3}	—	±0.5	±4	LSB	- ADCLK: 50 MHz - Sampling time: 10 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less - Digital filter: Sinc filter
			Gain error ^{*3}	—	±1	±4	LSB	
			DNL differential nonlinearity error ^{*1 *2}	—	-1 to +2	-1 to +4	LSB	
			INL integral nonlinearity error ^{*1}	—	±4	±8	LSB	
		Low-speed channels (AN016 to AN022)	Offset error ^{*3}	—	±0.5	±4	LSB	- ADCLK: 50 MHz - Sampling time: 20 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less - Digital filter: Sinc filter
			Gain error ^{*3}	—	±1	±4	LSB	
			DNL differential nonlinearity error ^{*1 *2}	—	-1 to +2	-1 to +4	LSB	
			INL integral nonlinearity error ^{*1}	—	±4	±12	LSB	
	Differential input	High-speed channels (AN000 to AN005) (AN006 to AN011)	Offset error ^{*3}	—	±0.25	±2	LSB	- ADCLK: 50 MHz - Sampling time: 3 ADCLK - Successive approximation time: 5 ADCLK - Signal source impedance: 50 Ω or less - Digital filter: Sinc filter
			Gain error (Single/Continuous mode) ^{*3}	—	±0.5	±2	LSB	
			Gain error (One-channel continuous mode) ^{*3}	—	±0.5	±2.5	LSB	
			DNL differential nonlinearity error ^{*1 *2}	—	-1 to +1.5	-1 to +2.0	LSB	
			INL integral nonlinearity error ^{*1}	—	±3	±6	LSB	

Note: These specification values are applicable when only one ADC16H is operating, DAC12 and ACMPHS are not operating, and there is no access to the external bus during A/D conversion.

If other ADC unit, DAC12, or ACMPHS is operating or bus access occurs during A/D conversion, values might not fall within the indicated ranges.

The use of ports 0 as digital outputs is not allowed when the ADC16H is used.

The characteristics apply when AVCC0, AVSS0, VREFH0, VREFH, VREFL0, VREFL, and ADC16H input voltage is stable.

Note 1. Test conditions: 0.2% to 99.8% of the analog input voltage range.

Note 2. DNL is measured using the Histogram Method, so the lower limit value is -1.

Note 3. This value is based on a 12-bit resolution.

Table 60.95 A/D conversion characteristics (Oversampling mode)

Conditions: AVCC: 1.62 to 2.7 V, VCC: 1.62 to 2.7 V, VREFH0/VREFH: 1.62 V to AVCC

Parameter				Min	Typ	Max	Unit	Test conditions
Oversampling mode	Single-ended input	High-speed channels (AN000 to AN005) (AN006 to AN011)	Offset error ^{*3}	—	±0.5	±4	LSB	- ADCLK: 50 MHz - Sampling time: 22 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less - Digital filter: Sinc filter
			Gain error (Single/Continuous mode) ^{*3}	—	±1	±4	LSB	
			Gain error (One-channel continuous mode) ^{*3}	—	±1	±5	LSB	
			DNL differential nonlinearity error ^{*1 *2}	—	-1 to +2	-1 to +2.5	LSB	
			INL integral nonlinearity error ^{*1}	—	±4	±8	LSB	
		Middle-speed channels (AN012 to AN015)	Offset error ^{*3}	—	±0.5	±4	LSB	- ADCLK: 50 MHz - Sampling time: 28 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less - Digital filter: Sinc filter
			Gain error ^{*3}	—	±1	±4	LSB	
			DNL differential nonlinearity error ^{*1 *2}	—	-1 to +2	-1 to +4	LSB	
			INL integral nonlinearity error ^{*1}	—	±4	±8	LSB	
		Low-speed channels (AN016 to AN022)	Offset error ^{*3}	—	±0.5	±4	LSB	- ADCLK: 50 MHz - Sampling time: 40 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less - Digital filter: Sinc filter
			Gain error ^{*3}	—	±1	±4	LSB	
			DNL differential nonlinearity error ^{*1 *2}	—	-1 to +2	-1 to +4	LSB	
			INL integral nonlinearity error ^{*1}	—	±4	±12	LSB	
	Differential input	High-speed channels (AN000 to AN005) (AN006 to AN011)	Offset error ^{*3}	—	±0.25	±2	LSB	- ADCLK: 50 MHz - Sampling time: 22 ADCLK - Successive approximation time: 10 ADCLK - Signal source impedance: 50 Ω or less - Digital filter: Sinc filter
			Gain error (Single/Continuous mode) ^{*3}	—	±0.5	±2	LSB	
			Gain error (One-channel continuous mode) ^{*3}	—	±0.5	±2.5	LSB	
			DNL differential nonlinearity error ^{*1 *2}	—	-1 to +2	-1 to +2.5	LSB	
			INL integral nonlinearity error ^{*1}	—	±3	±6	LSB	

Note: These specification values are applicable when only one ADC16H is operating, DAC12 and ACMPHS are not operating, and there is no access to the external bus during A/D conversion.

If other ADC unit, DAC12, or ACMPHS is operating or bus access occurs during A/D conversion, values might not fall within the indicated ranges.

The use of ports 0 as digital outputs is not allowed when the ADC16H is used.

The characteristics apply when AVCC0, AVSS0, VREFH0, VREFH, VREFL0, VREFL, and ADC16H input voltage is stable.

Note 1. Test conditions: 0.2% to 99.8% of the analog input voltage range.

Note 2. DNL is measured using the Histogram Method, so the lower limit value is -1.

Note 3. This value is based on a 12-bit resolution.

Table 60.96 A/D conversion characteristics (Hybrid mode)

Conditions: AVCC: 2.7 to 3.63 V, VCC: 2.7 to 3.63 V, VREFH0/VREFH: 2.7 V to AVCC

Parameter				Min	Typ	Max	Unit	Test conditions			
Hybrid mode	Without channel dedicated sample and hold circuits	Single-ended input	High-speed channels (AN000 to AN005) (AN006 to AN011)	Offset error *4	—	±0.5	±4	LSB	• ADCLK: 50 MHz • Sampling time: 8 ADCLK • Successive approximation time: 5 ADCLK • Signal source impedance: 50 Ω or less • Digital filter: Sinc filter		
				Gain error *4	—	±1	±5	LSB			
				DNL differential nonlinearity error*2 *3	—	-1 to +1.5	-1 to +2.5	LSB			
				INL integral nonlinearity error*2	—	±4	±8	LSB			
			Middle-speed channels (AN012 to AN015) *1	Offset error *4	—	±0.5	±4	LSB		• ADCLK: 50 MHz • Sampling time: 22 ADCLK • Successive approximation time: 5 ADCLK • Signal source impedance: 50 Ω or less • Digital filter: Sinc filter	
				Gain error *4	—	±1	±5	LSB			
				DNL differential nonlinearity error*2 *3	—	-1 to +2	-1 to +4	LSB			
				INL integral nonlinearity error*2	—	±4	±8	LSB			
		Low-speed channels (AN016 to AN022) *1	Offset error *4	—	±0.5	±4	LSB	• ADCLK: 50 MHz • Sampling time: 42 ADCLK • Successive approximation time: 5 ADCLK • Signal source impedance: 50 Ω or less • Digital filter: Sinc filter			
			Gain error *4	—	±1	±5	LSB				
			DNL differential nonlinearity error*2*3	—	-1 to +2	-1 to +4	LSB				
			INL integral nonlinearity error*2	—	±4	±12	LSB				
		Differential input	High-speed channels (AN000 to AN005) (AN006 to AN011)	Offset error *4	—	±0.25	±2	LSB	• ADCLK: 50 MHz • Sampling time: 8 ADCLK • Successive approximation time: 5 ADCLK • Signal source impedance: 50 Ω or less • Digital filter: Sinc filter		
				Gain error *4	—	±0.5	±2.5	LSB			
				DNL differential nonlinearity error*2 *3	—	-1 to +1.5	-1 to +2	LSB			
				INL integral nonlinearity error*2	—	±3	±6	LSB			
	With channel dedicated sample and hold circuits	Single-ended input	High-speed channels (AN000 to AN005) (AN006 to AN011)	Offset error *4	—	±0.5	±4	LSB	• ADCLK: 50 MHz • Sampling time of channel-dedicated sample-and-hold circuits: * ADCLK • Hold mode switching time of channel dedicated sample-and-hold circuits: * ADCLK • Sampling time: * ADCLK • Successive approximation time: * ADCLK • Signal source impedance: 50 Ω or less • Digital filter: Sinc filter		
				Gain error *4	—	±0.5	±4	LSB			
				DNL differential nonlinearity error*2 *3	—	±1	-1 to +2	LSB			
				INL integral nonlinearity error*2	—	±12	±16	LSB			
			Differential input	High-speed channels (AN000 to AN005) (AN006 to AN011)	Offset error *4	—	±0.5	±4		LSB	• ADCLK: 50 MHz • Sampling time of channel-dedicated sample-and-hold circuits: * ADCLK • Hold mode switching time of channel dedicated sample-and-hold circuits: * ADCLK • Sampling time: * ADCLK • Successive approximation time: * ADCLK • Signal source impedance: 50 Ω or less • Digital filter: Sinc filter
					Gain error *4	—	±0.5	±4		LSB	
					DNL differential nonlinearity error*2 *3	—	±1	-1 to +2		LSB	
					INL integral nonlinearity error*2	—	±4	±16		LSB	

Note: These specification values are applicable when only one ADC16H is operating, DAC12 and ACPHPS are not operating, and there is no access to the external bus during A/D conversion.

If other ADC unit, DAC12, or ACPHPS is operating or bus access occurs during A/D conversion, values might not fall within the indicated ranges.

The use of ports 0 as digital outputs is not allowed when the ADC16H is used.

The characteristics apply when AVCC0, AVSS0, VREFH0, VREFH, VREFL0, VREFL, and ADC16H input voltage is stable.

Note 1. Channel-dedicated sample-and-hold circuits are not available in these channels.

Note 2. Test conditions: 0.2% to 99.8% of the analog input voltage range.

Note 3. DNL is measured using the Histogram Method, so the lower limit value is -1.

Note 4. This value is based on a 12-bit resolution.

Table 60.97 A/D conversion characteristics (Hybrid mode)

Conditions: AVCC: 1.62 to 2.7 V, VCC: 1.62 to 2.7 V, VREFH0/VREFH: 1.62 V to AVCC

Parameter				Min	Typ	Max	Unit	Test conditions	
Hybrid mode	Without channel dedicated sample and hold circuits	Single-ended input	High-speed channels (AN000 to AN005) (AN006 to AN011)	Offset error *3	—	±0.5	±4	LSB	● ADCLK: 50 MHz ● Sampling time: 40 ADCLK ● Successive approximation time: 10 ADCLK ● Signal source impedance: 50 Ω or less ● Digital filter: Sinc filter
				Gain error *3	—	±1	±5	LSB	
				DNL differential nonlinearity error*1 *2	—	-1 to +2	-1 to +2.5	LSB	
				INL integral nonlinearity error*1	—	±4	±8	LSB	
		Middle-speed channels (AN012 to AN015)	Offset error *3	—	±0.5	±4	LSB	● ADCLK: 50 MHz ● Sampling time: 60 ADCLK ● Successive approximation time: 10 ADCLK ● Signal source impedance: 50 Ω or less ● Digital filter: Sinc filter	
			Gain error *3	—	±1	±5	LSB		
			DNL differential nonlinearity error*1 *2	—	-1 to +2	-1 to +4	LSB		
			INL integral nonlinearity error*1	—	±4	±8	LSB		
		Low-speed channels (AN016 to AN022)	Offset error *3	—	±0.5	±4	LSB	● ADCLK: 50 MHz ● Sampling time: 84 ADCLK ● Successive approximation time: 10 ADCLK ● Signal source impedance: 50 Ω or less ● Digital filter: Sinc filter	
			Gain error *3	—	±1	±5	LSB		
			DNL differential nonlinearity error*1 *2	—	-1 to +2	-1 to +4	LSB		
			INL integral nonlinearity error*1	—	±4	±12	LSB		
	Differential input	High-speed channels (AN000 to AN005) (AN006 to AN011)	Offset error *3	—	±0.25	±2	LSB	● ADCLK: 50 MHz ● Sampling time: 40 ADCLK ● Successive approximation time: 10 ADCLK ● Signal source impedance: 50 Ω or less ● Digital filter: Sinc filter	
			Gain error *3	—	±0.5	±2.5	LSB		
			DNL differential nonlinearity error*1 *2	—	-1 to +2	-1 to +2.5	LSB		
			INL integral nonlinearity error*1	—	±3	±6	LSB		

Note: These specification values are applicable when only one ADC16H is operating, DAC12 and ACPHPS are not operating, and there is no access to the external bus during A/D conversion.

If other ADC unit, DAC12, or ACPHPS is operating or bus access occurs during A/D conversion, values might not fall within the indicated ranges.

The use of ports 0 as digital outputs is not allowed when the ADC16H is used.

The characteristics apply when AVCC0, AVSS0, VREFH0, VREFH, VREFL0, VREFL, and ADC16H input voltage is stable.

Note 1. Test conditions: 0.2% to 99.8% of the analog input voltage range.

Note 2. DNL is measured using the Histogram Method, so the lower limit value is -1.

Note 3. This value is based on a 12-bit resolution.

Table 60.98 A/D internal reference voltage characteristics (1 of 2)

Parameter	Min	Typ	Max	Unit	Test conditions
A/D internal reference voltage	0.77	0.8	0.84	V	—

Table 60.98 A/D internal reference voltage characteristics (2 of 2)

Parameter	Min	Typ	Max	Unit	Test conditions
Sampling time	4.15	—	—	μs	—

Table 60.99 A/D conversion characteristics of D/A output

Parameter	Min	Typ	Max	Unit	Test conditions
Sampling time	1	—	—	μs	—

60.6 DAC12 Characteristics

Table 60.100 D/A conversion characteristics

Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
Resolution	—	—	—	12	Bits	—
INL	VREFH ≥ 2.7V	—	—	±2.0	LSB	—
	VREFH < 2.7V	—	—	±4.0		
DNL	VREFH ≥ 2.7V	—	—	±0.5	LSB	—
	VREFH < 2.7V	—	—	±1.0		
Conversion time	VREFH ≥ 2.7V	t _{DCONV1}	—	3.5	μs	—
	VREFH < 2.7V	t _{DCONV2}	—	6		
Output destination switching time	VREFH ≥ 2.7V	t _{DSLPU1}	—	3.5	μs	—
	VREFH < 2.7V	t _{DSLPU2}	—	6		
Buffer preparation time	VREFH ≥ 2.7V	t _{DISOUT}	—	3.5	μs	—
	VREFH < 2.7V	—	—	6		
Setup time	t _{SU}	—	—	4	ns	—
Resistive load	—	5	—	—	kΩ	—
Capacitive load	—	—	—	50	pF	—
Output voltage range	VREFH ≥ 2.7V	—	0.20	—	V	—
	VREFH < 2.7V	—	0.34	—		

60.7 TSN Characteristics

Table 60.101 TSN characteristics

Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
Relative accuracy	—	-1.0	—	1.0	°C	A/D converter error is not included.
Temperature slope	—	—	2.7	—	mV/°C	—
Output voltage (at 25 °C)	—	—	0.83	—	V	—
Temperature sensor stabilization time	t _{TSTBL}	—	—	30	μs	—
Comparator stabilization time	t _{RSTBL}	—	—	30	μs	—
Sampling time	—	4.15	—	—	μs	—

60.8 OSC Stop Detect Characteristics

Table 60.102 Oscillation stop detection circuit characteristics

Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
Detection time	t _{dr}	—	—	1	ms	Figure 60.126

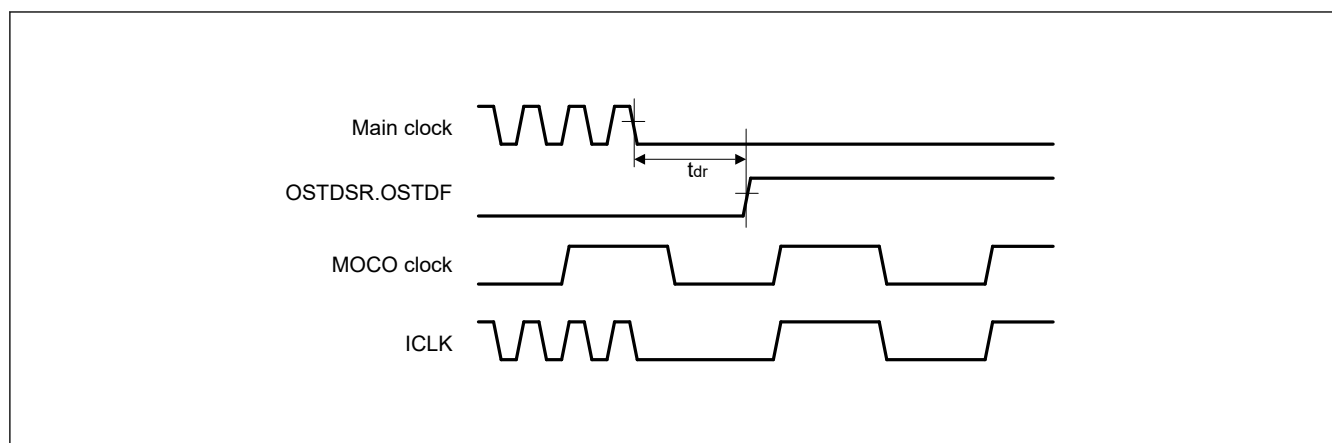


Figure 60.126 Oscillation stop detection timing

Table 60.103 Sub-clock oscillation stop detection circuit characteristics

Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
Wait time for startup	t_{sosdup}	100	—	—	μs	Figure 60.127
Detection time	t_{dr}	—	—	2	ms	Figure 60.128

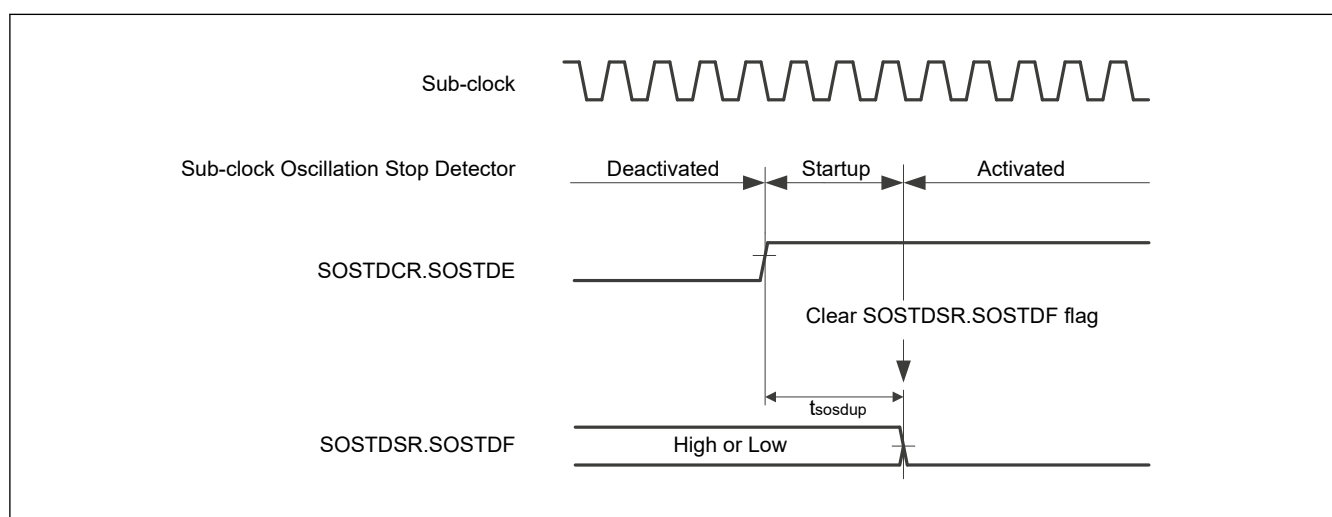


Figure 60.127 Sub-clock Oscillation Stop Detector start-up time

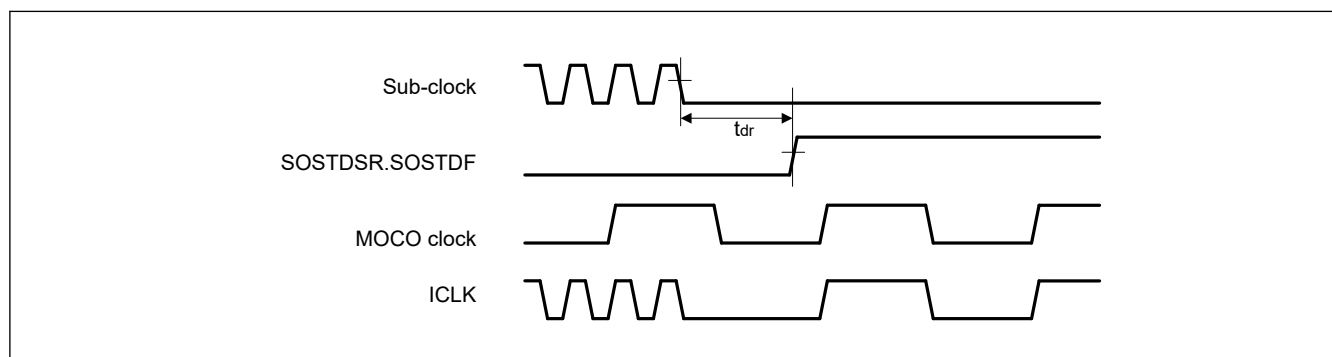


Figure 60.128 Sub-clock oscillation stop detection timing

60.9 POR and PVD Characteristics

Table 60.104 Power-on reset circuit and voltage detection circuit characteristics (1 of 2)

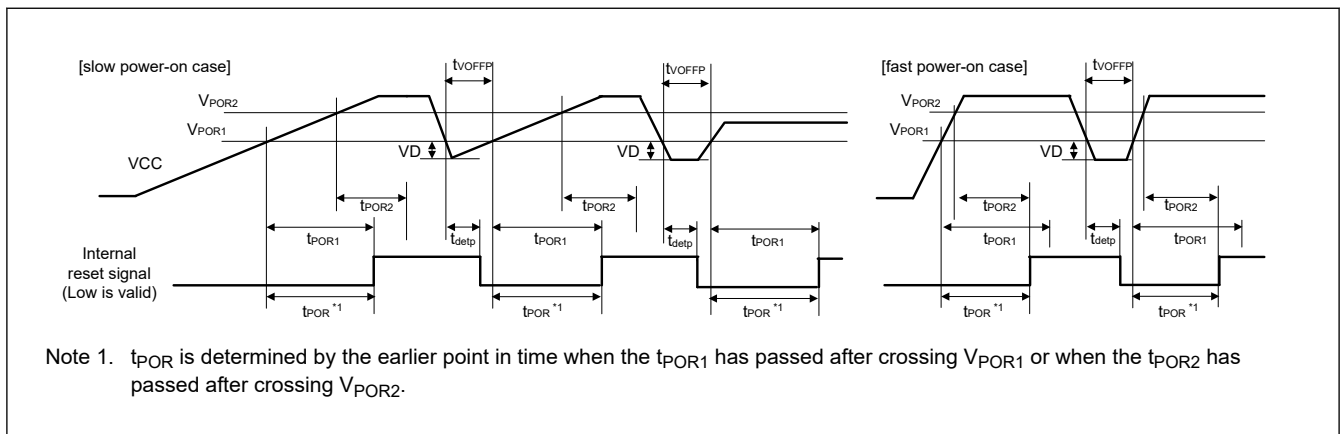
Parameter		Symbol	Min	Typ	Max	Unit	Test conditions
Voltage detection level	Power-on reset (POR)	V _{POR1}	1.52	1.56	1.61	V	Figure 60.129
		V _{POR2}	—	—	1.73		
	Voltage detection circuit (PVD0)	V _{det0_0}	2.76	2.85	2.94		Figure 60.130
		V _{det0_1}	2.50	2.58	2.66		
		V _{det0_2}	2.08	2.15	2.22		
		V _{det0_3}	1.93	2.00	2.07		
		V _{det0_4}	1.84	1.90	1.96		
		V _{det0_5}	1.74	1.80	1.86		
		V _{det0_6}	1.62	1.67	1.73		
		V _{det0_7}	1.51	1.56	1.61		
	Voltage detection circuit (PVDn) (n = 1, 2, 4, 5)	V _{detn_3_rise}	3.78	3.92	4.05		Figure 60.131
		V _{detn_3_fall}	3.72	3.86	3.99		
		V _{detn_4_rise}	3.09	3.20	3.30		
		V _{detn_4_fall}	3.03	3.14	3.24		
		V _{detn_5_rise}	3.05	3.16	3.26		
		V _{detn_5_fall}	2.99	3.10	3.20		
		V _{detn_6_rise}	3.03	3.14	3.24		
		V _{detn_6_fall}	2.97	3.08	3.18		
		V _{detn_7_rise}	2.81	2.91	3.00		
		V _{detn_7_fall}	2.75	2.85	2.94		
		V _{detn_8_rise}	2.79	2.89	2.98		
		V _{detn_8_fall}	2.73	2.83	2.92		
		V _{detn_9_rise}	2.76	2.86	2.95		
		V _{detn_9_fall}	2.70	2.80	2.89		
		V _{detn_10_rise}	2.58	2.67	2.75		
		V _{detn_10_fall}	2.53	2.62	2.70		
		V _{detn_11_rise}	2.30	2.38	2.46		
		V _{detn_11_fall}	2.25	2.33	2.41		
		V _{detn_12_rise}	1.88	1.94	2.00		
		V _{detn_12_fall}	1.84	1.90	1.96		
		V _{detn_13_rise}	1.84	1.90	1.96		
		V _{detn_13_fall}	1.80	1.86	1.92		
		V _{detn_14_rise}	1.72	1.78	1.84		
		V _{detn_14_fall}	1.68	1.74	1.80		
		V _{detn_15_rise}	1.66	1.72	1.77		
		V _{detn_15_fall}	1.62	1.68	1.73		

Table 60.104 Power-on reset circuit and voltage detection circuit characteristics (2 of 2)

Parameter		Symbol	Min	Typ	Max	Unit	Test conditions
Internal reset time* ¹	Power-on reset time	t_{POR1}	—	—	6.7	ms	Figure 60.129
		t_{POR2}	—	—	1.6		
	PVD0 reset time	t_{PVD0}	—	—	*1		Figure 60.129
	PVD1 reset time	t_{PVD1}	—	—	*1		Figure 60.130
	PVD2 reset time	t_{PVD2}	—	—	*1		
	PVD4 reset time	t_{PVD4}	—	—	*1		
	PVD5 reset time	t_{PVD5}	—	—	*1		
Minimum VCC downtime (POR)* ²	50mV < VD	t_{VOFFP}	900	—	—	μ s	Figure 60.129
	VD ≤ 50mV		2000	—	—		
Minimum VCC downtime (PVD)* ²	PVD0	t_{VOFF}	25	—	—	μ s	Figure 60.130
	PVD1, PVD2, PVD4, PVD5		25	—	—		
Response delay time (POR)	50mV < VD	t_{detp}	—	—	900	μ s	Figure 60.129
	VD ≤ 50mV		—	—	2000		
Response delay time (PVD)	PVD0	t_{det}	—	—	25	μ s	Figure 60.130, Figure 60.131
	PVD1, PVD2, PVD4, PVD5		—	—	25		
PVDn operation stabilization time (after PVD is enabled) (n = 1, 2, 4, 5)		$T_d(E-A)$	—	—	20	μ s	Figure 60.131

Note 1. The maximum value of t_{PVD0} , t_{PVD1} , t_{PVD2} , t_{PVD4} , t_{PVD5} are equal to t_{DSBY} because the internal reset time is maximized when returning from Deep Software Standby mode.

Note 2. The minimum VCC downtime indicates the time when VCC is below the minimum value of voltage detection levels V_{POR1} , V_{det0} , V_{det1} , V_{det2} , V_{det4} and V_{det5} for the POR / PVD.

**Figure 60.129 Power-on reset timing**

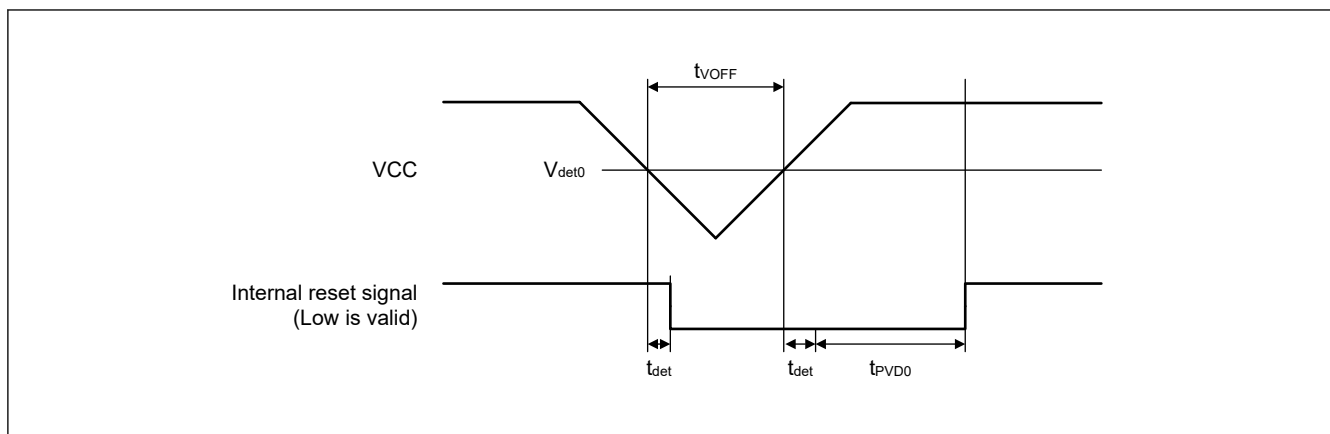


Figure 60.130 Voltage detection circuit timing (V_{det0})

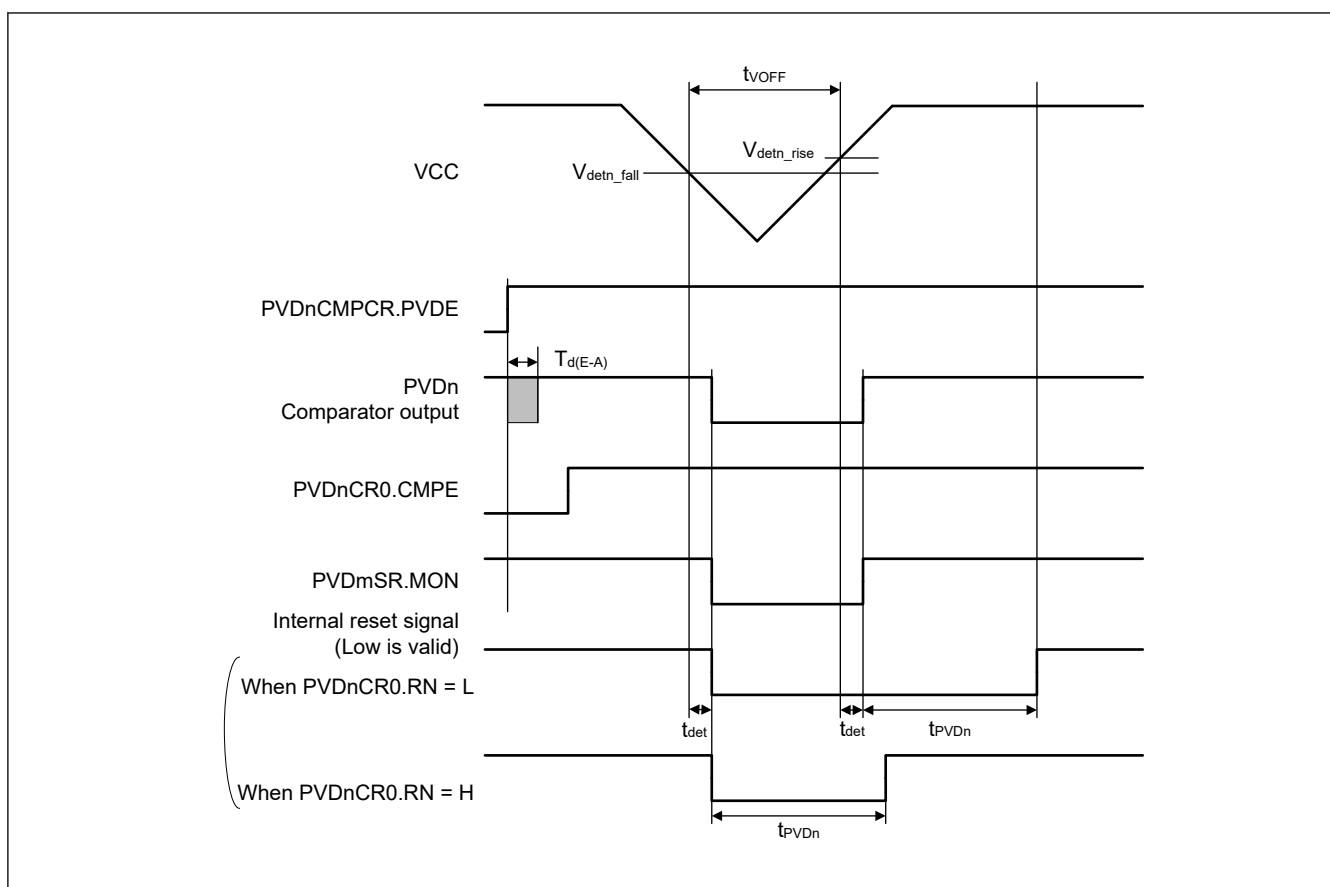


Figure 60.131 Voltage detection circuit timing (V_{detn}) ($n = 1, 2, 4, 5$)

60.10 External VDD Timing Characteristics

Table 60.105 External VDD timing characteristics

Parameter	Symbol	Min	Typ	Max	Unit	Test condition
Reset hold time at power up of external VDD with using RES pin	t_{EXTVRH}	600.00	—	—	μs	Figure 60.132 Figure 60.133
VDD rise time at power up of external VDD without using RES pin	t_{EXTVDD}	—	—	550	μs	

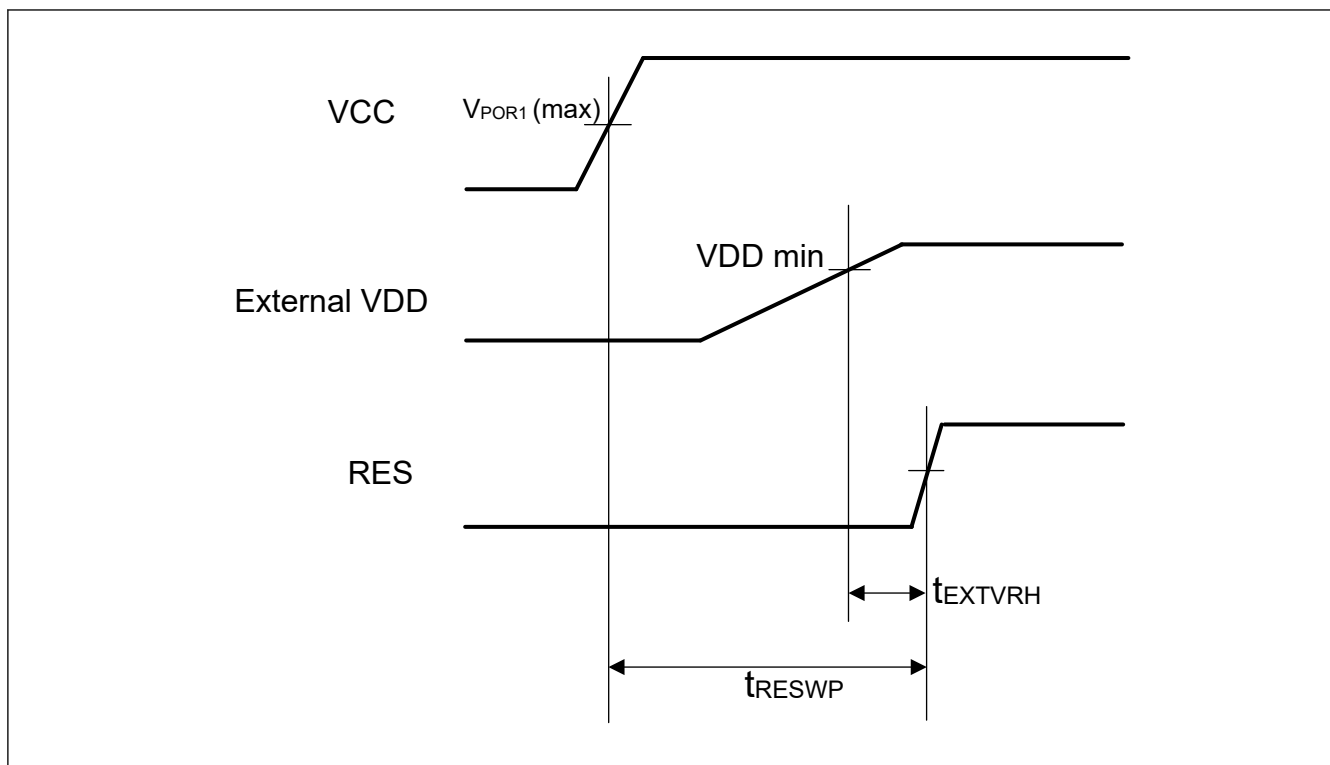


Figure 60.132 Power-up sequence of external VDD mode with using RES pin

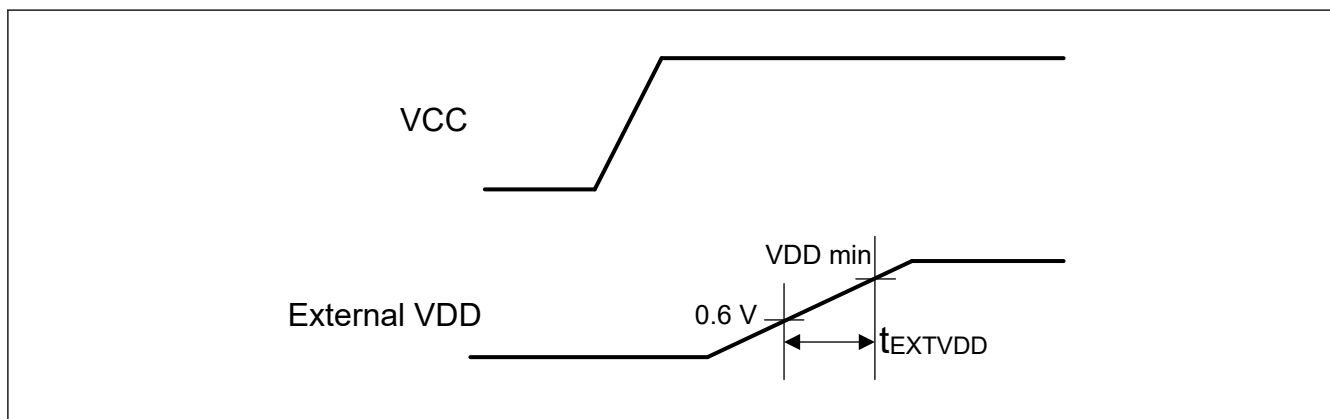


Figure 60.133 Power-up sequence of external VDD mode without using RES pin

60.11 Core Voltage Monitor Reset Characteristics

Table 60.106 Core voltage monitor reset characteristics

Parameter			Symbol	Min	Typ	Max	Unit	Test conditions
Voltage detection level	Core voltage monitor reset (CVMR)		V _{det_VDDH}	1.05	1.10	1.15	V	Figure 60.134
			V _{det_VDDL}	0.55	0.58	0.61		
Internal reset time	Core voltage monitor reset time	DCDC mode	t _{CVM}	—	—	0.18	ms	
		External VDD mode		—	—	2.6		
Minimum VDD down /up time (CVMR)			t _{CVMOFF}	45	—	—	μs	
Response delay time (CVMR)			t _{CVMdet}	—	—	45	μs	
Hysteresis width (CVMR)			V _{CVMH}	—	0.225	—	V	

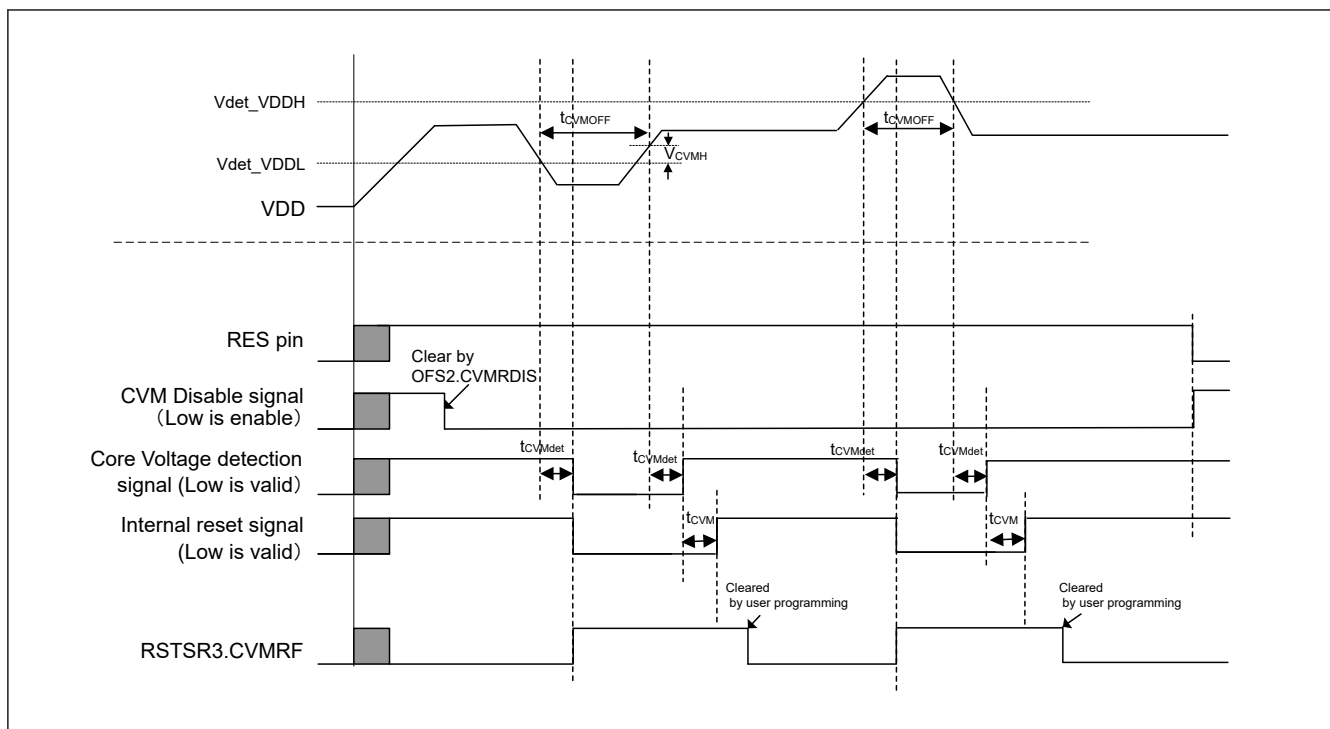


Figure 60.134 Timing of core voltage monitor reset

60.12 Temperature Monitor Reset Characteristics

Table 60.107 Temperature monitor reset characteristics

Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
High threshold temperature	$T_{J_detect_H}$	105	—	125	°C	Figure 60.135
Low threshold temperature	$T_{J_detect_L}$	-40	—	-20	°C	Figure 60.136

Note: Temperature monitor reset is not supported in 0 to 95 °C product (product group A).

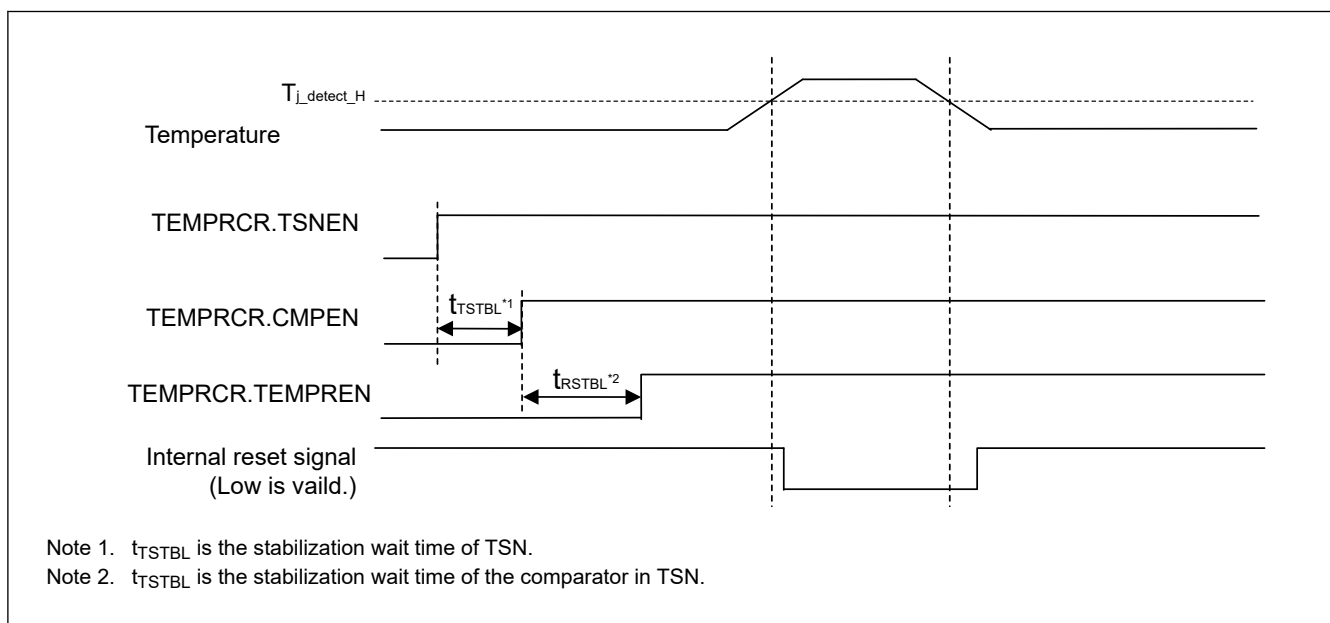


Figure 60.135 Timing of temperature monitor reset (High-temperature detection)

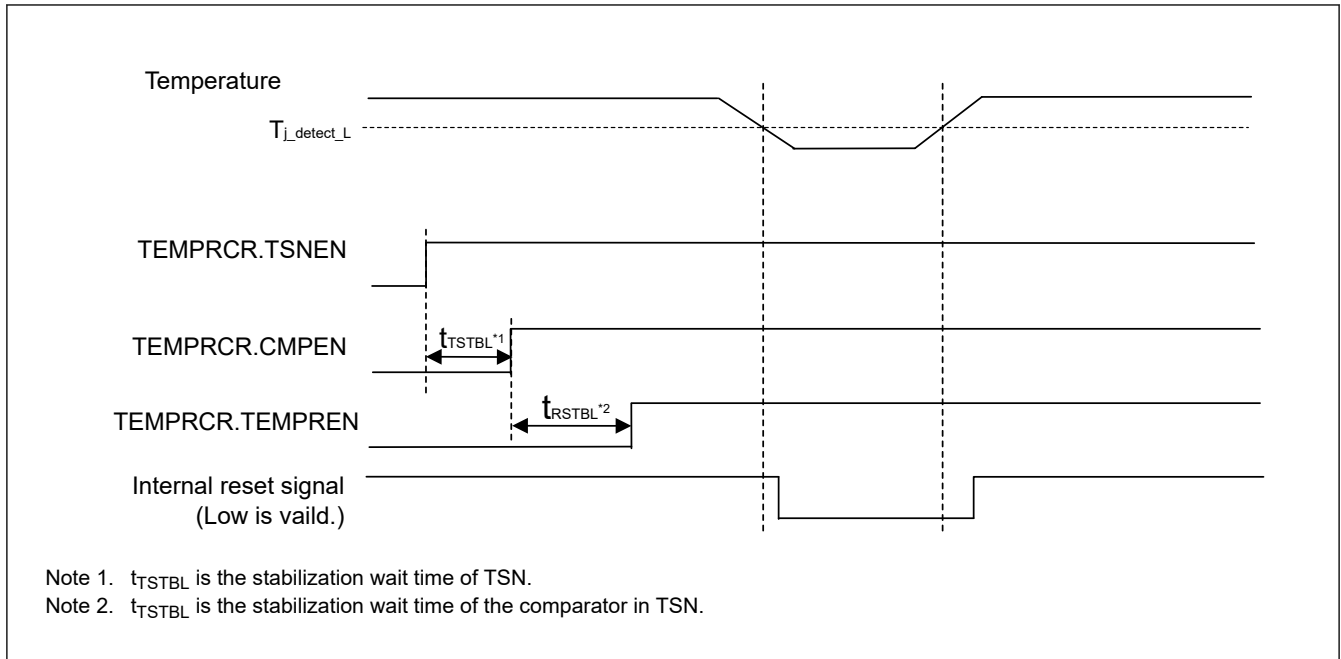


Figure 60.136 Timing of temperature monitor reset (Low-temperature detection)

60.13 VBATT Characteristics

Table 60.108 Battery backup function characteristics (1 of 2)

Conditions: VCC = VCC_DCDC = VCC_USB = 1.62 to 3.63 V, VBATT = 1.62 to 3.63 V

Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
Voltage level for switching to battery backup OFS1.PVDAS and PVDLPSEL are 0 in Deep Software Standby mode 1, 2 (VDETVATT_n follows VDSEL[2:0] setting for PVD0)	V _{DETBATT_0}	2.760	2.850	2.940	V	Figure 60.137
	V _{DETBATT_1}	2.500	2.580	2.660		
	V _{DETBATT_2}	2.080	2.150	2.220		
	V _{DETBATT_3}	1.935	2.000	2.065		
	V _{DETBATT_4}	1.840	1.900	1.960		
	V _{DETBATT_5}	1.740	1.800	1.860		
	V _{DETBATT_6}	1.620	1.670	1.730		
Voltage level for switching to battery backup (Other than above)	V _{DETBATT_0}	2.710	2.800	2.890	V	Figure 60.137
	V _{DETBATT_1}	2.450	2.530	2.610		
	V _{DETBATT_2}	2.030	2.100	2.170		
	V _{DETBATT_3}	1.885	1.950	2.015		
	V _{DETBATT_4}	1.790	1.850	1.910		
	V _{DETBATT_5}	1.690	1.750	1.810		
VCC drop detection stabilization wait time ^{*2}	t _{DETWT}	—	—	20	μs	—
Lower-limit VBATT voltage for power supply switching caused by VCC voltage drop	V _{BATTSW}	1.8	—	—	V	Figure 60.137
VCC-off period for starting power supply switching ^{*1} (OFS1.PVDAS and PVDLPSEL are 0 in Deep Software Standby mode 1, 2)	t _{VOFFBATT}	25	—	—	μs	
VCC-off period for starting power supply switching ^{*1} (Other than above)		25	—	—		

Table 60.108 Battery backup function characteristics (2 of 2)

Conditions: VCC = VCC_DCDC = VCC_USB = 1.62 to 3.63 V, VBATT = 1.62 to 3.63 V

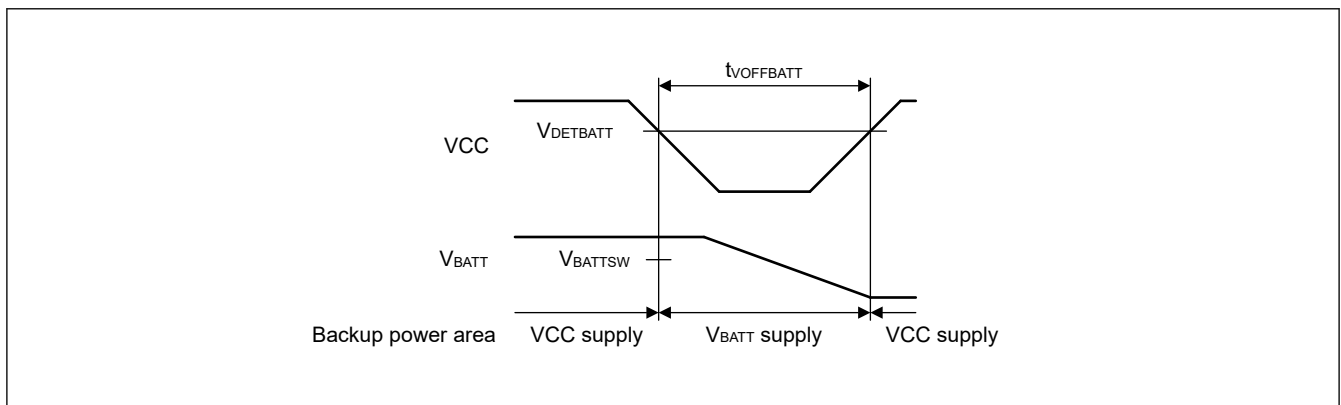
Parameter	Symbol	Min	Typ	Max	Unit	Test conditions
Backup domain power-down detection level	V _{PDR (BATR)}	1.43	1.47	1.52	V	Figure 60.138
Time delay in assertion of the reset signal for the backup domain*3	t _{p (PDRL)}	—	—	2000	μs	
Time delay in negation of the reset signal for the backup domain	t _{p (PDRH)}	—	—	3000	μs	
VBATT monitor operation stabilization time (after VBATTMNSLR.VBTMNSLR is changed to 1)	t _{MONWT}	—	—	4.2	μs	—
VBATT voltage monitor level	V _{MONBATT}	—	VBATT / 6	—	V	—
VBATT current increase (when VBATTMNSLR.VBTMNSLR is 1 compared to the case that VBATTMNSLR.VBTMNSLR is 0)	I _{VBATTSELB}	—	1.35	2.00	μA	—
VCC current increase (when VBATTMNSLR.VBTMNSLR is 1 compared to the case that VBATTMNSLR.VBTMNSLR is 0)	I _{VBATTSELC}	—	15	25	μA	—

Note 1. The VCC-off period for starting power supply switching indicates the period in which VCC is below the minimum value of the voltage level for switching to battery backup (V_{DET BATT}).

In addition, this period indicates the time t_{OFFBATT} when VCC is below the minimum value of voltage detection levels V_{POR1}.

Note 2. Stable time when VBTBPCR2.VDETLVL is changed or VBTBPCR1. BPWSWSTP is changed from 1 to 0.

Note 3. When the VBATT_R recovers within this period, the backup domain reset signal may not be generated.

**Figure 60.137 Battery backup function characteristics**

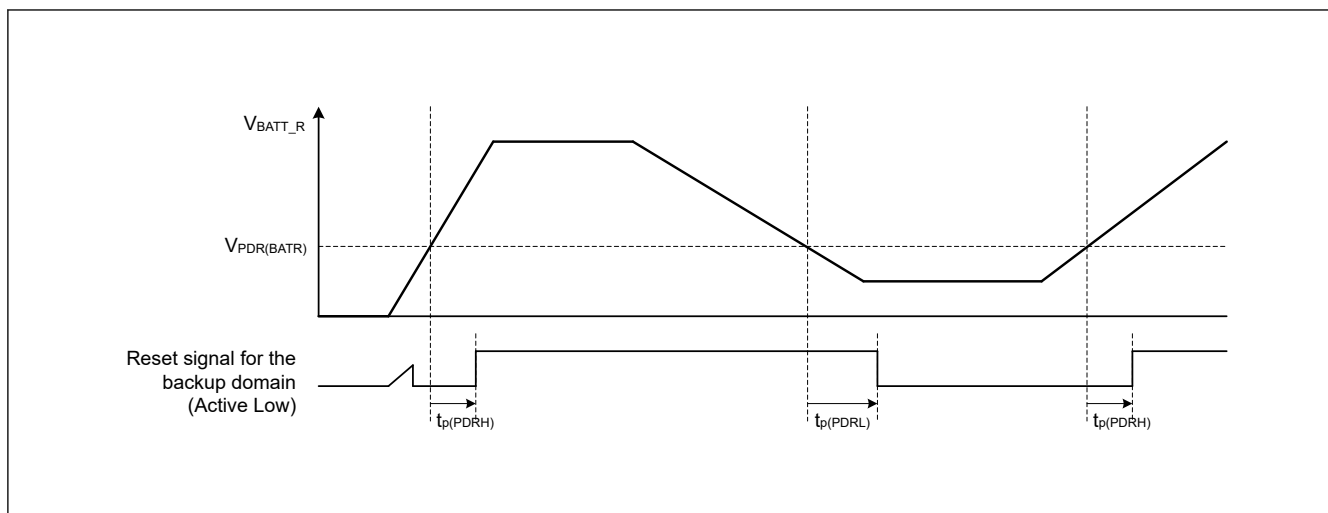


Figure 60.138 Backup domain reset characteristics

60.14 ACPHPS Characteristics

Table 60.109 ACPHPS

Parameter			Symbol	Min	Typ	Max	Unit	Test conditions
Reference voltage range			VREF	0	—	AVCC0	V	—
Input voltage range	ACMPHS0 , 1	IVCMP1 to IVCMP3	VI	0	—	AVCC0	V	—
		IVCMP0		0	—	AVCC0		VCC ≥ AVCC0
				0	—	VCC		VCC < AVCC0
	ACMPHS2, 3			0	—	AVCC0		—
Output delay*1			Td	—	50	100	ns	VI = VREF ± 100mV
Internal reference voltage			Vref	0.77	0.8	0.84	V	—

Note 1. This value is the internal propagation delay.

60.15 MRAM Characteristics

60.15.1 Code MRAM Characteristics

Table 60.110 Code MRAM Characteristics (1 of 2)

Parameter		Symbol	MRICLK = 250 MHz			MRICLK = 200 MHz			MRICLK = 150 MHz			MRICLK = 133 MHz			Unit	Test conditions
			Min	Typ*4	Max	Min	Typ*4	Max	Min	Typ*4	Max	Min	Typ*4	Max		
Programming time of 32-byte*7 *8	Normal program mode (MRPSC. MHSPEN = 0)	t _{PM} C	—	6.7*5 *6	83.3*6	—	6.74*5 *6	83.6*6	—	6.92*5 *6	85.6*6	—	7.09*5 *6	87.3*6	μs	—
	High speed program mode (MRPSC. MHSPEN = 1)	t _{PM} C	—	4.7*5 *6	81.3*6	—	4.74*5 *6	81.6*6	—	4.92*5 *6	83.6*6	—	5.09*5 *6	85.3*6	μs	—

Table 60.110 Code MRAM Characteristics (2 of 2)

Parameter	Symbol	MRICLK = 250 MHz			MRICLK = 200 MHz			MRICLK = 150 MHz			MRICLK = 133 MHz			Unit	Test conditions
		Min	Typ *4	Max	Min	Typ *4	Max	Min	Typ *4	Max	Min	Typ *4	Max		
Reprogramming cycle*9	N _{PC}	100000 *1	—	—	100000 *1	—	—	100000 *1	—	—	100000 *1	—	—	Times	—
Data hold time*2	t _{DRP}	10*2 *3	—	—	10*2 *3	—	—	10*2 *3	—	—	10*2 *3	—	—	Years	T _j = +125°C

Note 1. This is the minimum number of times to guarantee all the characteristics after reprogramming. The guaranteed range is from 1 to the minimum value.

Note 2. This indicates the minimum value of the characteristic when reprogramming is performed within the specified range.

Note 3. This result is obtained from reliability testing.

Note 4. The reference value at VCC = 3.3 V and room temperature.

Note 5. Perform rewrite 50% of bits at Typ. condition.

Note 6. If MRPClk < 125MHz, add the time of MRPClk 1cycle to the Program time.

Note 7. To calculate the updating time at other frequencies, using approximate formula below listed.

If MRPClk < 125MHz, add 1/FMRPClk [μs] to the formula. (F_{MRICLK}: frequency of MRICLK [MHz], F_{MRPCLK}: frequency of MRPCLK [MHz])

t_{PMC} (Typ) = 137.8/F_{MRICLK} + 6.452 [μs], t_{PMC} (Max) = 1879/F_{MRICLK} + 78.75 [μs] for normal program mode.

t_{PMC} (Typ) = 137.8/F_{MRICLK} + 4.452 [μs], t_{PMC} (Max) = 1879/F_{MRICLK} + 76.75 [μs] for high speed program mode.

Note 8. Read and program operations for the code MRAM cannot be executed simultaneously. This value is for independent program operation without arbitration between read and program.

Note 9. The reprogramming cycle is the number of programming per 32-byte code MRAM address space. Programming the same data to the same address also increases the reprogramming cycle count by one. For a reprogramming cycle of n times (n = 100,000), programming can be performed n times for every 32-byte code MRAM address space.

60.15.2 Option Setting Memory (configuration area) Characteristics

Table 60.111 Option Setting Memory (configuration area) characteristics

Parameter	Symbol	MRPClk = 125 MHz			MRPClk = 100 MHz			MRPClk = 75 MHz			MRPClk = 66 MHz			Unit	Test conditions
		Min	Typ *4	Max	Min	Typ *4	Max	Min	Typ *4	Max	Min	Typ *4	Max		
Command time for configuration set*5 *7	Normal program mode (MRPSC. MHSPEN = 0)	—	0.35	8.19 *6	—	0.35 6	8.3 *6	—	0.37 2	8.83 *6	—	0.39 9	9.42 *6	ms	—
	High speed program mode (MRPSC. MHSPEN = 1)	—	0.06 65	7.85 *6	—	0.07 25	7.96 *6	—	0.08 82	8.5 *6	—	0.09 78	9.06 *6	ms	—
Update cycle*8	N _{CU} PC	100000 *1	—	—	100000 *1	—	—	100000 *1	—	—	100000 *1	—	—	Times	—
Data hold time*2	t _{DRP}	10*2 *3	—	—	10*2 *3	—	—	10*2 *3	—	—	10*2 *3	—	—	Years	T _j = +125°C

Note 1. This is the minimum number of times to guarantee all the characteristics after update. The guaranteed range is from 1 to the minimum value.

Note 2. This indicates the minimum value of the characteristic when update is performed within the specified range.

Note 3. This result is obtained from reliability testing.

Note 4. The reference value at VCC = 3.3 V and room temperature.

Note 5. To calculate the updating time at other frequencies, using approximate formula below listed. (F_{MRPCLK}: frequency of MRPCLK [MHz])

$t_{PCFG} (Typ) = 6.146/F_{MRPCLK} + 0.3133 [msec]$, $t_{PCFG} (Max) = 266.5/F_{MRPCLK} + 6.331 [msec]$ for normal program mode.

$t_{PCFG} (Typ) = 5.184/F_{MRPCLK} + 0.02754 [msec]$, $t_{PCFG} (Max) = 267.5/F_{MRPCLK} + 6.025 [msec]$ for high speed program mode.

Note 6. Values in this table are calculated based on the times of 5-times programming (16 bytes). Timeout values of Figure 58.16 is about 1.1 times of the maximum value ($t_{PCFG} + t_{PMC} + \text{Read time (MRPCLK 40cycle)}$).

Note 7. Read and program operations for the extra MRAM cannot be executed simultaneously. This value is for independent program operation without arbitration between read and program.

Note 8. The update cycle is the number of a configuration set command that can be issued configuration area. Programming the same data to the same address also increases the update cycle count by one.

60.15.3 Option Setting Memory (OTP area with ECC) Characteristics

Table 60.112 Option setting memory (OTP area with ECC) characteristics

Parameter		Symbol	MRPCLK = 125 MHz			MRPCLK = 100 MHz			MRPCLK = 75 MHz			MRPCLK = 66 MHz			Unit	Test conditions
			Min	Typ*1	Max	Min	Typ*1	Max	Min	Typ*1	Max	Min	Typ*1	Max		
Programming time of 16-byte*4 *5	Normal speed write mode (MWMCR.MWM[1:0] = 00)	t_{OTPE}	—	8.05*2	113*3	—	8.05*2	113*3	—	8.05*2	113*3	—	8.05*2	113*3	ms	—
	High speed write mode 0 (MWMCR.MWM[1:0] = 01)		—	4.03*2	56.7*3	—	4.03*2	56.7*3	—	4.03*2	56.7*3	—	4.03*2	56.7*3	ms	—
	High speed write mode 1 (MWMCR.MWM[1:0] = 10)		—	0.22g*2	3.14*3	—	0.22g*2	3.14*3	—	0.22g*2	3.14*3	—	0.22g*2	3.14*3	ms	—

Note 1. The reference value at VCC = 3.3 V and room temperature.

Note 2. Perform rewrite 50% of bits at Typ. condition.

Note 3. Timeout values of Figure 58.13 is about 1.1 times of the maximum value. ($t_{OTPE} + t_{PMC} + \text{Read time (MRPCLK 40cycle)}$).

Note 4. To calculate the updating time at other frequencies, using approximate formula below listed. (F_{MRPCLK} : frequency of MRPCLK [MHz])

$t_{OTPE} (Typ) = 0.5065/F_{MRPCLK} + 8.123 [ms]$, $t_{OTPE} (Max) = 4.433/F_{MRPCLK} + 114.1 [ms]$ for normal speed mode.

$t_{OTPE} (Typ) = 0.3389/F_{MRPCLK} + 4.067 [ms]$, $t_{OTPE} (Max) = 2.428/F_{MRPCLK} + 57.08 [ms]$ for high speed mode 0.

$t_{OTPE} (Typ) = 0.1458/F_{MRPCLK} + 0.2312 [ms]$, $t_{OTPE} (Max) = 0.3904/F_{MRPCLK} + 3.166 [ms]$ for high speed mode 1.

Note 5. Read and program operations for the extra MRAM cannot be executed simultaneously. This value is for independent program operation without arbitration between read and program.

60.15.4 Option Setting Memory (OTP area without ECC) Characteristics

Table 60.113 Option Setting Memory (OTP area without ECC) Characteristics

Parameter	Symbol	MRPCLK = 125 MHz			MRPCLK = 100 MHz			MRPCLK = 75 MHz			MRPCLK = 66 MHz			Unit	Test conditions
		Min	Typ *1	Max	Min	Typ *1	Max	Min	Typ *1	Max	Min	Typ *1	Max		
Programming time of 16-byte ^{*4} ^{*5}	Normal speed write mode (MWMCR.MWM[1:0] = 00)	—	14.1 ^{*2}	200 ^{*3}	—	14.1 ^{*2}	200 ^{*3}	—	14.1 ^{*2}	200 ^{*3}	—	14.1 ^{*2}	200 ^{*3}	ms	—
	High speed write mode 0 (MWMCR.MWM[1:0] = 01)	—	7.07 ^{*2}	100 ^{*3}	—	7.07 ^{*2}	100 ^{*3}	—	7.07 ^{*2}	100 ^{*3}	—	7.07 ^{*2}	100 ^{*3}	ms	—
	High speed write mode 1 (MWMCR.MWM[1:0] = 10)	—	0.45 ^{8*2}	6.28 ^{*3}	—	0.45 ^{8*2}	6.28 ^{*3}	—	0.45 ^{8*2}	6.28 ^{*3}	—	0.45 ^{8*2}	6.28 ^{*3}	ms	—

Note 1. The reference value at VCC = 3.3 V and room temperature.

Note 2. Perform rewrite 50% of bits at Typ. condition.

Note 3. Timeout values of Figure 58.13 is about 1.1 times of the maximum value. ($t_{\text{OTPE}} + t_{\text{PMC}} + \text{Read time (MRPCLK 40cycle)}$).

Note 4. To calculate the updating time at other frequencies, using approximate formula below listed. (F_{MRPCLK} : frequency of MRPCLK [MHz])

$t_{\text{OTPE}} (\text{Typ}) = 0.8486/F_{\text{MRPCLK}} + 14.25$ [ms], $t_{\text{OTPE}} (\text{Max}) = 7.711/F_{\text{MRPCLK}} + 201.5$ [ms] for normal speed mode.

$t_{\text{OTPE}} (\text{Typ}) = 0.5479/F_{\text{MRPCLK}} + 7.133$ [ms], $t_{\text{OTPE}} (\text{Max}) = 4.148/F_{\text{MRPCLK}} + 100.8$ [ms] for high speed mode 0.

$t_{\text{OTPE}} (\text{Typ}) = 0.2571/F_{\text{MRPCLK}} + 0.4627$ [ms], $t_{\text{OTPE}} (\text{Max}) = 0.7401/F_{\text{MRPCLK}} + 6.333$ [ms] for high speed mode 1.

Note 5. Read and program operations for the extra MRAM cannot be executed simultaneously. This value is for independent program operation without arbitration between read and program.

60.15.5 MACI Command Characteristics

Table 60.114 MACI command Characteristics

Parameter	Symbol	MRPCLK = 125 MHz			MRPCLK = 100 MHz			MRPCLK = 75 MHz			MRPCLK = 66 MHz			Unit	Test conditions
		Min	Typ *1	Max	Min	Typ *1	Max	Min	Typ *1	Max	Min	Typ *1	Max		
Command time for forced stop command ^{*2}	t_{FS}	—	—	3.35 ^{*3}	—	—	3.38 ^{*3}	—	—	3.47 ^{*3}	—	—	3.7 ^{*3}	μs	—
Command time for increment counter ^{*2}	$t_{\text{INC C}}$	—	0.25 ²	1.61 ^{*4}	—	0.25 ²	1.61 ^{*4}	—	0.25 ²	1.61 ^{*4}	—	0.25 ²	1.61 ^{*4}	ms	—
Command time for read counter ^{*2}	$t_{\text{RD C}}$	—	—	0.15 ^{6*5}	—	—	0.18 ^{2*5}	—	—	0.24 ^{3*5}	—	—	0.27 ^{6*5}	μs	—

Note 1. The reference value at VCC = 3.3 V and room temperature.

Note 2. To calculate the updating time at other frequencies, using approximate formula below listed. (F_{MRPCLK} : frequency of MRPCLK [MHz])

$t_{\text{FS}} (\text{Max}) = 38.62/F_{\text{MRPCLK}} + 3.155$ [μs]

$t_{\text{INCC}} (\text{Typ}) = 0.3348/F_{\text{MRPCLK}} + 0.2533$ [msec], $t_{\text{INCC}} (\text{Max}) = 0.8698/F_{\text{MRPCLK}} + 1.62$ [msec]

$$t_{RDC}(\text{Max}) = 19.13/F_{MRPCLK} + 0.004099 [\mu\text{sec}]$$

Note 3. Timeout values of Figure 58.15 is about 1.1 times of the maximum value. (t_{FS} + Read time (MRPCLK 40cycle)).

Note 4. Values in this table are calculated based on the times of 2-time programming (32 bytes). Timeout values of Figure 58.17 is about 1.1 times of the maximum value ($t_{INCC} + t_{PMC}$ + Read time (MRPCLK 40cycle)).

Note 5. Counter read time is 100 cycles at MRPCLK. Timeout values of Figure 58.18 is about 1.1 times of the maximum value (t_{RDC} + Read time (MRPCLK 40cycle)).

60.15.6 Zeroizing of W-HUK

Table 60.115 Zeroizing of W-HUK Characteristics

Parameter	Symbol	MRPCLK = 125 MHz			MRPCLK = 100 MHz			MRPCLK = 75 MHz			MRPCLK = 66 MHz			Unit	Test conditions
		Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	Min	Typ	Max		
Action time of Zeroing of W-HUK ^{*1}	t_{ZW} H	—	—	793	—	—	793	—	—	793	—	—	793	ms	—

Note 1. To calculate the updating time at other frequencies, using approximate formula below listed. (FMRPCLK: frequency of MRPCLK [MHz])

$$t_{ZWH}(\text{Max}) = 34.52/F_{MRPCLK} + 799 [\mu\text{s}]$$

60.15.7 MRAM Magnetic Field Immunity Characteristics

Table 60.116 MRAM Magnetic field Immunity Characteristics

Parameter		Symbol	Min	Typ	Max	Unit	Test conditions
Operation	Write state	$G_{\text{max_wr}}$	—	—	200	Gauss	—
	Read state	$G_{\text{max_rd}}$	—	—	200	Gauss	—
	No access state	$G_{\text{max_noac}}$	—	—	500	Gauss	—
Storage		$G_{\text{max_stg}}$	—	—	500	Gauss	The applied temperature is T_{stg} .

60.16 Boundary Scan

Table 60.117 Boundary scan characteristics (1 of 2)

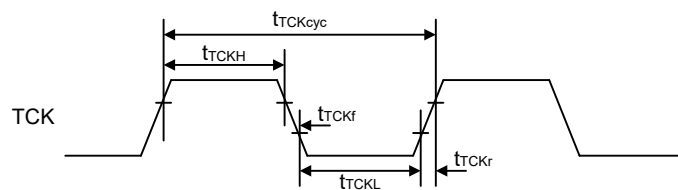
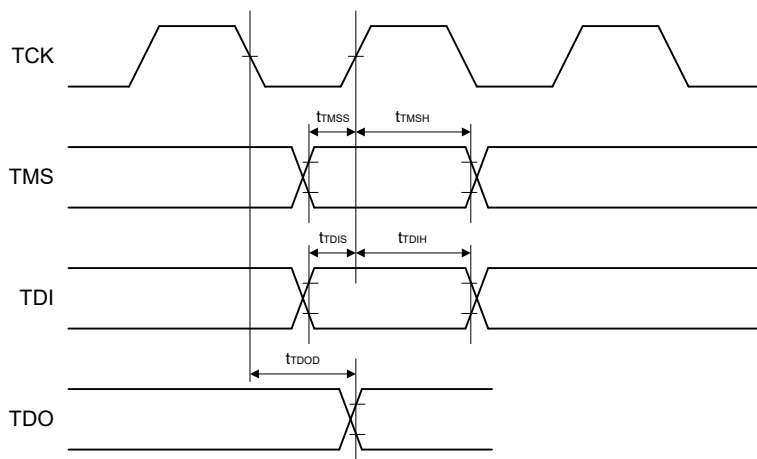
Parameter	VCC	Symbol	Min	Typ	Max	Unit	Test conditions
TCK clock cycle time	1.62 V or above	t_{TCKcyc}	100	—	—	ns	Figure 60.139
TCK clock high pulse width	1.62 V or above	t_{TCKH}	0.45	—	—	t_{TCKcyc}	
TCK clock low pulse width	1.62 V or above	t_{TCKL}	0.45	—	—	t_{TCKcyc}	
TCK clock rise time	1.62 V or above	t_{TCKr}	—	—	0.05^{*2}	t_{TCKcyc}	
TCK clock fall time	1.62 V or above	t_{TCKf}	—	—	0.05^{*2}	t_{TCKcyc}	

Table 60.117 Boundary scan characteristics (2 of 2)

Parameter	VCC	Symbol	Min	Typ	Max	Unit	Test conditions
TMS setup time	1.62 V or above	t_{TMSS}	20	—	—	ns	Figure 60.140
TMS hold time	1.62 V or above	t_{TMSH}	20	—	—	ns	
TDI setup time	1.62 V or above	t_{TDIS}	20	—	—	ns	
TDI hold time	1.62 V or above	t_{TDIH}	20	—	—	ns	
TDO data delay	1.62 V or above	t_{TDOD}	—	—	40	ns	
Capture register setup time	1.62 V or above	t_{CAPTS}	20	—	—	ns	Figure 60.141
Capture register hold time	1.62 V or above	t_{CAPTH}	20	—	—	ns	
Update register delay time	1.62 V or above	$t_{UPDATED}$	—	—	40	ns	
Boundary scan circuit startup time*1	1.62 V or above	T_{BSSTUP}	t_{RESWP}	—	—	—	Figure 60.142

Note 1. Boundary scan does not function until the power-on reset becomes negative.

Note 2. 1 μ s at the longest

**Figure 60.139 Boundary scan TCK timing****Figure 60.140 Boundary scan input/output timing (1)**

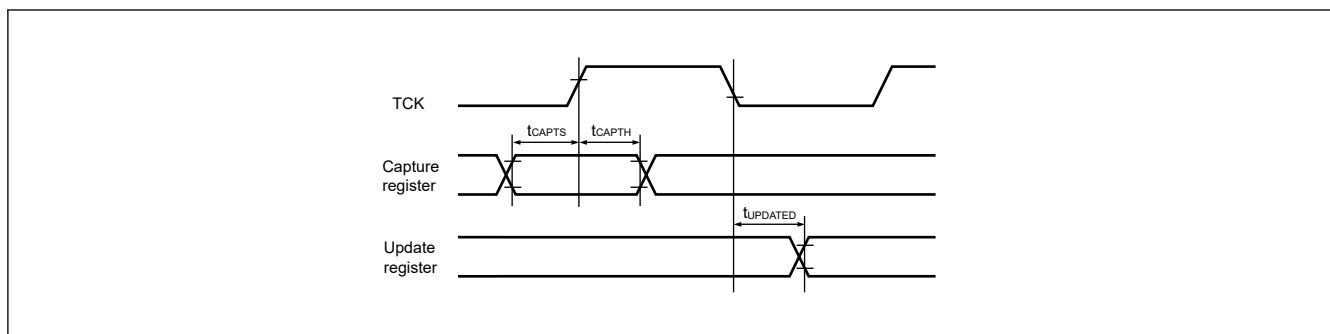


Figure 60.141 Boundary scan input/output timing (2)

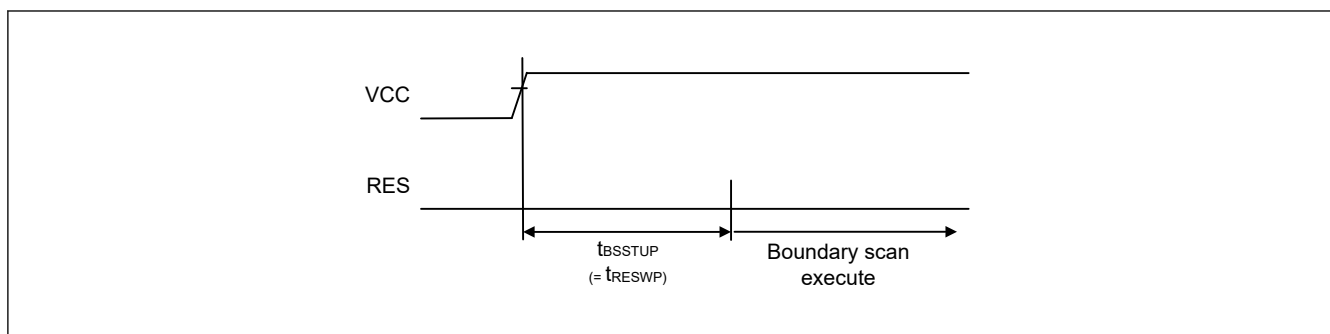


Figure 60.142 Boundary scan circuit startup timing

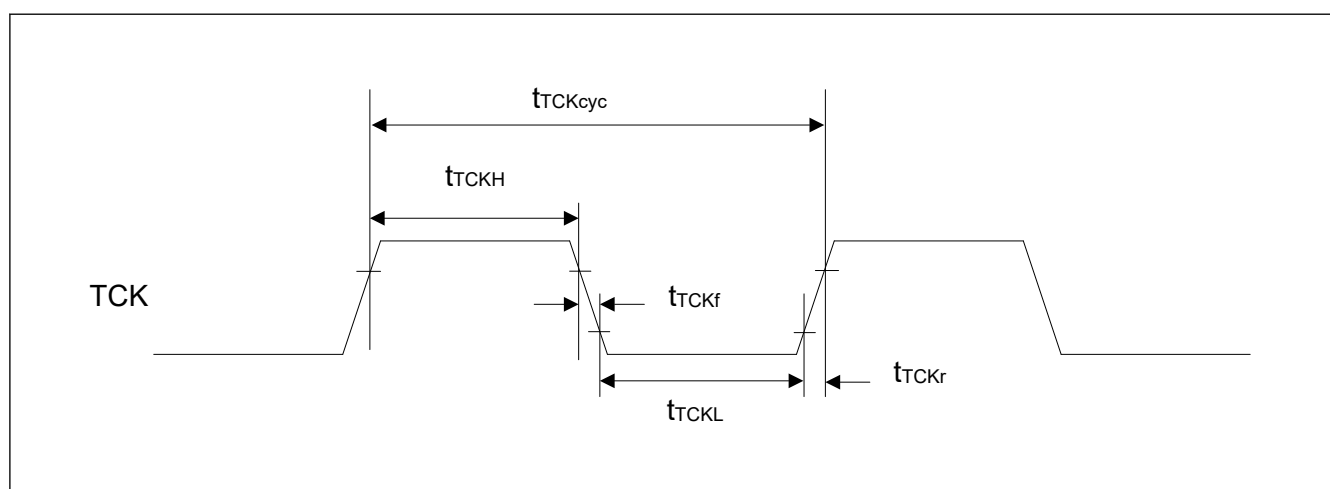
60.17 Joint European Test Action Group (JTAG)

Table 60.118 JTAG (1 of 2)

Parameter	VCC	Symbol	Min	Typ	Max	Unit	Test conditions
TCK clock cycle time	2.7 V or above	t_{TCKcyc}	40.0	—	—	ns	Figure 60.143
	1.62 V or above		40.0	—	—	ns	
TCK clock high pulse width	2.7 V or above	t_{TCKH}	0.375	—	—	t_{TCKcyc}	
	1.62 V or above		0.375	—	—	t_{TCKcyc}	
TCK clock low pulse width	2.7 V or above	t_{TCKL}	0.375	—	—	t_{TCKcyc}	
	1.62 V or above		0.375	—	—	t_{TCKcyc}	
TCK clock rise time	2.7 V or above	t_{TCKr}	—	—	0.125^{*1}	t_{TCKcyc}	
	1.62 V or above		—	—	0.125^{*1}	t_{TCKcyc}	
TCK clock fall time	2.7 V or above	t_{TCKf}	—	—	0.125^{*1}	t_{TCKcyc}	
	1.62 V or above		—	—	0.125^{*1}	t_{TCKcyc}	

Table 60.118 JTAG (2 of 2)

Parameter	VCC	Symbol	Min	Typ	Max	Unit	Test conditions
TMS setup time	2.7 V or above	t_{TMSS}	8.0	—	—	ns	Figure 60.144
	1.62 V or above		8.0	—	—	ns	
TMS hold time	2.7 V or above	t_{TMSH}	8.0	—	—	ns	
	1.62 V or above		8.0	—	—	ns	
TDI setup time	2.7 V or above	t_{TDIS}	8.0	—	—	ns	
	1.62 V or above		8.0	—	—	ns	
TDI hold time	2.7 V or above	t_{TDIH}	8.0	—	—	ns	
	1.62 V or above		8.0	—	—	ns	
TDO data delay time	2.7 V or above	t_{TDOD}	—	—	20.0	ns	
	1.62 V or above		—	—	28.0	ns	

Note 1. 1 μ s at the longest**Figure 60.143 JTAG TCK timing**

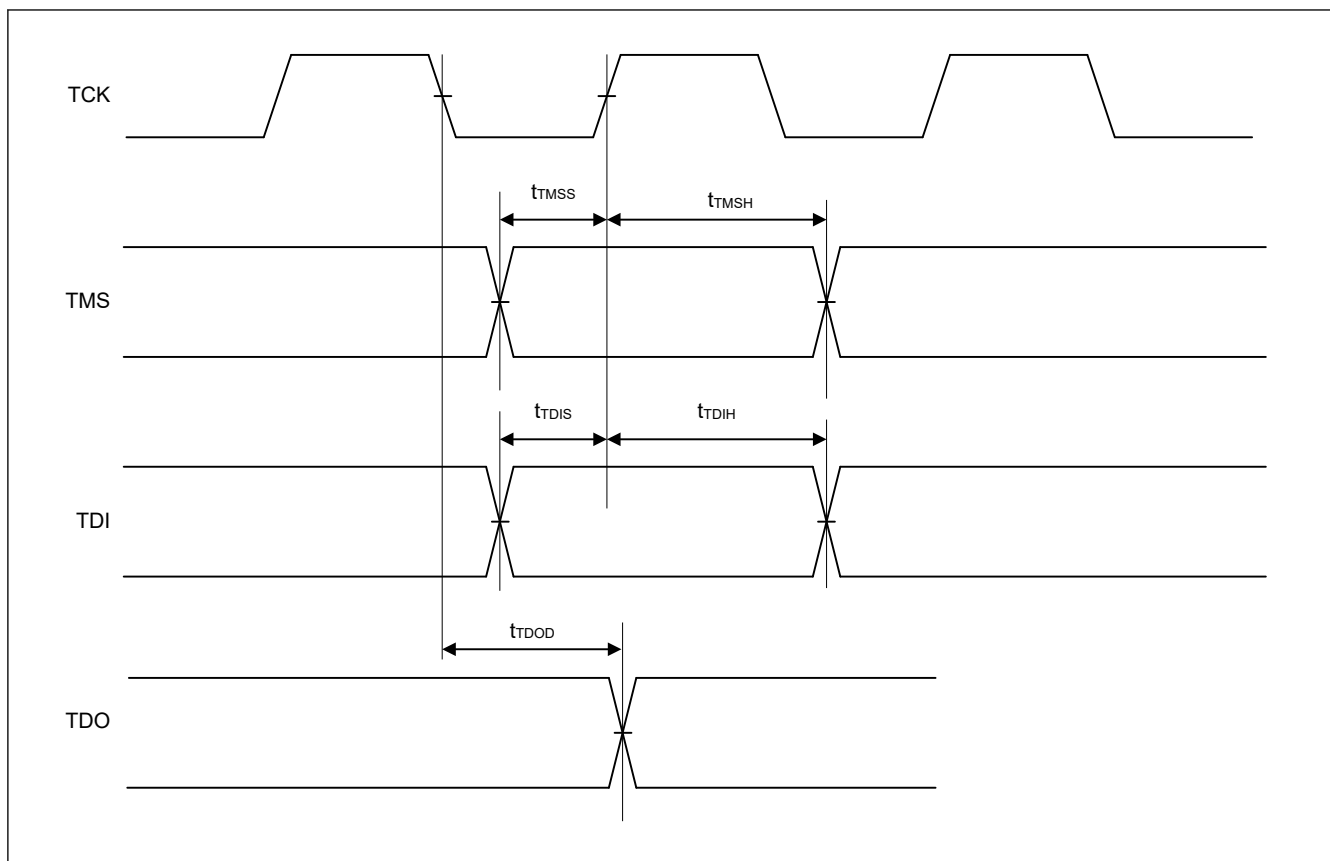


Figure 60.144 JTAG input/output timing

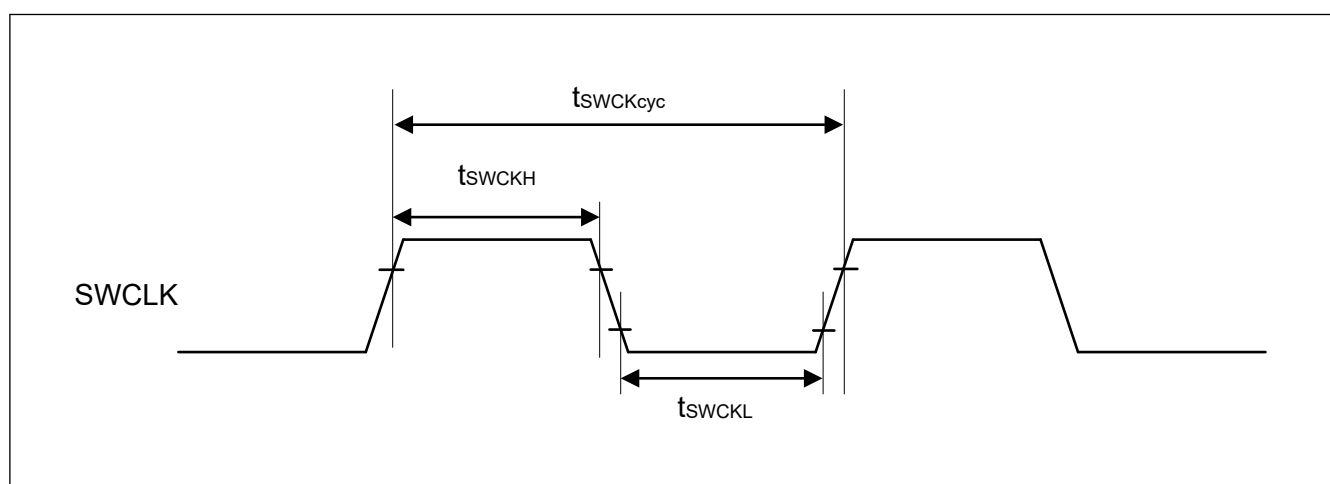
60.18 Serial Wire Debug (SWD)

Table 60.119 SWD (1 of 2)

Parameter	VCC	Symbol	Min	Typ	Max	Unit	Test conditions
SWCLK clock cycle time	2.7 V or above	$t_{SWCKcyc}$	40.0	—	—	ns	Figure 60.145
	1.62 V or above		40.0	—	—	ns	
SWCLK clock high pulse width	2.7 V or above	t_{SWCKH}	0.375	—	—	$t_{SWCKcyc}$	
	1.62 V or above		0.375	—	—	$t_{SWCKcyc}$	
SWCLK clock low pulse width	2.7 V or above	t_{SWCKL}	0.375	—	—	$t_{SWCKcyc}$	
	1.62 V or above		0.375	—	—	$t_{SWCKcyc}$	
SWCLK clock rise time	2.7 V or above	t_{SWCKr}	—	—	0.125^{*1}	$t_{SWCKcyc}$	
	1.62 V or above		—	—	0.125^{*1}	$t_{SWCKcyc}$	
SWCLK clock fall time	2.7 V or above	t_{SWCKf}	—	—	0.125^{*1}	$t_{SWCKcyc}$	
	1.62 V or above		—	—	0.125^{*1}	$t_{SWCKcyc}$	

Table 60.119 SWD (2 of 2)

Parameter	VCC	Symbol	Min	Typ	Max	Unit	Test conditions
SWDIO setup time	2.7 V or above	t_{SWDS}	8.0	—	—	ns	Figure 60.146
	1.62 V or above		8.0	—	—	ns	
SWDIO hold time	2.7 V or above	t_{SWDH}	8.0	—	—	ns	
	1.62 V or above		8.0	—	—	ns	
SWDIO data delay time	2.7 V or above	t_{SWDD}	2.0	—	28.0	ns	
	1.62 V or above		2.0	—	32.0	ns	

Note 1. 1 μ s at the longest**Figure 60.145 SWD SWCLK timing**

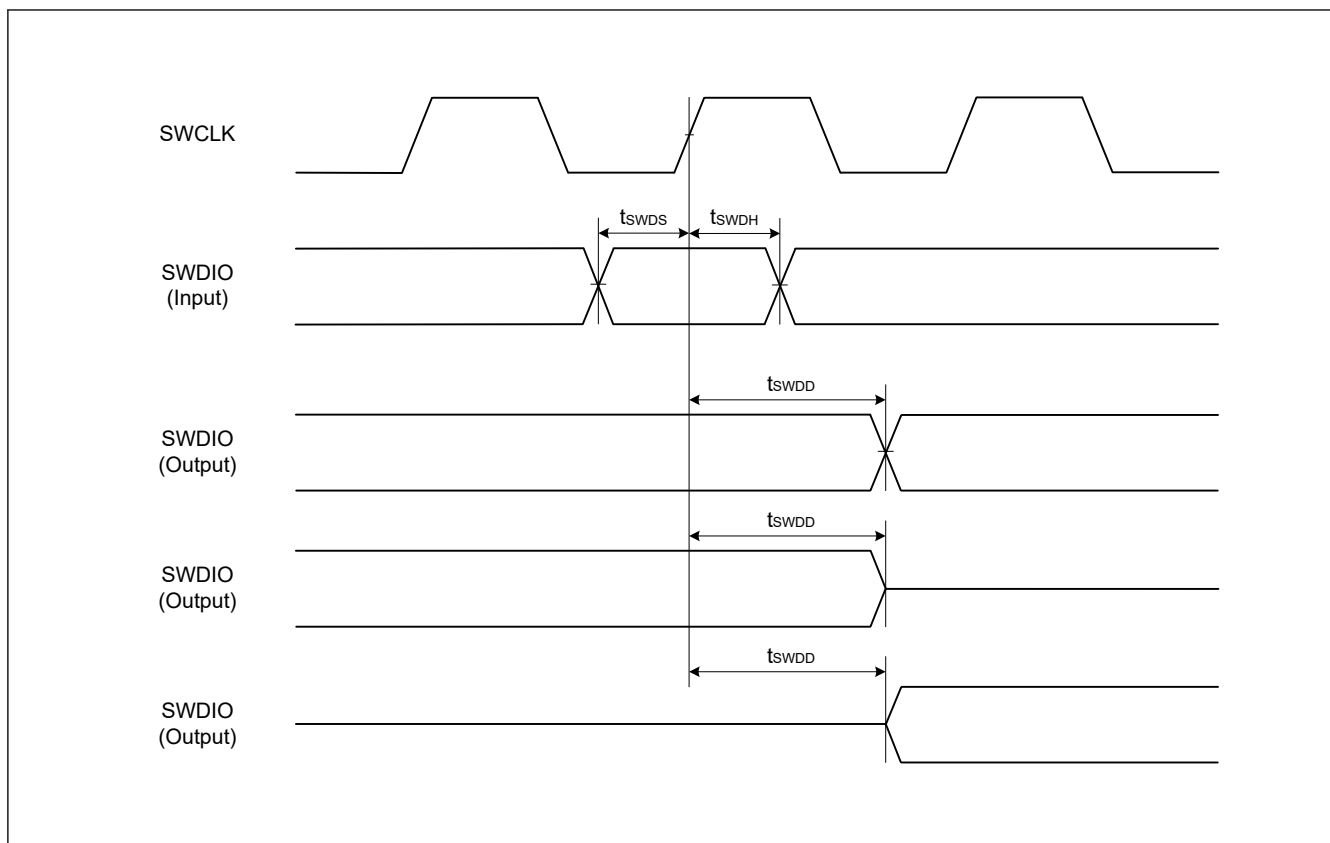


Figure 60.146 SWD input/output timing

60.19 Embedded Trace Macro Interface (ETM)

Table 60.120 ETM (1 of 2)

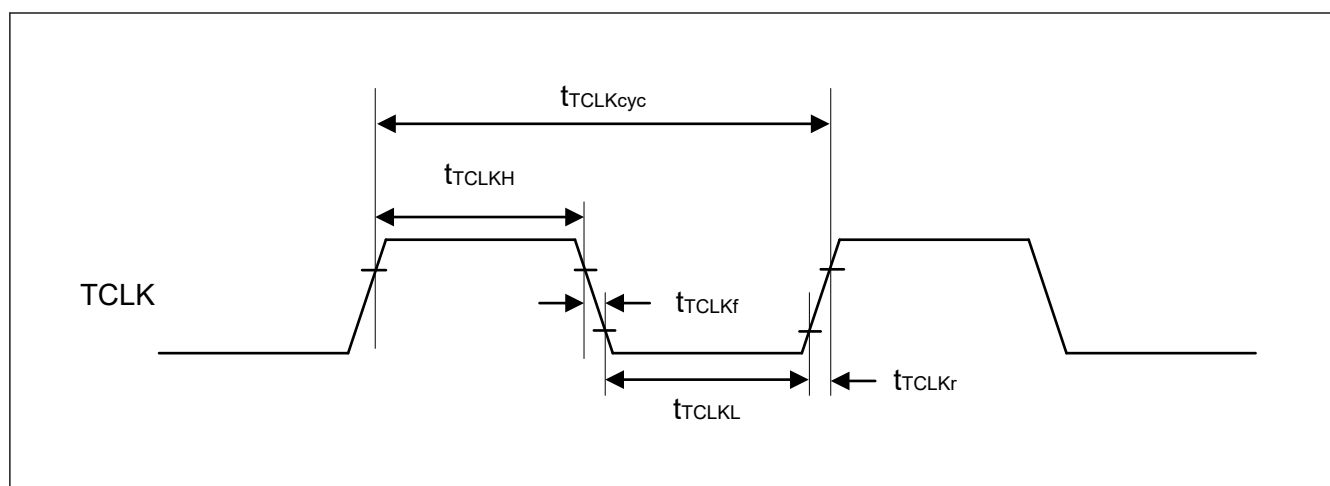
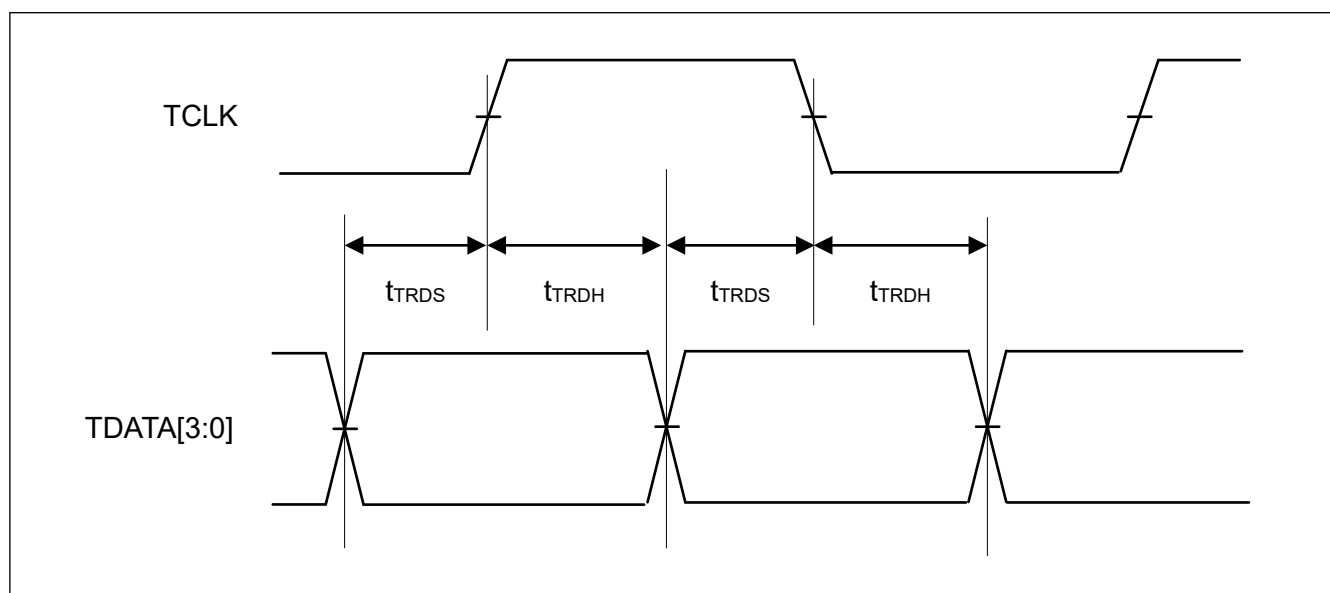
Conditions: High-speed high drive output is selected in the Port Drive Capability bit in the PmnPFS register.

Parameter	VCC	Symbol	Min	Typ	Max	Unit	Test conditions
TCLK clock cycle time	2.7 V or above	$t_{TCLKcyc}$	16	—	—	ns	Figure 60.147
	1.62 V or above		16	—	—	ns	
TCLK clock high pulse width	2.7 V or above	t_{TCLKH}	7	—	—	ns	
	1.62 V or above		6	—	—	ns	
TCLK clock low pulse width	2.7 V or above	t_{TCLKL}	7	—	—	ns	
	1.62 V or above		6	—	—	ns	
TCLK clock rise time	2.7 V or above	t_{TCLKr}	—	—	1.0	ns	
	1.62 V or above		—	—	2.0	ns	
TCLK clock fall time	2.7 V or above	t_{TCLKf}	—	—	1.0	ns	
	1.62 V or above		—	—	2.0	ns	

Table 60.120 ETM (2 of 2)

Conditions: High-speed high drive output is selected in the Port Drive Capability bit in the PmnPFS register.

Parameter	VCC	Symbol	Min	Typ	Max	Unit	Test conditions
TDATA[3:0] output valid time	2.7 V or above	t_{TRDV}	—	—	$t_{TCLKcyc}/4 + 1.5$	ns	Figure 60.148
	1.62 V or above		—	—	$t_{TCLKcyc}/4 + 1.5$	ns	
TDATA[3:0] output hold time	2.7 V or above	t_{TRDH}	1.5	—	—	ns	
	1.62 V or above		1.5	—	—	ns	

**Figure 60.147 ETM TCLK timing****Figure 60.148 ETM output timing**